Basics for Group Theory

Kaizhi Bai

November 3, 2025

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Preface

千里之行，始于足下。

It is from Albert Einstein that people begin to know symmetry determines dynamics. After Landau, people realize that symmetry and its spontaneous breaking also determine the phase of matter, the thermodynamics. Nowadays, symmetry helps us to go beyond the Landau paradigm, especially in the realm of the topological phase of matter and extraordinary quantum phase of matter.

The group theory is (part of) the (developing) language of symmetry, and it is the key to understand the modern physics. People who knows how language determines the boundary of thinking, will agree over how group theory determines the boundary of 'physics' (although physics is not actually bounded by a tool). And if someone really wants to gain some feelings, the group theory can be a best way to go.

But I shall still say that, such a theory is only a tool to understand the world, and it is not the world itself. After we figure out the way from symmetry to dynamics or thermodynamics, we shall still need to go back to the world, to the physics, which is closer to the world, itself. Do not worry too much about the abstract, and do not forget the reality, vice versa. Also do not worry too much if you realize to learn this language too late, because the world is always there, waiting for you to explore. Learn it when you need it, and learn it when you want it.

Chapter 1

Group Representation

The modern physics language is quantum mechanics based on the Hilbert space $\mathcal{H}$. And a symmetry G is a group action on the Hilbert space. Basically, we can not directly act the group element on the Hilbert space, and we need an intermediary to conduct the action. This intermediary is the representation of the group. From the postulate of quantum mechanics and Wigner theorem, we basically need a unitary operator to represent the group element

$$\forall g \in G \mapsto U(g) : \mathcal{H} \to \mathcal{H}, \tag{1.1}$$

and at many times (but not always), the symmetry acts in the same way as a physical symmetry, meaning that

$$U(g_1)U(g_2) = U(g_1 g_2).$$

For a special physical system, the Schördinger equation and Planck's relation tells that we relate the spectrum and eigenstates of the Hamiltonian to the physical energy and states, which in turn determines the measurable quantities of the system. The symmetry of the system is then reflected in the Hamiltonian, as the invariance of the Hamiltonian under the group action

$$U(g)HU(g)^{-1} = H \Leftrightarrow [H, U(g)] = 0. \tag{1.2}$$

The unlabeled equation above is essentially the definition of the representation of the group G.

1.1 Representations

Assign carrier space V, a vector space over a field κ, and denote $GL(V) : V \to V$ as the general linear group of V with all non-singular linear transformations (all invertible matrices). A representation of G on V is a group homomorphism

$$T : G \to GL(V), \tag{1.3}$$

that brings $g \mapsto T(g) : V \to V$. The dimension of the representation is thus the dimension of the carrier space V. We call the rep (T, V) or T or V for short.

Definition 1.1.1. *A linear transformation $A : V \to V'$ between representations of G is called an intertwiner if $\forall G$, the diagram*

$$\begin{array}{ccc} V & \xrightarrow{T(g)} & V \\ \downarrow{\scriptstyle A} & & \downarrow{\scriptstyle A} \\ V' & \xrightarrow{T'(g)} & V' \end{array}$$

commutes. Equivalently, $T'(g)A = AT(g)$ for all $g \in G$.

Two representations will thus be called equivalent if there exists an intertwiner as an **isomorphism** between them, i.e., A is invertible, or more explicitly,

$$T'(g) = AT(g)A^{-1}, \ \forall g \in G. \tag{1.4}$$

The algebraic laws between representations can be written as

- Direct sum: $\oplus : (T_1, V_1) \times (T_2, V_2) \to (T_1 \oplus T_2, V_1 \oplus V_2)$ such that

$$(T_1 \oplus T_2)(g) \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} T_1(g)v_1 \\ T_2(g)v_2 \end{pmatrix}.$$

- Tensor product: $\otimes : (T_1, V_1) \times (T_2, V_2) \to (T_1 \otimes T_2, V_1 \otimes V_2)$ such that

$$(T_1 \otimes T_2)(g)(v_1 \otimes v_2) = (T_1(g)v_1) \otimes (T_2(g)v_2).$$

- Faithful representation: T is faithful if it is injective, i.e., $\ker(T) = \{e\}$.

- Dual representation: given rep V we construct a dual rep on dual space $V^\vee$ by

$$\langle T^\vee(g)l, T(g)v\rangle = \langle l, v\rangle .$$

  Notice that generally we have $T^\vee(g) = T(g^{-1})^{\text{adj}}$, where adj denotes the adjoint operation, and we need to read the equation as

$$\langle T^\vee(g)l, T(g)v\rangle = \langle T(g^{-1})^{\text{adj}}l, T(g)v\rangle = \langle l, T(g^{-1})T(g)v\rangle = \langle l, v\rangle .$$

- Conjugate representation: a special case of dual representation, where V is a complex vector space, and the dual representation is $\bar{T}(g) \in GL(\bar{V})$.

- A real representation is one where $(\bar{T}, \bar{V})$ is equivalent to (T, V). The word equivalent refers to an isomorphism.

If V, W are both reps, then $\hom(V, W)$ is also a rep with possible representation

$$(T_A(g) \cdot A)(v) = \begin{cases} T_W(g) \cdot (Av) \\ A(T_V(g)^{-1}v) \\ T_W(g) \cdot (A(T_V(g)^{-1}v)) \end{cases} .$$

We need to be aware of the second type with a contravariant action, verified as

$$\begin{aligned} (T_A(g_1g_2) \cdot A)(v) &= A(T_V(g_1g_2)^{-1}v) = A(T_V(g_2)^{-1}T_V(g_1)^{-1}v) \\ &= T_A(g_2) \cdot A(T_V(g_1)^{-1}v) = (T_A(g_1) \cdot T_A(g_2) \cdot A)(v). \end{aligned}$$

1.1.1 Matrix Representation

A matrix representation appears when we assign an ordered basis (column vectors) $\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n\}$ for V, with identification $GL(V) = GL(n, \kappa)$. As we are physicists, we shall use the Dirac notation to denote the basis $\{|i\rangle\}_{i=1}^n$. As a linear transformation, $T(g)$ is represented as a $(1,1)$ type tensor, and we need to specify the basis for the dual space $V^\vee$ as row vectors $\{\boldsymbol{v}_1^\vee, \cdots, \boldsymbol{v}_n^\vee\}$, or $\{\langle i|\}_{i=1}^n$. Notice that $T(g)$ is not a matrix, but a linear transformation or an operator, and its matrix implementation is conducted through the basis

$$T(g)\boldsymbol{v}_i = \sum_{j=1}^n T^g_{ji}\boldsymbol{v}_j,\ T^g_{ji} = \langle \boldsymbol{v}_j^\vee, T(g)\boldsymbol{v}_i\rangle\,. \tag{1.5}$$

as an analogy to

$$\hat{O}\,|n\rangle = \sum_m |m\rangle\langle m|\,\hat{O}\,|n\rangle = \sum_m O_{mn}\,|m\rangle\,,\ O_{mn} = \langle m|\hat{O}|n\rangle\,.$$

When the basis itself is subjected to a linear transformation

$$\boldsymbol{v}_i = \sum_{j=1}^n S_{ji}\boldsymbol{v}'_j,\ \boldsymbol{v}_i^\vee = \sum_{j=1}^n (S^{-1})_{ij}\boldsymbol{v}'^\vee_j, \tag{1.6}$$

in corresponding to the form

$$(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n) = (\boldsymbol{v}'_1, \cdots, \boldsymbol{v}'_n)S,\ \begin{pmatrix}\boldsymbol{v}_1^\vee\\ \vdots\\ \boldsymbol{v}_n^\vee\end{pmatrix} = S^{-1}\begin{pmatrix}\boldsymbol{v}'^\vee_1\\ \vdots\\ \boldsymbol{v}'^\vee_n\end{pmatrix},$$

where the inverse action of S^{-1} on the dual basis preserves the inner product, we obtain the similarity transformation of the matrix representation

$$T'^g = ST^gS^{-1}, \tag{1.7}$$

similarly from

$$\begin{aligned}\hat{O} &= \sum_{ij} O_{ij}\,|i\rangle\langle j| = \sum_{mn}\sum_{ij} O_{ij}S_{mi}(S^{-1})_{jn}\,|m'\rangle\langle n'|\\ &= \sum_{mn} O'_{mn}\,|m'\rangle\langle n'|\,,\end{aligned}$$

to $O' = SOS^{-1}$.

- Due to the identification, the existence of the similarity transformation implies the equivalence of the representations.
- For rep T^g, the inverse and transpose $(T^g)^{-1}$, $(T^g)^T$ are also representations.
- The above defined real representation means that $\exists S \in GL(n, \mathbb{C})$ s.t. $(T^g)^* = ST^gS^{-1}$ for all $g \in G$.

We have many examples of matrix representations:

- Trivial representation: $T(g) = I$ for all $g \in G$.

- Fundamental representation of matrix groups: $GL(n,\kappa)$, $SL(n,\kappa)$, $O(n,\kappa)$, $U(n)$, $SU(n)$, $SO(n)$, $Sp(n)$, etc. They are matrix representations of themselves.
- Determinant representation of $GL(n,\kappa)$ is a one-dimensional representation with $T(A) = |\det(A)|^\mu$, $\mu \in \mathbb{R}$.
- A n-dimensional representation of symmetric group S_n: $T(\sigma) : e_i \mapsto e_{\sigma(i)}$ with $\{e_i\}$ the standard basis. The coordinate is transformed accordingly as

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mapsto \begin{pmatrix} x_{\sigma^{-1}(1)} \\ \vdots \\ x_{\sigma^{-1}(n)} \end{pmatrix},$$

  and the matrix representation is a permutation matrix with 0s and a single 1 in each row and column.
- Projection: for $\theta \in \mathbb{R}/\mathbb{Z}$ we can define representation of group $\mathbb{Z}$ as $T_\theta(n) = e^{2\pi i n\theta}$, and one sees that for $\theta = p/q$ rational, the representation has finite order q such the group is degenerated to $\mathbb{Z}_q$, while for irrational θ the representation is faithful. Another example is from the braid group to the permutation group.
- The finite-order Heisenberg group is a group generated by q, U, V with q central and $UV = qUV$

$$H = \langle q, U, V | qU = Uq, qV = Vq, UV = qVU, q^N = U^N = V^N = 1 \rangle,$$

  which can be described by the group extension of $\mathbb{Z}_N \times \mathbb{Z}_N$ by $\mathbb{Z}_N$ (see group extension). Its matrix representation can be given, by defining ω as the N-th root of unity, as

$$U = \mathrm{diag}(1, \omega, \cdots, \omega^{N-1}),\ V = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}_{N\times N},\ q = \omega,$$

  with U the 'clock operator' and V the 'shift operator'. Note that changing ω to ω^k will give N inequivalent representations. Also note that if we release the condition Nth power to be identity, we indeed obtain the position-momentum Heisenberg algebra, with U, V generated by x, p.

1.1.2 Unitary Representation

As we have claimed at the beginning, the representation of a group in quantum mechanics is usually unitary.

Definition 1.1.2. *Let V be an inner product space, a representation $T : G \to GL(V)$ is unitary if*

$$\langle T(g)v, T(g)w \rangle = \langle v, w \rangle,\ \forall v, w \in V. \tag{1.8}$$

To read this equation we write

$$\langle T(g)v, T(g)w \rangle = \langle v, T(g)^{\mathrm{adj}} T(g)w \rangle \Rightarrow T(g)^{\mathrm{adj}} = T(g^{-1}).$$

The unitary representation is equivalent to the existence of a unitary matrix representation, i.e., $U(g)U(g)^\dagger = U(g)^\dagger U(g) = I$, where the adjoint operation is the conjugate transpose. A representation is unitarizable if it is equivalent to a unitary representation.

Invariant Integration

For a unitary representation, we can define the invariant integration on the group G. The integration of a group is related to averaging over a finite group, where one defines a group function f with group as its domain and write

$$f \to \frac{1}{|G|}\sum_{g\in G} f(g).$$

When the group is continuous/topological, integration of a function $f : G \to \mathbb{C}$ is defined as

$$\int f(g)\mathrm{d}g,$$

with $\mathrm{d}g$ the left-invariant Haar measure on the group

$$\int f(hg)\mathrm{d}g = \int f(g)\mathrm{d}g,\ \forall h \in G.$$

Notice that we can also define the right-invariant Haar measure, while for a compact group the two measures coincide. The measure change $g \to hg$ can be made similarly with the usual integration $x \to x + a$, but generally non-Abelian. We can give examples:

- $\int_{G=U(1)} f(g)\mathrm{d}g = \int_0^{2\pi} \frac{\mathrm{d}\theta}{2\pi} f(\theta)$
- $\int_{G=SU(2)} f(g)\mathrm{d}g = \frac{1}{16\pi^2}\int_0^{2\pi}\mathrm{d}\theta\int_0^{\pi}\mathrm{d}\phi\int_0^{4\pi}\mathrm{d}\psi f(\theta,\phi,\psi)$
- $\int_{G=\mathbb{Z}} f(g)\mathrm{d}g = \sum_n f(n)$
- $\int_{G=\mathbb{R}} f(g)\mathrm{d}g = \int_{-\infty}^{\infty}\mathrm{d}x f(x)$

Unitarizable Representation

We claim that a rep of a compact group G over a finite-dimensional vector space is unitarizable. Given any inner product $\langle .,. \rangle_1$, we can always construct a new inner product

$$\langle v, w\rangle_2 = \frac{1}{|G|}\sum_{g\in G}\langle T(g)v, T(g)w\rangle_1,$$
$$\langle v, w\rangle_2 = \int_G \langle T(g)v, T(g)w\rangle_1 \mathrm{d}g,$$

which always makes the representation unitary. In the second case, we require the measure to be both-sides invariant protected by the compactness of the group. We may take the first case as an illustration

$$\begin{aligned}\langle T(g)v, T(g)w\rangle_2 &= \frac{1}{|G|}\sum_{h\in G}\langle T(h)T(g)v, T(h)T(g)w\rangle_1 \\ &= \frac{1}{|G|}\sum_{h\in G}\langle T(hg)v, T(hg)w\rangle_1 \\ &= \frac{1}{|G|}\sum_{l=hg\in G}\langle T(l)v, T(l)w\rangle_1 = \langle v, w\rangle_2.\end{aligned}$$

Symmetry on Schrödinger Equation

The symmetry of the Hamiltonian under the group action is reflected in the Schrödinger equation, say with a central potential

$$\left[-\frac{\hbar^2}{2m}\Delta + V(r)\right]\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}),$$

the system is clearly invariant under $O(3)$. Let us define the solution space

$$V_E = \{\psi : \psi(\boldsymbol{r}) \text{ solves the equation and is normalizable}\}.$$

Here, normalizable means square-integrable, i.e., $\int |\psi(\boldsymbol{r})|^2 \mathrm{d}^3\boldsymbol{r} < \infty$. And we have the representation $T : O(3) \to GL(V_E)$ as

$$[T(g)\psi](\boldsymbol{r}) = \psi(g^{-1}\boldsymbol{r}),$$

where g is fundamental representation of $O(3)$ acting on $\boldsymbol{r}$. We mainly need to verify that the Laplacian is invariant under $SO(3)$, where $\boldsymbol{r} = g\boldsymbol{r}'$ and

$$\begin{aligned}\Delta' &= \sum_{i=1}^{3} \frac{\partial}{\partial x_i'}\frac{\partial}{\partial x_i'} = \sum_{ijk} \frac{\partial}{\partial x_j}\frac{\partial x_j}{\partial x_i'}\frac{\partial}{\partial x_k}\frac{\partial x_k}{\partial x_i'} \\ &= \sum_{ijk} \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_k} g_{ji}g_{ki} = \sum_{jk} \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_k}\delta_{jk} = \Delta.\end{aligned}$$

Generally, an equation is equipped with some symmetry, like the Helmhotz equation $[\Delta + k^2]\psi(\boldsymbol{r}) = 0$, where its solution space is a representation of $E(3)$, the Euclidean group in 3D.

1.1.3 Induced Group Action

Consider X and Y be arbitrary sets with mutual function space $\mathrm{Map}(X, Y)$. Now given $\psi \in \mathrm{Map}(X, Y)$ and $g \in G$ we claim that the action of G over $\mathrm{Map}(X, Y)$ can be induced as $\phi(g, \Psi)$ given that X or Y (or both) is a G-set.

For X a G-set, we have the action $g \cdot x$ or $x \cdot g$, i.e., a left or right action for $x \in X$. There are then four ways to define the induced action:

- For G left-action on X, $\phi(g, \psi)(x) = \psi(g^{-1} \cdot x)$ is a left action on $\mathrm{Map}(X, Y)$.
- For G left-action on X, $\phi(g, \psi)(x) = \psi(g \cdot x)$ is a right action on $\mathrm{Map}(X, Y)$.
- For G right-action on X, $\phi(g, \psi)(x) = \psi(x \cdot g)$ is a left action on $\mathrm{Map}(X, Y)$.
- For G right-action on X, $\phi(g, \psi)(x) = \psi(x \cdot g^{-1})$ is a right action on $\mathrm{Map}(X, Y)$.

To illustrate them, we have correspondingly

$$\begin{aligned}\text{LL: } &\phi(g_1g_2, \Psi)(x) = (g_1 \cdot g_2 \cdot \Psi)(x) = (g_2 \cdot \Psi)(g_1^{-1} \cdot x) = \Psi((g_1g_2)^{-1} \cdot x), \\ \text{LR: } &\phi(g_1g_2, \Psi)(x) = (\Psi \cdot g_1 \cdot g_2)(x) = (\Psi \cdot g_1)(g_2 \cdot x) = \Psi(g_1g_2 \cdot x), \\ \text{RL: } &\phi(g_1g_2, \Psi)(x) = (g_1 \cdot g_2 \cdot \Psi)(x) = (g_2 \cdot \Psi)(x \cdot g_1) = \Psi(x \cdot g_1g_2), \\ \text{RR: } &\phi(g_1g_2, \Psi)(x) = (\Psi \cdot g_1 \cdot g_2)(x) = (\Psi \cdot g_1)(x \cdot g_2^{-1}) = \Psi(x \cdot (g_1g_2)^{-1}).\end{aligned}$$

Example: We consider the scalar field that maps to $\kappa = \mathbb{R}, \mathbb{C}$ on Minkowski spacetime $\mathcal{S} = \mathbb{M}^{1,d-1}$, then the induced left action of the Poincaré group with left action $O(1,d-1)$: $g^{-1} \cdot x = \Lambda x + v$ on the scalar field $\Psi \in \mathrm{Map}(\mathcal{S}, \kappa)$ is $(g \cdot \Psi)(x) = \Psi(\Lambda x + v)$.

Similarly, for Y a G-set, we have the action $g \cdot y$ or $y \cdot g$, and there are also four ways to define the induced action:

- For G left-action on Y, $\phi(g, \psi)(x) = g \cdot \psi(y)$ is a left action on $\mathrm{Map}(X, Y)$.
- For G left-action on Y, $\phi(g, \psi)(x) = g^{-1} \cdot \psi(y)$ is a right action on $\mathrm{Map}(X, Y)$.
- For G right-action on Y, $\phi(g, \psi)(x) = \psi(y) \cdot g^{-1}$ is a left action on $\mathrm{Map}(X, Y)$.
- For G right-action on Y, $\phi(g, \psi)(x) = \psi(y) \cdot g$ is a right action on $\mathrm{Map}(X, Y)$.

To illustrate them, we have correspondingly

$$\begin{aligned}
\text{LL: } & \phi(g_1 g_2, \Psi)(x) = (g_1 \cdot g_2 \cdot \Psi)(x) = g_1 \cdot [(g_2 \cdot \Psi)(x)] = g_1 \cdot g_2 \cdot \Psi(x), \\
\text{LR: } & \phi(g_1 g_2, \Psi)(x) = (\Psi \cdot g_1 \cdot g_2)(x) = g_2^{-1} \cdot [(g_1 \cdot \Psi)(x)] = (g_1 g_2)^{-1} \cdot \Psi(x), \\
\text{RL: } & \phi(g_1 g_2, \Psi)(x) = (g_1 \cdot g_2 \cdot \Psi)(x) = [(g_2 \cdot \Psi)(x)] \cdot g_1^{-1} = \Psi(x) \cdot (g_1 g_2)^{-1}, \\
\text{RR: } & \phi(g_1 g_2, \Psi)(x) = (\Psi \cdot g_1 \cdot g_2)(x) = \Psi(g_1 \cdot x) \cdot g_2 = \Psi(x) \cdot g_1 \cdot g_2.
\end{aligned}$$

These actions can be combined as a $G_1 \times G_2$ action over $\mathrm{Map}(X, Y)$ with X a G_1-set, Y a G_2-set and all left actions,

$$\phi((g_1, g_2), \Psi)(x) = g_2 \cdot \Psi(g_1^{-1} \cdot x).$$

Notice that like the group action over the homomorphism space between representations above, here we can not write $g_2 \cdot (\Psi(g_1 \cdot x))$ as it does not generally admit a group action. In most cases, we identify $G_1 = G_2$ and specialize to the diagonal action

$$\phi(g, \Psi)(x) = g \cdot \Psi(g^{-1} \cdot x).$$

Example: Following the above example, now we want to give a non-trivial action over the field configuration itself. Still let $\mathcal{S} = \mathbb{M}^{1,d-1}$, and G is the Poincaré group. We consider $Y = V$ a finite dimensional carrier space of G with representation $\rho(g)$, then by $\Psi : \mathcal{S} \to V$, we have

$$g \cdot \Psi(x) = \rho(g) \Psi(g^{-1} \cdot x),$$

which gives the usual general field transformation under the a group. More concretely, we consider spin-1/2 field $\Psi(\boldsymbol{x}) = \begin{pmatrix} \psi_+(\boldsymbol{x}) \\ \psi_-(\boldsymbol{x}) \end{pmatrix}$ under the spatial rotation. Since we can not construct a globally well-defined homomorphism from $SO(3)$ to $SU(2)$, we may take $G = SU(2)$, with its action on space is defined by its projection homomorphism $\pi(u) = R$ where $u \in SU(2)$ and $R \in SO(3)$ as $u\boldsymbol{x} \cdot \boldsymbol{\sigma} u^{-1} = (R\boldsymbol{x}) \cdot \boldsymbol{\sigma}$. Then the induced action is

$$u \cdot \Psi(\boldsymbol{x}) = u \begin{pmatrix} \psi_+(R^{-1}\boldsymbol{x}) \\ \psi_-(R^{-1}\boldsymbol{x}) \end{pmatrix}.$$

Notice that u is already in its fundamental representation.

1.1.4 Regular Representation

The induced group action offers us a powerful tool to learn a special representation of the group, called the regular representation, from which we can learn all the representations.

The important thing is that the group G is itself a G-set, and by letting $X = G$ and $Y = \mathbb{C}$, we have the carrier space $\mathcal{F}_{G,\mathbb{C}} = \text{Map} : G \to \mathbb{C}$, also denoted as R_G. The regular representation is then defined by a left or right action:

- LRR: $\phi \mapsto L(g) \cdot \phi$ defined by $(L(g) \cdot \phi)(h) = \phi(g^{-1}h)$.
- RRR: $\phi \mapsto R(g) \cdot \phi$ defined by $(R(g) \cdot \phi)(h) = \phi(hg)$.

There are several properties:

- $L(g_1)L(g_2) = L(g_1g_2)$ and $R(g_1)R(g_2) = R(g_1g_2)$, and furthermore, we have $R(g)L(h) = L(h)R(g)$ as they act by different sides, which indicates that R_G is a rep of $G \times G = G_{\text{left}} \times G_{\text{right}}$.
- R_G can be somehow treated as the dual space of the group, and is faithful.
- For $|G| < \infty$, R_G is a finite-dimensional vector space, with vector as complex-valued function over group, which can be denoted as $f(g)$. For instance, for $G = \mathbb{Z}_2$, we can let $R_{\mathbb{Z}_2} \cong \mathbb{C}^2$, with any function labelled by vector $(f(1), f(\sigma))$. Quite generally, we may introduce the delta-basis function δ_g as

  $$\delta_g(g') = \begin{cases} 1 & g = g' \\ 0 & g \neq g' \end{cases},$$

  and δ_g spans R_G as $f = \sum_g f(g)\delta_g$, verified by $f(g) = \sum_{g'} f(g')\delta_{g'}(g) = f(g)$. We can utilize the delta-basis to define the matrix equivalent form of the regular representation, as

  $$(L(h))_{gg'} = (L(h) \cdot \delta_g)(g') = \delta_g(h^{-1}g') = \delta_{hg}(g'),$$
  $$(R(h))_{gg'} = (R(h) \cdot \delta_g)(g') = \delta_g(g'h) = \delta_{gh^{-1}}(g'),$$

  which are permutation matrices. We can also extract from the above formula that

  $$L(h) \cdot \delta_g = \delta_{hg},$$
  $$R(h) \cdot \delta_g = \delta_{gh^{-1}}.$$

- We may now recognize the matrix representation element $g \to T_{ij}(g) \in R_G$.
- As claimed, the regular rep is like a dual space, and we can define the inner product as (and since it is defined over complex space, the adjoint of the function is just complex conjugate)

  $$\langle \phi_1, \phi_2 \rangle = \frac{1}{|G|} \sum_g \phi_1^*(g)\phi_2(g). \tag{1.9}$$

Now let R_G to be rep of $G \times G$, meaning that

$$((g_1, g_2) \cdot \Psi)(h) = \Psi(g_1^{-1} h g_2),\ \Psi \in R_G. \tag{1.10}$$

Consider another carrier space V and homomorphism representation $\rho : G \to GL(V)$. We can consider the vector space of linear transformation $\mathrm{End}(V)$ of V to itself, which can also be a rep of $G \times G$ by the induced left action

$$(g_1, g_2) \cdot T = \rho(g_1) \circ T \circ \rho(g_2^{-1}),\ T \in \mathrm{End}(V), \tag{1.11}$$

as

$$\begin{aligned}(g_1, g_2) \cdot (g_1', g_2') \cdot T &= \rho(g_1) \circ ((g_1', g_2') \cdot T) \circ \rho(g_2^{-1}) \\ &= \rho(g_1) \circ \rho(g_1') \circ T \circ \rho(g_2'^{-1}) \circ \rho(g_2^{-1}) \\ &= \rho(g_1 g_1') \circ T \circ \rho(g_2'^{-1} g_2) = (g_1 g_1', g_2' g_2) \cdot T.\end{aligned}$$

We now claim that there exists an inclusion map $\iota : \mathrm{End}(V) \to \mathrm{Map}(G, \mathbb{C})$ by $\iota : T \mapsto \Psi_T$ defined by

$$\Psi_T(g) = \mathrm{Tr}\big(T\rho(g^{-1})\big). \tag{1.12}$$

Indeed, by explicitly choosing the matrix rep of $\rho(g)$ as $D(g)$ by

$$\rho(g) \cdot \omega_\nu = \sum_\mu D(g)_{\mu\nu} \omega_\mu,$$

with $\{\omega_\nu\}$ the basis, we see that for $T = e_{\nu\mu} = \omega_\nu \omega_\mu^*$ (anything unclear, try it in bracket) we have

$$\begin{aligned}\Psi_{e_{\nu\mu}}(g) &= \mathrm{Tr}\big(e_{\nu\mu}\rho(g^{-1})\big) = \mathrm{Tr}\left(e_{\nu\mu} \sum_{mn} D(g^{-1})_{mn} \omega_m \omega_n^*\right) \\ &= \sum_{mn} D(g^{-1})_{mn} \omega_n^* e_{\nu\mu} \omega_m = D(g^{-1})_{\mu\nu}.\end{aligned}$$

Thus, Ψ_T are linear combinations of representation matrices of G. There exists furthermore a commutative diagram

$$\begin{array}{ccc} \mathrm{End}(V) & \xrightarrow{\iota} & \mathrm{Map}(G, \mathbb{C}) \\ \downarrow{\scriptstyle \rho_{\mathrm{End}(V)} \times \rho_{\mathrm{End}(V)}} & & \downarrow{\scriptstyle \rho_{R_G} \times \rho_{R_G}} \\ GL(V) & \xrightarrow{\iota} & \mathrm{Map}(G, \mathbb{C}) \end{array},$$

meaning that

$$(g_1, g_2) \cdot \Psi_T = \Psi_{(g_1, g_2) \cdot T}, \tag{1.13}$$

verified as

$$\begin{aligned}((g_1, g_2) \cdot \Psi_T)(h) &= \Psi_T(g_1^{-1} h g_2) = \mathrm{Tr}\big(T\rho(g_2^{-1})\rho(h^{-1})\rho(g_1)\big) \\ &= \mathrm{Tr}\big(\rho(g_1) T \rho(g_2^{-1})\rho(h^{-1})\big) = \Psi_{(g_1, g_2) \cdot T}(h).\end{aligned}$$

1.1.5 Induced Representation

In an abstract manner, induced representation is about lifting a representation from a subgroup to the whole group, from a section to the whole bundle. This section is rather abstract, and people confused should refer to a little extension and category theories.

Suppose $H \subset G$ is a subgroup, equipped with exact sequence

$$1 \to H \to G \xrightarrow{\pi} G/H \to 1,$$

which can be read as a principle H bundle over base space G/H, forming a fiber bundle G. Also suppose a rep of H over V as $\rho : H \to GL(V)$, and an induced rep wants an exact sequence

$$1 \to V \to G \times_H V \xrightarrow{\pi} G/H \to 1,$$

such that $\mathrm{Ind}_H^G(V) = \Gamma(G \times_H V)$ is defined. Basically, we can identify $G \times_H V$ as $(G \times V)/H$, meaning that the equivalent class is given by

$$[g, v] = \{(gh, \rho(h^{-1})v) | g \in G, v \in V, h \in H\}.$$

The projection is defined by $\pi(g, v) = gH$, and the equivalent class is preserved by $\pi(gh, \rho(h^{-1})v) = ghH = gH$. In this sense, we may also write

$$1 \to H \to G \times V \xrightarrow{\pi} G \times_H V \to 1.$$

Formally, we define the induced representation as follows:

Definition 1.1.3. *For (V, ρ) rep of H, we have carrier space $\mathrm{Ind}_H^G(V)$ as the set of all smooth functions $\psi : G \to V$ such that*

$$\psi(gh) = \rho(h^{-1})\psi(g), \ \forall g \in G, h \in H. \tag{1.14}$$

Denote rep of G as T, we then have induced action over $\mathrm{Ind}_H^G(V)$ as

$$(T(g) \cdot \psi)(g') = \psi(g^{-1}g'),$$

which is a left one since

$$\begin{aligned}(T(g_1)T(g_2) \cdot \psi)(g') &= (T(g_1) \cdot T(g_2) \cdot \psi)(g') = T(g_2) \cdot \psi(g_1^{-1}g') \\ &= \psi(g_1^{-1}g_2^{-1}g') = \psi((g_1g_2)^{-1}g') = (T(g_1g_2) \cdot \psi)(g').\end{aligned}$$

Note that the group action is compatible with the equivariance condition, i.e.,

$$(T(g) \cdot \psi)(g'h) = \psi(g^{-1}g'h) = \rho(h^{-1})\psi(g^{-1}g') = \rho(h^{-1})((T(g) \cdot \psi)(g')).$$

In this way, we construct an equivariant vector bundle with left group action.

Now it is clear that $\mathrm{Ind}_H^G(V) \subset R_G$, the regular representation, and we expect to decompose the induced rep into irreducible ones, but not that big as R_G. To do this, we introduce category $\mathrm{Rep}(G)$ as the set of all finite-dimensional representations of G, with morphisms between objects $W_1, W_2 \in \mathrm{Rep}(G)$ as G-intertwiners, i.e., linear maps $T : W_1 \to W_2$ such that $TW_1(g)T^{-1} = W_2(g)$ for all $g \in G$. Let us denote the vector space of morphisms as $\mathrm{Hom}_G(W_1, W_2)$. The induced rep is then a functor

$$\mathrm{Ind} \equiv \mathrm{Ind}_H^G : \mathrm{Rep}(H) \to \mathrm{Rep}(G), \tag{1.15}$$

that we induce any rep of H to a rep of G. On the other hand, given any rep of G, we can naturally restrict it to a rep of H by the restriction functor Res_H^G.

$$R \equiv \mathrm{Res}_H^G : \mathrm{Rep}(G) \to \mathrm{Rep}(H). \tag{1.16}$$

Now since that we have such bidirectional functors, the statement of Frobenius reciprocity is that the two functors are adjoint to each other, i.e.,

$$\mathrm{Hom}_G(W, \mathrm{Ind}(V)) = \mathrm{Hom}_H(\mathrm{R}(W), V). \tag{1.17}$$

Now consider W_α as irreps of G, then by Schur's lemma, we decompose

$$\mathrm{Ind}_H^G(V) \cong \oplus_\alpha W_\alpha \otimes \mathrm{Hom}_G(W_\alpha, \mathrm{Ind}_H^G(V)) = \oplus_\alpha W_\alpha \otimes \mathrm{Hom}_H(R(W_\alpha), V), \tag{1.18}$$

where for the first equality one recognize we utilize (dimension of) $\mathrm{Hom}_G(W_\alpha, \mathrm{Ind}_H^G(V))$ to represent the copy number of W_α in $\mathrm{Ind}_H^G(V)$. This stands since the morphism vector space is composed of intertwiners, which is non-zero only when W_α is a sub-rep of $\mathrm{Ind}_H^G(V)$. To prove the reciprocity or the second equality, we note that it points to an equal counting of copies of irreps, and from Eq. (1.18), this is translated into the counting of characters

$$\chi(g) = \sum_{x \in G/H} \hat{\chi}(x^{-1} g x), \tag{1.19}$$

where χ, $\hat{\chi}$ are characters of G over $\mathrm{Ind}_H^G(V)$ and H over V respectively, and $\hat{\chi}$ is zero outside H ($\chi(g) = \mathrm{Tr}(T(g))$, $\hat{\chi}(h) = \mathrm{Tr}(\rho(h))$). We come to this since character of a rep equals sum of its irreps with copies counted, thus the $\oplus_\alpha$ can be absorbed somehow as $\mathrm{Hom}_H(\oplus_\alpha W_\alpha \otimes R(W_\alpha), V)$, and we then use conjugate relation as intertwiners to find equivalent copies in V.

To see how this works, let us use an intuitive geometric thought of the fiber bundle $G \times_H V$ over G/H as a collection of fibers V over each point $gH \in G/H$, and the fiber is a rep of H. Then for $g_j \in G$ as a representative of $g_i H$, we may write

$$g \cdot g_j = g_{g \cdot j} h(g, j),$$

i.e., g translates g_j to $g_{g \cdot j}$ with an additional H-action $h(g, j)$, counting the shift in H. Since we want to see the trace, we need a basis $\{v_a\}$ for V such that vectors in Ind_H^G as equivariant functions are given by

$$\psi_{i,a}(g_j) = v_a \delta_{i,j},$$

with equivariance $\psi_{i,a}(g_j h) = \rho(h^{-1}) \psi_{i,a}(g_j)$. Then we calculate how the rep acts on the function space, as

$$\begin{aligned}(T(g) \cdot \psi_{i,a})(g_j) &= \psi_{i,a}(g^{-1} g_j) = \psi_{i,a}(g_{g^{-1} \cdot j} h(g^{-1}, j)) \\ &= \delta_{i, g^{-1} \cdot j} \rho(h(g^{-1}, j)^{-1}) v_a,\end{aligned}$$

and by taking the trace, only 'eigen' states, or 'fixed points' contribute, thus $i = j$, or simply diagonal elements. The RHS, on the other hand, requires that $i = g^{-1} j$. Requiring both to stand means that we have

$$g g_i = g_i h(g, i),$$

i.e., translation of g over g_i making it not escaping $g_i H$, and hence $g^{-1} g_i = g_i h(g, i)^{-1}$, thus we obtain

$$h(g^{-1}, i) = h(g, i)^{-1},$$

and this solves $h(g, i) = g_i^{-1} g g_i$, corresponding the RHS of Eq. (1.19).

A geometric way to view induced rep is to consider all $\mathrm{Map}(G, V)$ functions Ψ, left-acted by $G \times H$ as

$$((g, h) \cdot \Psi)(g_0) = \rho(h) \cdot \Psi(g^{-1} g_0 h),$$

and we consider the invariant subspace fixed by action of $1 \times H$, i.e.,

$$\rho(h) \cdot \Psi(g_0 h) = \Psi(g_0) \Rightarrow \Psi(gh) = \rho(h^{-1})\Psi(g).$$

In this way, we choose the functions that change in the same way under two kinds of left-H actions ('equal' variant), which we denote as $\mathrm{Ind}_H^G(V)$. When the rep (ρ, V) is trivial as $\rho(h) \equiv 1$ for all $h \in H$, we see that the equivariant condition states that $\Psi(gh) = \Psi(g)$, i.e., the function is invariant under H-action, and we have $\mathrm{Ind}_H^G(V) \cong \mathrm{Map}(G/H, V)$. On the other hand, for non-trivial rep, we view $\mathrm{Ind}_H^G(V)$ as vector bundle over G/H with data determined by (ρ, V).

1.1.6 Reducible and Irreducible Representation

Reducing a bigger rep to smaller reps is closely analogous to diagonalizing a matrix.

Definition 1.1.4. *Let $W \subset V$ be a sub-carrier space, then W is an invariant subspace of V under T if $T(g)W \subset W$ for all $g \in G$.*

- The rep (T, V) is thus restricted to (T, W).
- If T is unitary on V then it is unitary on W.
- $\{0\}$ and $\{V\}$ are always invariant subspaces.

Definition 1.1.5. *A rep (T, V) is reducible if there exists a non-trivial invariant subspace $W \subset V$, i.e., $W \neq \{0\}, V$.*

- V is irreducible if it is not reducible, i.e., the only invariant subspaces are $\{0\}$ and V, for short 'irrep'.
- $\forall v \in V$, the linear span of $\{T(g)v\}$ is an invariant subspace. For an irrep, the linear span is V.
- If W is an sub-rep of V then T descends to rep on V/W.

For the last point, we consider that for T reducible, we choose basis $\{v_1, \cdots, v_k\}$ for W and $\{v_1, \cdots, v_k, v_{k+1}, \cdots, v_n\}$ for V. Then the matrix rep of T is $T(g) = \begin{pmatrix} T_1(g) & T_{12}(g) \\ & T_2(g) \end{pmatrix}$, where $T_1(g) \in \mathrm{Mat}_{k \times k}$. Then as it is easy to verify, both T_1 and T_2 form reps, on W and V/W respectively. Notice that the quotient space $V/W = \{[v] = v + W\}$, composed of equivalent classes of vectors by W, is different from the compensate space $W^\perp$, which is the orthogonal complement of W, and it is not a subspace of V, thus T_2 is not a sub-rep. Note W is invariant subspace, while $W^\perp$ is not, since

$$T(g)v_W = \begin{pmatrix} T_1(g) & T_{12}(g) \\ & T_2(g) \end{pmatrix} \begin{pmatrix} v_W \\ 0 \end{pmatrix} = \begin{pmatrix} T_1(g)v_W \\ 0 \end{pmatrix} \in W,$$

$$T(g)v_{W^\perp} = \begin{pmatrix} T_1(g) & T_{12}(g) \\ & T_2(g) \end{pmatrix} \begin{pmatrix} 0 \\ v_{W^\perp} \end{pmatrix} = \begin{pmatrix} T_{12}(g)v_{W^\perp} \\ T_2(g)v_{W^\perp} \end{pmatrix} \notin W^\perp,$$

as long as $T_{12} \neq 0$.

Definition 1.1.6. *A rep T is completely reducible if it is a direct sum of irreps, i.e., $V = \bigoplus_i^n W_i$ with W_i irreps. In matrix form,*

$$T(g) = \begin{pmatrix} T_1(g) & & \\ & \ddots & \\ & & T_n(g) \end{pmatrix}. \tag{1.20}$$

- $G = S_2 \cong \mathbb{Z}_2$: $1 \to \sigma_0$, $\sigma \to \sigma_x$ is completely reducible on $\mathbb{R}^2$, as Pauli matrix σ_x is diagonalizable. Viewing higher, the average vector $\{(x, x)\} \in \mathbb{R}^2$ is always a 1-dim invariant subspace.
- The idea can be generalized to S_n on $\mathbb{R}^n$. The 1-dim subspace $L = \{(x, \cdots, x)\}$ is always invariant (a 1D sub-rep), and $L^\perp = \{(x_1, \cdots, x_n) | \sum_i x_i = 0\}$ is a $(n-1)$-dim sub-rep.
- Invariant subspace of R_G: for any n-dim mat-rep T_{ij} of G, the linear span $\mathcal{R}_i = \text{Span}\{T_{ij}\}_{j=1,\cdots,n}$ is an invariant subspace of R_G under RRR.

For the last point, we consider

$$(R(g) \cdot T_{ij})(h) = T_{ij}(hg) = \sum_{s=1}^{n} T_{is}(h) T_{sj}(g),$$

i.e.,

$$R(g) \cdot T_{ij} = \sum_{s=1}^{n} T_{sj}(g) T_{is} \in \mathcal{R}.$$

Similarly, we have $\mathcal{L}_j = \text{Span}\{T_{ij}\}_{i=1,\cdots,n}$ as an invariant subspace of R_G under LRR,

$$\begin{aligned} (R(g) \cdot T_{ij})(h) &= T_{ij}(g^{-1}h) = \sum_{s=1}^{n} T_{is}(g^{-1}) T_{sj}(h) \\ \implies R(g) \cdot T_{ij} &= \sum_{s=1}^{n} T_{is}(g^{-1}) T_{sj} \in \mathcal{L}. \end{aligned} \tag{1.21}$$

Sometimes, like T_{12} appeared above, we can not diagonalize the matrix, and the rep is not completely reducible, or it is indecomposable. The natural question is when the rep is completely reducible. Let us use unitary rep as an example.

Claim 1. *For (T, V) unitary rep over inner product space V, if $W \subset V$ is an invariant subspace, then $W^\perp$ is also invariant.*

Proof. For $y \in W^\perp$, we have that $\langle y, x \rangle = 0$ for all $x \in W$, and thus

$$\langle T(g)y, x \rangle = \langle y, T(g)^\dagger x \rangle = 0,$$

since $T(g)^\dagger x \in W$. Thus $T(g)y \in W^\perp$. □

There are some corollaries and parallels:

- Finite-dim unitary rep is completely reducible.

- Finite-dim reps of finite groups are completely reducible.
- Finite-dim reps of compact Lie groups are completely reducible.
- For a finite group, the regular rep is completely reducible.

For V completely decomposable, we write

$$V \cong \oplus_{\mu=1}^{r} \oplus_{i=1}^{a_\mu} T^{(\mu)} = \oplus_{\mu=1}^{r} V^{(\mu)}, \tag{1.22}$$

where μ labels different irreps, and i labels the degeneracy of the μ-th irrep. We may write (with abbreviation) the isotypical component as

$$V^{(\mu)} = \oplus_{i=1}^{a_\mu} T^{(\mu)} = a_\mu T^{(\mu)} = \mathbb{C}^{a_\mu} \otimes T^{(\mu)}.$$

1.2 Schur's Lemmas and Orthogonal Theorem

1.2.1 Schur's Lemmas

Lemma 1.2.1. *(Schur's First Lemma) If $A : V \to V'$ is an intertwiner between two irreducible representations of a group G, then A is either zero or an isomorphism.*

Proof. The key is both $\ker A$ and $\operatorname{Im} A$ are invariant subspaces, then A is either zero or an isomorphism. Let us peel this into pieces. Recall that an intertwiner means that $T'(g)A = AT(g)$ for any g.

Assume $v_k \in \ker A$, meaning that $Av_k = 0$, then $AT(g)v_k = T'(g)Av_k = 0 \Rightarrow T(g)v_k \in \ker A$, and $\ker A$ is invariant under $T(g)$.

Similarly, for $v_i' \in \operatorname{Im} A$, we have $\exists v \in V$, s.t. $Av = v_i'$, then $T'(g)v_i' = T'(g)Av = AT(g)v \Rightarrow T'(g)v_i' \in \operatorname{Im} A$, and $\operatorname{Im} A$ is invariant under $T'(g)$.

Consider that both V, V' are irreps, then $\ker A = 0$ or V, and $\operatorname{Im} A = 0$ or V', and we have two choices:

- $\ker A = 0$, then $\operatorname{Im} A = V'$, and A is an isomorphism. Note that $V \cong V'$ here.
- $\ker A = V$, then $\operatorname{Im} A = 0$, and A is zero.

Vice choice versa. □

Lemma 1.2.2. *(Schur's Second Lemma) If A is a complex matrix commuting with an irreducible matrix representation, then A is a scalar multiple of the identity.*

Proof. The key is as a complex matrix, A has at least one eigenvector, which makes $\ker A \neq 0$ thus falls into the second choice.

Let A be a complex matrix, and T be an irreducible matrix representation, then $T(g)A = AT(g)$ for any g. Thus A in an intertwiner from T to itself, and by the first lemma, A is either zero (identity) or an isomorphism. Since A is complex, then the characteristic polynomial $\det(A - \lambda I) = 0$ has full solutions and leads to at least one non-zero eigenvector, i.e., $Av = \lambda v$ which makes $\ker A \neq 0$, and A is zero, meaning it is identity times a scalar. □

- The lemmas tell that there is no non-zero isomorphism between two different irreps, and there is no non-trivial automorphism for an irreducible representation.

- Now consider the isotypical component $V^{(\mu)} = \mathbb{C}^{a_\mu} \otimes T^{(\mu)}$, then by Schur's lemma, the intertwiners from $V^{(\mu)}$ to itself are of the form $K \otimes 1^{(\mu)}$, where $K \in \mathrm{End}(\mathbb{C}^{a_\mu})$.
- In quantum mechanics, we often meet symmetry group $g \in G$ with $[U(g), H] = 0$, and the Hamiltonian H becomes an intertwiner of unitary rep U. Then by decomposing the Hilbert space into (carrier spaces of) irreps of G, then H must be a scalar multiple of the identity in each isotypical component. This fact leads to the selection rules. For instance, for ΔH commuting with G, then its matrix elements between different states belonging to different carrier spaces of irreps must be zero.

1.2.2 Orthogonal Relations

Below we shall introduce the orthogonal relation for matrix elements of irreps, to do which let us introduce the notations: $T^{(\mu)},\ \mu = 1, 2, \cdots, r$ to be inequivalent irreps, and $n_\mu = \dim(V^{(\mu)})$ the dimension of the carrier space of $T^{(\mu)}$. Consider that we have made matrix equivalence irreps of $T^{(\mu)} \in \mathrm{Mat}_{n_\mu \times n_\mu}$. Now let us consider $B \in \mathrm{Mat}_{n_\mu \times n_\nu}$, then we have $A \in \mathrm{Mat}_{n_\mu \times n_\nu}$ defined by

$$A = \frac{1}{|G|} \sum_{g \in G} T^{(\mu)}(g) B T^{(\nu)}(g^{-1}), \tag{1.23}$$

which we claim to be an intertwiner from $T^{(\mu)}$ to $T^{(\nu)}$: $T^{(\mu)}(h)A = AT^{(\nu)}(h)$. This is easily proved by the rearrangement

$$\begin{aligned} T^{(\mu)}(h)A &= \frac{1}{|G|} \sum_{g \in G} T^{(\mu)}(h) T^{(\mu)}(g) B T^{(\nu)}(g^{-1}) \\ &= \frac{1}{|G|} \sum_{g \in G} T^{(\mu)}(hg) B T^{(\nu)}(g^{-1}) \\ &= \frac{1}{|G|} \sum_{g' \in G} T^{(\mu)}(g') B T^{(\nu)}((h^{-1}g')^{-1}) \\ &= \frac{1}{|G|} \sum_{g' \in G} T^{(\mu)}(g') B T^{(\nu)}(g'^{-1}) T^{(\nu)}(h) \\ &= AT^{(\nu)}(h). \end{aligned}$$

Then simply, from Schur's lemma, we have

$$A = \lambda \delta_{\mu\nu} \mathbb{I}_{n_\mu}, \tag{1.24}$$

where the scalar λ depends on μ, B.

The above general relation degrades into the orthogonal relation for matrix elements of irreps by letting $B = e_{lm}$ as matrix basis ($1 \le l \le n_\mu$, $1 \le m \le n_\nu$), where e_{lm} is the matrix with 1 at (l, m) and 0 elsewhere. Then we have the orthogonal relation for matrix elements of irreps:

$$\sum_{g \in G} T^{(\mu)}_{il}(g) T^{(\nu)}_{mj}(g^{-1}) = |G| \lambda \delta_{\mu\nu} \delta_{ij},$$

with λ a function of μ, l, m. To make it certain, we let $\mu = \nu$, $i = j$ and sum over i, which leads to

$$\begin{aligned}\sum_i \sum_{g\in G} T^{(\mu)}_{il}(g) T^{(\mu)}_{mi}(g^{-1}) &= \sum_{g\in G} T^{(\mu)}_{ml}(g^{-1}g) = |G|\delta_{ml} \\ &= \sum_i |G|\lambda = |G| n_\mu \lambda,\end{aligned}$$

which gives $\lambda(\mu, l, m) = \frac{\delta_{ml}}{n_\mu}$. We thus finally have

Theorem 1.2.3. *The orthogonal relation for matrix elements of irreps:*

$$\frac{1}{|G|}\sum_{g\in G} T^{(\mu)}_{il}(g)\hat{T}^{(\nu)}_{jm}(g) = \frac{1}{n_\mu}\delta_{\mu\nu}\delta_{ij}\delta_{lm}, \tag{1.25a}$$

$$\int_G T^{(\mu)}_{il}(g)\hat{T}^{(\nu)}_{jm}(g)dg = \frac{1}{n_\mu}\delta_{\mu\nu}\delta_{ij}\delta_{lm}, \tag{1.25b}$$

where the Haar measure for a compact group is normalized to unit volume.

Notice that $\hat{T}^{(\nu)}_{jm}(g) \equiv T^{(\nu)}_{mj}(g^{-1}) = (T^{(\nu)}(g))^{-1}_{mj} = (T^{(\nu)}(g))^{-1,T}_{jm}$ is the matrix representation of the dual representation $\hat{T}^{(\nu)}$. Especially, for $T^{(\mu)}$ a unitary rep, we have $\hat{T}^{(\mu)}(g) = T^{(\mu)}(g)^*$, and by defining $\phi^{(\mu)}_{il} = n_\mu^{1/2} T^{(\mu)}_{il}$, we have

$$\frac{1}{|G|}\sum_{g\in G} \phi^{(\mu)}_{il}(g)\phi^{(\nu)}_{jm}(g)^* = \delta_{\mu\nu}\delta_{ij}\delta_{lm}, \tag{1.26a}$$

$$\int_G \phi^{(\mu)}_{il}(g)\phi^{(\nu)}_{jm}(g)^* dg = \delta_{\mu\nu}\delta_{ij}\delta_{lm}, \tag{1.26b}$$

and we can interpret $\phi^{(\mu)}_{il}(g)$ as the orthogonal set of vectors in R_G.

1.2.3 Decomposition of the Regular Representation

With orthogonal relations, we can state more about the regular representations. Recall irrep over complex field, and its matrix element $T^{(\mu)}_{ml}$ can be viewed asa a complex function on G, i.e., a vector in R_G. Also recall that we have constructed RRR and LRR, and for now we focus on the first $\mathcal{R}^\mu_i = \text{Span}\{T^{(\mu)}_{ij}\}_{j=1,\cdots,n_\mu}$, and invariant subspace of R_G indicated by

$$R(g)\cdot T^{(\mu)}_{ij} = \sum_{s=1}^{n_\mu} T^{(\mu)}_{sj}(g) T^{(\mu)}_{is}.$$

Thus $\mathcal{R}^\mu_i$ is also a carrier space of G, and especially, its vector follows the action induced by $T^{(\mu)}$, and thus by definition, we have

$$\mathcal{R}^\mu_i \cong T^{(\mu)}, \tag{1.27}$$

independent of i, thus since $i = 1, \cdots, n_\mu$, R_G contains 'at least' n_μ copies of $T^{(\mu)}$.

To remove the ambiguous word 'at least', we consider

$$W = \text{Span}\{T^{(\mu)}_{ij}\} = \oplus_\mu \oplus_{i=1}^{n_\mu} \mathcal{R}^{(\mu)}_i \cong \oplus_\mu n_\mu T^{(\mu)} \subset R_G, \tag{1.28}$$

where one notices that it is the orthogonal relation of these matrix element vectors that leads to the orthogonal decomposition. Then clearly $R_G \cong W \oplus W^\perp$, and since R_G, W

are unitary reps, so does $W^\perp$, which should then also be a direct sum of irreps $T^{(\mu)}$. Nevertheless, by supposing $\text{Span}\{f_i\}_{i=1,\cdots,n_\mu} \subset W^\perp$ follows from transformation by $T^{(\mu)}$ under RRR, we have

$$\begin{aligned} f_i(gh) = (R(h)\cdot f_i)(g) &= \sum_s T^{(\mu)}_{si}(h) f_s(g) \\ \Longrightarrow f_i(g') = \sum_s T^{(\mu)}_{si}(g^{-1}g') f_s(g) &= \sum_{s,k} T^{(\mu)}_{sk}(g^{-1}) T^{(\mu)}_{ki}(g') f_s(g) \\ \Longrightarrow f_i = \sum_k \alpha_k T^{(\mu)}_{ki},\ &\text{where } \alpha_k = \sum_s T^{(\mu)}_{sk}(g^{-1}) f_s(g),\ \forall g \in G. \end{aligned}$$

This indicates that $f_i \in W$, as its linear combination of $T^{(\mu)}_{ki}$. Thus we have $W^\perp = 0$, and $R_G \cong W$, i.e.,

$$R_G \cong \oplus_\mu n_\mu T^{(\mu)}. \tag{1.29}$$

We can go a step further to remove the somehow ambiguous n_μ. The LRR $\mathcal{L}^\mu_j = \text{Span}\{T^{(\mu)}_{ij}\}_{i=1,\cdots,n_\mu}$ is also a carrier space of G, and invariant under LRR,

$$R(g)\cdot T^{(\mu)}_{ij} = \sum_{s=1}^{n_\mu} T^{(\mu)}_{is}(g^{-1}) T^{(\mu)}_{sj} = \sum_{s=1}^{n_\mu} \hat{T}^{(\mu)}_{si}(g) T^{(\mu)}_{sj},$$

i.e., we have $\mathcal{L}^\mu_j \cong \hat{T}^{(\mu)}$, the dual rep. Similarly we have

$$R(g) \cong \oplus_\mu n_\mu \hat{T}^{(\mu)}. \tag{1.30}$$

Now if we combine both RRR and LRR, with similar deduction by combining the two procedures, we see that $\text{Span}\{T^{(\mu)}_{ij}\}_{i,j=1,\cdots,n_\mu}$ is isomorphic to $\hat{T}^{(\mu)} \otimes T^{(\mu)}$ of $G_L \times G_R$, as

$$R(g)\cdot T^{(\mu)}_{ij} = \sum_{s,k} \hat{T}^{(\mu)}_{si}(g) T^{(\mu)}_{kj}(g) T^{(\mu)}_{sk}. \tag{1.31}$$

We can further identify $\hat{T}^{(\mu)} \otimes T^{(\mu)} \cong \text{End}(T^{(\mu)})$ by action

$$T(g_L, g_R)\cdot T = T^{(\mu)}(g_L) T T^{(\mu)}(g_R^{-1}).$$

Thus we have the decomposition of the regular representation:

$$R_G \cong \oplus_\mu \hat{T}^{(\mu)} \otimes T^{(\mu)} \cong \oplus_\mu \text{End}(T^{(\mu)}). \tag{1.32}$$

Since R_G takes basis to be map from G to $\mathbb{C}$, we have $\dim(R_G) = |G|$ for a finite group, and thus we have

$$\sum_\mu n_\mu^2 = |G|. \tag{1.33}$$

1.2.4 Fourier Analysis as Rep Theory

For an Abelian group G, due to that $T(g)T(g_0) = T(g_0)T(g)$ for any g, then by Schur's lemma, $T(g_0) = \lambda(g_0)\mathbb{I}$, i.e., any rep T comes down to be determined by a function $\lambda(g)$, and all (finite-dim complex) irreps are 1-dim, or from the simple fact that for G Abelian, any (decomposable) finite-dim rep matrices commute and can be simultaneously diagonalized. Again notice that reducible is not equivalent to decomposable.

Character group

For *any* group G, the space of all 1-dim reps is a group under tensor product, i.e., multiplication,

$$T_1 \otimes T_2(g) = T_1(g)T_2(g),$$

with identity trivial rep $T_0(g) \equiv 1\ \forall g$ and inverse by $T^{-1}(g) = T(g)^{-1}$.

The subspace of the above group, consists of all 1-dim unitary irreps, is called the *unitary dual* $\hat{G}$ of G, with inverse specified by complex conjugation $\chi^*(g) = \chi(g)^{-1}$. When G is Abelian, $\hat{G}$ is also a group, called the *character group* or *Pontryagin dual* of G.

- For $G = \mathbb{Z}_n$ with generator $\omega = \exp(i2\pi/n)$, its irreps are $T^{(s)}(\omega) = \omega^s = \exp(i2\pi s/n)$ with $s = 0, 1, \cdots, n-1$. To see this, let us assume arbitrary s, and the homomorphism requirement $1 = T^{(s)}(1) = T^{(s)}(\omega^n) = \exp(i2\pi s)$ requires that s is an integer. Then further observation says that $T^{(s)} \otimes T^{(s')} = T^{(s+s')}$ and $T^{(s+n)} = T^{(s)}$, and the dual group is another copy of $\mathbb{Z}_n$: $\hat{\mathbb{Z}}_n \cong \mathbb{Z}_n$. Notice we have found all irreps in since that $n = \sum_{i=1}^{n} 1^2$.
- For $G = \mathbb{R}$, we have for any momentum $k \in \mathbb{R}$, rep $T^{(k)}(x) = e^{ikx}$, and the dual group is $\hat{\mathbb{R}} \cong \mathbb{R}$.
- For $G = \mathbb{Z}$, we again have $T^{(\theta)}(n) = e^{in\theta}$ with notice $T^{(\theta)} \otimes T^{(\theta')} = T^{(\theta+\theta')}$ and a periodicity $T^{(\theta+2\pi)} = T^{(\theta)}$. We thus identify that $\hat{\mathbb{Z}} \cong U(1) \cong S^1$.
- For $G = U(1)$, conversely by labelling elements as $e^{i\theta}$, we have $T^{(n)}(e^{i\theta}) = e^{in\theta}$ with $n \in \mathbb{Z}$, with reason that $1 = T^{(n)}(e^{i2\pi}) = \exp(i2\pi n)$, and the dual group is $\hat{U}(1) \cong \mathbb{Z}$.
- For $G = \Lambda \subset \mathbb{R}^d$ as a lattice, we have $T^{(\boldsymbol{k})}(\boldsymbol{n}) = \exp(i2\pi \boldsymbol{k}\cdot\boldsymbol{n})$ for $\boldsymbol{k} \in \mathbb{R}^d$, with group structure $T^{(\boldsymbol{k})} \otimes T^{(\boldsymbol{k}')} = T^{(\boldsymbol{k}+\boldsymbol{k}')}$ and equivalence relation $T^{(\boldsymbol{k}+\boldsymbol{b})} = T^{(\boldsymbol{k})}$, where $\boldsymbol{b} \in \Lambda^* \subset \mathbb{R}^d$ is the dual lattice defined by $\Lambda^* = \{\boldsymbol{b} \in \mathbb{R}^d | \forall \boldsymbol{n} \in \Lambda, \boldsymbol{b}\cdot\boldsymbol{n} \in \mathbb{Z}\}$. We thus have $\hat{\Lambda} \cong \mathbb{R}^d/\Lambda^* \cong U(1)^{\times d}$, the d-dim torus, the Brillouin zone. This can be viewed as a generalization of the above example of $\mathbb{Z}$.

Generally, we have $\hat{\hat{G}} \cong G$ for Abelian groups.

Fourier duality

Let us see some examples summarizing the above information:

- $G = \mathbb{Z}_2 = \{1, \sigma\}$. Then vector in carrier space $R_G = \mathrm{Map}(G, \mathbb{C}) \cong \mathbb{C}^2$ is specified by $(\psi_+, \psi_-) = (\Psi(1), \Psi(\sigma)) \in \mathbb{C}^2$. The decomposition follows as two 1-dim irreps $V_\pm$ with $\rho(\sigma) = \pm 1$. The corresponding matrix elements are

$$T^+(1) = 1,\ T^+(\sigma) = 1,$$
$$T^-(1) = 1,\ T^-(\sigma) = -1,$$

  which satisfies the orthogonal relation:

$$\frac{1}{2}\sum_{g=\{1,\sigma\}} T^{\pm,*}(g)T^{\pm}(g) = 1,$$
$$\frac{1}{2}\sum_{g=\{1,\sigma\}} T^{\pm,*}(g)T^{\mp}(g) = 0.$$

The completeness follows as any vector in R_G can be decomposed as

$$\Psi = \frac{\psi_+ + \psi_-}{2} T^+ + \frac{\psi_+ - \psi_-}{2} T^-,$$

as easily verified by calculating $\Psi(1) = \psi_+$ and $\Psi(\sigma) = \psi_-$.

- Let $G = \mathbb{Z}_n$ with generator $\omega = e^{2\pi i/n}$. As G is Abelian, all irreps are 1-dim, and the matrix elements are given by $T^{(k)}(g) = \omega^{kg}$ with $k = 0, 1, \cdots, n-1$. We have orthogonal relation as

$$\frac{1}{n}\sum_{g=0}^{n-1} T^{(k)*}(g) T^{(l)}(g) = \frac{1}{n}\sum_{g=0}^{n-1} \omega^{(l-k)g} = \delta_{kl},$$

which is the basis for discrete Fourier transform. According to the decomposition of the regular representation, we can decompose any function over $\mathbb{Z}_n$ as

$$f = \sum_{k=0}^{n-1} \hat{f}_k T^{(k)},$$

where by RRR,

$$\hat{f}_k = \frac{1}{n}\sum_s f(\omega^s)(T^{(k)}(\omega^{-s})) = \frac{1}{n}\sum_s f(\omega^s) e^{-i2\pi sk/n},$$

which we recognize as the discrete Fourier transform. One can prove this in another direct way by

$$\begin{aligned} f(\omega^p) &= \sum_{k=0}^{n-1} \hat{f}_k T^{(k)}(\omega^p) = \frac{1}{n}\sum_{k=0}^{n-1}\sum_{s=0}^{n-1} f(\omega^s) T^{(k)}(\omega^{-s}) T^{(k)}(\omega^p) \\ &= \frac{1}{n}\sum_{k=0}^{n-1}\sum_{s=0}^{n-1} f(\omega^s) \hat{T}^{(s)}(\omega^k)(\hat{T}^{(p)}(\omega^k))^{-1} \\ &= \sum_{s=0}^{n-1} f(\omega^s)\delta_{sp} = f(\omega^p), \end{aligned}$$

where the dual rep is given by $\hat{T}^{(k)}(\omega^s) = (T^{(s)}(\omega^k))^{-1} = \omega^{-sk}$, and the orthogonal relation follows.

- $G = \mathbb{R}$ with irreps $T^{(k)}(x) = \frac{1}{\sqrt{2\pi}} e^{i2\pi kx}$, and the orthogonal relation is given by Fourier transform:

$$\int_{\mathbb{R}} \mathrm{d}x T^{(k)}(x)(T^{(l)}(x))^* = \delta(k-l),$$

and we have

$$\begin{aligned} f(x) &= \int_{\hat{\mathbb{R}}} \mathrm{d}k \hat{f}(k) T^{(k)}(x), \\ \hat{f}(k) &= \int_{\mathbb{R}} \mathrm{d}x f(x) \hat{T}^{(x)}(k), \end{aligned}$$

where $\hat{T}^{(k)}(x) = (T^{(k)}(x))^{-1} = \frac{1}{\sqrt{2\pi}} e^{-i2\pi kx}$.

- $G = U(1)$ with $z = e^{i\theta}$, and again all irreps are 1-dim by Abelian nature with $T^{(k)}(z) = z^k$ with $k \in \mathbb{Z}$. The orthogonal relation is given by Fourier transform:

$$\frac{1}{2\pi}\int_0^{2\pi} T^{(k)*}(z)T^{(l)}(z)d\theta = \frac{1}{2\pi}\int_0^{2\pi} e^{-i(k-l)\theta}d\theta = \delta_{kl},$$

where 2π comes from normalized norm. In other words, any L^2 function $\Psi(\theta)$ defined on the circle can be decomposed as Fourier series:

$$\Psi = \sum_{k=-\infty}^{\infty} \hat{\Psi}_k T^{(k)}, \ \hat{\Psi}_k = \frac{1}{2\pi}\int_0^{2\pi} \hat{T}^{(k)}(\theta)\Psi(\theta)\mathrm{d}\theta.$$

The proof follows as

$$\begin{aligned}\Psi(\theta) &= \sum_{k=-\infty}^{\infty} \hat{\Psi}_k T^{(k)}(\theta) \\ &= \sum_{k=-\infty}^{\infty} \frac{1}{2\pi}\int_0^{2\pi} T^{(k)*}(\theta')T^{(k)}(\theta)\Psi(\theta')\mathrm{d}\theta' \\ &= \int_0^{2\pi} \mathrm{d}\theta'\Psi(\theta')\delta(\theta-\theta') \\ &= \Psi(\theta),\end{aligned}$$

where in the third equality we utilized the orthogonal relation for $\mathbb{Z}$ group with its irreps $\tilde{T}^{(\theta)}(k) = e^{-ik\theta} = (T^{(k)}(\theta))^{-1}$, and

$$\sum_k \tilde{T}^{(\theta')*}(k)\tilde{T}^{(\theta)}(k) = 2\pi\delta(\theta-\theta').$$

We see in the above examples a duality-like relation introduced, and that is the Fourier duality. It comes from the RRR and orthogonal relation, and can be stated as

Theorem 1.2.4. *For G locally compact, a natural isomorphism $L^2(G) \cong L^2(\hat{G})$ exists that expands any function over G in terms of unitary irreps pf G as*

$$f(g) = \int_{\hat{G}} \mathrm{d}\chi\chi(g)^*\hat{f}(\chi), \tag{1.34}$$

with isometry

$$\int_G \mathrm{d}g f_1^* f_2 = \int_{\hat{G}} \mathrm{d}\chi \hat{f}_1^*\hat{f}_2. \tag{1.35}$$

This is easily proved by a direct calculation with orthogonal relation,

$$\begin{aligned}\int_G \mathrm{d}g f_1^* f_2 &= \int_G \mathrm{d}g \int_{\hat{G}} \mathrm{d}\chi\mathrm{d}\chi' \ \chi(g)\chi'^*(g)\hat{f}_1^*(\chi)\hat{f}_2(\chi') \\ &= \int_{\hat{G}} \mathrm{d}\chi\mathrm{d}\chi' \ \delta(\chi-\chi')\hat{f}_1^*(\chi)\hat{f}_2(\chi') \\ &= \int_{\hat{G}} \mathrm{d}\chi \hat{f}_1^*\hat{f}_2.\end{aligned}$$

Poisson summation formula

The Poisson summation formula is a special case of the Fourier duality for lattice Λ and dual lattice Λ^*, and can be stated as

Theorem 1.2.5. *For $f \in \mathcal{S}(\Lambda)$ (the Schwartz space of rapidly decreasing functions), we have*

$$\sum_{\boldsymbol{r}\in\Lambda} f(\boldsymbol{r}) = \sum_{\boldsymbol{b}\in\Lambda^*} \hat{f}(\boldsymbol{b}), \tag{1.36}$$

where the Fourier transform is defined in $\mathbb{R}^d$ as

$$\hat{f}(\boldsymbol{b}) = \int_{\mathbb{R}^d} \mathrm{d}^d x f(\boldsymbol{x}) e^{-i2\pi \boldsymbol{b}\cdot\boldsymbol{x}}.$$

This can be proved simply by orthogonal relation, where we write the 1D irrep of Λ as $T^{(\boldsymbol{k})}(\boldsymbol{r}) = e^{2\pi i \boldsymbol{k}\cdot\boldsymbol{r}}$, and the orthogonal relation follows as

$$\sum_{\boldsymbol{r}\in\Lambda} e^{2\pi i \boldsymbol{k}\cdot\boldsymbol{r}} = \sum_{\boldsymbol{r}\in\Lambda} T^{(\boldsymbol{k})}(\boldsymbol{r})(T^{(\boldsymbol{b})}(\boldsymbol{r}))^* = \sum_{\boldsymbol{b}\in\Lambda^*} \delta^d(\boldsymbol{k}-\boldsymbol{b}),$$

where $\boldsymbol{b} \in \Lambda^*$ is a dual lattice vector satisfying $\boldsymbol{b}\cdot\boldsymbol{r} \in \mathbb{Z}$ for any $\boldsymbol{r} \in \Lambda$. We then integrate $f(\boldsymbol{x})$ over $\mathbb{R}^d$ with the above relation, and we obtain the Poisson summation formula.

Here, we give a 1D example by letting $f(t) = \exp(-\pi a t^2 + 2\pi i b t)$, and we have

$$\begin{aligned}
\sum_{n\in\mathbb{Z}} e^{-\pi a n^2 + 2\pi i b n} &= \sum_{n\in\mathbb{Z}} f(n) = \sum_{m\in\mathbb{Z}} \hat{f}(m) \\
&= \sum_{m\in\mathbb{Z}} \int_{-\infty}^{+\infty} \mathrm{d}t\ f(t) e^{-2\pi i m t} \\
&= \sum_{m\in\mathbb{Z}} \int_{-\infty}^{+\infty} \mathrm{d}t\ e^{-\pi a t^2 + 2\pi i (b-m) t} \\
&= \sum_{m\in\mathbb{Z}} \frac{1}{\sqrt{a}} e^{-\frac{\pi (b-m)^2}{a}}.
\end{aligned}$$

1.2.5 Characters and Decomposition of Irreps

Given the concept of irreps, one meets with the problem of how to construct the irreps, how to identify the irreps, and how to decompose a rep into irreps. Let us focus on the last two.

Definition 1.2.1. *Let (T, V) be a rep of G, then a special kind of function in* $\mathrm{Map}(G, \mathbb{C})$ *defined by*

$$\chi_T(g) \equiv \mathrm{Tr}_V(T(g)) = \sum_{i=1}^{\dim(V)} T_{ii}(g), \tag{1.37}$$

is called the characters of the rep T.

- χ_T is independent of basis of V.
- $\chi_T = \chi_{T'}$ if $T \cong T'$.

- $\chi \in R_G$.
- For T unitary, $\chi(g^{-1}) = \chi(g)^*$.

By introducing conjugacy class of g as

$$C(g) = \{h = kgk^{-1}, \forall k \in G\},$$

we see that χ is constant on conjugacy classes, thus allowing us to write $\chi(C)$ as *class function*.

Theorem 1.2.6. *All χ_μ of irreps $T^{(\mu)}$ form an orthogonal basis of the class function basis, a subspace of R_G.*

Proof. Since $T_{ij}^{(\mu)}$ form an orthogonal basis of R_G, we have

$$F(g) = \sum_{\mu,i,j} \hat{F}_{ij}^{\mu} T_{ij}^{(\mu)}(g),$$

with $\hat{F}_{ij}^{\mu} = \frac{1}{|G|} \sum_{g\in G} F(g) T_{ij}^{(\mu)*}(g)$, or in the sense of

$$\begin{pmatrix} T_{11}^{(1)}(g_1) & \cdots & T_{n_r n_r}^{(r)}(g_1) \\ \vdots & \ddots & \vdots \\ T_{11}^{(1)}(g_{|G|}) & \cdots & T_{n_r n_r}^{(r)}(g_{|G|}) \end{pmatrix} \begin{pmatrix} \hat{F}_{11}^{(1)} \\ \vdots \\ \hat{F}_{n_r n_r}^{(r)} \end{pmatrix} = \begin{pmatrix} F(g_1) \\ \vdots \\ F(g_{|G|}) \end{pmatrix},$$

with $\sum_{\mu=1}^{r} n_\mu^2 = |G|$. Then let us compute the class-average as

$$\begin{aligned} (T_{ij}^{(\mu)})^{\text{ave}}(g) &= \frac{1}{|G|} \sum_{h\in G} T_{ij}^{(\mu)}(hgh^{-1}) \\ &= \frac{1}{|G|} \sum_{h\in G} \sum_{mn} T_{im}^{(\mu)}(h) T_{nj}^{(\mu)}(h^{-1}) T_{mn}^{(\mu)}(g) \\ &= \sum_{mn} T_{mn}^{(\mu)}(g) \frac{1}{n_\mu} \delta_{ij} \delta_{mn} \\ &= \frac{\delta_{ij}}{n_\mu} \chi_\mu(g), \end{aligned}$$

with orthonormal relation used. While for class function, we expect

$$F^{\text{ave}}(g) = \frac{1}{|G|} \sum_{h\in G} F(hgh^{-1}) = \frac{1}{|G|} \sum_{h\in G} F(C(g)) = F(g),$$

and thus we obtain

$$F(g) = \sum_{\mu,i,j} \hat{F}_{ij}^{\mu} \frac{\delta_{ij}}{n_\mu} \chi_\mu(g) = \sum_{\mu,i} \hat{F}_{ii}^{\mu} \frac{1}{n_\mu} \chi_\mu(g),$$

and thus χ_μ for a complete basis of class functions.

For orthogonality, we start from the orthogonal relation of matrix elements as

$$\frac{1}{|G|} \sum_{g\in G} T_{ij}^{(\mu)}(g) T_{lk}^{(\nu)}(g^{-1}) = \frac{1}{n_\mu} \delta_{\mu\nu} \delta_{ik} \delta_{jl},$$

and by letting $i = j$, $k = l$ and summing over i and k, we have

$$\begin{aligned}\langle \chi_\nu, \chi_\mu \rangle &\equiv \frac{1}{|G|} \sum_{g\in G} \chi_\mu(g)\chi_\nu(g)^{-1} = \frac{1}{|G|} \sum_{i,k} \sum_{g\in G} T^{(\mu)}_{ii}(g) T^{(\nu)}_{kk}(g^{-1}) \\ &= \sum_{i,k} \frac{1}{n_\mu} \delta_{\mu\nu}\delta_{ik}\delta_{ik} = \delta_{\mu\nu}.\end{aligned}$$

□

- Assuming distinct conjugacy classes $C_1, \cdots, C_r$, we can define r class function basis as $\delta_{C_i}(C_j) = \delta_{ij}$. Thus r here equals the number of irreps. In this since, we have character table

$$\begin{array}{cccccc} & m_1C_1 & m_2C_2 & \cdots & m_rC_r \\ \chi_1 & \chi_1(C_1) & \chi_1(C_2) & \cdots & \chi_1(C_r) \\ \chi_2 & \chi_2(C_1) & \chi_2(C_2) & \cdots & \chi_2(C_r), \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \chi_r & \chi_r(C_1) & \chi_r(C_2) & \cdots & \chi_r(C_r) \end{array}$$

  where $m_i = |C_i|$ is order of the conjugacy class C_i.

- Any finite dim rep T is completely reducible as

$$T \cong \oplus_\mu a_\mu T^{(\mu)}$$

  with $a_\mu = \langle \chi_T, \chi_\mu \rangle$, as easy to verify.

- For unitary rep, the orthogonal relation can be re-written as

$$\frac{1}{|G|} \sum_{C_i} m_i \chi_\mu(C_i)\chi_\nu(C_i)^* = \delta_{\mu\nu}, \tag{1.38}$$

  from which we can define $r\times r$ unitary matrix $S_{\mu i} = \sqrt{\frac{m_i}{|G|}}\chi_\mu(C_i)$ such that $\sum_{i=1}^r S_{\mu i}S^*_{\nu i} = \delta_{\mu\nu}$ as $SS^\dagger = \mathbb{I}$, and $S^\dagger S = \mathbb{I}$ similarly leads to

$$\sum_\mu \chi_\mu(C_i)^* \chi_\mu(C_j) = \frac{|G|}{m_i}\delta_{ij}, \tag{1.39}$$

  as the other orthogonal relation.

This relation can be proved in another way by computing $\mathrm{Tr}_{R_G}(L(g_1) \otimes R(g_2))$ in two bases. Firstly consider normalized matrix element basis $\phi^{(\mu)}_{ij}$, and

$$L(g_1) \otimes R(g_2) \cdot \phi^{(\mu)}_{ij} = \sum_{i',j'} T^{(\mu)}_{j'j}(g_2) T^{(\mu)}_{i'i}(g_1^{-1}) \phi^{(\mu)}_{i'j'},$$

and the trace is given by

$$\begin{aligned}\mathrm{Tr}_{R_G}(L(g_1) \otimes R(g_2)) &= \sum_{\mu ij} \langle \phi^{(\mu)}_{ij}, L(g_1) \otimes R(g_2) \cdot \phi^{(\mu)}_{ij} \rangle \\ &= \sum_{\mu ij} \sum_{i',j'} T^{(\mu)}_{j'j}(g_2) T^{(\mu)}_{i'i}(g_1^{-1}) \langle \phi^{(\mu)}_{ij}, \phi^{(\mu)}_{i'j'} \rangle \\ &= \sum_{\mu ij} T^{(\mu)}_{jj}(g_2) T^{(\mu)}_{ii}(g_1^{-1}) = \sum_\mu \chi^{(\mu)}(g_1^{-1})\chi^{(\mu)}(g_2).\end{aligned}$$

Secondly, we use delta-function basis as

$$L(g_1) \otimes R(g_2) \cdot \delta_g = \delta_{g_1 g g_2^{-1}},$$

and the trace counts only when $g = g_1 g g_2^{-1}$, or $g_2 = g^{-1} g_1 g$. Note that by letting centralizer $Z(g_1) = \{g_c | g_c^{-1} g_1 g_c = g_1\}$, we see that $C(g_1) = G/Z(g_1)$, and especially $\#\{g | g_2 = g^{-1} g_1 g\} = |Z(g_1)|$, thus we have

$$\mathrm{Tr}_{R_G}(L(g_1) \otimes R(g_2)) = |Z(g_1)| = \frac{|G|}{m_1}.$$

Example: S_3

The permutation/symmetric group S_n is the group of all permutations of n objects, with order $n!$. Its group elements is usually spoken in cyclic notation, e.g., (132) means that we give a map $g : \{a, b, c\} \to \{c, a, b\}$ as $g(a) = `3' = c$, $g(b) = `1' = a$, $g(c) = `2' = b$. Multiple cycles compose a type of certain disjoint cycle like (12)(345) is of type $(2, 3)$, and a key fact is that the conjugacy class of S_n corresponds to the cycle type. Its generators are neighboring transportations σ_i that swap i and $i + 1$ objects, with relations $\sigma_i^2 = 1$, $\sigma_i \sigma_j = \sigma_j \sigma_i$ for $|i - j| > 1$, and $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$. The difference between S_n and braiding group B_n is that the latter memorizes the permutation process, thus forming an infinite discrete group.

For S_3 which is of order 6, we have three conjugacy classes $C_1 = \{1\}$, $C_2 = \{(12), (13), (23)\}$, and $C_3 = \{(123), (132)\}$. It has three irreps:

- $\mathbf{1}_+$ is the trivial rep with all elements mapped to 1.
- $\mathbf{1}_-$ is the sign rep with $1 \to 1$, $(ij) \to -1$, and $(ijk) \to 1$, homomorphic to $\mathbb{Z}_2$.
- T_3 is the symmetry of the equilateral triangle, with matrix elements

$$1 \to \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \ (23) \to \begin{pmatrix} -1 & \\ & 1 \end{pmatrix}, \ (123) \to R(2\pi/3) = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}.$$

Since we have $6 = 1^2 + 1^2 + 2^2$, these are all irreps, and the character table is given by

$$\begin{array}{c|ccc} & C_1 & C_2 & C_3 \\ \hline \mathbf{1}_+ & 1 & 1 & 1 \\ \mathbf{1}_- & 1 & -1 & 1 \\ T_3 & 2 & 0 & -1 \end{array}.$$

Using this table, we can decompose any rep T of S_3 into irreps. Let us consider the $V = \mathbb{R}^3$ rep

$$1 \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \ (12) \to \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \ (123) \to \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

with $\chi_V(C_1) = 3$, $\chi_V(C_2) = 1$, and $\chi_V(C_3) = 0$. The orthogonality relation tells us the decomposition coefficients as

$$a_{\mathbf{1}_+} = \frac{1}{6}\sum_{C_i} m_i \chi_V(C_i)\chi_{\mathbf{1}_+}(C_i) = \frac{1}{6}(1\cdot 1\cdot 3 + 3\cdot 1\cdot 1 + 2\cdot 1\cdot 0) = 1,$$
$$a_{\mathbf{1}_-} = \frac{1}{6}\sum_{C_i} m_i \chi_V(C_i)\chi_{\mathbf{1}_-}(C_i) = \frac{1}{6}(1\cdot 1\cdot 3 - 3\cdot 1\cdot 1 + 2\cdot 1\cdot 0) = 0,$$
$$a_{T_3} = \frac{1}{6}\sum_{C_i} m_i \chi_V(C_i)\chi_{T_3}(C_i) = \frac{1}{6}(1\cdot 2\cdot 3 + 3\cdot 0\cdot 1 - 2\cdot 1\cdot 0) = 1,$$

and therefore $\chi_V = \chi_{\mathbf{1}_+} + \chi_{T_3}$ as $V \cong \mathbf{1}_+ \oplus T_3$.

We may also consider tensorial carrier space like $V^3 = V \otimes V \otimes V$ with $\dim(V) = d$, and basically the rep is given by

$$\rho(1)(e_1 \otimes e_2 \otimes e_3) = e_1 \otimes e_2 \otimes e_3,$$
$$\rho((12))(e_1 \otimes e_2 \otimes e_3) = e_2 \otimes e_1 \otimes e_3,$$
$$\rho((123))(e_1 \otimes e_2 \otimes e_3) = e_2 \otimes e_3 \otimes e_1,$$

with e_i being the basis of V. The character thus reads $\chi_{V^3}(C_1) = d^3$, $\chi_{V^3}(C_2) = d^2$, and $\chi_{V^3}(C_3) = d$, say

$$\chi_{V^3}(C_2) = \mathrm{Tr}(\rho((12))) = \sum_{ijk} \langle e_i \otimes e_j \otimes e_k, e_j \otimes e_i \otimes e_k\rangle = \sum_{i,j,k} \delta_{ij} = d^2.$$

Thus we can compute

$$a_{\mathbf{1}_+} = \frac{1}{6}\sum_{C_i} m_i \chi_{V^3}(C_i)\chi_{\mathbf{1}_+}(C_i) = \frac{1}{6}(1\cdot 1\cdot d^3 + 3\cdot 1\cdot d^2 + 2\cdot 1\cdot d) = \frac{d(d+1)(d+2)}{6},$$
$$a_{\mathbf{1}_-} = \frac{1}{6}\sum_{C_i} m_i \chi_{V^3}(C_i)\chi_{\mathbf{1}_-}(C_i) = \frac{1}{6}(1\cdot 1\cdot d^3 - 3\cdot 1\cdot d^2 + 2\cdot 1\cdot d) = \frac{d(d-1)(d-2)}{6},$$
$$a_{T_3} = \frac{1}{6}\sum_{C_i} m_i \chi_{V^3}(C_i)\chi_{T_3}(C_i) = \frac{1}{6}(1\cdot 2\cdot d^3 + 3\cdot 0\cdot d^2 - 2\cdot 1\cdot d) = \frac{d(d^2-1)}{3},$$

and

$$V^3 \cong \frac{d(d+1)(d+2)}{6}\mathbf{1}_+ \oplus \frac{d(d-1)(d-2)}{6}\mathbf{1}_- \oplus \frac{d(d^2-1)}{3}T_3.$$

One can verify the decomposition is complete by checking the dimension:

$$\dim(V^3) = d^3 = 1\cdot a_{\mathbf{1}_+} + 1\cdot a_{\mathbf{1}_-} + 2\cdot a_{T_3}.$$

Example: D_n

The dihedral group D_n is the symmetry group of regular n-gon, with order $2n$. It is generated by a mirror reflection x and n-fold rotation y satisfying $x^2 = 1$, $y^n = 1$, and $xyx = y^{-1}$. We can write 2-dim rep $T^{(j)}$ as

$$x \to \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ y \to \begin{pmatrix} \omega^j & 0 \\ 0 & \omega^{-j} \end{pmatrix},$$

with $\omega = e^{2\pi i/n}$. Such rep is real since $(T^{(j)})^* = T^{(-j)}$, making $T^{(j)}$ and $T^{(n-j)}$ equivalent.

For j odd, $T^{(j)}$ is irreducible and inequivalent for $j = 1, \cdots, (n-1)/2$. For $j = 0$ we clearly can decompose it into $\mathbf{1}_\pm$ by $x \to \pm 1$, $y \to 1$. Thus altogether we have

$$2n = \frac{n-1}{2} \cdot 2^2 + 1^2 + 1^2.$$

For j even, $T^{(j)}$ is irreducible and inequivalent for $j = 1, \cdots, n/2 - 1$, while for $j = 0$ and $j = n/2$, we have $\mathbf{1}_{\pm\pm}$ with $x, y \to \pm 1$, thus totally

$$2n = (n/2 - 1) \cdot 2^2 + 4 \cdot 1^2.$$

We can list the character table for, say n is odd as

	$[1]$	$[x]$	$[y]$	$\cdots$	$[y^k]$
$\mathbf{1}_+$	1	1	1	$\cdots$	1
$\mathbf{1}_-$	1	-1	1	$\cdots$	1
$\mathbf{2}^{(j)}$	2	0	$\omega^j + \omega^{-j}$	$\cdots$	$\omega^{jk} + \omega^{-jk}$

,

with $k = (n-1)/2$ and totally $k + 2$ irreps.

1.2.6 Fusion Rules

In quantum mechanics, we meet the problem of angular momentum addition, where the so-called Clebsch-Gordon coefficients appear; in topological quantum field theory, we meet the problem of how to fuse two anyons, where the so-called fusion rules appear. Basically, each angular momentum, or anyon type, is an irrep of some group, and the 'addition' or 'fusion' describes how to decompose the tensor product of two irreps into irreps.

More generally, we consider the fusion of two (completely decomposable) reps, where one representative reads

$$V \cong \bigoplus_{\mu=1}^{r} \bigoplus_{i=1}^{a_\mu} T^{(\mu)} \cong \bigoplus_{\mu=1}^{r} \mathbb{C}^{a_\mu} \otimes T^{(\mu)} \equiv \bigoplus_{\mu=1}^{r} a_\mu T^{(\mu)}, \tag{1.40}$$

where a_μ counts for the degeneracy of the irrep $T^{(\mu)}$ with $\mathbb{C}^{a_\mu}$ understood as a a_μ-dim trivial rep. Each isotypical component $V^{(\mu)} = \mathbb{C}^{a_\mu} \otimes T^{(\mu)}$ has intertwiners to itself in form $K \otimes \mathbb{I}^{(\mu)}$ with $K \in \mathrm{End}(\mathbb{C}^{a_\mu})$. The last equality refers to an abbreviation.

Now given $V_1 = \oplus a_\mu T^{(\mu)}$ and $V_2 = \oplus b_\nu T^{(\nu)}$, we can consider the tensor product $V_1 \otimes V_2$ as

$$V_1 \otimes V_2 = \bigoplus_{\mu,\nu} a_\mu b_\nu T^{(\mu)} \otimes T^{(\nu)},$$

and we are left with the *fusion rules* that tells us how to decompose $T^{(\mu)} \otimes T^{(\nu)}$ into irreps.

Definition 1.2.2. *Fusion coefficients $N_{\mu\nu}^\lambda$ are defined as the number of times an irrep $T^{(\lambda)}$ appears in the decomposition of $T^{(\mu)} \otimes T^{(\nu)}$:*

$$T^{(\mu)} \otimes T^{(\nu)} = \bigoplus_\lambda \overbrace{T^{(\lambda)} \oplus T^{(\lambda)} \oplus \cdots \oplus T^{(\lambda)}}^{N_{\mu\nu}^\lambda \text{ times}} \equiv \bigoplus_\lambda N_{\mu\nu}^\lambda T^{(\lambda)}. \tag{1.41}$$

This gives a fusion algebra structure on the set of irreps, with multiplication given by the fusion rules. This is an extremely important concept in all quantum theories. Now since

$$\chi_{T_1\otimes T_2}(g) = \mathrm{Tr}(T_1(g)\otimes T_2(g)) = \sum_{i,j}\langle e_i\otimes e_j, T_1(g)\otimes T_2(g)(e_i\otimes e_j)\rangle$$
$$= \sum_{i,j}\langle e_i, T_1(g)e_i\rangle\langle e_j, T_2(g)e_j\rangle = \chi_{T_1}(g)\chi_{T_2}(g),$$

we take trace over the fusion rule formula and obtain

$$\chi_\mu(g)\chi_\nu(g) = \sum_\lambda N_{\mu\nu}^\lambda \chi_\lambda(g),\ \forall g\in G, \tag{1.42}$$

and the orthogonal relation reads

$$N_{\mu\nu}^\lambda = \langle\chi_\lambda, \chi_\mu\chi_\nu\rangle = \frac{1}{|G|}\sum_{g\in G}\chi_\mu(g)\chi_\nu(g)\chi_\lambda(g^{-1}). \tag{1.43}$$

When we are dealing with unitary reps, we recall the unitary S-matrix

$$S_{\mu i} = \sqrt{\frac{m_i}{|G|}}\chi_\mu(C_i),\ \mu, i = 1,\cdots,r,$$

and re-write the fusion coefficients as

$$N_{\mu\nu}^\lambda = \sum_{i=1}^r \frac{S_{\mu i}S_{\nu i}S_{\lambda i}^*}{S_{0i}}, \tag{1.44}$$

where $S_{0i} = \sqrt{m_i/|G|}$ refers to the trivial rep. This celebrated formula is known as the Verlinde formula, and it is a key result in conformal field theory and topological quantum field theory. Note that by definition, $N_{\mu\nu}^\lambda$ are non-negative integers, and there are some elementary properties:

- Commutativity: $N_{\mu\nu}^\lambda = N_{\nu\mu}^\lambda$, this corresponds to say that fusion is independent of the order of the two reps, or by denoting anyons as μ and ν, we have $\mu\times\nu = \nu\times\mu$.
- Time-reversal: $N_{\mu\nu}^\lambda = (N_{\mu\nu}^\lambda)^* = N_{\bar\mu\bar\nu}^{\bar\lambda}$ with $\bar\mu$ refers to complex conjugation of the irrep $T^{(\mu)}$.
- Trivial fusion: $N_{\mu 0}^\lambda = \delta_{\lambda\mu}$, and this follows from the unitarity, explained as unitary evolution of a quantum state.
- Uniqueness of inverse: $N_{\mu\nu}^0 = \delta_{\mu\bar\nu}$ with $S_{\bar\mu i} = (S_{\nu i})^*$ understood, and this is explained as the anti-particle of anyon is unique.
- Associativity: $\sum_\gamma N_{\mu\nu}^\gamma N_{\gamma\delta}^\lambda = \sum_\gamma N_{\nu\delta}^\gamma N_{\mu\gamma}^\lambda$, and this means that fusion result does not depend on the order of fusion, or by denoting anyons as μ, ν, δ, we have $(\mu\times\nu)\times\delta = \mu\times(\nu\times\delta)$.

The proofs of the first four properties are straightforward, while the last one is a bit more involved as

$$\sum_\gamma N_{\mu\nu}^\gamma N_{\gamma\delta}^\lambda = \sum_\gamma\sum_{ij}\frac{S_{\mu i}S_{\nu i}S_{\gamma i}^*}{S_{0i}}\frac{S_{\gamma j}S_{\delta j}S_{\lambda j}^*}{S_{0j}}$$
$$= \sum_{ij}\frac{S_{\mu i}S_{\nu i}}{S_{0i}}\frac{S_{\delta j}S_{\lambda j}^*}{S_{0j}}\delta_{ij} = \sum_i\frac{S_{\mu i}S_{\nu i}S_{\delta i}}{(S_{0i})^2},$$

which is the same for RHS.

1.3 $SU(2)$ Representation

The regular representation of the $SU(2)$ is decomposed as

$$L^2(SU(2)) = \bigoplus_{j\in\frac{1}{2}\mathbb{Z}^+} \bar{D}^j \otimes D^j, \tag{1.45}$$

with D^j irrep of $SU(2)$ with matrix dimension $d_j = 2j+1$ and $\bar{D}^j$ the dual representation. The $j = \frac{1}{2}$ irrep is the spin-$\frac{1}{2}$ representation, which is the fundamental representation of $SU(2)$, and is given by

$$g = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \ \det(g) = |\alpha|^2 + |\beta|^2 = 1, \tag{1.46}$$

with the inverse given by $g^{-1} = g^\dagger = \begin{pmatrix} \bar{\alpha} & \bar{\beta} \\ -\beta & \alpha \end{pmatrix}$.

1.3.1 Homogeneous Polynomials

A concrete example to grasp $SU(2)$ is to consider its induced group action over the homogenous polynomials in two variables u, v of degree k:

$$\mathcal{H}_k = \{\psi : \psi(\lambda u, \lambda v) = \lambda^k \psi(u, v)\},$$

with basis functions given by the monomials $\psi_l(u, v) = u^l v^{k-l}$, with $l = 0, 1, \ldots, k$. The two variable space spanned by (u, v) naturally carries an $SU(2)$ rep, which allows us to write down the induced (left-)group action. However, for now we only consider a more general $GL(2, \mathbb{C})$ action, which reads

$$\begin{aligned}(g \cdot \psi_l)(u, v) &= \psi_l \left(g^{-1} \begin{pmatrix} u \\ v \end{pmatrix} \right) \\ &= \psi_l \left(\det(g)^{-1} (g_{22}u - g_{12}v, -g_{21}u + g_{11}v) \right) \\ &= \det(g)^{-k} (g_{22}u - g_{12}v)^l (-g_{21}u + g_{11}v)^{k-l}.\end{aligned}$$

Note that by expanding the polynomial into the monomial basis, we get the representation of $GL(2, \mathbb{C})$ over $\mathcal{H}_k$, a $k+1$ dimensional vector space. In spirit of induced rep, the subgroup generating the whole rep can be defined as those with eigen-action over $\mathcal{H}_k$, which turns out to be

$$H = \left\{ h \middle| h = \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix}, \ a \in \mathbb{C}^* \right\}.$$

This gives that

$$(h \cdot \psi_l)(u, v) = a^{k-2l} \psi_l(u, v),$$

and we identify ψ_l as eigenvectors of H with eigenvalues a^{k-2l}. This fulfils the requirement of induced rep as we can identify $(g, \psi_l) \sim (gh, h^{-1} \cdot \psi_l)$.

Now we restrict $\mathrm{Ind}_H^{GL(2,\mathbb{C})}(\mathcal{H}_k)$ to that of $SU(2)$, and since $SU(2) \cap H = U(1)$, we are led to $\mathrm{Ind}_{U(1)}^{SU(2)}(\mathcal{H}_k)$. Let us now choose $k = 2j$, and basis $\tilde{f}_m(u, v) = u^{j+m} v^{j-m}$, with $m = -j, -j+1, \ldots, j$. For $g \in SU(2)$, we have

$$\begin{aligned}(g \cdot \tilde{f}_m)(u, v) &= \tilde{f}_m(\bar{\alpha}u + \bar{\beta}v, -\beta u + \alpha v) \\ &= (\bar{\alpha}u + \bar{\beta}v)^{j+m} (-\beta u + \alpha v)^{j-m} \\ &\equiv \sum_{m'} \tilde{D}^j_{m'm}(g) \tilde{f}_{m'}(u, v),\end{aligned} \tag{1.47}$$

or we extract the matrix group action over the carrier space $\mathcal{H}_k$ as

$$g \cdot \tilde{f}_m = \sum_{m'} \tilde{D}^j_{m'm}(g) \tilde{f}_{m'}, \tag{1.48}$$

with explicitly by binomial expansion

$$\tilde{D}^j_{m'm}(g) = \sum_{s+t=j+m'} \binom{j+m'}{s} \binom{j-m}{t} \bar{\alpha}^s \alpha^{j-m-t} \bar{\beta}^{j+m-s} (-\beta)^t. \tag{1.49}$$

Notice that j is an extra label for different irreps. The induced rep of this type can correspond to a charged particle bounded to a magnetic monopole.

Using equivariant group action, the carrier space of the induced representation can be written as

$$\psi(g e^{-i\chi\sigma_3}) = e^{-ik\chi}\psi(g), \tag{1.50}$$

where the group function $\psi : SU(2) \to \mathbb{C}$, and the $U(1)$ subgroup is generated by σ_3. The explicit form of the induced representation is given by

$$\begin{aligned}
\langle j, m_L | T^{(j)}(g) | j, m_R \rangle &\equiv D^j_{m_L m_R}(\phi, \theta, \psi) \\
&= \langle j, m_L | \, T^{(j)}(e^{-i\frac{1}{2}\phi\sigma_3}) T^{(j)}(e^{-i\frac{1}{2}\theta\sigma_2}) T^{(j)}(e^{-i\frac{1}{2}\psi\sigma_3}) \, | j, m_R \rangle \\
&= e^{-im_L\phi} P^j_{m_L m_R}(\cos\theta) e^{im_R\psi},
\end{aligned}$$

with $P^j_{m_L m_R}(\cos\theta)$ some associated Legendre polynomial known as the (reduced) Wigner (small) d-matrix, and D here is proportional to $\tilde{D}$ above. The j-rep is then spanned by the D-functions, constructing a subspace of equivariant functions with $k = -2m_R$. And one orthonormal basis can be chosen as $D^j_{m_L,-k/2}$, with $m_L = -j, -j+1, \ldots, j$. For this to stand, one recognizes

$$\mathrm{Ind}^{SU(2)}_{U(1)}(\mathcal{H}_k) \cong \bigoplus_{j \geq |k|/2} \bar{D}^j.$$

The conjugate is introduced since under LRR

$$\begin{aligned}
(L(g_0) \cdot D^j_{m_L m_R})(g) = D^j_{m_L m_R}(g_0^{-1} g) &= \langle j, m_L | T^{(j)}(g_0^{-1}) T^{(j)}(g) | j, m_R \rangle \\
&= \sum_{m'} (D^j(g_0^{-1}))_{m_L m'} D^j_{m' m_R}(g) = \sum_{m'} \left(D^j_{m' m_L}(g_0)\right)^* D^j_{m' m_R}(g).
\end{aligned}$$

Especially, for $k = 0$, we identify a trivial bundle without magnetic monopole, and the section is directly composed of complex-valued L^2 functions on S^2, which corresponds to the spherical harmonics $D^l_{m,0}(\phi, \theta, \psi) = Y_{l,m}(\theta, \phi)$, and ψ does not appear in the spherical harmonics.

1.3.2 Matrix reps of $\mathfrak{su}(2)$

We can deduce the matrix representation of $\mathfrak{su}(2)$ (Lie algebra) from the $SU(2)$ representation (Lie group). The idea is to consider the infinitesimal action of $SU(2)$ on the space of homogeneous polynomials $\mathcal{H}_k$. Assuming exponential map $\exp : \mathfrak{su}(2) \to SU(2)$, we let $g = \exp(\epsilon X)$, and as $\epsilon \to 0$,

$$T(e^{\epsilon X}) \cdot f = T(1 + \epsilon X) \cdot f = (T(1) + \epsilon T(X)) \cdot f = f + \epsilon (T(X) \cdot f). \tag{1.51}$$

Note that the third expression directly comes from the beginning, where we have used the fact that exponential map is a group homomorphism and commutes with representation, i.e., $T(\exp(X)) = \exp(dT(X))$, where $dT(X)$ is the differential

$$dT(X) = \frac{\mathrm{d}}{\mathrm{d}t} T(\exp(tX))\bigg|_{t=0},$$

which is basically the $T(X)$ above. To see this, let us calculate

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t} T(\exp(tX)) &= \lim_{h\to 0} \frac{T(\exp((t+h)X)) - T(\exp(tX))}{h} \\ &= T(\exp(tX)) \lim_{h\to 0} \frac{\exp(hX) - 1}{h} \\ &= T(\exp(tX)) dT(X),\end{aligned}$$

where the homomorphism property of T is used in the second line. This gives us a differential equation that solves

$$T(\exp(tX)) = \exp(t dT(X)),$$

and QED.

Now the above formula applies for any $X \in \mathfrak{su}(2)$, and firstly we choose $X = X_3 = \frac{i\sigma_3}{2}$, which leads to

$$\begin{aligned}T(1+\epsilon X_3)\cdot f(u,v) &= f\left((1-\epsilon X_3)\begin{pmatrix}u\\v\end{pmatrix}\right) = f(u - i\frac{1}{2}\epsilon u, v + i\frac{1}{2}\epsilon v) \\ &= f - \epsilon\frac{i}{2}\left(u\frac{\partial}{\partial u} - v\frac{\partial}{\partial v}\right) f(u,v) = f + \epsilon(T(X_3)\cdot f),\end{aligned}$$

which leads to

$$T(X_3)\cdot f = \frac{i}{2}\left(u\frac{\partial}{\partial u} - v\frac{\partial}{\partial v}\right) f. \tag{1.52}$$

In particular, for $f = \tilde{f}_m(u,v)$, we have

$$iT(X_3)\cdot \tilde{f}_m(u,v) = \frac{i}{2}\Big((j+m)\tilde{f}_m(u,v) - (j-m)\tilde{f}_m(u,v)\Big) = m\tilde{f}_m(u,v),$$

i.e., $T(X_3)$ is diagonal in the $\tilde{f}_m$ basis, with eigenvalues m. Similarly, we can calculate the action of $X_{1,2} = i\sigma_{1,2}/2$ by

$$\begin{aligned}T(X_1)\cdot f(u,v) &= f\left((1-\epsilon X_1)\begin{pmatrix}u\\v\end{pmatrix}\right) = f(u - i\frac{1}{2}\epsilon v, v - i\frac{1}{2}\epsilon u) \\ &= f - \epsilon\frac{i}{2}\left(v\frac{\partial}{\partial u} + u\frac{\partial}{\partial v}\right) f = f + \epsilon(T(X_1)\cdot f), \\ T(X_2)\cdot f(u,v) &= f\left((1-\epsilon X_2)\begin{pmatrix}u\\v\end{pmatrix}\right) = f(u - \frac{1}{2}\epsilon v, v + \frac{1}{2}\epsilon u) \\ &= f - \epsilon\frac{1}{2}\left(v\frac{\partial}{\partial u} - u\frac{\partial}{\partial v}\right) f = f + \epsilon(T(X_2)\cdot f),\end{aligned}$$

which gives us the action of $T(X_{1,2})$ as

$$T(X_1)\cdot f = -\frac{i}{2}\left(v\frac{\partial}{\partial u} + u\frac{\partial}{\partial v}\right) f, \tag{1.53a}$$

$$T(X_2)\cdot f=-\frac{1}{2}\left(v\frac{\partial}{\partial u}-u\frac{\partial}{\partial v}\right)f. \tag{1.53b}$$

By defining $J^{\pm}=iT(X_1)\pm T(X_2)$, one c an verify that

$$J^{+}\cdot\tilde{f}_m=(j-m)\tilde{f}_{m+1},$$
$$J^{-}\cdot\tilde{f}_m=(j+m)\tilde{f}_{m-1},$$

or in other words, $J^{\pm}$ are raising and lowering operators, and in matrix form, they are given by

$$J^{+}=\begin{pmatrix}0&1&0&\cdots&0\\0&0&2&\cdots&0\\\vdots&\vdots&\ddots&\ddots&\vdots\\0&0&\cdots&0&2j\\0&0&\cdots&0&0\end{pmatrix},\ J^{-}=\begin{pmatrix}0&0&\cdots&0&0\\2j&0&\cdots&0&0\\\vdots&\ddots&\ddots&\vdots&\vdots\\0&0&\cdots&0&0\\0&0&\cdots&1&0\end{pmatrix},$$

which differs from the standard rep of $\mathfrak{su}(2)$, as this is not a unitary representation.

1.4 Lorentz Group

Now we directly introduce the Lorentz group in d dimensional spacetime as $O(1,d-1)$, with natural representation as Lorentz transformations on the Minkowski space $\mathbb{M}^{1,d-1}$, which is the vector space of d-dimensional vectors with one time-like and $d-1$ space-like components. The Lorentz group consists of all linear transformations that preserve the Minkowski metric, which can be expressed as:

$$\Lambda^T\eta\Lambda=\eta, \tag{1.54}$$

where $\eta=\mathrm{diag}(1,-\mathbf{1})$. In the convention $x^\mu=(t,\boldsymbol{x})$.

We may introduce a left group action over scalar fields (i.e., functions with a single component) on the Minkowski space $\psi:\mathbb{M}\to\kappa$, where κ is a number field, as

$$\phi(\Lambda,\psi)(x)\equiv\tilde{\psi}(x)=\psi(\Lambda^{-1}x). \tag{1.55}$$

If we consider more complicate fields with inner structure, like vector or spinor fields with multi-components ψ^α, we are lifting the representation of base Minkowski space to a bundle with fibers as vectors or spinors, thus forcing us to consider induced rep as

$$\phi(\Lambda,\psi^\alpha)(x)\equiv\tilde{\psi}^\alpha(x)=D(\Lambda)^\alpha{}_\beta\psi^\beta(\Lambda^{-1}x), \tag{1.56}$$

where we exploit the matrix rep $D(\Lambda)$ of the Lorentz group, which is a representation of the Lorentz group in the space of vectors or spinors.

To proceed, we claim that for V a carrier space of G, and $\mathcal{O}\in\mathrm{End}(V)$ an invariant operator satisfying $\rho(g)\mathcal{O}=\mathcal{O}\rho(g)$ for all $g\in G$, then any eigenspace like $\{v\in V|\mathcal{O}v=\lambda v\}$ (and their direct sum) composes a sub-rep of G. This is visible by block-diagonalize $\mathcal{O}$ and all $\rho(g)$ simultaneously, and in each eigenspace $\mathcal{O}$ acts as a scalar (λ times identity) by Schur lemma, thus can commute with all $\rho(g)$.

For the Lorentz group, one such most important operator is the d'Alembertian operator form by derivatives $\partial_\mu=(\partial_t,\nabla)$,

$$\partial^\mu\partial_\mu=\partial_t^2-\nabla^2,$$

which acts on a scalar field and gives the Klein-Gordon equation

$$(\partial^\mu \partial_\mu + m^2)\psi(x) = 0. \tag{1.57}$$

The solution of the equation is obtained by Fourier transformation

$$\psi(x) = \int \mathrm{d}^d k \ \tilde{\psi}(k) e^{-ik\cdot x},$$

which leads to the dispersion relation

$$(k^2 - m^2)\tilde{\psi}(k) = 0, \tag{1.58}$$

where we understand $k^2 = k^\mu k_\mu = k_0^2 - \boldsymbol{k}^2$.

In physics, the orbit with $k_0 > 0$ is called the mass-shell. At the same time, we assume $m^2 > 0$, and the basic solution is

$$\tilde{\psi}(k) = \bar{a}\delta(k^2 - m^2),$$

where $\bar{a}$ is the amplitude of the solution orbit, and δ the Dirac delta function. The general solution is then

$$\psi(x) = \int \mathrm{d}^d k \ \delta(k^2 - m^2)\bar{a}e^{ik\cdot x} = \int_{\mathbb{R}^{d-1}} \frac{\mathrm{d}^{d-1}\boldsymbol{k}}{2\sqrt{\boldsymbol{k}^2 + m^2}} \ \bar{a}e^{i\sqrt{\boldsymbol{k}^2+m^2}t}e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}, \tag{1.59}$$

where we have performed the integral by letting $f(k_0) = k_0^2 - \boldsymbol{k}^2 - m^2$, and

$$\int \mathrm{d}k_0 \delta(f(k_0)) = \frac{1}{|f'(k_0)|}\bigg|_{f(k_0)=0} = \frac{1}{2k_0}\bigg|_{k_0=\sqrt{\boldsymbol{k}^2+m^2}} = \frac{1}{2\sqrt{\boldsymbol{k}^2+m^2}}.$$

Note that both $\mathrm{d}^d k \ \delta(k^2 - m^2)$ and $\mathrm{d}^{d-1}\boldsymbol{k} \ \delta(\sqrt{\boldsymbol{k}^2 + m^2} - k_0)$ are Lorentz invariant measures, and the solution is Lorentz invariant.

Now assume that we have two solutions $\psi_1(x)$ and $\psi_2(x)$, and for physical effects or operations, we expect an inner product well-defined (i.e., Lorentz invariant, or the action of Lorentz group over solution space is unitary) between them. For this, we define an inter-current

$$j_\mu(\psi_1, \psi_2) = -i(\psi_1^* \partial_\mu \psi_2 - (\partial_\mu \psi_1)^* \psi_2), \tag{1.60}$$

which satisfies the continuity/conservation equation

$$\begin{aligned} \partial^\mu j_\mu &= -i\left[(\partial^\mu \psi_1)^* \partial_\mu \psi_2 + \psi_1^* \partial^\mu \partial_\mu \psi_2 - (\partial^\mu \partial_\mu \psi_1)^* \psi_2 - (\partial_\mu \psi_1)^* (\partial^\mu \psi_2)\right] \\ &= -i\left(\psi_1^* m^2 \psi_2 - m^2 \psi_1^* \psi_2\right) = 0, \end{aligned}$$

or equivalently,

$$\partial_t \rho(\psi_1, \psi_2) + \nabla \cdot \boldsymbol{j}(\psi_1, \psi_2) = 0, \tag{1.61}$$

or in integral form by Stokes' theorem (or Gauss/divergence theorem here)

$$\partial_t Q(\psi_1, \psi_2) = -\int_{\partial V} \boldsymbol{j}(\psi_1, \psi_2) \cdot \mathrm{d}\boldsymbol{S} = 0, \tag{1.62}$$

which means that the total charge Q is conserved in time. Thus, we define the inner product as

$$\begin{aligned} (\psi_1, \psi_2) &= \int_{\mathbb{R}^{d-1}} \mathrm{d}^{d-1}\boldsymbol{x} \ j^0(\psi_1, \psi_2) = -i \int_{\mathbb{R}^{d-1}} \mathrm{d}^{d-1}\boldsymbol{x} \ (\psi_1^* \partial_t \psi_2 - (\partial_t \psi_1)^* \psi_2) \\ &= \int_{\mathbb{R}^{d-1}} \frac{\mathrm{d}^{d-1}\boldsymbol{k}}{2\sqrt{\boldsymbol{k}^2 + m^2}} \ (\bar{a}_1(\boldsymbol{k}))^* \bar{a}_2(\boldsymbol{k}). \end{aligned} \tag{1.63}$$

Note that more generally, we can define the inner product as

$$(\psi_1, \psi_2) = \int_\Sigma \text{vol } n^\mu j_\mu, \tag{1.64}$$

where Σ is a spacelike hypersurface with unit normal vector n^μ, and vol is the volume element on Σ, and the above definition equals to choose $\Sigma = \mathbb{R}^{d-1} \times \{t = 0\}$ with $n^\mu = (1, 0, \ldots, 0)$. The general choice then corresponds to a Lorentz transformation of the hypersurface, a change of coordinates in the Minkowski space.

To get more insight, let us classify the orbits/solutions of the Klein-Gordon equation by the mass m.

- For $m^2 > 0$, we define particle mass-shell

  $$\mathcal{O}^+(m) = \{k | k^2 - m^2 = 0 \ \& \ \text{sgn}(k_0) = \text{sgn}(m)\}, \tag{1.65}$$

  and since the stabilizer subgroup of this orbit (by considering a point $k^\mu = (m, \mathbf{0})$) is $SO(d-1)$, we identify

  $$\mathcal{O}^+(m) \cong SO(1, d-1)/SO(d-1).$$

  By considering the other branch with $\text{sgn}(k_0) = -\text{sgn}(m)$ (particle-hole or chiral conjugate), we obtain full dispersion as the massive Dirac cone.

- For $m^2 < 0$ we let $\mu^2 = -m^2 > 0$ and define

  $$\mathcal{O}^-(\mu) = \{k | k^2 + \mu^2 = 0\}, \tag{1.66}$$

  and since the stabilizer subgroup of this orbit (by considering a point $k^\mu = (0, 0, \cdots, 0, \mu)$) is $SO(1, d-2)$, we identify

  $$\mathcal{O}^-(\mu) \cong SO(1, d-1)/SO(1, d-2).$$

  In this case, there is only one branch of the dispersion relation as a hyperboloid, with its minimal radius as μ.

- For $m^2 = 0$, we have

  $$\mathcal{O}^\pm = \{k | k^2 = 0 \ \& \ \text{sgn}(k_0) = \pm 1\}. \tag{1.67}$$

  We note that vectors in this orbit are null vectors $k = |n|(\pm 1, \hat{n})$ with $\hat{n} \in S^{d-2}$ parametrizing the direction of light rays. The stabilizer subgroup of one light ray is $\mathcal{I} = \mathbb{E}^{d-2}$, the Euclidean group in $d-2$ dimensions, and we identify the orbit as

  $$\mathcal{O}^\pm \cong SO(1, d-1)/\mathcal{I}.$$

  The dispersion gives the massless Dirac cone but without the Dirac point.

- The final orbit is the $\{k = 0\}$ point in the $m^2 = 0$ case, the Dirac point.

A special case is the $d = 2$ spacetime with one spatial dimension, where by denoting mass-shell $k = m(\cosh\theta, \sinh\theta)$, we have

$$\begin{aligned}\psi(x) &= \int_{\mathbb{R}} \frac{\mathrm{d}k^1}{2\sqrt{(k^1)^2+m^2}}\ \bar{a}(k^1)e^{i\sqrt{(k^1)^2+m^2}t}e^{-ik^1\cdot x^1} \\ &= \frac{1}{2}\int_{-\infty}^{\infty}\mathrm{d}\theta\ \bar{a}(\theta)e^{im(\cosh\theta t - m\sinh\theta x^1)},\end{aligned}$$

where we have brought $k^1 = m\sinh\theta$ into the integral. It is clear that the solution is identified with the orbit of the Lorentz group $SO(1,1)$. Furthermore, by writing

$$\partial^\mu\partial_\mu = \partial_0^2 - \partial_1^2 = 4\partial_+\partial_-,$$

where $\partial_\pm = \partial/\partial x^\pm = \frac{1}{2}(\partial_0 \mp \partial_1)$, we identify that the massless Klein-Gordon equation contains two decoupled equations

$$\begin{aligned}\partial_+\psi(x) = 0,\\ \partial_-\psi(x) = 0,\end{aligned}$$

which enlightens us to write the general solution as

$$\Psi(x) = \psi_L(x^+) + \psi_R(x^-),$$

where ψ_L and ψ_R are left-moving and right-moving modes, respectively. Such on-shell scalar fields are called **chiral scalars** or **chiral bosons**, and they are the basic building blocks of the chiral algebra in 2D conformal field theory (CFT).

1.5 Group Algebra

1.5.1 Algebra

Definition 1.5.1. *An algebra over a field κ is a vector space V equipped with a bilinear map $V \times V \to V$ known as multiplication $v_1 \cdot v_2 \in V$ (vector space with ring structure).*

- The algebra has a ring structure compatible with scalar multiplication by the field

$$\begin{aligned}&1.\ (v_1+v_2)\cdot v_3 = v_1\cdot v_3 + v_2\cdot v_3\\ &2.\ v_1\cdot(v_2+v_3) = v_1\cdot v_2 + v_1\cdot v_3\\ &3.\ \alpha(v_1\cdot v_2) = (\alpha v_1)\cdot v_2 = v_1\cdot(\alpha v_2),\ \text{where } \alpha\in\kappa.\end{aligned}$$

- A unital algebra means there exists an element $1_V \in V$ such that $1_V \cdot v = v \cdot 1 = v$ for all $v \in V$.
- An associative algebra means $(v_1 \cdot v_2)\cdot v_3 = v_1\cdot(v_2\cdot v_3)$ for all $v_1, v_2, v_3 \in V$.
- Generally, if $\{v_i\}$ is a basis for the algebra, then the structure constants c_{ij}^k are defined by

$$v_i \cdot v_j = \sum_k c_{ij}^k v_k. \tag{1.68}$$

A representation of an algebra is a linear map $T : V \to \mathrm{End}(W)$, where W is a vector space, such that

$$T(\alpha v + \beta w) = \alpha T(v) + \beta T(w), \tag{1.69a}$$
$$T(v \cdot w) = T(v)T(w). \tag{1.69b}$$

For A an algebra, then it is a module (replace field κ by a ring) over itself via LRR $a \to L(a)$ where $L(a) \cdot b = a \cdot b$. And if we choose $\{a_i\}$ as basis with structure constants c_{ij}^k, then $L(a)$ has a matrix rep with elements $(L(a_i))_j{}^k = c_{ij}^k$. This is done by considering the for an operator acting on a vector that $L(a_i)a_j = \sum_k L(a_i)_j{}^k a_k$.

1.5.2 Group Algebra

To give group algebra, we first give the *group ring* $\mathbb{Z}[G]$ as

$$\mathbb{Z}[G] = \{x = \sum_{g \in G} x_g g : x_g \in \mathbb{Z}\}. \tag{1.70}$$

The addition is purely given by the addition of coefficients, and for multiplication we define

$$g_1 *_2 g_2 = g_1 \cdot g_2, \ x_{g_1} *_2 x_{g_2} = x_{g_1} x_{g_2}. \tag{1.71}$$

Note that $*_2$ is the multiplication law on ring $\mathbb{Z}[G]$, but degrades into the group multiplication law on G and usual multiplication law on $\mathbb{Z}$, and their product is considered as a tensorial one. Thus totally, we have

$$\begin{aligned} x *_2 y &= \left(\sum_g x_g g\right) *_2 \left(\sum_h y_h h\right) = \sum_{g,h} (x_g y_h)(g \cdot h) \\ &= \sum_k \left(\sum_{g \cdot h = k} x_g y_h\right) k. \end{aligned}$$

Now we replace $\mathbb{Z}$ by complex field $\mathbb{C}$, and we have the *group algebra* $\mathbb{C}[G]$. Basically, it is a vector space over $\mathbb{C}$ with basis G, but it has a ring structure with multiplication defined as above.

Now consider the carrier space of regular rep $R_G = \mathrm{Map}(G \to \mathbb{C})$, which is also an algebra over $\mathbb{C}$ with natural multiplication defined as

$$(f_1 *_1 f_2)(g) = f_1(g) f_2(g).$$

Naturally, we see that R_G is the dual space of $\mathbb{C}[G]$ by inner product

$$\langle f, x \rangle = \sum_g x_g f(g), \tag{1.72}$$

and the correspondingly, there should be an isomorphism between them. The element correspondence is given by $\delta_g \leftrightarrow g$ and takes $f = \sum_g f(g)\delta_g \in R_G$ to $x_f = \sum_g f(g) g$. However, for multiplicative law, we find that if we transit $*_1$ from R_G to $\mathbb{C}[G]$, we have

$$\begin{aligned} f_1 *_1 f_2(g) &= f_1(g) f_2(g) \\ \Longrightarrow \delta_{g_1} *_1 \delta_{g_2} &= \delta_{g_1, g_2} \delta_{g_1} \\ \Longrightarrow g_1 *_1 g_2 &= \delta_{g_1, g_2} g_1, \end{aligned}$$

and clearly, this is not the same as $g_1 *_2 g_2 = g_1 \cdot g_2$, and is unnatural. Thus, if we truly want the isomorphism, we should transit $*_2$ from $\mathbb{C}[G]$ to R_G indeed, and that corresponds to the *convolution product* defined as

$$(f_1 *_2 f_2)(g) = \frac{1}{|G|}\sum_{h\in G} f_1(h)f_2(h^{-1}g). \tag{1.73}$$

To see how this works, again we consider the basis

$$\begin{aligned}(\delta_{g_1} *_2 \delta_{g_2})(g) &= \sum_h \delta_{g_1}(h)\delta_{g_2}(h^{-1}g) = \sum_h \delta_{g_1,h}\delta_{g_2,h^{-1}g}\\ &= \delta_{g_2,g_1^{-1}g} = \delta_{g_1g_2,g},\end{aligned}$$

which is the same as $g_1 *_2 g_2 = g_1 \cdot g_2$.

For instance, for $G = \mathbb{Z}_{\not=} = \{1, \sigma\}$, we have

$$R_{\mathbb{Z}_2} \cong \mathbb{C}[\mathbb{Z}_2] = \{a + b\sigma, a, b \in \mathbb{C}\},$$

with ring structure

$$(a_1 + b_1\sigma) *_2 (a_2 + b_2\sigma) = a_1a_2 + b_1b_2 + (a_1b_2 + a_2b_1)\sigma.$$

Center of group algebra and subalgebra as class functions

As generally, group multiplication is not commutative, the group algebra $\mathbb{C}[G]$ generated by $*_2$ is not commutative followed. The isomorphism between $\mathbb{C}[G]$ and R_G then tells that the convolution product $*_2$ is not commutative either.

There is a subalgebra of $\mathbb{C}[G]$ that is commutative, which is the *center* of $\mathbb{C}[G]$ denoted as $Z[\mathbb{C}[G]]$. Identifying $*_2$ with group multiplication, this requires commutation of the center element with all elements of $\mathbb{C}[G]$, i.e., with all group elements. The expression satisfying this is the sum of group elements in a conjugacy class $c_i = \sum_{g\in C_i} g$, since any conjugate operation $g' = hgh^{-1} \in C_i$ and is unique for each $h \in G$ (but note that this is a bad notation, as there is no multiplicative inverse in group algebra, we should write $hc_i = c_ih$).

Correspondingly, there exists an algebra center $Z[R_G]$ isomorphic to $Z[\mathbb{C}[G]]$, which consists of class functions on G. To see this we expect that $(f *_2 \chi)(g) = (\chi *_2 f)(g)$ for all $g \in G$, and

$$\begin{aligned}\text{LHS} &= \frac{1}{|G|}\sum_{h\in G} f(h)\chi(h^{-1}g) = \frac{1}{|G|}\sum_{k\in G} f(gk^{-1})\chi(k) = \frac{1}{|G|}\sum_i \chi(C_i)\sum_{k\in C_i} f(gk^{-1}),\\ \text{RHS} &= \frac{1}{|G|}\sum_{h\in G} \chi(h)f(h^{-1}g) = \frac{1}{|G|}\sum_i \chi(C_i)\sum_{h\in C_i} f(h^{-1}g)\\ &\overset{k^{-1}=g^{-1}h^{-1}g\in C_i}{=} \frac{1}{|G|}\sum_i \chi(C_i)\sum_{k\in C_i} f(gk^{-1}) = \text{LHS},\end{aligned}$$

where property of class function and conjugacy class is used. The fusion rules of class functions under convolution product $*_2$ is given by

$$\chi_\mu *_2 \chi_\nu = \frac{\delta_{\mu\nu}}{n_\mu}\chi_\mu, \tag{1.74}$$

as

$$(\chi_\mu *_2 \chi_\nu)(g) = \frac{1}{|G|}\sum_{h\in G}\chi_\mu(h)\chi_\nu(h^{-1}g) = \frac{1}{|G|}\sum_h\sum_{ijk}T^{(\mu)}_{ii}(h)T^{(\nu)}_{jk}(h^{-1})T^{(\nu)}_{kj}(g)$$
$$= \frac{1}{n_\mu}\delta_{\mu\nu}\sum_{ijk}\delta_{ij}\delta_{ik}T^{(\nu)}_{kj}(g) = \frac{\delta_{\mu\nu}}{n_\mu}\chi_\mu(g).$$

While in this case, the basis product of class functions is given by

$$\delta_{C_i} *_2 \delta_{C_j} = \sum_k M^k_{ij}\delta_{C_k}, \tag{1.75}$$

where the structure constants

$$M^k_{ij} = \frac{S_{0i}S_{0j}}{S_{0k}}\sum_\mu \frac{S^*_{\mu i}S^*_{\mu j}S_{\mu k}}{n_\mu}, \tag{1.76}$$

with $S_{\mu i} = \sqrt{\frac{m_i}{|G|}}\chi_\mu(C_i)$.

Let us firstly accept this result, and compare it with the pairwise product of class functions in the usual sense

$$\chi_\mu *_1 \chi_\nu = \sum_\lambda N^\lambda_{\mu\nu}\chi_\lambda, \tag{1.77a}$$
$$\delta_{C_i} *_1 \delta_{C_j} = \delta_{ij}\delta_{C_i}, \tag{1.77b}$$

and we see that under two kinds of product, the 'diagonal' function and 'fusion' function interchange their roles. This reminds us of the commutation relation between of position and momentum spaces, together with the Fourier transform. Thus, similarly here, we propose that χ_μ and δ_{C_i} are related by a (discrete) Fourier transform generated by the unitary matrix $S_{\mu i}$, i.e.,

$$\chi_\mu = \sum_i \sqrt{\frac{|G|}{m_i}}S_{\mu i}\delta_{C_i}, \tag{1.78a}$$
$$\delta_{C_i} = \sum_\mu \sqrt{\frac{m_i}{|G|}}S^*_{\mu i}\chi_\mu. \tag{1.78b}$$

The proof follows simply as

$$\chi_\mu(C_i) = \sum_j \sqrt{\frac{|G|}{m_j}}S_{\mu j}\delta_{C_j}(C_i) = \sum_j \chi_\mu(C_j)\delta_{ij} = \chi_\mu(C_i),$$
$$\delta_{C_i}(C_j) = \sum_\mu \sqrt{\frac{m_i}{|G|}}S^*_{\mu i}\chi_\mu(C_j) = \frac{m_i}{|G|}\sum_\mu \chi_\mu(C_j)^*\chi_\mu(C_i) = \delta_{ij}.$$

Thus we have

$$M^k_{ij} = (\delta_{C_i} *_2 \delta_{C_j})(C_k) = \frac{1}{|G|}\sum_{h\in G}\delta_{C_i}(h)\delta_{C_j}(h^{-1}C_k) = \frac{1}{|G|}\sum_l \delta_{C_i}(C_l)\sum_{h\in C_l}\delta_{C_j}(h^{-1}C_k)$$
$$= \frac{1}{|G|}\sum_{h\in C_i}\delta_{C_j}(h^{-1}C_k) = \frac{1}{|G|}\sum_{h\in C_i}\sum_\mu \sqrt{\frac{m_j}{|G|}}S^*_{\mu j}\chi_\mu(h^{-1}C_k)$$
$$= \frac{1}{|G|}\sum_\mu \sqrt{\frac{m_j}{|G|}}S^*_{\mu j}\,\mathrm{Tr}\left(T^{(\mu)}(C_k)\sum_{h\in C_i}T^{(\mu)}(h^{-1})\right),$$

and by noticing that $\sum_{h\in C_i} T^{(\mu)}(h^{-1})$ is over the conjugacy class and is thus G-invariant (over the (μ) rep), then by Schur's lemma, it is proportional to the identity operator, i.e.,

$$\sum_{h\in C_i} T^{(\mu)}(h^{-1}) = \frac{m_i}{n_\mu}\chi_\mu(C_i)^*\mathbb{I},$$

where the coefficients is easily seen by taking the trace of both sides. Thus we have

$$\begin{aligned} M_{ij}^k &= \frac{1}{|G|}\sum_\mu \sqrt{\frac{m_j}{|G|}} S_{\mu j}^* \,\mathrm{Tr}\left(T^{(\mu)}(C_k)\frac{m_i}{n_\mu}\chi_\mu(C_i)\mathbb{I}\right) \\ &= \frac{1}{|G|}\frac{m_i}{n_\mu}\sum_\mu \sqrt{\frac{m_j}{|G|}} S_{\mu j}^*\chi_\mu(C_i)^*\chi_\mu(C_k) \\ &= \frac{S_{0i}S_{0j}}{S_{0k}}\sum_\mu \frac{S_{\mu i}^* S_{\mu j}^* S_{\mu k}}{n_\mu}. \end{aligned}$$

Note that since $\chi_\mu *_1 \chi_\nu = \sum_\lambda N_{\mu\nu}^\lambda \chi_\lambda$, then by LRR, we have

$$L(\chi_\mu)_\nu{}^\lambda = N_{\mu\nu}^\lambda,$$

by general algebra. Since pairwise product of class functions is commutative, we expect that the matrix $L(\chi_\mu)$ is simultaneously diagonalizable for different μ by the unitary matrix $S_{\mu i}$, i.e.,

$$D_\mu = S^\dagger L(\chi_\mu) S = \mathrm{diag}(\chi_\mu(C_1), \cdots, \chi_\mu(C_r)). \tag{1.79}$$

To see this, we note that

$$(S^\dagger N_\mu S)_{ij} = \sum_{\nu\lambda}(S^\dagger)_{i\nu}N_{\mu\nu}^\lambda S_{\lambda j} = \sum_{\nu\lambda}\sum_k \frac{S_{\nu i}^* S_{\mu k} S_{\nu k} S_{\lambda k}^* S_{\lambda j}}{S_{0k}} = \delta_{ij}\frac{S_{\nu i}}{S_{0i}} = \delta_{ij}\chi_\mu(C_i).$$

1.5.3 Group projector

The group algebra $\mathbb{C}[G]$ serves as a carrier space for the rep of $G_{\text{left}} \times G_{\text{right}}$, meaning that

$$T(g_1, g_2)\cdot x = g_1 x g_2^{-1},\ \forall g_1, g_2 \in G,\ x \in \mathbb{C}[G]. \tag{1.80}$$

This rep is highly reducible (as it is isomorphic to R_G), as we will see later.

Let us define the projection operator $P_{ij}^\mu \in \mathbb{C}[G]$ with $1 \le i, j \le n_\mu$ as

$$P_{ij}^\mu = \frac{n_\mu}{|G|}\sum_{g\in G}\hat{T}_{ij}^{(\mu)}(g)g, \tag{1.81}$$

where the dual rep satisfies $\hat{T}_{ij}^{(\mu)}(g) = T_{ji}^{(\mu)}(g^{-1})$. They are orthogonal projectors on group algebra by

$$P_{ij}^\mu P_{mn}^\nu = \delta_{\mu\nu}\delta_{jm}P_{in}^\mu, \tag{1.82}$$

followed by the orthonormal relation as

$$\begin{aligned}
P^{\mu}_{ij}P^{\nu}_{mn} &= \frac{n_\mu n_\nu}{|G|^2}\sum_{g,h}\hat{T}^{(\mu)}_{ij}(g)\hat{T}^{(\nu)}_{mn}(h)(g\cdot h)\\
&= \frac{n_\mu n_\nu}{|G|^2}\sum_{g,k}\hat{T}^{(\mu)}_{ij}(g)\hat{T}^{(\nu)}_{mn}(g^{-1}k)k\\
&= \frac{n_\mu n_\nu}{|G|^2}\sum_{g,k}\sum_{l}T^{(\mu)}_{ji}(g^{-1})T^{(\nu)}_{nl}(k^{-1})T^{(\nu)}_{lm}(g)k\\
&= \frac{n_\mu}{|G|}\sum_{k}\sum_{m}\delta_{\mu\nu}\delta_{jm}\delta_{il}T^{(\nu)}_{nl}(k^{-1})k\\
&= \delta_{\mu\nu}\delta_{jm}\frac{n_\mu}{|G|}\sum_{k}\hat{T}^{(\mu)}_{in}(k)k = \delta_{\mu\nu}\delta_{jm}P^{\mu}_{in}.
\end{aligned}$$

And we further have that

$$g\cdot P^{\mu}_{ij} = \sum_{s}T^{(\mu)}_{si}(g)P^{\mu}_{sj}, \tag{1.83a}$$

$$P^{\mu}_{ij}\cdot g^{-1} = \sum_{s}\hat{T}^{(\mu)}_{sj}(g)P^{\mu}_{is}. \tag{1.83b}$$

Similarly deduced by

$$\begin{aligned}
g\cdot P^{\mu}_{ij} &= \frac{n_\mu}{|G|}\sum_{h}T^{(\mu)}_{ji}(h^{-1})gh = \frac{n_\mu}{|G|}\sum_{k}T^{(\mu)}_{ji}(k^{-1}g)k\\
&= \frac{n_\mu}{|G|}\sum_{k}\sum_{s}T^{(\mu)}_{js}(k^{-1})T^{(\mu)}_{si}(g)k = \sum_{s}T^{(\mu)}_{si}(g)P^{\mu}_{sj},\\
P^{\mu}_{ij}\cdot g^{-1} &= \frac{n_\mu}{|G|}\sum_{h}T^{(\mu)}_{ji}(h^{-1})hg = \frac{n_\mu}{|G|}\sum_{k}T^{(\mu)}_{ji}(gk^{-1})k\\
&= \frac{n_\mu}{|G|}\sum_{k}\sum_{s}T^{(\mu)}_{sj}(g)T^{(\mu)}_{si}(k^{-1})k = \sum_{s}\hat{T}^{(\mu)}_{sj}(g)P^{\mu}_{is}.
\end{aligned}$$

And one sees how the projectors P^{μ}_{ij} are related to the representation $T^{(\mu)}$.

Decomposition of rep into irreps

The orthonormal relation of characters tells how many irreps are contained in a rep, while on the ground that $T^{(\mu)}_{ij}$ are known, we can decompose a rep into irreps by using the projectors P^{μ}_{ij}.

Now consider an arbitrary rep (T, V), we have linear operators from the projectors

$$\mathcal{P}^{\mu}_{ij} = T(P^{\mu}_{ij}) = \frac{n_\mu}{|G|}\sum_{g\in G}\hat{T}^{(\mu)}_{ij}(g)T(g), \tag{1.84}$$

with similar properties as P^{μ}_{ij}, i.e.,

$$\mathcal{P}^{\mu}_{ij}\mathcal{P}^{\nu}_{mn} = \delta_{\mu\nu}\delta_{jm}\mathcal{P}^{\mu}_{in}, \tag{1.85a}$$

$$T(g)\cdot\mathcal{P}^{\mu}_{ij} = \sum_{s}T^{(\mu)}_{si}(g)\mathcal{P}^{\mu}_{sj}, \tag{1.85b}$$

$$\mathcal{P}^{\mu}_{ij} \cdot T(g^{-1}) = \sum_s \hat{T}^{(\mu)}_{sj}(g)\mathcal{P}^{\mu}_{is}. \tag{1.85c}$$

Thus, we have that $\mathcal{P}^{\mu}_{jk}$ is a projector on V for each fixed μ, k, meaning that $\{\boldsymbol{\omega}_j\} = \{\mathcal{P}^{\mu}_{jk}\boldsymbol{v}\}_{j=1,\cdots,n_\mu}$ form an invariant subspace of V for any $\boldsymbol{v} \in V$ satisfying $\mathcal{P}^{\mu}_{kk}\boldsymbol{v} \neq 0$. This is easily seen by

$$T(g)\cdot\boldsymbol{\omega}_j = T(g)\cdot\mathcal{P}^{\mu}_{jk}\boldsymbol{v} = \sum_s T^{(\mu)}_{sj}(g)\mathcal{P}^{\mu}_{sk}\boldsymbol{v} = \sum_s T^{(\mu)}_{sj}(g)\boldsymbol{\omega}_s,$$

thus $T(g)$ maps $\boldsymbol{\omega}_j$ to the linear combination of $\{\boldsymbol{\omega}_s\}$, showing the invariance. Especially, this subspace transforms under $T(g)$ according to the irrep $T^{(\mu)}$, forming the μth carrier space.

For instance, let us decompose the 3-dim rep of S_3 in this way. Recall that the irreps of S_3 is given by

$$\begin{aligned}
&\mathbf{1}_+ : g \to 1\\
&\mathbf{1}_- : (1) \to 1,\ (12),(13),(23) \to -1,\ (123),(132) \to 1\\
&T^2 : (1) \to \begin{pmatrix}1&0\\0&1\end{pmatrix},\ (12) \to \begin{pmatrix}1/2&\sqrt{3}/2\\\sqrt{3}/2&-1/2\end{pmatrix},\ (13) \to \begin{pmatrix}1/2&-\sqrt{3}/2\\-\sqrt{3}/2&-1/2\end{pmatrix},\\
&\quad (23) \to \begin{pmatrix}-1&0\\0&1\end{pmatrix},\ (123) \to \begin{pmatrix}-1/2&-\sqrt{3}/2\\\sqrt{3}/2&-1/2\end{pmatrix},\ (132) \to \begin{pmatrix}-1/2&\sqrt{3}/2\\-\sqrt{3}/2&-1/2\end{pmatrix}.
\end{aligned}$$

And the 3-dim rep is given by

$$\begin{aligned}
T^3 : (1) &\to \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix},\ (12) \to \begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix},\ (13) \to \begin{pmatrix}0&0&1\\0&1&0\\1&0&0\end{pmatrix},\\
(23) &\to \begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix},\ (123) \to \begin{pmatrix}0&0&1\\1&0&0\\0&1&0\end{pmatrix},\ (132) \to \begin{pmatrix}0&1&0\\0&0&1\\1&0&0\end{pmatrix}.
\end{aligned}$$

Thus, we have the projectors

$$\begin{aligned}
\mathcal{P}^{\mathbf{1}_+}_{11} &= \frac{n_{\mathbf{1}_+}}{|S_3|}\sum_g \hat{T}^{\mathbf{1}_+}_{11}(g)T^3(g) = \frac{1}{6}\sum_g T^3(g) = \frac{1}{3}\begin{pmatrix}1&1&1\\1&1&1\\1&1&1\end{pmatrix},\\
\mathcal{P}^{\mathbf{1}_-}_{11} &= 0,\\
\mathcal{P}^{T_2}_{11} &= \frac{n_{T_2}}{|S_3|}\sum_g \hat{T}^{2}_{11}(g)T^3(g) = \begin{pmatrix}0&0&0\\0&1/2&-1/2\\0&-1/2&1/2\end{pmatrix},\\
\mathcal{P}^{T_2}_{21} &= \frac{n_{T_2}}{|S_3|}\sum_g \hat{T}^{2}_{21}(g)T^3(g) = \begin{pmatrix}0&1/\sqrt{3}&-1/\sqrt{3}\\0&-1/2\sqrt{3}&1/2\sqrt{3}\\0&-1/2\sqrt{3}&1/2\sqrt{3}\end{pmatrix}.
\end{aligned}$$

Therefore, we have the $\mathbf{1}_+$ invariant subspace spanned by

$$\mathcal{P}^{\mathbf{1}_+}_{11}\begin{pmatrix}x\\y\\z\end{pmatrix} = \frac{x+y+z}{3}\begin{pmatrix}1\\1\\1\end{pmatrix},$$

thus with $V_{\mathbf{1}_+} = \text{Span}\{\boldsymbol{\omega}_1 = (1,1,1)\} = \{(x,x,x) : x \in \mathbb{R}\}$. Together with the T_2 invariant subspace spanned by

$$\mathcal{P}_{11}^{T_2}\begin{pmatrix}x\\y\\z\end{pmatrix} = \frac{y-z}{2}\begin{pmatrix}0\\1\\-1\end{pmatrix},$$
$$\mathcal{P}_{21}^{T_2}\begin{pmatrix}x\\y\\z\end{pmatrix} = \frac{\sqrt{3}(y-z)}{2}\begin{pmatrix}2\\-1\\-1\end{pmatrix},$$

and we have $V_{T_2} = \text{Span}\{\boldsymbol{\omega}_1 = (0,1,-1), \boldsymbol{\omega}_2 = (2,-1,-1)\} = \{(x,y,z) : x+y+z = 0\}$. As a coincidence, since we have a 3-dim rep, the basis we find should be equivalent to the character table matrix S, which is also 3×3.

The above method is powerful but too tedious, and it would be better if there is a more straightforward way. In general $V \cong \oplus_\mu a_\mu T^{(\mu)}$, and for unitary rep, we have

$$\mathcal{P}_{ij}^{\mu} = \frac{n_\mu}{|G|}\sum_{g\in G}(T_{ij}^{(\mu)}(g))^* T(g),$$

and $(\mathcal{P}_{ji}^{\mu})^\dagger = \mathcal{P}_{ij}^{\mu}$ follows. Thus by defining

$$\mathcal{P}^{\mu} = \sum_{i=1}^{n_\mu}\mathcal{P}_{ii}^{\mu} = \frac{n_\mu}{|G|}\sum_{g\in G}\chi_\mu(g)^* T(g), \tag{1.86}$$

we get a set of orthogonal, Hermitian projectors $(\mathcal{P}^{\mu})^2 = \mathcal{P}^{\mu}$, $(\mathcal{P}^{\mu})^\dagger = \mathcal{P}^{\mu}$ and $\mathcal{P}^{\mu}\mathcal{P}^{\nu} = 0$ for $\mu \neq \nu$.

1.5.4 Block diagonalization of Hermitian operators

Assuming that the Hamiltonian is represented as $N \times N$ matrix acting on $\mathbb{C}^N$, and its symmetry group G is represented as $(T, \mathbb{C}^N)$, meaning that $[T(g), H] = 0\ \forall g \in G$. Then accordingly we have $[\mathcal{P}_{ij}^{\mu}, H] = 0\ \forall \mu, i, j$ and $[\mathcal{P}^{\mu}, H] = 0\ \forall \mu$. In this case, by reducing $\mathbb{C}^N$ into irreps of G, say by projecting onto them, we have partially (as there are possibly other symmetries) block-diagonalized H.

To do that, let $\{\psi_a\}$ the collection of functions that transform under G according to T, possibly the eigen-wavefunctions, and we can thus define projected basis functions as

$$\psi_{ij,a}^{\mu}(x) = \mathcal{P}_{ij}^{\mu}\cdot\psi_a(x) = \frac{n_\mu}{|G|}\sum_{h\in G}(T_{ij}^{\mu}(h))^*\psi_a(h^{-1}x), \tag{1.87}$$

with a left action. A direct calculation shows that by fixing μ, j, a, the indexed $i = 1, \cdots, n_\mu$ functions span an irrep subspace

$$\begin{aligned} T(g)\cdot\psi_a(x) = \psi_{ij,a}^{\mu}(g^{-1}x) &= \frac{n_\mu}{|G|}\sum_{h\in G}(T_{ij}^{\mu}(h))^*\psi_a(h^{-1}g^{-1}x) \\ &= \frac{n_\mu}{|G|}\sum_{h\in G}(T_{ij}^{\mu}(g^{-1}h))^*\psi_a(h^{-1}x) \\ &= \sum_{s=1}^{\nu} T_{si}^{\mu}(g)\psi_{sj,a}^{\mu}(x). \end{aligned}$$

As an example, let us consider $G = \mathbb{Z}_2$ acting on $\mathbb{R}$ with non-trivial element $\sigma : x \to -x$, thus $\mathcal{P}^{\mathbf{1}_+} = \frac{1}{2}(\mathbb{I} + \sigma)$, and $\mathcal{P}^{\mathbf{1}_-} = \frac{1}{2}(\mathbb{I} - \sigma)$, leading to

$$\psi_+(x) = \mathcal{P}^{\mathbf{1}_+} \cdot \psi(x) = \frac{1}{2}(\mathbb{I} + \sigma) \cdot \psi(x) = \frac{1}{2}(\psi(x) + \psi(-x)),$$
$$\psi_-(x) = \mathcal{P}^{\mathbf{1}_-} \cdot \psi(x) = \frac{1}{2}(\mathbb{I} - \sigma) \cdot \psi(x) = \frac{1}{2}(\psi(x) - \psi(-x)).$$

This symmetry analysis works not only for quantum system. For instance, consider classical system with DoFs $\boldsymbol{q} = (q_1, \cdots, q_n)$, as a generalized harmonic oscillator with

$$T = \frac{1}{2} m_{ij} \dot{q}_i \dot{q}_j, \ V = \frac{1}{2} U_{ij} q^i q^j,$$

with summation indicated, where m_{ij} is the mass matrix and U_{ij} is the potential matrix. The classical EoM is given by

$$0 = \frac{\delta L}{\delta q_i} = m_{ij} \dot{q}_j^2 + U_{ij} q^j,$$

and by $q^i(t) = \gamma^i e^{i\omega t}$, we obtain the linear set of equations

$$(-\omega^2 m_{ij} + U_{ij})\gamma^j = 0.$$

Now we assume that the system has symmetry group G, meaning that m_{ij} and U_{ij} are invariant under G action, i.e.,

$$[T(g), \mathbf{m}] = [T(g), \mathbf{U}] = 0,$$

making the effective Hamiltonian $H = -\omega^2 m + U$ invariant under G action. Concretely, we choose $m_{ij} = m\delta_{ij}$, $U_{ij} = k(2\delta_{ij} - \delta_{i,j\pm1})$ modulo n, meaning that we are considering a ring of coupled oscillators with nearest-neighbor coupling as

$$L = \sum_i \left(\frac{1}{2} m \dot{q}^i \dot{q}^i - \frac{1}{2} k (q^i - q^{i+1})^2 \right),$$

and the normal mode equation reads

$$(-m\omega^2 + kA)\gamma = 0, \ A = \begin{pmatrix} 2 & -1 & 0 & \cdots & -1 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \cdots & 2 \end{pmatrix}.$$

An obvious symmetry is given by $\mathbb{Z}_n$ with generator

$$T(\omega) : q^1 e_1 + \cdots q^n q_n \to q^1 e_2 + \cdots q^{n-1} e_n + q^n e_1,$$

i.e., a shift of basis $T(\omega^l)e_i = e_{i+l}$ modulo n, with $\omega = \exp(2\pi i/n)$. The irreps of $\mathbb{Z}_n$ are 1-dim and reads $T^{(\mu)}(\omega^j) = \omega^{j\mu}$, $\mu = 0, \cdots, n-1$. The projectors are thus

$$\mathcal{P}^\mu = \frac{1}{n} \sum_{l=0}^{n-1} \omega^{-\mu l} T(\omega^l) = \frac{1}{n} \sum_{i,l=0}^{n-1} \omega^{-\mu l} e_{i+l,i},$$

where we denote $e_{i+l,i}$ the shift operator which is basically matrix unity. Similarly,

$$A = 2 - \sum_i (e_{i,i+1} + e_{i+1,i}).$$

Now we project onto irreps, in the sense that $\mathcal{P}^\mu$ can somehow diagonalize A:

$$\begin{aligned}
A\mathcal{P}^\mu &= 2\mathcal{P}^\mu - \frac{1}{n}\sum_{i,j,l} \omega^{-\mu l}(e_{i,i+1} + e_{i+1,i})e_{j+l,j} \\
&= 2\mathcal{P}^\mu - \frac{1}{n}\sum_{i,l} \omega^{-\mu l}(e_{i,i+1-l} + e_{i+1,i-l}) \\
&= 2\mathcal{P}^\mu - \frac{1}{n}\sum_{i,l} \omega^{-\mu l}(e_{i+(l-1),i} + e_{i+(l+1),i}) \\
&= 2\mathcal{P}^\mu - \frac{1}{n}\sum_{i,l} \omega^{-\mu l}(e^{-\mu}e_{i+l,i} + e^{\mu}e_{i+l,i}) \\
&= (2 - \omega^{-\mu} - \omega^{\mu})\mathcal{P}^\mu = 4\sin^2(\pi\mu/n)\mathcal{P}^\mu,
\end{aligned}$$

and $\mathcal{P}^\mu$ project out basis wavefunctions/normal modes over $\boldsymbol{q}$ as

$$\mathcal{P}^\mu \boldsymbol{q} = \frac{1}{n}\sum_{i,l,s=0}^{n-1} \omega^{-\mu l} e_{i+l,i} q^s e_s = \frac{1}{n}\sum_{l,s}^{n-1} q_s \omega^{-\mu l} e_{s+l} = \frac{1}{n}\sum_s \omega^{\mu s} q^s \left(\sum_j \omega^{-\mu j} e_j\right),$$

thus separately give us eigenvalues/eigen-frequencies $\omega_\mu^2 = 4\dfrac{k}{m}\sin^2(\pi\mu/n)$ and eigenmodes $\boldsymbol{q}_\mu = \alpha_\mu e^{i\omega_\mu t}\sum_j \omega^{-\mu j} e_j$. Note the degeneracy $\omega_\mu = \omega_{n-\mu}$, which hints that we have a further symmetry to explain this, and here it is the dihedral group D_n. With a similar analysis, the two modes are classified according to their mirror-reflection properties.

1.6 Permutation Group

Different conjugate classes of S_n are labelled by partitions of n, that a cycle decomposition has the form

$$(1)^{\nu_1}(2)^{\nu_2}\cdots(n)^{\nu_n},$$

where ν_i stands for the appearance time (≥ 0) for cycle with length i, and since S_n acts on n objects, we have the constraint

$$n = \nu_1 + 2\nu_2 + \cdots + n\nu_n = \sum_{i=1}^n i\nu_i.$$

Clearly, $\nu_n \leq 1$, and there are definitely many zeros among ν_i's. Meanwhile, such an equation has finite solutions, which we can give a set of label $\{\lambda_i\}_{i=1}^n$ as

$$\begin{aligned}
\lambda_1 &= \nu_1 + \nu_2 + \cdots + \nu_n, \\
\lambda_2 &= \nu_2 + \nu_3 + \cdots + \nu_n. \\
\lambda_3 &= \nu_3 + \nu_4 + \cdots + \nu_n. \\
&\cdots\cdots \\
\lambda_n &= \nu_n,
\end{aligned}$$

satisfying that

- $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$
- $\sum_{i=1}^{n} \lambda_i = n$
- $\nu_i = \lambda_i - \lambda_{i+1}$ for $1 \leq i < n$ and $\nu_n = \lambda_n$.

1.6.1 Young diagram

(This is more likely a list of results rather than a proof, to be completed)

The irreps of S_n can be effectively found using the Young diagrams, which find the projection operators onto the carrier space group algebra $\mathbb{C}[G]$. Basically, let Young tableau T be a Young diagram in which we use $1, 2, \cdots, n$ to label and order the objects of S_n, and an English standard Young tableau is one in which the numbers increase from left to right in each row and from top to bottom in each column.

With T, we define two subgroups of S_n:

- The row-stabilizer R_T is the subgroup that permutes the numbers within each row of the tableau.
- The column-stabilizer C_T is the subgroup that permutes the numbers within each column of the tableau.

We can also define several elements in the group ring $\mathbb{Z}[S_n]$:

$$P = \sum_{p \in R(T)} p, \tag{1.88a}$$

$$Q = \sum_{q \in C(T)} \epsilon(q) q, \tag{1.88b}$$

$$P(T) = PQ = \sum_{p \in R(T), q \in C(T)} \epsilon(q) pq, \tag{1.88c}$$

where $\epsilon(q)$ is the sign of the permutation q. The followed statements stand:

- $P(T)^2 = c_T P(T)$ with c_T some integer.
- $P(T)P(T') = 0$ for T and T' distinct standard Young tableaus (either the same partition or not).
- $P(T)$ projects onto an irrep of S_n, that $\mathbb{C}[S_n] \cdot P(T)$ transforms under the irrep. Reps for different partitions are inequivalent, while two reps for the same partition are equivalent.
- $c_T = \dfrac{n!}{\dim(\mathbb{C}[S_n] \cdot P(T))}$, and the dim of irrep equals to the number of standard Young tableaus of the same shape (partition).
- $\dim(\mathbb{C}[S_n] \cdot P(T)) = \dfrac{n!}{\prod \text{HookLengths}}$, where HookLengths are the hook lengths of each square, referring to the number of squares to the right and below, including itself once.

For example, let us consider S_3 with possible partitions $(\nu_1, \nu_2, \nu_3) = \{(0,0,1), (1,1,0), (3,0,0)\}$, corresponding to Young diagrams

$$T1 : \begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \end{array}, \quad T2 : \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1} \end{array}, \quad T3 : \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1} \end{array}, \quad T4 : \begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array},$$

respectively. Notice that there are two standard Young tableaus for the partition $(1,1,0)$, meaning that the corresponding irrep is two-dimensional, and this is checked as $3! = 6 = 1^2 + 2^2 + 1^2$.

- For T_1, $P(T_1) = \sum_{p \in S_3} p$ with $P(T_1)^2 = 6P(T_1)$, and clearly $x \cdot P(T_1) = \kappa P(T_1)$ with $x \in \mathbb{C}[S_3]$ and $\kappa \in \mathbb{C}$, making $\mathbb{C}[S_3] \cdot P(T_1) \cong \mathbb{C}$ thus trivial rep.
- For T_4, $P(T_4) = \sum_{p \in S_3} \epsilon(p) p$ with $P(T_4)^2 = 6P(T_4)$, and $g \cdot P(T_4) = \pm P(T_4)$ depending on the sign of $g \in S_3$, thus spanning the sign rep.
- for T_2 and T_3, this gives us two equivalent 2-dim irreps with $P(T_{2/3})^2 = 3P(T_{2/3})$ but $P(T_2)P(T_3) = 0$, where

$$\begin{aligned} P(T_2) &= (1 + (12))(1 - (13)) = 1 + (12) - (13) - (123), \\ P(T_3) &= (1 + (13))(1 - (12)) = 1 + (13) - (12) - (132). \end{aligned}$$

  Now we have

$$\begin{aligned} (12) \cdot P(T_2) &= P(T_2) \equiv v_1, \\ (13) \cdot P(T_3) &= -1 + (13) - (23) + (132) \equiv v_2, \\ (23) \cdot P(T_2) &= -(12) + (23) - (132) + (123) = -v_1 - v_2, \\ (123) \cdot P(T_2) &= -v_1 - v_2, \\ (132) \cdot P(T_2) &= v_2. \end{aligned}$$

  And (v_1, v_2) spans the irrep carrier space with action $(12) \cdot v_2 = -v_1 - v_2$, $(13) \cdot v_2 = v_1$, $(23) \cdot v_2 = v_2$, thus representing

$$\rho((12)) = \begin{pmatrix} 1 & -1 \\ & -1 \end{pmatrix}, \ \rho((13)) = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}, \ \rho((23)) = \begin{pmatrix} -1 & \\ -1 & 1 \end{pmatrix}.$$

  One can similarly construct irrep by T_3 and see the equivalence.

1.6.2 Schur-Weyl duality

Consider $V = \kappa^d$ with $\kappa = \mathbb{R}, \mathbb{C}$. Then $V^{\otimes n} = \overbrace{V \otimes \cdots \otimes V}^{n}$ is a rep of S_n by permuting the tensor factors

$$\sigma \cdot (w_1 \otimes \cdots \otimes w_n) = w_{\sigma(1)} \otimes \cdots \otimes w_{\sigma(n)},$$

with w_i being the vector of V (or one uses σ^{-1}). On the other hand, by treating V as a carrier space (such as the fundamental rep) of $GL(d, \kappa)$, $V^{\otimes n}$ is also a rep of $GL(d, \kappa)$ by

$$\rho(g) \cdot (w_1 \otimes \cdots \otimes w_n) = (g \cdot w_1) \otimes \cdots \otimes (g \cdot w_n).$$

Permutation acts on the tensor product structure, while $GL(d,\kappa)$ acts on each space factor.

Concretely, let $\{v_i\}_{i=1}^d$ be a basis of V with $GL(d,\kappa)$ in its fundamental rep, and the typical element is written as

$$v = t^{i_1,i_2,\cdots,i_n} v_{i_1} \otimes v_{i_2} \otimes \cdots \otimes v_{i_n},$$

with summation over dummy indices. Then counting on the basis transformation $g \cdot v_i = g_{ji} v_j$, we have

$$g \cdot v = t^{i_1,i_2,\cdots,i_n} g_{j_1 i_1} g_{j_2 i_2} \cdots g_{j_n i_n} v_{j_1} \otimes v_{j_2} \otimes \cdots \otimes v_{j_n}, \tag{1.89a}$$

$$\sigma \cdot v = t^{i_1,i_2,\cdots,i_n} v_{\sigma(i_1)} \otimes v_{\sigma(i_2)} \otimes \cdots \otimes v_{\sigma(i_n)}. \tag{1.89b}$$

Importantly, the two actions commute since they act on different structures, or as deduced from the action on the basis

$$\begin{aligned}
\sigma \cdot (g \cdot v_{i_1} \otimes \cdots \otimes v_{i_n}) &= \sigma \cdot g_{j_1 i_1} \cdots g_{j_n i_n} v_{j_1} \otimes \cdots \otimes v_{j_n} \\
&= g_{j_1 i_1} \cdots g_{j_n i_n} v_{\sigma(j_1)} \otimes \cdots \otimes v_{\sigma(j_n)} = g_{\sigma^{-1}(j_1) i_1} \cdots g_{\sigma^{-1}(j_n) i_n} v_{j_1} \otimes \cdots \otimes v_{j_n}. \\
g \cdot (\sigma \cdot v_{i_1} \otimes \cdots \otimes v_{i_n}) &= g \cdot v_{\sigma(i_1)} \otimes \cdots \otimes v_{\sigma(i_n)} = g_{j_1 \sigma(i_1)} \cdots g_{j_n \sigma(i_n)} v_{j_1} \otimes \cdots \otimes v_{j_n},
\end{aligned}$$

and we are left to compare $g_{\sigma^{-1}(j_1) i_1} \cdots g_{\sigma^{-1}(j_n) i_n}$ and $g_{j_1 \sigma(i_1)} \cdots g_{j_n \sigma(i_n)}$, which are the same by constructing correspondence $g_{\sigma^{-1}(j_m) i_m} = g_{j_n \sigma(i_n)}$ with $\sigma(i_n) = i_m$. If this is not clear enough, one considers the generator with permutating two indices.

Now $V^{\otimes n}$ is a rep of $GL(d,\kappa) \times S_n$, and we thus have complete decomposition

$$V^{\otimes n} \cong \bigoplus_\lambda \mathcal{V}_\lambda \otimes \mathcal{S}_\lambda, \tag{1.90}$$

where $\mathcal{S}_\lambda$ is irrep of S_n labelled by partition λ, and we have abbreviated carrier space and rep in a mixed way. By Schur's lemma, the multiplicity space

$$\mathcal{V}_\lambda = \mathrm{Hom}_{S_n}(\mathcal{S}_\lambda, V^{\otimes n}), \;\; \dim(\mathcal{V}_\lambda) = \left\langle \chi_{\mathcal{S}_\lambda}, \chi_{T(V^{\otimes n})} \right\rangle.$$

As $GL(d,\kappa)$ and S_n reps commute, $\mathcal{V}_\lambda$ is also a rep of $GL(d,\kappa)$. Now the *Schur-Weyl duality* states that $\mathcal{V}_\lambda$ indeed carries an irrep of $GL(d,\kappa)$ (up to a power of det rep), and all irreps of $GL(d,\kappa)$ can be obtained in this way. The statement holds also for $U(d)$.

Proof. A sketch of the idea is as follows.

Note that the center of algebra endomorphism $\mathrm{End}(W)$ is just κ, and by considering the group-invariant sub-algebra we have

$$\mathrm{End}_{S_n}(V^{\otimes n}) = ((\mathrm{End}(V))^{\otimes n})^{S_n}.$$

And here is a claim that as $GL(d,\kappa)$ is a dense open set in $\mathrm{End}(V)$, and rep of g on $V^{\otimes n}$ commutes with S_n action, we have that $\mathrm{End}_{S_n}(V^{\otimes n})$ is generated by the action of $GL(d,\kappa)$.

This leads to the conclusion that operators commute with both $GL(d,\kappa)$ and S_n actions, i.e., $\mathrm{End}_{GL(d,\kappa)\times S_n}(V^{\otimes n})$, must lie in the center of $\mathrm{End}_{S_n}(V^{\otimes n})$, while we also have

$$\mathrm{End}_{S_n}(V^{\otimes n}) = \bigoplus_\lambda \mathrm{End}(\mathcal{V}_\lambda),$$

$$\mathrm{End}_{GL(d,\kappa)\times S_n}(V^{\otimes n}) = \bigoplus_\lambda \mathrm{End}_{GL(d,\kappa)}(\mathcal{V}_\lambda),$$

thus making $\mathrm{End}_{GL(d,\kappa)}(\mathcal{V}_\lambda)$ in center of $\mathrm{End}(\mathcal{V}_\lambda)$, i.e., κ, and by Schur's lemma $\mathcal{V}_\lambda$ is irreducible under $GL(d,\kappa)$ action. □

Chapter 2

Lie Groups and Lie Algebras

Sometimes (and indeed at many times), a perfect cartesian coordinate system (a linear system) is unable to handle the realistic problem with non-linear nature. Nevertheless, at the cost of reducing applicability of domains, we can always find a local cartesian coordinate system to reduce the non-linear problem.

In context of group theory, we have Lie groups as curved manifolds, and a reduction transforms the multiplication law over points in the manifold to a linear operation over linear algebra, the Lie algebra.

2.1 Geometric approach to Lie algebra

A Lie group is a group with continuous or even smooth parameter-dependence, thus itself forming a manifold, too. A **manifold** is a topological space that locally resembles Euclidean space near each point. More precisely, an n-dimensional manifold M is a second-countable, Hausdorff topological space where every point $p \in M$ has a neighborhood U homeomorphic to an open subset of $\mathbb{R}^n$.

This local Euclidean structure is described by a pair (U, φ) called a **chart**, where:

- $U \subset M$ is an open set
- $\varphi : U \to \varphi(U) \subset \mathbb{R}^n$ is a homeomorphism

A collection of charts whose domains cover M is called an **atlas**. For a **differentiable manifold**, we require that the **transition maps**:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are smooth (infinitely differentiable) whenever two charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) overlap.

2.1.1 Sophus Lie's approach

Now consider group G as a manifold with local coordinate system $\varphi(g) = (x^1, x^2, \cdots, x^n)$ in neighborhood $\mathcal{U}$ of identity $1_G : x = 0$ with $n = \dim G$ in fundamental rep. The group element $g(x)$ then contains a smooth coordinate dependence. The continuity principle implies that the product of two such elements near 1_G still lies inside $\mathcal{U}' \subset \mathcal{U}$, allowing us to write the multiplication law as

$$g(x) \cdot g(y) = g(\phi(x, y)), \tag{2.1}$$

where $\phi = (\phi^1, \phi^2, \cdots, \phi^n)$ is a multi-variable smooth function defined on $\mathcal{U}' \times \mathcal{U}' \to \mathcal{U}'$. The group laws are indicated correspondingly as

- Associativity: $\phi(x, \phi(y, z)) = \phi(\phi(x, y), z)$
- Identity: $\phi(x, 0) = x$ and $\phi(0, x) = x$
- Inverses: $\exists$ unique x s.t. $\phi(x, x_0) = 0\ \forall x_0 \in \mathcal{U}'$

To gain insight, we differentiate the associativity relation with respect to z as

$$\frac{\partial \phi^i(x, \phi(y, z))}{\partial \phi^j(y, z)} \frac{\partial \phi^j(y, z)}{\partial z^k} = \frac{\partial \phi^i(\phi(x, y), z)}{\partial z^k},$$

with chain rule and summation convention. Now by defining

$$u^i{}_k(x) = \left.\frac{\partial \phi^i(x, y)}{\partial y^k}\right|_{y=0},$$

which is an $n \times n$ matrix with single variable, we can set $z = 0$ and obtain

$$\frac{\partial \phi^i(x, y)}{\partial y^j} u^j{}_k(y) = u^i{}_k(\phi(x, y)),$$

which is equivalent to the first-order differential equation system

$$u^j{}_k(y) \frac{\partial}{\partial y^j} = u^i{}_k(\phi(x, y)) \frac{\partial}{\partial \phi^i},$$

as easily seen by applying both sides over $\phi^i(x, y)$. Note that here x is arbitrary, and we thus expect $\phi(x, y)$ to run over the whole $\mathcal{U}'$ as x does. Thus, the equation can be re-written as

$$u^j{}_k(y) \frac{\partial}{\partial y^j} = u^i{}_k(\phi) \frac{\partial}{\partial \phi^i}, \tag{2.2}$$

with both sides depend on arbitrary variable. Now if we write

$$I_k(x) = u^j{}_k(x) \frac{\partial}{\partial x^j},\ \ \frac{\partial}{\partial x^j} = (u^{-1}(x))^k{}_j I_k(x), \tag{2.3}$$

we have

$$\begin{aligned}
[I_m(x), I_n(x)] &= [u^j{}_m(x) \frac{\partial}{\partial x^j}, u^j{}_n(x) \frac{\partial}{\partial x^j}] \\
&= (u^j{}_m \partial_j u^l{}_n - u^j_n \partial_j u^l{}_m) \partial_l \\
&= (u^j{}_m \partial_j u^l{}_n - u^j_n \partial_j u^l{}_m)(u^{-1})^p{}_l I_p(x) \equiv f^p_{mn}(x) I_p(x).
\end{aligned}$$

Now by the differential equation, we have that $f^p_{mn}(y) = f^p_{mn}(\phi)$, and thus $f^p_{mn}(x)$ is a constant, independent of x, and we can thus write

$$[I_m(x), I_n(x)] = f^l_{mn}(x) I_l(x). \tag{2.4}$$

Now $\{I_k(x)\}_{k=1}^n$ spans a vector space at each point $x \in \mathcal{U}'$, and the commutator operation $[,]$ defines a bilinear map over the vector space that can is closed and can be considered as a multiplication law. Thus $\{I_k(x)\}_{k=1}^n$ indeed spans an algebra space, the Lie algebra $\mathfrak{g} = L(G)$ of G with $\dim \mathfrak{g} = n$ also. The Lie algebra is thus endowed with a formal definition as follows:

Definition 2.1.1. *An abstract Lie algebra* $\mathfrak{g}$ *over a field* κ *is a vector space over* κ *with a bilinear product map* $(v_1, v_2) \to [v_1, v_2] \in \mathfrak{g}$*, s.t.* $\forall\ v_1, v_2, v_3 \in \mathfrak{g}$ *and* $\alpha, \beta \in \kappa$*, we have*

- $[v_1, v_2] = -[v_2, v_1]$
- $[\alpha v_1 + \beta v_2, v_3] = \alpha[v_1, v_3] + \beta[v_2, v_3]$
- $[[v_1, v_2], v_3] + [[v_2, v_3], v_1] + [[v_3, v_1], v_2] = 0$

The last property, aka the Jacobi identity, follows directly from the property of brackets:

$$\begin{aligned}
[[I_l, I_m], I_n] &= I_l I_m I_n - I_m I_l I_n - I_n I_l I_m + I_n I_m I_l \\
[[I_m, I_n], I_l] &= I_m I_n I_l - I_n I_m I_l - I_l I_m I_n + I_l I_n I_m \\
[[I_n, I_l], I_m] &= I_n I_l I_m - I_l I_n I_m - I_m I_n I_l + I_m I_l I_n,
\end{aligned}$$

and one is easily to see that the sum of the three terms vanishes. The identity gives us constraints over the *structure constants* f_{ij}^k as

$$\begin{aligned}
[[I_l, I_m], I_n] &= f_{lm}^i [I_i, I_n] = f_{lm}^i f_{in}^j I_j \\
[[I_m, I_n], I_l] &= f_{mn}^i [I_i, I_l] = f_{mn}^i f_{il}^j I_j \\
[[I_n, I_l], I_m] &= f_{nl}^i [I_i, I_m] = f_{nl}^i f_{im}^j I_j,
\end{aligned}$$

and thus leads to

$$f_{lm}^i f_{in}^j + f_{mn}^i f_{il}^j + f_{nl}^i f_{im}^j = 0. \tag{2.5}$$

2.1.2 Differential geometry's approach

Tangent vectors over a general manifold M are directional derivatives along paths. If we think M to be embedded in equal/higher dimensional Euclidean space, those tangent vectors span a 'tangent plane' at each point $p \in M$, called the tangent space T_pM. Concretely, let $C^1(p)$ be functions defined on a neighborhood of p that are at least continuously differentiable. And let $\gamma : \mathbb{R} \to M$ be a differentiable/smooth curve with $\gamma(0) = p$, then we can define a tangent vector $\dot{\gamma} : C^1(p) \to \mathbb{R}$ at p by specifying its action on $f \in C^1(p)$ as

$$\dot{\gamma}(f) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} (f \circ \gamma)(t) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t)). \tag{2.6}$$

In the local coordinate system constructed by φ, we can express the tangent vector in terms of the coordinates $(x^1, x^2, \cdots, x^n)$ as

$$\dot{\gamma}(f) = \frac{\partial f}{\partial x^i}\frac{\partial x^i}{\partial t}\bigg|_{t=0} = v^i \frac{\partial}{\partial x^i} f, \quad v^i = \frac{\partial x^i}{\partial t}\bigg|_{t=0}. \tag{2.7}$$

Note that there are abbreviations like $x^i(t) = (\varphi^i \circ \gamma)(t)$, and $f(x) \equiv (f \circ \varphi^{-1})(x^1, x^2, \cdots, x^n)$. In this sense, we have tangent vector

$$v = v^i \frac{\partial}{\partial x^i}, \tag{2.8}$$

with ∂_i specifies the basis of the tangent space T_pM at point p, and v^i are the components of the tangent vector in this basis. Note that each v corresponds to a $\dot{\gamma}$, i.e., an extending

direction of a curve. If we organize tangent vectors at all points $p \in M$ specified in a way that they are continuous, we obtain a tangent field V over M. If we organize all tangent spaces at all points $p \in M$, we obtain the tangent bundle $TM = \bigsqcup_{p\in M} T_pM$, and a tangent field V can be viewed as a section of the tangent bundle TM, with M the base space, and T the fiber.

Now by giving $\phi : M_1 \to M_2$ a map between two manifolds, we can transfer structures between them with certain order. The functions are contravariant in the sense that they are *pullback*

$$\phi^* : C^1(M_2) \to C^1(M_1), \quad \phi^*(f) = f \circ \phi, \ f \in C^1(M_2). \tag{2.9}$$

The operation indicates that we pull function $f \in C^1(M_2)$ back to M_1, such that $\phi^*(f)$ directly acts on point $p \in M_1$, and its output is set to be equal to $f(\phi(p))$, i.e., we firstly map $p \in M_1$ to $\phi(p) \in M_2$, and then evaluate f at this point. On the other hand, the vector-fields are covariant and are *pushforward*

$$\phi_* : T_pM_1 \to T_{\phi(p)}M_2, \quad \phi_*(v) \circ f = v \circ \phi^*(f), \ f \in C^1(M_2), \tag{2.10}$$

in a similar indication. In expression of curves, we have

$$\phi_*(\dot\gamma)(f) = \dot\gamma(\phi^*(f)) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} (f \circ \phi \circ \gamma)(t). \tag{2.11}$$

In expression of local coordinate system, we assume $v = v^i \frac{\partial}{\partial x^i}$ and $\phi_*(v) = w^i \frac{\partial}{\partial y^i}$, and

$$\begin{aligned}(\phi_*(v) \circ f)(\phi(p)) &= w^i \frac{\partial f(\phi(p))}{\partial y^i}, \\ (v \circ \phi^*(f))(p) &= v^i \frac{\partial f(\phi(p))}{\partial x^i} = v^i \frac{\partial f}{\partial y^j} \frac{\partial y^j(\phi(p))}{\partial x^i},\end{aligned}$$

and we obtain the coordinate transformation

$$w^i = \frac{\partial y^i(\phi(p))}{\partial x^j} v^j. \tag{2.12}$$

Note that when pushing forward commutator, we have

$$\phi_*[v_1, v_2] = [\phi_* v_1, \phi_* v_2].$$

We can also in a sense 'pushforward' a function when ϕ is invertible, and this 'pushforward' is just a pullback by ϕ^{-1}

$$\phi_* : C^1(M_1) \to C^1(M_2), \quad \phi_*(f) \equiv (\phi^{-1})^* f = f \circ \phi^{-1}, \ f \in C^1(M_1).$$

Lie group as manifold

Now let us consider a Lie group G as a manifold, and we can define maps from G to itself by left and right multiplications

$$L_g : G \to G, \quad L_g(h) = g \cdot h, \tag{2.13a}$$
$$R_g : G \to G, \quad R_g(h) = h \cdot g^{-1}. \tag{2.13b}$$

This is verified to be right by

$$L_{g_1} \circ L_{g_2}(h) = L_{g_1}(g_2 \cdot h) = g_1 \cdot (g_2 \cdot h) = (g_1 \cdot g_2) \cdot h = L_{g_1 \cdot g_2}(h),$$
$$R_{g_1} \circ R_{g_2}(h) = R_{g_1}(h \cdot g_2^{-1}) = (h \cdot g_2^{-1}) \cdot g_1^{-1} = h \cdot (g_1 \cdot g_2)^{-1} = R_{g_1 \cdot g_2}(h).$$

These two maps belong to indeed diffeomorphisms from G to itself.

Now we have invariant functions and vector fields under these maps. A function $f \in C^1(G)$ is left-invariant if

$$L_g^*(f) = f, \ \forall g \in G, \tag{2.14}$$

and since $L_g^*(f)(h) = f \circ L_g(h) = f(gh) = f(h)$, then by the arbitrariness of h, we see f is a constant function. The same is true for right-invariant functions.

Now a vector field $\xi \in \mathrm{Vec}(G)$ is left (LIVF) or right (RIVF) invariant if

$$\begin{aligned} (L_{g_0})_* \xi_g &= \xi_{g_0 g}, \ \forall g_0, g \in G, \\ (R_{g_0})_* \xi_g &= \xi_{g g_0^{-1}}, \ \forall g_0, g \in G, \end{aligned} \tag{2.15}$$

meaning that we can either pushforward the vector field at g by left/right multiplication, or directly evaluate the vector field at the new point $g_0 g$ or $g g_0^{-1}$. To give a picture for this, let us consider

$$(L_{g_0})_*(\xi_g)(f) = \xi_g(L_{g_0}^* f) = \xi_g(f \circ L_{g_0}),$$

and left invariance indicates that this is equal to $\xi_{g_0 g}(f)$. Notice that by definition, $\xi_g(f) = (\xi(f))(g)$, meaning that ξ maps f to a function $\xi(f) \in C^1(G)$, and then evaluates it at point g. Thus, the left invariance indicates that

$$(\xi(f \circ L_{g_0}))(g) = (\xi(f))(g_0 g), \ \forall g_0, g \in G, \tag{2.16}$$

or equivalently

$$\xi(f \circ L_{g_0}) = \xi(f) \circ L_{g_0}, \ \forall g_0 \in G, \tag{2.17}$$

which means that ξ commutes with left multiplication when acting on functions. We know above that we have defined Lie algebra as vector fields $I_k(y) = u^j{}_k(y)\partial_j$ with identity $I_k(y) = I_k(\phi)$, and we claim that these are indeed LIVF. To see this, let us assign local coordinates y at point g, and ϕ at point $g_0 g$. Then we have, by the above deduction of pushforward,

$$(L_{g_0})_* I_k(y)\Big|_g = u^j{}_k(y) \frac{\partial \phi^i(g_0 g)}{\partial y^j(g)} \frac{\partial}{\partial \phi^i}\Big|_{g_0 g},$$

while on the other hand, we have

$$I_k(\phi)\Big|_{g_0 g} = u^i{}_k(\phi) \frac{\partial}{\partial \phi^i}\Big|_{g_0 g}.$$

Notice that from the very beginning, we have defined $u^i{}_k(\phi) = \frac{\partial \phi^i(x,y)}{\partial y^k}$, and by setting $x = g_0$ and $y = g$, we have the equivalence. Also consider their commutator, we have

$$[I_m(\phi), I_n(\phi)] = f_{mn}^l I_l(\phi) = f_{mn}^l (L_{g_0})_* I_l(y) = (L_{g_0})_* [I_m(y), I_n(y)], \tag{2.18}$$

meaning that the commutator of two LIVF is still a LIVF, and thus the Lie algebra $\mathfrak{g} = L(G)$ is closed under commutator. We now have an alternative definition of Lie algebra as follows:

Definition 2.1.2. *The Lie algebra $\mathfrak{g} = L(G)$ of a Lie group G is the set of all left-invariant vector fields on G, with the Lie bracket (commutator) as the Lie product.*

Now suppose ξ is a LIVF, then $\xi_g = (L_g)_*(\xi_1)$, and thus a LIVF is completely determined by its value at $g = 1$. We can identify $\mathfrak{g} = L(G)$ as the tangent space at identity as $L(G) \cong T_1(G)$. Inversely, given $X \in T_1(G)$, we can associate a curve $\gamma_X(t)$ through $g = 1$ with its tangent vector at $t = 0$ being X, and then define a LIVF $\xi(X)$ by $\gamma_{X,g}(t) = g \cdot \gamma_X(t) \in T_g(G)$. To see this, we can consider

$$\begin{aligned}(L_{g_0})_*\dot{\gamma}_{X,g}(f) = \dot{\gamma}_{X,g}((L_{g_0})^* f) &= \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}(f \circ L_{g_0} \circ \gamma_{X,g})(t) \\ &= \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}(f \circ L_{g_0} g\gamma_X)(t) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}(f \circ g_0 g\gamma_X)(t) = \dot{\gamma}_{X,g_0 g}(f).\end{aligned}$$

Thus identifying $(L_{g_0})_*\dot{\gamma}_{X,g} = \dot{\gamma}_{X,g_0 g}$, or equivalently $\xi_g(X) = (L_g)_*\xi_1(X) = (L_g)_*X$. Those LIVF $\xi(X)$ are called the **fundamental vector fields** generated by $X \in T_1(G)$, and we have a one-to-one correspondence between $X \in T_1(G)$ and $\xi_X \in L(G)$, thus giving

$$[\xi(X), \xi(Y)] = \xi([X, Y]),\ \forall X, Y \in T_1(G). \tag{2.19}$$

This defines a Lie bracket over $T_1(G)$, and thus $T_1(G)$ is also a Lie algebra isomorphic to $L(G)$. Moreover, $\xi : T_1(G) \to \mathrm{Vec}(G)$ is a homomorphism.

For instance, consider $G = U(1)$, then generally $\xi = f(\theta)\partial_\theta$, and the LIVF condition indicates that $f(\theta + \theta_0) = f(\theta)\ \forall\theta_0$, and thus $f(\theta)$ is a constant, and we have $\mathfrak{g} = L(U(1)) = \{c\partial_\theta | c \in \mathbb{R}\}$.

For a more cooperative example, we let $G = GL(n, \kappa)$ and choose global coordinates on the manifold as the matrix elements g_{ij} with $g \in G$. The $n \times n$ vector fields are given by

$$\left(\frac{\partial}{\partial g}\right)_{ij} = \frac{\partial}{\partial g_{ij}},$$

with its meaning that $\left(\frac{\partial}{\partial g_{ij}}\right)_{g_0}$ is the directional derivative at point g_0 along the direction $g_0 + te_{ij}$, where e_{ij} is the matrix with only one non-zero element 1 at position (i, j). Now we compute, by assigning $f_{kl}(g) = g_{kl}$ as

$$\begin{aligned}(L_{g_0})_*\left(\frac{\partial}{\partial g}\right)_{ij}\Bigg|_g (f_{kl}) = \left(\frac{\partial}{\partial g}\right)_{ij}\Bigg|_g (f_{kl} \circ L_{g_0}(g)) &= \left(\frac{\partial}{\partial g}\right)_{ij}\Bigg|_g f_{kl}(g_0 g) \\ &= \sum_{p,q}\left.\frac{\partial f_{kl}}{\partial h_{pq}}\right|_{h=g_0 g}\frac{\partial (g_0 g)_{pq}}{\partial g_{ij}} = \sum_{p,q,r}(g_0)_{pr}\delta_{ri}\delta_{qj}\left.\frac{\partial f_{kl}}{\partial h_{pq}}\right|_{h=g_0 g} \\ &= \sum_p (g_0)_{pi}\left(\frac{\partial f_{kl}}{\partial g}\right)_{pj}\Bigg|_{g_0 g},\end{aligned}$$

where one should be careful that due to argument difference, the derivative must be evaluated at $h = g_0 g$ instead of g. Thus we have

$$(L_{g_0})_*\left.\frac{\partial}{\partial h_{ij}}\right|_{h=g} = (g_0)_{ki}\left.\frac{\partial}{\partial h_{kj}}\right|_{h=g_0 g}, \tag{2.20}$$

where summation over k is implied. Equivalently we have the matrix form $(L_{g_0})_* \partial_h \big|_{h=g} = g_0^T \partial_h \big|_{h=g_0 g}$. Now by considering a new set of vector fields defined by

$$\xi_{ij}^g = \left(g^T \frac{\partial}{\partial g} \right)_{ij} = g_{ki} \frac{\partial}{\partial g_{kj}}, \tag{2.21}$$

we claim them as LIVF, since

$$(L_{g_0})_* \xi_{ij}^g = g_{ki} (L_{g_0})_* \frac{\partial}{\partial g_{kj}} = g_{ki} (g_0)_{lk} \frac{\partial}{\partial h_{lj}} \bigg|_{h=g_0 g} = \left(h_{li} \frac{\partial}{\partial h_{lj}} \right) \bigg|_{h=g_0 g} = \xi_{ij}^{g_0 g}.$$

Now those LIVF constructs $\mathfrak{gl}(n, \kappa)$, and we can compute their commutators as

$$\begin{aligned}
[\xi_{ij}^g, \xi_{kl}^g] &= \left[g_{mi} \frac{\partial}{\partial g_{mj}}, g_{nk} \frac{\partial}{\partial g_{nl}} \right] \\
&= g_{mi} \frac{\partial g_{nk}}{\partial g_{mj}} \frac{\partial}{\partial g_{nl}} - g_{nk} \frac{\partial g_{mi}}{\partial g_{nl}} \frac{\partial}{\partial g_{mj}} \\
&= g_{mi} \delta_{mn} \delta_{jk} \frac{\partial}{\partial g_{nl}} - g_{nk} \delta_{nm} \delta_{li} \frac{\partial}{\partial g_{mj}} \\
&= \delta_{jk} g_{ni} \frac{\partial}{\partial g_{kl}} - \delta_{li} g_{mk} \frac{\partial}{\partial g_{ij}} = \delta_{jk} \xi_{il}^g - \delta_{li} \xi_{kj}^g.
\end{aligned}$$

Or compactly,

$$[\xi_{ij}^g, \xi_{kl}^g] = \delta_{jk} \xi_{il}^g - \delta_{li} \xi_{kj}^g. \tag{2.22}$$

Utilizing the idea of curves $\gamma_{ij}(t) = \exp(t e_{ij})$ for e_{ij}, we calculate

$$\begin{aligned}
\dot{\gamma}_{ij}(f_{sk}) &= \frac{\mathrm{d}}{\mathrm{d}t} \bigg|_{t=0} f_{sk}(\exp(t e_{ij})) = \frac{\mathrm{d}}{\mathrm{d}t} \bigg|_{t=0} (\exp(t e_{ij}))_{sk} \\
&= \frac{\mathrm{d}}{\mathrm{d}t} \bigg|_{t=0} \begin{cases} 1 + t \delta_{si} \delta_{kj}, i \neq j \\ e^t \delta_{si} \delta_{kj}, i = j \end{cases} = \delta_{si} \delta_{kj} = \frac{\partial f_{sk}}{\partial g_{ij}} \bigg|_{g=I},
\end{aligned}$$

where we have utilized that

$$e_{ij}^2 = e_{ii} \delta_{ij} \Longrightarrow e^{t e_{ij}} = \begin{cases} 1 + t e_{ij}, i \neq j \\ e^t e_{ii}, i = j \end{cases}.$$

And correspondingly, we have ξ_{ij}^g the tangent vector of $\gamma_{ij,g}(t) = g \gamma_{ij}(t) = g e^{t e_{ij}}$ as

$$\begin{aligned}
\dot{\gamma}_{ij,g}(f_{sk}) &= g \dot{\gamma}_{ij}(f_{sk}) = \frac{\mathrm{d}}{\mathrm{d}t} \bigg|_{t=0} f_{sk}(g e^{t e_{ij}}) = \frac{\mathrm{d}}{\mathrm{d}t} \bigg|_{t=0} g_{sm} (e^{t e_{ij}})_{mk} \\
&= g_{sm} \delta_{mi} \delta_{kj} = g_{si} \delta_{kj}, \\
\xi_{ij}^g(f_{sk}) &= g_{mi} \frac{\partial f_{sk}}{\partial g_{mj}} = g_{si} \delta_{kj},
\end{aligned}$$

which are the same, meaning that $\xi_{ij}^g = \dot{\gamma}_{ij,g}$. This indeed leads to

$$\xi_{ij}^g = g_{mi} \frac{\partial}{\partial g_{mj}} = \frac{\mathrm{d}}{\mathrm{d}t} \bigg|_{t=0} g e^{t e_{ij}}. \tag{2.23}$$

We can examine, for instance, that

$$\xi([e_{ij}, e_{i'j'}]) = \xi(\delta_{ji'}e_{ij'} - \delta_{j'i}e_{i'j}) = \delta_{ji'}\xi_{ij'} - \delta_{j'i}\xi_{i'j},$$
$$[\xi(e_{ij}), \xi(e_{i'j'})] = [\xi_{ij}, \xi_{i'j'}] = \delta_{ji'}\xi_{ij'} - \delta_{j'i}\xi_{i'j},$$

which is consistent with the deduction above.

Note that when $G \subset GL(n, \kappa)$ is a subgroup, the Lie algebra $\mathfrak{g} = L(G)$ is a subalgebra of $\mathfrak{gl}(n, \kappa)$, and generally there will be constraints over the matrix elements g_{ij} that define the subgroup, and these constraints will lead to constraints over the tangent vectors, making its dimension smaller than n^2.

2.1.3 Exponential map

The exponential map $\exp : \mathfrak{g} = L(G) \to G$ gives the correspondence between the Lie algebra and the Lie group, through the one-parameter subgroups of G generated by elements of $L(G)$.

We firstly introduce the exp-map for matrix Lie groups, i.e., $G \subset GL(n, \kappa)$. We consider a homomorphism $f : \mathbb{R} \to GL(n, \mathbb{R})$ satisfying that

$$\frac{\mathrm{d}}{\mathrm{d}t}f(t) = \lim_{h\to 0}\frac{f(t+h) - f(t)}{h} = \lim_{h\to 0}\frac{f(h) - \mathbb{I}}{h}f(t) = Af(t),$$

where homomorphism property $f(t + h) = f(t)f(h)$ is utilized, and $A = \frac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0}f(t)$ is a constant matrix. With boundary condition $f(0) = \mathbb{I}$, we have the unique solution

$$f(t) = e^{tA} = \sum_{m=0}^{\infty}\frac{(tA)^m}{m!}. \tag{2.24}$$

This is checked as

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0}f(t) &= \lim_{h\to 0}\frac{f(h) - \mathbb{I}}{h} = \lim_{h\to 0}h^{-1}\left(\sum_{n=0}^{\infty}\frac{h^nA^n}{n!} - \mathbb{I}\right) \\
&= \lim_{h\to 0}\sum_{n=1}^{\infty}\frac{h^{n-1}A^n}{n!} = A\lim_{h\to 0}\sum_{m=0}^{\infty}\frac{h^mA^m}{(m+1)!} \\
&= A\sum_{m=0}^{\infty}\lim_{h\to 0}\frac{h^mA^m}{(m+1)!} = A, \\
\frac{\mathrm{d}}{\mathrm{d}t}f(t) &= \lim_{h\to 0}\sum_{n=0}^{\infty}\frac{(t+h)^n - t^n}{h}\frac{A^n}{n!} = \sum_{n=0}^{\infty}\lim_{h\to 0}\frac{(t+h)^n - t^n}{h}\frac{A^n}{n!} \\
&= \sum_{n=0}^{\infty}nt^{n-1}\frac{A^n}{n!} = A\sum_{m=0}^{\infty}t^m\frac{A^m}{m!} = Af(t),
\end{aligned}$$

where we have utilized the uniform convergence of the series to exchange limit and summation.

Thus, for $A \in \mathfrak{g}$, we can define a one-parameter subgroup $f(t) = e^{tA} \in G$ with $f(0) = \mathbb{I}$ and $\frac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0}f(t) = A$. Generally, this gives $\exp : M_n(\mathbb{R}) \to GL(n, \mathbb{R})$, and we can restrict the image to G by restricting the domain to $\mathfrak{g}$. In a sufficiently small neighborhood of

$\mathbb{I} \in G$, the exp-map is bijective, and we can define its inverse as the logarithm map $\log : G \to \mathfrak{g}$ by

$$\log(g) = -\sum_{n=1}^{\infty} \frac{(\mathbb{I} - g)^n}{n}, \ \|g - \mathbb{I}\| < 1.$$

For a general Lie group G, we can also define the exp-map through one-parameter subgroups.

Theorem 2.1.1. *If G is a Lie group, then there exists a one-to-one correspondence between the tangent space at identity $T_1(G)$ and the set of group homomorphisms (one-parameter subgroups) $f : \mathbb{R} \to G$.*

Proof. Formally considering the above differential equation tells us that each homomorphism f determines a unique $X \in T_1(G)$ by $X = \frac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} f(t)$.

Inversely, for $X \in T_1(G)$, we can associate a curve $\gamma_X(t)$ through $g = 1$ with its tangent vector at $t = 0$ being X, and then define a LIVF $\xi(X)$ by $\gamma_{X,g}(t) = g \cdot \gamma_X(t) \in T_g(G)$. Now we consider the integral curve of $\xi(X)$ passing through $g = 1$ at $t = 0$, denoted as

$$\dot{\gamma}_X(t) = \xi(X)\big|_{\gamma_X(t)}, \quad \gamma_X(0) = 1.$$

This is verified to be a one-parameter subgroup by

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\gamma_X(t + t_0) &= \xi(X)\big|_{\gamma_X(t+t_0)} = \xi(X)\big|_{\gamma_X(t)\gamma_X(t_0)} = (L_{\gamma_X(t_0)})_* \xi(X)\big|_{\gamma_X(t)} = (L_{\gamma_X(t_0)})_* \dot{\gamma}_X(t) \\ &= \gamma_X(t_0) \cdot \dot{\gamma}_X(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\gamma_X(t_0) \cdot \gamma_X(t)\right),\end{aligned}$$

where we have utilized the LIVF property of $\xi(X)$. Thus by uniqueness of solution of ODE, and by boundary condition $\gamma_X(0) = 1$, we have $\gamma_X(t + t_0) = \gamma_X(t_0) \cdot \gamma_X(t)$, which corresponds to a group homomorphism $f_X(t) = \gamma_X(t)$. This gives the way to evaluate the exp-map $\exp : T_1(G) \to G$ as

$$\exp(X) = f_X(1) = \gamma_X(1) = \left(\gamma_X(t/n)\right)^n.$$

□

- $\exp : \mathbb{R} \to U(1)$ by $\exp(t) = e^{it}$, with $\exp(2\pi) = 1$. The exp-map is generally not injective.
- $\exp : \mathfrak{so}(3) \to SO(3)$ by $\exp(A) = e^A$, with $A \in \mathfrak{so}(3)$ a 3×3 antisymmetric matrix. The exp-map is surjective but not injective.
- Generally, for compact Lie group G, the exp-map is surjective but not injective; for non-compact G, the exp-map may be neither surjective nor injective.
- Finite dimensional Lie algebras can be exponentiated to Lie groups, but infinite dimensional ones may not.

2.1.4 BCH formula

Up to now, we have established the picture about the correspondence between Lie algebra $\mathfrak{g} = L(G)$ as the tangent space at identity $T_1(G)$ of a Lie group G. Each vector X in the Lie algebra can generate a one-parameter subgroup through the exponential map $\exp : \mathfrak{g} \to G$, that uniquely associates to a curve $g(t) = \gamma_X(t) = \exp(tX)$ in G passing through identity, with $\dot{\gamma}_X(t) = \xi(X)\big|_{g(t)}$. Since we have multiplication law in $\mathfrak{g}$ as Lie bracket $[\cdot, \cdot]$, and multiplication law in G as group multiplication $\cdot$, it is natural to ask how these two multiplications are related. The Baker-Campbell-Hausdorff (BCH) formula answers this question by giving an expression of the product of two exponentials in G as a single exponential:

$$C(A, B) = \ln e^A e^B = A+B+\frac{1}{2}[A, B]+\frac{1}{12}([A, [A, B]]+[B, [B, A]])-\frac{1}{24}[B, [A, [A, B]]]+\cdots .$$

Note that generally the exp map is neither injective nor surjective, so the logarithm $\ln : G \to \mathfrak{g}$, as an inverse function to $e^C = e^A e^B$ is well-defined only locally around identity.

To begin with, by assuming A and B near 0 in $\mathfrak{g}$, we can write Taylor series

$$C(A, B) = A + B + \frac{1}{2}b(A, B) + \cdots ,$$

and since $C(A, 0) = A$, $C(0, B) = B$, we see that $b(A, 0) = b(0, B) = 0$, and b should not contain A^2 or B^2, and since b is of second order, it should only contain AB and BA, and in the sense of completeness, we expect

$$b : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}, \quad b(A, B) = \alpha AB + \beta BA.$$

On the other hand, $e^{-A} = (e^A)^{-1}$, thus the equality $C(-B, -A) = -C(A, B)$ leads to $b(A, B) = -b(-B, -A) = -b(B, A)$, making b antisymmetric, and hence $\alpha = -\beta$, meaning that $b(A, B) = \alpha(AB - BA) = \alpha[A, B]$ is a Lie bracket. Now our claim is that the entire BCH series can be expressed as a linear combination of Lie brackets.

Now to introduce the BCH formula, we need to introduce the adjoint action and several lemmas.

Definition 2.1.3. *For $A \in M_n(\kappa)$, we define the adjoint action $\mathrm{Ad}_A : M_n(\kappa) \to M_n(\kappa)$ as*

$$\begin{aligned} \mathrm{Ad}_A(B) &= [A, B] = AB - BA, \\ (\mathrm{Ad}_A)^j(B) &= \overbrace{[A, [A, \cdots, [A, B]\cdots]]}^{j \text{ times}}. \end{aligned} \tag{2.25}$$

Lemma 2.1.2. *For A constant matrix, we have*

$$\frac{\mathrm{d}}{\mathrm{d}t}e^{tA} = Ae^{tA} = e^{tA}A. \tag{2.26}$$

Proof. Easy to verify by Taylor expansion.

$$\frac{\mathrm{d}}{\mathrm{d}t}e^{tA} = \frac{\mathrm{d}}{\mathrm{d}t}\sum_{n=0}^{\infty}\frac{(tA)^n}{n!} = \sum_{n=1}^{\infty}\frac{(tA)^{n-1}}{(n-1)!}A = Ae^{tA}.$$

□

Lemma 2.1.3.

$$\begin{aligned} e^A B e^{-A} = e^{\mathrm{Ad}_A}(B) &= \sum_{n=0}^{\infty} \frac{(\mathrm{Ad}_A)^n(B)}{n!} \\ &= B + [A,B] + \frac{1}{2!}[A,[A,B]] + \frac{1}{3!}[A,[A,[A,B]]] + \cdots . \end{aligned} \tag{2.27}$$

Proof. Let $B(t) = e^{tA} B e^{-tA}$, then

$$\begin{aligned} \dot{B}(t) &= A e^{tA} B e^{-tA} - e^{tA} B e^{-tA} A = [A, B(t)] = \mathrm{Ad}_A(B(t)), \\ \left(\frac{\mathrm{d}}{\mathrm{d}t}\right)^j B(t) &= (\mathrm{Ad}_A)^j(B(t)). \end{aligned}$$

Thus by Taylor expansion at $t = 0$, we have

$$e^A B e^{-A} = B(1) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\mathrm{d}}{\mathrm{d}t}\right)^j B(t)\Big|_{t=0} = \sum_{n=0}^{\infty} \frac{(\mathrm{Ad}_A)^n(B)}{n!}.$$

□

Lemma 2.1.4. *By letting*

$$f(z) = \frac{e^z - 1}{z} = \sum_{n=0}^{\infty} \frac{z^n}{(n+1)!} = 1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots ,$$

we have

$$-\left(\frac{\mathrm{d}}{\mathrm{d}t} e^{A(t)}\right) e^{-A(t)} = e^{A(t)} \left(\frac{\mathrm{d}}{\mathrm{d}t} e^{-A(t)}\right) = -f(\mathrm{Ad}_{A(t)}) \dot{A}(t).$$

Proof. Let $B(s,t) = e^{sA(t)} \left(\frac{\mathrm{d}}{\mathrm{d}t} e^{-sA(t)}\right)$, then

$$\begin{aligned} \frac{\partial B}{\partial s} &= A(t) e^{sA(t)} \left(\frac{\mathrm{d}}{\mathrm{d}t} e^{-sA(t)}\right) - e^{sA(t)} \left(\frac{\mathrm{d}}{\mathrm{d}t} e^{-sA(t)} A(t)\right) \\ &= \mathrm{Ad}_{A(t)} B(s,t) - \dot{A}(t), \\ \left(\frac{\partial}{\partial s}\right)^j B(s,t) &= (\mathrm{Ad}_{A(t)})^j B(s,t) - (\mathrm{Ad}_{A(t)})^{j-1} \dot{A}(t). \end{aligned}$$

Since $B(0,t) = 0$, by Taylor expansion at $s = 0$, we have

$$B(s,t) = \sum_{n=0}^{\infty} \frac{s^n}{n!} \left(\frac{\partial}{\partial s}\right)^j B(s,t)\Big|_{s=0} = \sum_{n=0}^{\infty} \frac{s^n}{n!} (-(\mathrm{Ad}_{A(t)})^{n-1} \dot{A}(t)) = -f(s\mathrm{Ad}_{A(t)}) \dot{A}(t).$$

□

By noticing that $f(z) = \int_0^1 e^{sz} \mathrm{d}s$, we can write

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} e^{-A(t)} &= -e^{-A(t)} f(\mathrm{Ad}_{A(t)}) \dot{A}(t) = -e^{-A(t)} \int_0^1 e^{s\mathrm{Ad}_{A(t)}} \mathrm{d}s \dot{A}(t) \\ &= -e^{-A(t)} \int_0^1 e^{sA(t)} \dot{A}(t) e^{-sA(t)} \mathrm{d}s = -\int_0^1 e^{-(1-s)A(t)} \dot{A}(t) e^{-sA(t)} \mathrm{d}s. \end{aligned}$$

This is called the **Duhamel's formula**, and gives us an intuitive picture about the derivative of matrix exponential, as generally,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[M_1(t)M_2(t)\cdots M_n(t)\right]=\sum_{j=1}^{n}M_1(t)\cdots M_{j-1}(t)\dot{M}_j(t)M_{j+1}(t)\cdots M_n(t),$$

and now we can write

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}e^{A(t)}&=\lim_{N\to\infty}\frac{\mathrm{d}}{\mathrm{d}t}\left[e^{\frac{A(t)}{N}}e^{\frac{A(t)}{N}}\cdots e^{\frac{A(t)}{N}}\right]\\&=\lim_{N\to\infty}\sum_{j=1}^{N}e^{A(t)\left(1-\frac{j}{N}\right)}\dot{A}(t)\frac{1}{N}e^{A(t)\frac{j}{N}}\\&=\int_0^1 e^{(1-s)A(t)}\dot{A}(t)e^{sA(t)}\mathrm{d}s.\end{aligned}$$

Theorem 2.1.5. *(Baker-Campbell-Hausdorff formula) Let*

$$g(z)=\frac{\ln z}{z-1}=\sum_{n=0}^{\infty}\frac{(1-z)^n}{n+1},$$

then for $A,B\in M_n(\kappa)$ near 0, we have

$$\begin{aligned}C=\ln\left(e^Ae^B\right)&=A+\int_0^1 g(e^{-s\mathrm{Ad}_B}e^{-\mathrm{Ad}_A})B\mathrm{d}s\\&=B+\int_0^1 g(e^{s\mathrm{Ad}_A}e^{\mathrm{Ad}_B})A\mathrm{d}s,\end{aligned}\tag{2.28}$$

with first several terms

$$C=A+B+\frac{1}{2}[A,B]+\frac{1}{12}([A,[A,B]]+[B,[B,A]])-\frac{1}{24}[B,[A,[A,B]]]+\cdots.$$

Before proving, let us see how to get the series expansion, and for simplicity, we only consider the first expression. For $j=0$, we have $g^{(0)}=1$ and thus $C^{(0)}=B+A$; for $j=1$, we have $g^{(1)}(z)=\frac{1-z}{2}$, and thus

$$\begin{aligned}C^{(1)}&=\int_0^1\frac{1-e^{s\mathrm{Ad}_A}e^{\mathrm{Ad}_B}}{2}A\mathrm{d}s=\frac{A}{2}-\frac{1}{2}\int_0^t e^{s\mathrm{Ad}_A}\sum_{n=0}^{\infty}\frac{(\mathrm{Ad}_B)^n}{n!}A\mathrm{d}s\\&=\frac{A}{2}-\frac{1}{2}\int_0^t e^{s\mathrm{Ad}_A}\left(A+[B,A]+O(A^3)\right)\mathrm{d}s\\&=\frac{A}{2}-\frac{A}{2}-\frac{1}{2}\int_0^t\sum_{n=0}^{\infty}\frac{s^n(\mathrm{Ad}_A)^n}{n!}[B,A]\mathrm{d}s+O(A^3)\\&=-\frac{1}{2}\int_0^t\left([B,A]+s[A,[B,A]]+O(A^3)\right)\mathrm{d}s+O(A^3)=-\frac{1}{2}[B,A]+O(A^3),\end{aligned}$$

and thus we have $C=A+B+\frac{1}{2}[A,B]+\cdots$. Notice that $e^{s\mathrm{Ad}_A}e^{\mathrm{Ad}_B}\approx 1+s\mathrm{Ad}_A+\mathrm{Ad}_B$, and for higher order terms, we have $(-s\mathrm{Ad}_A+\mathrm{Ad}_B)^n$ and do not affect the lowest order terms.

Proof. Let $e^{C(t)} = e^{tA}e^{B}$ with $C(0) = B$, then by lemma above we have

$$e^{C(t)}\frac{\mathrm{d}}{\mathrm{d}t}e^{-C(t)} = -f(\mathrm{Ad}_{C(t)})\dot{C}(t),$$

while a direct calculation gives

$$e^{C(t)}\frac{\mathrm{d}}{\mathrm{d}t}e^{-C(t)} = e^{tA}e^{B}\frac{\mathrm{d}}{\mathrm{d}t}e^{-B}e^{-tA} = -A,$$

thus we have

$$f(\mathrm{Ad}_{C(t)})\dot{C}(t) = A,$$

or formally

$$\dot{C}(t) = f(\mathrm{Ad}_{C(t)})^{-1}A = g(e^{\mathrm{Ad}_{C(t)}})A.$$

This holds since

$$f(z)g(e^z) = \frac{e^z - 1}{z}\cdot\frac{\ln e^z}{e^z - 1} = 1.$$

Notice that

$$e^{\mathrm{Ad}_{C(t)}}O = e^{C(t)}Oe^{-C(t)} = e^{tA}e^{B}Oe^{-B}e^{-tA} = e^{\mathrm{Ad}_{tA}}e^{\mathrm{Ad}_{B}}O,$$

leads to

$$e^{\mathrm{Ad}_{C(t)}} = e^{\mathrm{Ad}_{tA}}e^{\mathrm{Ad}_{B}},$$

and hence

$$\dot{C}(t) = g(e^{\mathrm{Ad}_{tA}}e^{\mathrm{Ad}_{B}})A.$$

Integrating from 0 to 1 gives the first expression. The second expression can be proved similarly by letting $e^{C(t)} = e^{A}e^{tB}$. □

Now given $g_1 = e^{t_1A_1}$, $g_2 = e^{t_2A_2}$, there group commutator is defined as

$$\begin{aligned} g_1g_2g_1^{-1}g_2^{-1} &= e^{t_1A_1}e^{t_2A_2}e^{-t_1A_1}e^{-t_2A_2} \\ &= e^{t_1A_1+t_2A_2+\frac{1}{2}t_1t_2[A_1,A_2]+O(t^4)}e^{-t_1A_1-t_2A_2+\frac{1}{2}t_1t_2[A_1,A_2]+O(t^4)} \\ &= e^{t_1t_2[A_1,A_2]+O(t^4)} = 1 + t_1t_2[A_1,A_2] + O(t^4). \end{aligned}$$

This shows that the Lie bracket in Lie algebra corresponds to the group commutator in Lie group, at least to the second order. And a Lie algebra is said to be **abelian** if $[A, B] = 0$ for all $A, B \in \mathfrak{g}$, and this passes through the exponential map to the Lie group, making G abelian as well. Meanwhile, as $A_i = \frac{\mathrm{d}}{\mathrm{d}t}g_i(t)\big|_{t=0}$, we have

$$g_{12}(t) \equiv g_1(\sqrt{t})g_2(\sqrt{t})g_1^{-1}(\sqrt{t})g_2^{-1}(\sqrt{t}) = 1 + t[A_1,A_2] + O(t^2) = e^{t[A_1,A_2]+O(t^2)},$$

which is a one-parameter subgroup with tangent vector $[A_1, A_2]$ at identity.

Heisenberg algebra

Suppose $[A, B] = z$ with $[z, A] = [z, B] = 0$, then the BCH formula gives

$$e^A e^B = e^{A+B+\frac{1}{2}z} = e^z e^B e^A. \tag{2.29}$$

This is called the Weyl form of canonical commutation relation (CCR), and A and B are said to be in **Heisenberg relation**. For instance, in quantum mechanics, we have $[\hat{x}, \hat{p}] = i\hbar$ with $[\hat{x}, i\hbar] = [\hat{p}, i\hbar] = 0$, then the translational operators $e^{a\hat{p}}$ and $e^{b\hat{x}}$ satisfy

$$e^{a\hat{p}} e^{b\hat{x}} = e^{-i\hbar ab} e^{b\hat{x}} e^{a\hat{p}},$$

or by writing $\exp\left(\boldsymbol{A} \cdot \hat{\boldsymbol{P}}\right) = \exp(A_p \hat{p} + A_x \hat{x})$, we have

$$e^{\boldsymbol{A}\cdot\hat{\boldsymbol{P}}} e^{\boldsymbol{B}\cdot\hat{\boldsymbol{P}}} = e^{\frac{i\hbar}{2}\boldsymbol{A}^T\cdot\Lambda\cdot\boldsymbol{B}} e^{(\boldsymbol{A}+\boldsymbol{B})\cdot\hat{\boldsymbol{P}}},$$

with $\Lambda = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ the symplectic matrix. This gives the geometric view of quantization.

Another example of Heisenberg relation appears in the problem of an electron confined to a two-dimensional plane in a perpendicular magnetic field B. Basically, the Hamiltonian is given by

$$H = \frac{1}{2m}(\boldsymbol{p} - e\boldsymbol{A})^2,$$

with $\boldsymbol{A}$ the vector potential satisfying $\nabla \times \boldsymbol{A} = B\hat{z}$. By defining the kinetic momentum $\boldsymbol{\pi} = \boldsymbol{p} - e\boldsymbol{A}$, we have $[\pi_x, \pi_y] = i\hbar eB$, and the Hamiltonian can be written as

$$H = \frac{1}{2m}(\pi_x^2 + \pi_y^2).$$

Now we can define the guiding center coordinates as

$$R_x = x - \frac{\pi_y}{eB}, \quad R_y = y + \frac{\pi_x}{eB},$$

which satisfy $[R_x, R_y] = -\frac{i\hbar}{eB}$, and commute with π_x and π_y. Thus the guiding center coordinates are also in Heisenberg relation, and we have two sets of Heisenberg algebras, one from kinetic momentum and the other from guiding center coordinates. In the symmetric gauge $\boldsymbol{A} = \frac{B}{2}(-y, x, 0)$, we have

$$\pi_x = p_x + \frac{eB}{2}y, \quad \pi_y = p_y - \frac{eB}{2}x,$$

and we can define the magnetic translation operators as

$$\tilde{\boldsymbol{\pi}} = \boldsymbol{p} + e\boldsymbol{A} = (\tilde{\pi}_x, \tilde{\pi}_y) = (p_x - \frac{eB}{2}y, p_y + \frac{eB}{2}x),$$

which commute with π_x and π_y, but satisfy $[\tilde{\pi}_x, \tilde{\pi}_y] = -i\hbar eB$. Thus the magnetic translation operators also form a Heisenberg algebra, and generates the magnetic translation group.

Infinitesimal A

Now consider $A = \epsilon$ infinitesimal, then by BCH formula we have

$$\begin{aligned} C &= B + \int_0^1 g(e^{s\mathrm{Ad}_\epsilon} e^{\mathrm{Ad}_B})\epsilon \mathrm{d}s = B + \int_0^1 g(e^{\mathrm{Ad}_B})\epsilon \mathrm{d}s + O(\epsilon^2) \\ &= B + \frac{\mathrm{Ad}_B}{e^{\mathrm{Ad}_B} - 1}\epsilon + O(\epsilon^2) \\ &= B + \epsilon - \frac{1}{2}[B, \epsilon] + \frac{1}{12}[B, [B, \epsilon]] - \frac{1}{720}[B, [B, [B, [B, \epsilon]]]] + \cdots + O(\epsilon^2). \end{aligned}$$

Note that we have utilized the expansion

$$\begin{aligned} \frac{x}{e^x - 1} &= \frac{x}{\sum_{n=1}^{\infty} \frac{x^n}{n!}} = \frac{1}{1 + \frac{x}{2!} + \frac{x^2}{3!} + \frac{x^3}{4!} + \cdots} \\ &= 1 - \left(\frac{x}{2!} + \frac{x^2}{3!} + \cdots\right) + \left(\frac{x}{2!} + \frac{x^2}{3!} + \cdots\right)^2 - \left(\frac{x}{2!} + \frac{x^2}{3!} + \cdots\right)^3 + \cdots \\ &= 1 - \frac{x}{2} + \frac{x^2}{12} - \frac{x^4}{720} + \cdots \\ &= 1 + \sum_{n=1}^{\infty} (-1)^n \left(\sum_{m=1}^{\infty} \frac{x^m}{(m+1)!}\right)^n \\ &= 1 + \sum_{l=1}^{\infty} \frac{B_l}{l!} x^l, \end{aligned}$$

where the lth Bernoulli number B_l is defined as

$$B_l = l! \sum_{mn=l} (-1)^n \left(\frac{1}{(m+1)!}\right)^n.$$

2.1.5 Abstract Lie Algebra

Definition 2.1.4. *An abstract Lie algebra $\mathfrak{g}$ over a field κ is a vector space over κ with a bilinear product map $(v_1, v_2) \to [v_1, v_2] \in \mathfrak{g}$, s.t. $\forall\ v_1, v_2, v_3 \in \mathfrak{g}$ and $\alpha, \beta \in \kappa$, we have*

- $[v_1, v_2] = -[v_2, v_1]$
- $[\alpha v_1 + \beta v_2, v_3] = \alpha[v_1, v_3] + \beta[v_2, v_3]$
- $[[v_1, v_2], v_3] + [[v_2, v_3], v_1] + [[v_3, v_1], v_2] = 0$.

This is a repeat of definition in previous section, with remarks:

- Generically, we can not write $[v_1, v_2] = v_1 \cdot v_2 - v_2 \cdot v_1$, because there is no associative product defined in $\mathfrak{g}$.
- Any associate algebra A can be made into a Lie algebra by defining $[a, b] = ab - ba$, $\forall a, b \in A$. This is called the commutator Lie algebra associated to A. Typical examples are matrix algebras.
- Lie algebra is a special case of non-associative algebra.

- A sub-Lie algebra $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is a linear subspace of $\mathfrak{g}$ that is closed under the Lie bracket.
- An ideal $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is a sub-Lie algebra that satisfies $[v, h] \in \mathfrak{h}$, $\forall v \in \mathfrak{g}$ and $h \in \mathfrak{h}$.
- A direct sum of two Lie algebras $\mathfrak{g}_1$ and $\mathfrak{g}_2$ is a Lie algebra $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$, with the Lie bracket defined as $[(v_1, v_2), (u_1, u_2)] = ([v_1, u_1], [v_2, u_2])$, $\forall v_1, u_1 \in \mathfrak{g}_1$ and $v_2, u_2 \in \mathfrak{g}_2$.
- There is no tensor product of Lie algebras.
- If a Lie algebra has a unit then it is trivial.

For the last point, a unit e means $[e, v] = v$ for any $v \in \mathfrak{g}$, then by letting $v_3 = e$ in Jacobi identity, we have $[v_1, v_2] = 0$ for any $v_1, v_2 \in \mathfrak{g}$, thus $[e, v] = 0$ and $v = 0$.

Definition 2.1.5. *A homomorphism between two Lie algebras $\mu : \mathfrak{g}_1 \to \mathfrak{g}_2$ is a linear map that preserves the Lie bracket, i.e. $\mu([v_1, v_2]) = [\mu(v_1), \mu(v_2)]$, $\forall v_1, v_2 \in \mathfrak{g}_1$.*

- As usual, we can define isomorphism, automorphism, endomorphism, kernel and image of Lie algebra homomorphisms.
- The kernel of a Lie algebra homomorphism is an ideal.
- The image of a Lie algebra homomorphism is a sub-Lie algebra.
- If μ is surjective, then $\mathfrak{g}_1 / \ker(\mu) \cong \mathfrak{g}_2$.

For the second point, the kernel means $\mu(v_1) = 0$ for any $v_1 \in \ker(\mu) \subset \mathfrak{g}_1$, thus for any $v_2 \in \mathfrak{g}_1$, we have $\mu([v_1, v_2]) = [\mu(v_1), \mu(v_2)] = 0$, thus $[v_1, v_2] \in \ker(\mu)$.

For example, in one dimension, real Lie algebra $= \mathbb{R}$, with $\{e_1\}$ as the only basis, we have $[e_1, e_1] = 0$. The Lie group reads $(\mathbb{R}^*, \times) \cong (\mathbb{R}, +)$ ($\mathbb{R}^*$ means non-zero real numbers).

In two dimensions, the vector space of Lie algebra is $\mathbb{R}^2$, with $\{e_1, e_2\}$ as a basis with:
(a) Abelian Lie algebra: $[e_1, e_2] = 0$. The Lie group reads $(\mathbb{R}^* \times \mathbb{R}^*, \times) \cong (\mathbb{R}^2, +)$.
(b) Non-Abelian Lie algebra: $[e_1, e_2] = e_1$.
Notice that these are the only two possibilities (up to isomorphism) in two dimensions.

Proof. To show this, we consider a non-Abelian 2D Lie algebra, then by assuming $a, b \in \mathfrak{g}$ linearly independent, we let $[a, b] = e_1$, then for any two vectors $w_1 = \alpha_1 a + \beta_1 b$, $w_2 = \alpha_2 a + \beta_2 b$, we have

$$[w_1, w_2] = (\alpha_1\beta_2 - \alpha_2\beta_1)[a, b] = (\alpha_1\beta_2 - \alpha_2\beta_1)e_1,$$

we can thus choose a new basis e_1, k with $[e_1, k] = ce_1$, $c \neq 0$, and by $k = ce_2$ we have $[e_1, e_2] = e_1$. □

To concretely build the group, we can consider the matrix Lie group

$$e_1 \to \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad e_2 \to \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

with $e_1e_2 = e_1$ and $e_2e_1 = 0$. Then generally let $X = x_1e_1 + x_2e_2 = \begin{pmatrix} 0 & x_1 \\ 0 & x_2 \end{pmatrix}$, and by exponential map we have

$$g_X(t) = e^{tX} = \sum_{n=0}^{\infty} \frac{t^n}{n!} X^n = 1 + \sum_{n=1}^{\infty} \frac{t^n}{n!} \begin{pmatrix} 0 & x_1 x_2^{n-1} \\ 0 & x_2^n \end{pmatrix} = \begin{pmatrix} 1 & \frac{x_1}{x_2}(e^{tx_2} - 1) \\ 0 & e^{tx_2} \end{pmatrix}.$$

We see that the group is just the $ax + b$ group, i.e., the group of affine transformations on the real line, with $a = e^{tx_2} \in \mathbb{R}^+$ and $b = \frac{x_1}{x_2}(e^{tx_2} - 1) \in \mathbb{R}$. Put simply, the group is

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} | a \in \mathbb{R}^+, b \in \mathbb{R} \right\},$$

with the group multiplication

$$\begin{pmatrix} a_1 & b_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a_1a_2 & a_1b_2 + b_1 \\ 0 & 1 \end{pmatrix}.$$

By similar arguments, we have totally six 3D Lie algebras (up to isomorphism):

- Abelian Lie algebra: $[e_i, e_j] = 0$, $\forall i, j = 1, 2, 3$.
- Heisenberg Lie algebra: $[e_1, e_2] = e_3$, other brackets zero.
- $[e_1, e_2] = e_2$, other brackets zero.
- $[e_1, e_2] = e_2$, $[e_1, e_3] = e_3$, other brackets zero.
- $[e_1, e_2] = e_2$, $[e_1, e_3] = -e_3$, other brackets zero.
- $[e_1, e_2] = e_3$, $[e_2, e_3] = e_1$, $[e_3, e_1] = e_2$ (isomorphic to $\mathfrak{so}(3)$).

Derivation and Semi-direct Sum

Let us consider also the semi-direct sum between $\mathfrak{g}$ and $\mathfrak{h}$. To define this, we introduce the notion of derivation.

Definition 2.1.6. *A derivation of a Lie algebra $\mathfrak{g}$, called* $\mathrm{Der}(\mathfrak{g})$*, is a linear map $D : \mathfrak{g} \to \mathfrak{g}$ that satisfies the Leibniz rule:*

$$D([v_1, v_2]) = [D(v_1), v_2] + [v_1, D(v_2)], \quad \forall v_1, v_2 \in \mathfrak{g}. \tag{2.30}$$

Thus, we consider the homomorphism map $D : \mathfrak{g} \to \mathrm{Der}(\mathfrak{h})$ by $Y \to D_Y \in \mathrm{Der}(\mathfrak{h})$. Then the semi-direct sum $\mathfrak{g} \ltimes_D \mathfrak{h}$ is defined on the vector space $\mathfrak{g} \oplus \mathfrak{h}$ with the Lie bracket

$$[(X_1, Y_1), (X_2, Y_2)] = ([X_1, X_2], D_{X_1}(Y_2) - D_{X_2}(Y_1) + [Y_1, Y_2]), \quad \forall X_1, X_2 \in \mathfrak{g}, Y_1, Y_2 \in \mathfrak{h}.$$

This results in a new Lie algebra, and the skew-symmetry and linear properties are obvious. To see the Jacobi identity, we exploit the homomorphism property of D, i.e., $D_{[X_1,X_2]} = [D_{X_1}, D_{X_2}]$, thus we can check by one term

$$\begin{aligned}
&[[(X_1, Y_1), (X_2, Y_2)], (X_3, Y_3)] + \text{cyclic perm.} \\
=&[([X_1, X_2], [Y_1, Y_2] + D_{X_1}(Y_2) - D_{X_2}(Y_1)), (X_3, Y_3)] + \text{cyclic perm.} \\
=&([[X_1, X_2], X_3], [[Y_1, Y_2] + D_{X_1}(Y_2) - D_{X_2}(Y_1), Y_3] + D_{[X_1,X_2]}(Y_3) \\
&- D_{X_3}([Y_1, Y_2]) - D_{X_3}(D_{X_1}(Y_2)) + D_{X_3}(D_{X_2}(Y_1))),
\end{aligned}$$

and sees the cyclic nature of the expression.

The derivations themselves form a Lie algebra under the commutator bracket, i.e., $[D_1, D_2] = D_1D_2 - D_2D_1$, $\forall D_1, D_2 \in \mathrm{Der}(\mathfrak{g})$. To see this, we check the Leibniz rule:

$$\begin{aligned}
[D_1, D_2](v_1v_2) &= (D_1D_2 - D_2D_1)(v_1v_2) \\
&= D_1(D_2(v_1)v_2 + v_1D_2(v_2)) - (1 \leftrightarrow 2) \\
&= D_1D_2(v_1)v_2 + D_2(v_1)D_1(v_2) + D_1(v_1)D_2(v_2) + v_1D_1D_2(v_2) - (1 \leftrightarrow 2) \\
&= D_1D_2(v_1)v_2 - D_2D_1(v_1)v_2 + v_1(D_1D_2(v_2) - D_2D_1(v_2)) \\
&= [D_1, D_2](v_1)v_2 + v_1[D_1, D_2](v_2).
\end{aligned}$$

Thus the homomorphism $D : \mathfrak{g} \to \mathrm{Der}(\mathfrak{h})$ is well-defined as a Lie algebra homomorphism.

For an associative algebra $\mathcal{A}$, we can define an inner derivation $D : \mathcal{A} \to \mathrm{Der}(\mathcal{A})$ by $a \to D_a$ with $D_a(b) = [a, b] = ab - ba$, $\forall a, b \in \mathcal{A}$. This is a derivation since the Leibniz rule is satisfied:

$$D_a(bc) = abc - bca = abc - bac + bac - bca = [a, b]c + b[a, c] = D_a(b)c + bD_a(c).$$

Moreover, the inner derivation preserves the Lie bracket:

$$[D_{a_1}, D_{a_2}] = D_{[a_1,a_2]}, \quad \forall a_1, a_2 \in \mathcal{A}.$$

This is because

$$\begin{aligned}
[D_{a_1}, D_{a_2}](b) &= D_{a_1}(D_{a_2}(b)) - D_{a_2}(D_{a_1}(b)) \\
&= D_{a_1}([a_2, b]) - D_{a_2}([a_1, b]) \\
&= [a_1, [a_2, b]] - [a_2, [a_1, b]] \\
&= [[a_1, a_2], b] \quad \text{(by Jacobi identity)} \\
&= D_{[a_1,a_2]}(b).
\end{aligned}$$

Thus, the inner derivation is a commutator algebra homomorphism from the commutator algebra associated to $\mathcal{A}$ to $\mathrm{Der}(\mathcal{A})$ (not necessary smooth thus not Lie in general). Indeed, the kernel of this homomorphism is the center of $\mathcal{A}$, i.e., $\ker(D) = \{a | D_a(b) = 0, \forall b\} = \{a \in \mathcal{A} | ab = ba, \forall b \in \mathcal{A}\}$.

Now if we prompt $\mathcal{A}$ to $\mathfrak{g}$ as a Lie algebra, then the inner derivation is still well-defined, and gives a true Lie algebra homomorphism from $\mathfrak{g}$ to $\mathrm{Der}(\mathfrak{g})$. This is a Lie algebra derivation as

$$\begin{aligned}
D_y([x_1, x_2]) &= [y, [x_1, x_2]] \\
&= yx_1x_2 - yx_2x_1 - x_1x_2y + x_2x_1y \\
&= yx_1x_2 - x_1yx_2 + x_1yx_2 - x_1x_2y - (yx_2x_1 - x_2yx_1 + x_2yx_1 - x_2x_1y) \\
&= [y, x_1]x_2 + x_1[y, x_2] - (x_2[y, x_1] + [y, x_2]x_1) \\
&= [[y, x_1], x_2] + [x_1, [y, x_2]] \\
&= [D_y(x_1), x_2] + [x_1, D_y(x_2)].
\end{aligned}$$

Furthermore, the inner derivation is indeed an ideal of $\mathrm{Der}(\mathfrak{g})$. To see this, we consider an arbitrary derivation $D \in \mathrm{Der}(\mathfrak{g})$ and an inner derivation $D_y \in \mathrm{Der}(\mathfrak{g})$, then we have

$$\begin{aligned}
[D, D_y](x) &= D(D_y(x)) - D_y(D(x)) \\
&= D([y, x]) - [y, D(x)] \\
&= [D(y), x] + [y, D(x)] - [y, D(x)] \\
&= [D(y), x] = D_{D(y)}(x),
\end{aligned}$$

i.e., $[D, D_y] = D_{D(y)}$ is still an inner derivation.

Structure Constants

Now let us come back again to the structure constants, that by choosing $\{e_i\}$ as a basis of $\mathfrak{g}$, we have

$$[e_i, e_j] = \sum_k f_{ij}^k e_k, \tag{2.31}$$

where f_{ij}^k are the structure constants under the basis $\{e_i\}$. It is easy to see that $f_{ij}^k = -f_{ji}^k$ by the skew-symmetry of Lie bracket, and by the Jacobi identity we have

$$\sum_m (f_{ij}^m f_{mk}^n + f_{jk}^m f_{mi}^n + f_{ki}^m f_{mj}^n) = 0, \tag{2.32}$$

as we have derived above.

Although there is no symmetry relation generally between upper and lower indices of f_{ij}^k, we can still lower the upper index by a non-degenerate symmetric bilinear form $h : \mathfrak{g} \times \mathfrak{g} \to \kappa$, which is an invariant form by imposing

$$h(\mathrm{Ad}_X Y, Z) + h(Y, \mathrm{Ad}_X Z) = 0, \quad \forall X, Y, Z \in \mathfrak{g}. \tag{2.33}$$

Thus by defining $h_{ij} = h(e_i, e_j)$, we have

$$f_{ijk} = h_{il} f_{jk}^l = h(e_i, [e_j, e_k]) = h([e_i, e_j], e_k), \tag{2.34}$$

which is totally antisymmetric, i.e., $f_{ijk} = f_{[ijk]}$. To see this, we firstly notice that $f_{ijk} = -f_{ikj}$ by the skew-symmetry of Lie bracket, then by the symmetric property of h, we have $h_{ij} = h_{ji}$, thus

$$\begin{aligned} f_{jik} &= h(e_j, [e_i, e_k]) = -h([e_i, e_j], e_k) = -f_{ijk}, \\ f_{kji} &= h(e_k, [e_j, e_i]) = h([e_j, e_i], e_k) = -f_{jik} = f_{ijk}. \end{aligned}$$

Rep of Lie Algebra

Just as we define group representation as a homomorphism from group G to $\mathrm{GL}(V)$, we can define Lie algebra representation as a homomorphism from Lie algebra $\mathfrak{g}$ to $\mathrm{End}(V)$, the set of endomorphisms of a vector space V.

Definition 2.1.7. *A representation of a Lie algebra $\mathfrak{g}$ on a vector space V is a Lie algebra homomorphism $T : \mathfrak{g} \to \mathrm{End}(V)$, i.e., $T([X, Y]) = [T(X), T(Y)] = T(X)T(Y) - T(Y)T(X)$, $\forall X, Y \in \mathfrak{g}$. We also say that V is a $\mathfrak{g}$-module.*

Concretely, we have a vector space V with a $\mathfrak{g}$ action $\mathfrak{g} \times V \to V$ defined by $(X, v) \to T(X) \cdot v$, which is bilinear and satisfies $T([X, Y]) \cdot v = (T(X)T(Y) - T(Y)T(X)) \cdot v$, $\forall X, Y \in \mathfrak{g}$ and $v \in V$. With a basis for V we get a matrix rep of $\mathfrak{g}$.

Given any mat-rep T of Lie group G, we automatically get a mat-rep of its Lie algebra $\mathfrak{g} = L(G)$, that if $X = \frac{\mathrm{d}}{\mathrm{d}t} g(t)\big|_{t=0} \in \mathfrak{g}$, we have

$$T(X) = \frac{\mathrm{d}}{\mathrm{d}t} T(g(t))\bigg|_{t=0}. \tag{2.35}$$

This is a Lie algebra rep since we can exponential near identity $g(t) = e^{tX}$, thus $T(g(t)) = e^{tT(X)}$. Then by the BCH formula we have

$$\begin{aligned}
[T(X), T(Y)] &= \frac{\mathrm{d}^2}{\mathrm{d}t\mathrm{d}s}(T(e^{tX})T(e^{sY}) - T(e^{sY})T(e^{tX}))\Big|_{t,s=0} \\
&= \frac{\mathrm{d}^2}{\mathrm{d}t\mathrm{d}s}\left(T(e^{tX}e^{sY}) - T(e^{sY}e^{tX})\right)\Big|_{t,s=0} \\
&= \frac{\mathrm{d}^2}{\mathrm{d}t\mathrm{d}s}T\left(e^{tX+sY+\frac{1}{2}ts[X,Y]+\cdots} - e^{sY+tX+\frac{1}{2}st[Y,X]+\cdots}\right)\Big|_{t,s=0} \\
&= T(\frac{1}{2}[X,Y] - \frac{1}{2}[Y,X]) = T([X,Y]).
\end{aligned}$$

There exists a canonical rep with $V = \mathfrak{g}$ of any Lie algebra $\mathfrak{g}$, called the adjoint rep $\mathrm{Ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$, defined by $\mathrm{Ad}_X(Y) = [X, Y]$, $\forall X, Y \in \mathfrak{g}$. This is a Lie algebra rep as

$$\begin{aligned}
[\mathrm{Ad}_X, \mathrm{Ad}_Y](Z) &= \mathrm{Ad}_X(\mathrm{Ad}_Y(Z)) - \mathrm{Ad}_Y(\mathrm{Ad}_X(Z)) \\
&= \mathrm{Ad}_X([Y,Z]) - \mathrm{Ad}_Y([X,Z]) \\
&= [X,[Y,Z]] - [Y,[X,Z]] \\
&= [[X,Y],Z] \quad \text{(by Jacobi identity)} \\
&= \mathrm{Ad}_{[X,Y]}(Z),
\end{aligned}$$

i.e.,

$$[\mathrm{Ad}_X, \mathrm{Ad}_Y] = \mathrm{Ad}_{[X,Y]}. \tag{2.36}$$

This is similar to the inner derivation defined above, and indeed the inner derivation is just the adjoint rep of $\mathfrak{g}$ on itself. Furthermore, giving basis $\{e_i\}_{i=1}^n$ of $\mathfrak{g}$, we see that

$$\mathrm{Ad}_{e_i}(e_j) = [e_i, e_j] = \sum_k f_{ij}^k e_k, \tag{2.37}$$

or more explicitly in matrix form

$$\mathrm{Ad}_{e_i}\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} f_{i1}^1 & f_{i1}^2 & \cdots & f_{i1}^n \\ f_{i2}^1 & f_{i2}^2 & \cdots & f_{i2}^n \\ \vdots & \vdots & \ddots & \vdots \\ f_{in}^1 & f_{in}^2 & \cdots & f_{in}^n \end{pmatrix}\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix},$$

meaning that the matrix elements of adjoint rep are just the structure constants: $(\mathrm{Ad}_{e_i})_j{}^k = f_{ij}{}^k$.

- The kernel of the adjoint rep is the center of $\mathfrak{g}$, i.e., $\ker(\mathrm{Ad}) = \{X \in \mathfrak{g} | [X, Y] = 0, \forall Y \in \mathfrak{g}\}$, which is just the definition of center. Thus Ad is faithful iff $\mathfrak{g}$ has a trivial center.
- For $\mathfrak{g}$ Abelian, we have $\mathrm{Ad}_X = 0$ for any $X \in \mathfrak{g}$, thus the adjoint rep is trivial.
- With the adjoint rep, we can define the Killing form $K : \mathfrak{g} \times \mathfrak{g} \to \kappa$ by $K(X, Y) = \mathrm{Tr}_{\mathfrak{g}}(\mathrm{Ad}_X \mathrm{Ad}_Y)$, $\forall X, Y \in \mathfrak{g}$.

The Killing form is clearly a symmetric bilinear form, i.e., $K(X,Y) = K(Y,X)$ and $K(\alpha X_1+\beta X_2, Y) = \alpha K(X_1,Y)+\beta K(X_2,Y)$, $\forall X, X_1, X_2, Y \in \mathfrak{g}$ and $\alpha, \beta \in \kappa$. Moreover, it is an invariant form as

$$\begin{aligned} K([Z,X],Y) + K(X,[Z,Y]) &= \mathrm{Tr}_{\mathfrak{g}}(\mathrm{Ad}_{[Z,X]}\mathrm{Ad}_Y) + \mathrm{Tr}_{\mathfrak{g}}(\mathrm{Ad}_X\mathrm{Ad}_{[Z,Y]}) \\ &= \mathrm{Tr}_{\mathfrak{g}}([\mathrm{Ad}_Z, \mathrm{Ad}_X]\mathrm{Ad}_Y) + \mathrm{Tr}_{\mathfrak{g}}(\mathrm{Ad}_X[\mathrm{Ad}_Z, \mathrm{Ad}_Y]) \\ &= \mathrm{Tr}_{\mathfrak{g}}(\mathrm{Ad}_Z\mathrm{Ad}_X\mathrm{Ad}_Y - \mathrm{Ad}_X\mathrm{Ad}_Z\mathrm{Ad}_Y + \mathrm{Ad}_X\mathrm{Ad}_Z\mathrm{Ad}_Y - \mathrm{Ad}_X\mathrm{Ad}_Y\mathrm{Ad}_Z) \\ &= 0, \end{aligned}$$

$\forall X, Y, Z \in \mathfrak{g}$. Given a basis $\{e_i\}$ of $\mathfrak{g}$, we have $K_{ij} = K(e_i, e_j) = \mathrm{Tr}_{\mathfrak{g}}(\mathrm{Ad}_{e_i}\mathrm{Ad}_{e_j})$, or more explicitly

$$K_{ij} = f_{im}{}^n f_{jn}{}^m, \tag{2.38}$$

known as the Cartan-Killing metric.

Although each Lie algebra has its adjoint rep, it may be not faithful. There exists a strange theorem that

Theorem 2.1.6 (Ado's Theorem)**.** *Every finite-dimensional Lie algebra over a field of characteristic zero has a finite-dimensional faithful representation.*

The main idea is to construct a rep V that is faithful on center $Z(\mathfrak{g})$, then $V \oplus \mathfrak{g}$, or $Z(\mathfrak{g}) \oplus \mathfrak{g}/Z(\mathfrak{g})$ is faithful on $\mathfrak{g}$.

Lie's theorem

Given any Lie group G, we can get its Lie algebra $\mathfrak{g} = L(G)$, and any mat-rep of G gives a mat-rep of $\mathfrak{g}$. The Lie's theorem tries to answer the reverse question, that given a mat-rep of $\mathfrak{g}$, can we get a mat-rep of G? intuitively, this is always possible by exponential map, but the problem arises from the BCH formula:

- The BCH series may not converge, thus the exponential map may not be defined globally.
- Even if the BCH series converges, the exponential map may not be surjective, thus we can not get all elements of G by exponentiating elements of $\mathfrak{g}$.
- Even if the exponential map is surjective, it may not be injective, thus we can not get a unique element of G by exponentiating an element of $\mathfrak{g}$.

Theorem 2.1.7 (Lie's theorem)**.** *(a) Every finite-dimensional Lie algebra $\mathfrak{g}$ arises from a unique (up to isomorphism) simply connected Lie group G.*

(b) Under this correspondence, Lie group homomorphisms $f : G_1 \to G_2$ are in one-to-one correspondence with Lie algebra homomorphisms $\mu : \mathfrak{g}_1 \to \mathfrak{g}_2$.

- We need the simply connected condition to ensure the uniqueness of G, since locally isomorphic Lie groups always share the same Lie algebra.
- Following the theorem, if $\mathfrak{h}$ is a sub-Lie algebra of $\mathfrak{g}$, then there exists a unique simply connected Lie subgroup H of G such that $\mathfrak{h} = L(H)$. Moreover, there is

a homomorphism from $H \to G$, but H need not to be an embedded submanifold/topological subgroup of G. For instance, take $G = SU(n)$ with $n > 2$, and we firstly consider its subgroup $K = U(1) \times U(1)$ belonging to the maximal torus,

$$K = \mathrm{diag}(\lambda, \mu, (\lambda\mu)^{-1}, 1, \cdots, 1), \ \lambda, \mu \in U(1).$$

The Lie algebra of K originates from 2D Abelian Lie algebra $\mathfrak{k} = \mathbb{R} \oplus \mathbb{R}$, by Lie algebra homomorphism

$$\nu : (x, y) \to \mathrm{diag}(2\pi i x, 2\pi i y, -2\pi i(x+y), 0, \cdots, 0), \quad x, y \in \mathbb{R}.$$

Now for $H = (\mathbb{R}, +)$ (simply connected), and its Lie algebra $\mathfrak{h} = \mathbb{R}$, we can assign an embedding map $\mathfrak{h} \hookrightarrow \mathfrak{k}$ by

$$\iota : x \mapsto \mathrm{diag}(2\pi i x, 2\pi i \alpha x, -2\pi i(1+\alpha)x, 0, \cdots, 0), \quad x, \alpha \in \mathbb{R},$$

and the Lie group homomorphism $H \to K \hookrightarrow G$ is given by

$$x \mapsto \mathrm{diag}(e^{2\pi i x}, e^{2\pi i \alpha x}, e^{-2\pi i(1+\alpha)x}, 1, \cdots, 1).$$

This establishes the correspondence between Lie group homomorphism $H \to G$ and Lie algebra homomorphism $\mathfrak{h} \to \mathfrak{g}$. However, the image of H in G is not an embedded submanifold/topological subgroup of G if α is irrational, since the image is dense in a 2-torus in G (for α rational, the image is isomorphic to $U(1)$, while for α irrational, the image fills K).

- Lie's theorem is indeed a functor between categories of Lie algebras and Lie groups, i.e., $L(G) \to G$, with relations corresponding homomorphisms.

Proof. Given a Lie group, its Lie algebra is always well-defined, thus we only need to show the reverse direction.

We shall use Ado's theorem here to get a faithful mat-rep of $\mathfrak{g}$, making $\mathfrak{g} \subset M_n(\mathbb{R})$ a sub Lie-algebra of $M_n(\mathbb{R})$, which basically corresponds to a matrix Lie group $G \subset GL(n, \mathbb{R})$ by exponential map.

Now we build the Lie group by considering the first homotopy group of Lie algebra, whose basic element is a loop

$$\gamma : [0, 1] \to \mathfrak{g}, \ \gamma(0) = \gamma(1) = 0.$$

To such a loop, we associate a group element by solving the ODE

$$\frac{\mathrm{d}}{\mathrm{d}t} g_\gamma(t) = \gamma(t) g_\gamma(t),$$

or equivalently the holonomy

$$g_\gamma = \mathcal{P} \exp\left(\int_0^1 \gamma(t) \mathrm{d}t \right),$$

with path-ordering operator $\mathcal{P}$. For bounded $\gamma(t)$, the solution $g_\gamma(t)$ always exists and is unique, thus we have a well-defined map from loop γ to group element g_γ. With this correspondence, a closed loop in Lie algebra gives an open path in $GL(n, \mathbb{R})$.

Now we want to show that $\{g_\gamma\}$ form a subgroup. We would like to add a constraint that all derivatives of $\gamma(t)$ vanish at the endpoints, i.e., $\frac{\mathrm{d}^m}{\mathrm{d}t^m}\gamma(t)\big|_{t=0,1} = 0$, $\forall m \geq 1$. In this way, we can join two loops/paths smoothly as

$$\gamma_1 \circ \gamma_2(t) = \begin{cases} 2\gamma_2(2t), & t \in [0, \frac{1}{2}] \\ 2\gamma_1(2t-1), & t \in [\frac{1}{2}, 1] \end{cases},$$
$$g_{\gamma_1\circ\gamma_2}(t) = \begin{cases} g_{\gamma_2}(2t), & t \in [0, \frac{1}{2}] \\ g_{\gamma_1}(2t-1)g_{\gamma_2}(1), & t \in [\frac{1}{2}, 1] \end{cases}.$$

The factor 2 is to ensure the continuity of $\gamma_1 \circ \gamma_2(t)$ at $t = \frac{1}{2}$, and the continuity of $g_{\gamma_1\circ\gamma_2}(t)$ at $t = \frac{1}{2}$ is guaranteed by the ODE. Thus we have

$$g_{\gamma_1\circ\gamma_2} = g_{\gamma_1} g_{\gamma_2}.$$

In this way, we can define $G' = \{g_\gamma\} \subset GL(n, \mathbb{R})$ the subgroup with constrained elements. The identity element is given by the trivial loop $\gamma(t) = 0$, and the inverse element is given by the reversed loop $\gamma^{-1}(t) = \gamma(1-t)$, since $g_{\gamma^{-1}} = g_\gamma^{-1}$ by the ODE. However, as we have seen above, G' may not be an embedded submanifold of $GL(n, \mathbb{R})$.

To solve this, we firstly define

$$\mathcal{P} = \{\gamma | \gamma : [0,1] \to \mathfrak{g}, \gamma(0) = \gamma(1) = 0, \frac{\mathrm{d}^m}{\mathrm{d}t^m}\gamma(t)\big|_{t=0,1} = 0, \forall m \geq 0\}.$$

This can be considered as an infinite-dimensional manifold, and we can consider its tangent space by considering a path in $\mathcal{P}$ going through some $\gamma_*(t)$, i.e., a path $\gamma(s,t)$ with $\gamma(0,t) = \gamma_*(t)$. Then the tangent vector at $\gamma_*(t)$ is given by $T_{\gamma_*(t)} \cong \mathcal{P}$. The group law in $\mathcal{P}$ is given by the concatenation $\circ$ defined above, but it is not a group as if we define $\gamma_0(t) \equiv 0, \forall t$, then $\gamma \circ \gamma_0$ is only homotopic to γ instead of equal to γ. However, this enlightens us to consider the homotopy classes of $\mathcal{P}$, and the group for our purpose is obtained by imposing

$$\gamma_1 \sim \gamma_2 \Leftrightarrow g_{\gamma_1} = g_{\gamma_2},$$

and the quotient group $G = \mathcal{P}/\sim$ is a finite-dimensional manifold (as $GL(n, \mathbb{R})$ is finite-dimensional), locally isomorphic to $\mathfrak{g}$, and it is indeed a simply-connected Lie group as $\mathcal{P}$ is geometrically contractible ($\mathcal{P}$ is contractible as $\mathfrak{g}$ is contractible). We mention that this is a flat (gauge-equivalent) homotopy rather than a general geometric homotopy, as the latter possibly leads to a trivial group.

To define Lie algebra of G, consider neighborhood $\tilde{U}$ of the zero path $\gamma_0(t) \equiv 0$ in $\mathcal{P}$ where the exponential map is well-defined. We consider the corresponding neighborhood U of identity in $GL(n, \mathbb{R})$, where $g_\gamma(t) = e^{\eta_\gamma(t)}$ should be well-defined. Note that generally $\eta_\gamma(t) \neq \gamma(t)$ and especially $\eta_\gamma(1) \neq 0$, thus making $\eta_\gamma(t)$ generally an open path in $\mathfrak{gl}(n, \mathbb{R})$. But we claim that $\eta_\gamma(1) \in \mathfrak{g}$, since

$$\gamma(t) = \left(\frac{\mathrm{d}}{\mathrm{d}t} g_\gamma(t)\right) g_\gamma(t)^{-1} = \frac{e^{\mathrm{Ad}_{\eta_\gamma(t)}} - 1}{\mathrm{Ad}_{\eta_\gamma(t)}} \dot{\eta}_\gamma(t),$$

by the Lemma in BCH formula, and one expects to solve $\eta_\gamma(t)$ from $\gamma(t)$ perturbatively. Since $\gamma(t) \in \mathfrak{g}$, we have $\eta_\gamma(t) \in \mathfrak{g}$ by induction, thus $\eta_\gamma(1) \in \mathfrak{g}$.

Since in U, the exp-map is injective, and

$$g_{\gamma_1} = g_{\gamma_2} \Leftrightarrow \eta_{\gamma_1}(1) = \eta_{\gamma_2}(1).$$

We have the local neighborhood of identity in G signaled by $\mathcal{K} \equiv \pi(\tilde{U})$ as

$$\mathcal{K} \cong \{\eta_\gamma : [0,1] \to \mathfrak{g} | \eta(0) = 0\} / \sim,$$

where the equivalence relation is given by $\eta_{\gamma_1} \sim \eta_{\gamma_2} \Leftrightarrow \eta_{\gamma_1}(1) = \eta_{\gamma_2}(1)$. Thus $\mathcal{K}$ is completed determined by $\eta_\gamma(1) \in \mathfrak{g}$, i.e., $\mathcal{K} \cong \mathfrak{g}$. This gives the Lie algebra of G as $L(G) \cong \mathfrak{g}$. This proves part (a).

For part (b), again we only need to show the reverse direction. Suppose that we have $\mu : \mathfrak{g}_1 \to \mathfrak{g}_2$, and with (a) we have $\mathfrak{g}_1 \cong T_1(G_1)$ and $\mathfrak{g}_2 \cong T_1(G_2)$, with $G_{1/2}$ simply connected Lie groups. We define $f(g)$ by writing $g = e^X$ and taking $f(g) = e^{\mu(X)}$, and if this is well-defined, then it is a Lie group homomorphism as

$$f(g_1 g_2) = f(e^{X_1} e^{X_2}) = f(e^{X_1 + X_2 + \frac{1}{2}[X_1, X_2] + \cdots}) = e^{\mu(X_1) + \mu(X_2) + \frac{1}{2}[\mu(X_1), \mu(X_2)] + \cdots} = f(g_1) f(g_2).$$

The problem is that g may have multiple logarithms, i.e., $g = e^{X_1} = e^{X_2}$ with $X_1 \neq X_2$, thus generally $f(g)$ may depend on the choice of logarithm.

Instead we can define $f(g)$ again by considering a path $g(t)$ in G_1 with $g(0) = 1$ and $g(1) = g$, and a corresponding path $\gamma(t)$ in $\mathfrak{g}_1$ by

$$\gamma(t) = \left(\frac{\mathrm{d}}{\mathrm{d}t} g(t) \right) g(t)^{-1} \in \mathfrak{g}_1.$$

This is mapped into also a path in $\mathfrak{g}_2$ by $\tilde{\gamma}(t) = \mu(\gamma(t))$, and we can solve a path $\tilde{g}(t)$ in G_2 by

$$\frac{\mathrm{d}}{\mathrm{d}t} \tilde{g}(t) = \tilde{\gamma}(t) \tilde{g}(t), \quad \tilde{g}(0) = 1.$$

Then we try to define $f(g) = \tilde{g}(1)$.

By path concatenation, it is easy to see that $f(g)$ is a Lie group homomorphism. The only problem is that $f(g)$ may depend on the choice of path $g(t)$, which is solved by the simply-connected condition of G_1. Now suppose that we have two paths $g_0(t)$ and $g_1(t)$ with the same endpoints, i.e., $g_0(0) = g_1(0) = 1$ and $g_0(1) = g_1(1) = g$. By the simply-connected condition, there exists a homotopy $g(s,t)$ with $s \in [0,1]$ such that $g(0,t) = g_0(t)$ and $g(1,t) = g_1(t)$. We then define

$$\gamma(s,t) = \frac{\partial}{\partial t} g(s,t) \cdot g(s,t)^{-1} \in \mathfrak{g}_1, \tag{2.39a}$$

$$\eta(s,t) = \frac{\partial}{\partial s} g(s,t) \cdot g(s,t)^{-1} \in \mathfrak{g}_1. \tag{2.39b}$$

these satisfy the Maurer-Cartan condition

$$\frac{\partial}{\partial s} \gamma(s,t) - \frac{\partial}{\partial t} \eta(s,t) = [\gamma(s,t), \eta(s,t)], \tag{2.40}$$

as

$$\begin{aligned}
\frac{\partial}{\partial s} \gamma(s,t) &= \frac{\partial^2}{\partial s \partial t} g(s,t) \cdot g(s,t)^{-1} + \frac{\partial}{\partial t} g(s,t) \cdot \frac{\partial}{\partial s} g(s,t)^{-1} \\
&= \frac{\partial^2}{\partial t \partial s} g(s,t) \cdot g(s,t)^{-1} - \gamma(s,t) \eta(s,t), \\
\frac{\partial}{\partial t} \eta(s,t) &= \frac{\partial^2}{\partial t \partial s} g(s,t) \cdot g(s,t)^{-1} + \frac{\partial}{\partial s} g(s,t) \cdot \frac{\partial}{\partial t} g(s,t)^{-1} \\
&= \frac{\partial^2}{\partial s \partial t} g(s,t) \cdot g(s,t)^{-1} - \eta(s,t) \gamma(s,t),
\end{aligned}$$

where we have used that

$$\partial(AA^{-1}) = (\partial A)A^{-1} + A\partial(A^{-1}) = 0 \Rightarrow \partial(A^{-1}) = -A^{-1}(\partial A)A^{-1}.$$

Note that given γ, η, the Maurer-Cartan condition is a constraint rather than an identity, that leads to the existence of $g(s,t)$.

The Lie algebra homomorphism μ maps $\gamma(s,t)$ and $\eta(s,t)$ into $\tilde{\gamma}(s,t) = \mu(\gamma(s,t))$ and $\tilde{\eta}(s,t) = \mu(\eta(s,t))$ in $\mathfrak{g}_2$, which also satisfy the Maurer-Cartan condition as μ preserves Lie bracket. Thus we can solve $\tilde{g}(s,t)$ in G_2 by

$$\frac{\partial}{\partial t}\tilde{g}(s,t) = \tilde{\gamma}(s,t)\tilde{g}(s,t), \quad \frac{\partial}{\partial s}\tilde{g}(s,t) = \tilde{\eta}(s,t)\tilde{g}(s,t),$$

with $\tilde{g}(s,0) = 1$. Now $g(s,1) = g$ is independent of s, thus $\frac{\partial}{\partial s}g(s,1) = 0$, meaning that $\eta(s,1) = 0$, thus $\tilde{\eta}(s,1) = 0$ and $\frac{\partial}{\partial s}\tilde{g}(s,1) = 0$. This means that $\tilde{g}(s,1)$ is also independent of s, thus $\tilde{g}(0,1) = \tilde{g}(1,1)$, or $f(g)$ is independent of the choice of path $g(t)$. This finishes the proof.

□

• The Maurer-Cartan form $A = \gamma \mathrm{d}t + \eta \mathrm{d}s$ gives a flat connection on the trivial principal bundle $G_1 \times [0,1]^2 \to [0,1]^2$, as its curvature $F = \mathrm{d}A + A \wedge A = 0$ is just the Maurer-Cartan condition. The holonomy of this flat connection gives the Lie group homomorphism $f : G_1 \to G_2$ by solving the ODE above. This is similar to the construction of Wilson line in gauge theory. This is seen as

$$\begin{aligned} F &= \mathrm{d}A + A \wedge A \\ &= \partial_s \gamma \mathrm{d}s \wedge \mathrm{d}t + \partial_t \eta \mathrm{d}t \wedge \mathrm{d}s + \gamma\eta \mathrm{d}t \wedge \mathrm{d}s + \eta\gamma \mathrm{d}s \wedge \mathrm{d}t \\ &= (\partial_s \gamma - \partial_t \eta + [\gamma, \eta])\mathrm{d}s \wedge \mathrm{d}t \\ &= 0. \end{aligned}$$

Chapter 3

Homotopy Group

The language physicists utilize has been evolving from an innocent stage, passing through the analytical jungle feeding on calculus berries, sliding on geometrical mirrors but coming full circle, picking up shining shells and tying them as knot on a rope nowadays, by which I personally denote as new world with algebra. As a branch of algebraic topology, homotopy theory has entered physics a long time ago, so does its application in condensed matter physics, like classification of soliton solutions and topological insulators. Here in the essay, we try to figure out its application of topological defects in ordered media, which mainly follows the review of Mermin in 1979.

Our discussion is mainly separated and organized along a progressive path: firstly, we give some real physical objects as standard examples to reveal the central idea on classification; then we relate the most basic kind of defects to the abstract fundamental group and figure out its property; next we put this forward to the second homotopy group and finally to any order of that, and that is also when we propose the exact sequence formed by different kinds of homotopy groups and re-establish fundamental theorem as key to classification.

3.1 Ordered Media and Topological Defects

Our research is based on the continuous and uniform media, by which I in fact mean that our d-dimensional medium M contains total symmetry from corresponding Euclidean group $\mathbb{E}(d) \cong T(d) \times O(d)$, i.e., full translational and rotational symmetries (and reflection). This might be a bit ideal, but succeeds in avoiding dealing with sundry mistakes for crystalline systems caused by broken translational symmetry, where dislocation together with additional group structure need to be introduced and makes the problem tricky. For clearness, one may keep in mind that topological stability is not always equivalent to physical stability, where the former is a purely mathematical concept while the latter is a physical one depending on the free energy, which we also have no intention to discuss in the essay.

By an *ordered medium* we refer to a medium with a *(local) order parameter* $f(\boldsymbol{r})$ assigned to every point, and formally, this gives us a map from Euclidean space to the *order parameter space* R as

$$f : \mathbb{R}^d \longrightarrow \mathbb{R}^n \text{ or } \mathbb{C}^n, \tag{3.1}$$

where the values of f compose R which is a subset (or imbedded in) n-dimensional real or complex space. From a physical point of view, f must vary continuously throughout

the medium, e.g., a *uniform* medium where f is a constant as a trivial instance, except for some singular regions of lower dimension as *defects*, e.g., singular point in 2-D and 3-D, singular line in 3-D, etc., which are of our central interest.

To illustrate how topology describes singular defects, we use *planar spin* as an example. This kind of order parameter refers to a classical unit vector constrained in two-dimensional plane, and could be represented as

$$f(\boldsymbol{r}) = (u(\boldsymbol{r}), v(\boldsymbol{r})) = \hat{u}\cos(\varphi(\boldsymbol{r})) + \hat{v}\sin(\varphi(\boldsymbol{r})),\ \hat{u}\cdot\hat{v} = 0, \tag{3.2}$$

which is geometrically equivalent to point on 1-sphere S^1. By n-sphere now, we mean that it is the surface of a $(n+1)$-unit ball sitting at the origin and could be represented as

$$S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\},\ |x|^2 \equiv \sum_{i=1}^{n+1} x_i^2. \tag{3.3}$$

And now we could let the order parameter space of planar spin be $R = S^1 \cong SO(2)$ living in $\mathbb{R}^2$, where the isomorphic group relation now happens between the two-dimensional proper rotational group and circle group

$$S^1 = T = \{z \in \mathbb{C} : |z| = 1\}$$

in $\mathbb{C}$, which could also be that for superfluid ^{4}He

$$\psi(\boldsymbol{r}) = \psi_0 e^{i\varphi(\boldsymbol{r})}$$

with branch-cut $\varphi \in [0, 2\pi)$ and a real-constant ψ_0.

Now for defects for planar spins, one may notice that f is not well-defined when $\cos\varphi = 0 = \sin\varphi$, at some $\boldsymbol{r}$ as point P, since now 'vector' losses its meaning, i.e., we

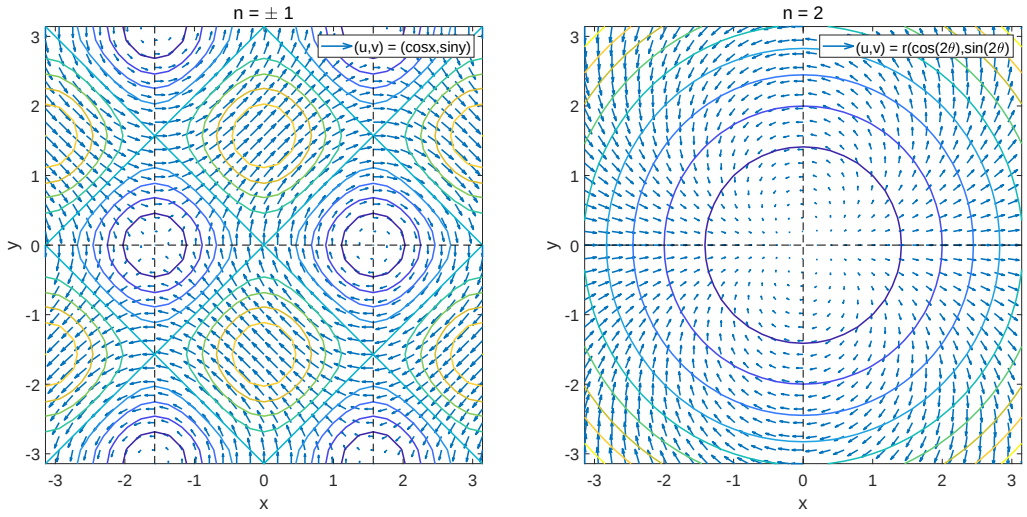


Figure 3.1: Point singularities (at crossing of black dash lines) of planar spins in two dimensions with winding numbers ±1 and 2.

meet a singularity. Now, imagine that we surround a closed contour C counter-clockwise with P as the only singularity inside, then as mentioned, now f on C should be continuous, and we are able to trace how the direction of the vector varies throughout C. Here one

map appears from $\mathbb{R}^2$ as real space to order parameter space $R = S^1$, where domain is C and values compose also a loop in R due to continuity, i.e.,

$$f : C \longrightarrow \{f\} \cong S^1.$$

However, two 1-spheres are very different, as mapped $\{f\}$ assigned on C could circle (also counter-clockwise) integer n times which corresponds to total $\Delta\varphi = n \cdot 2\pi$ changes, while R is just living space set for order parameter and only contains 2π angle. Then by division we get an integer n as how many times order parameter vector winds about P throughout C, which is known as *winding number*, as shown in Fig. 3.1 for instance. And this winding is what we desire, then totally we have a map like

$$\begin{array}{ccc} \mathbb{R}^2 & \longrightarrow & R = S^1 \\ \cap & f & \cap \\ C & \longrightarrow & \{f\} = S^1 \end{array} \quad \pi_1 \quad , \tag{3.4}$$

inside which the latter subset relation measures the time a loop winds on another loop, and could be described by one of the main character in this essay, the *first homotopy group*, aka *fundamental group* $\pi_1(R)$, where lower index refers to dimension of sphere composed by $\{f\}$. Nevertheless, we need to clarify some stuffs before proceeding.

Such singularities ($n \neq 0$) are topologically stable. To illustrate this we need to firstly introduce the concept of *local surgery*. Just as the name claims, now we need to bring

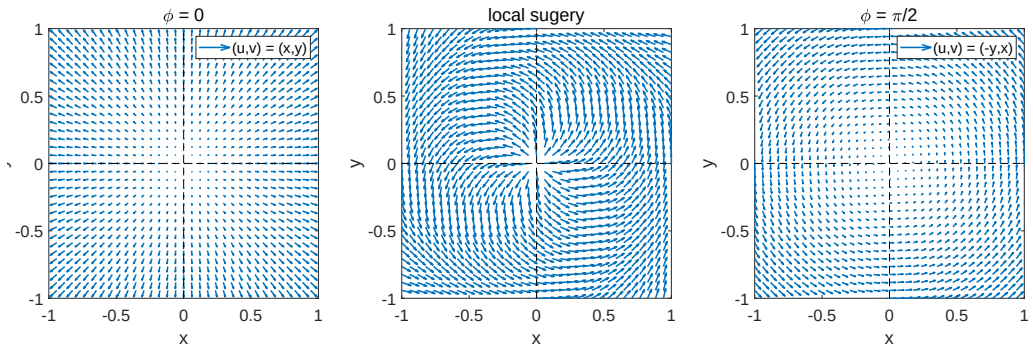


Figure 3.2: Equivalence of two $n = 1$ configurations by local surgery.

some change to the configuration of order parameter, which is not arbitrarily done but obeys the continuity principle, i.e., the changing operation on the configuration must also be continuous except at P. Come back to our example of planar spin, it is always possible to expand near P as

$$\varphi = n\theta + \phi, \ n \in \mathbb{Z}, \tag{3.5}$$

where θ is polar coordinate describing C and ϕ is some constant angle on C, and this holds from

$$\oint_C \mathrm{d}\boldsymbol{l} \cdot \nabla\varphi \sim \frac{n}{r} \oint_C \mathrm{d}s = \frac{n}{r} \cdot 2\pi r = 2\pi n.$$

As shown in left and right panels of Fig. 3.2, these two pictures correspond to rather different configurations. However, by local surgery, one could connect them by increasing ϕ from 0 at $r = 0$ to $\phi = \pi/2$ at $r = r_0$ continuously, as shown in the middle, where

the core of P correspond to $\phi = 0$ while the far-field configuration consist with $\phi = \pi/2$. And in this way we say that two configurations could be transformed into each other by local surgery without alterations outside neighborhood of P. Two configurations could be connected in this way if and only if they have the same winding number, because configuration changes continuously while n is discrete. In this way singularities with $n \neq 0$ are topologically stable since they can not be altered in continuous manner.

By discussion above, we have transformed the problem of classifying singularities by winding numbers to classifying loops around one singularity into topologically distinct classes, where loops in the same class are promised to be connected with each other under some 'local surgery', or more formally, they could continuously deform into each other as *homotopic*, which brings us the concept of *homotopy*. Now given two loop configurations $f_0(\boldsymbol{r})$ and $f_1(\boldsymbol{r})$, they are homotopic if and only if there exists a homotopy $h_t(\boldsymbol{r})$ as a continuous one-parameter family of mappings, s.t. $h_0 = f_0$ and $h_1 = f_1$.

One may notice that under our definition, there could be trivial singularity with $n = 0$, which is in fact topologically unstable as removable. One simple picture for this recalls our starting point for 'defect', as what we are interested in is far field property, say a homotopy $h_r(\theta)$ under polar coordinate from origin P, and since $n = 0$, any loop gives trivial winding and we could just ignore the trivial singularity.

Though not the main theme, we may consider defects combination law for planar spins as shown in Fig. 3.3, where one just uses a big loop to surround two defects, which is equivalent two small loops with two inverse paths connected as shown on the right by continuous deformation. By letting gap between two lines go to zero, they cancel

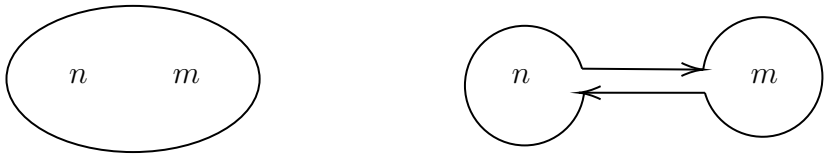


Figure 3.3: Combination of defects.

each other on winding contribution and only two small loops surrounding in the same direction are left, which immediately yields Abelian combination law as addition: $m + n$. Two descriptions are equivalent.

Now let us consider some other standard examples. Basically speaking, the order parameter space R of certain order parameter f could be described by either a manifold where f lives on, or a topological group which describes the symmetry of R, whose element translates f to some f', but one needs to take out elements that leave chosen representative f invariant. Re-speaking more mathematically to this idea, we conclude that R is bijective to a special coset space of a topological group G whose acts transitively on R.

Theorem 3.1.1. *If $\forall g \in G$, $\exists f_1, f_2 \in R$, s.t. $gf_1 = f_2$, then we have*

$$R = G/H, \tag{3.6}$$

where $H = H_f = \{g : gf = f\}$ is the isotropy subgroup of the representative element f or the fixer of f or the little group of f.

Proof. We did not use first isomorphism theorem since map from G to $R \subset G$ as it is not necessarily a homomorphism. And one could also see that H needs not to be

a normal subgroup, since if $f_2 = gf_1$, then one has $H_{f_2}f_2 = gH_{f_1}f_1 = gH_{f_1}g^{-1}f_1$ and thus $H_{f_2} = gH_{f_1}g^{-1}$, while H_{f_2} needs not to equal H_{f_1} generally. One example may be subgroup of $SO(3)$ as $SO(2)$ but around different axes.

Now for real proof, we show the bijective relation, from $G/H_f = \{gH_f\}$ to $R = \{f\}$. Onto: $\forall f' \in R$, $\exists g' \in G$, s.t. $f' = g'f$, then one could assign coset $g'H_f$ to f' as

$$g'H_f f = g'f = f'.$$

One to one: for $g_1 f = f_1 \neq f_2 = g_2 f$, clearly $g_1 H_f \neq g_2 H_f$ otherwise $f_1 = f_2$. □

What is more, the continuity of G naturally induces that on its coset space G/H, as

$$\lim_{n\to+\infty} g_n = g \Rightarrow \lim_{n\to+\infty} g_n H = gH,$$

and also,

$$\lim_{n\to+\infty} e_n = e \Rightarrow \lim_{n\to+\infty} e_n f = f,$$

or inversely,

$$\lim_{n\to+\infty} f_n = f \Rightarrow \exists \lim_{n\to+\infty} e_n = e,\ \lim_{n\to+\infty} e_n H = H.$$

Physically, such coset space description is related to *spontaneous symmetry breaking* (SSB). When chosen suitably, G will just describe unbroken disordered phase with $\langle f\rangle_G = 0$; when parameter changes so that system slides into broken ordered phase, H just describes residual symmetry with some chosen f, as $H_f f = f$.

Ordinary Spin. Just a 3-unit-vector $f(\boldsymbol{r}) = s_u(\boldsymbol{r})\hat{u}+s_v(\boldsymbol{r})\hat{v}+s_w(\boldsymbol{r})\hat{w}$ with $s_u^2+s_v^2+s_w^2 = 1$, and $R = S^2$. On the other hand, such a vector could be viewed as a rotating axis in 3-D with arbitrary angle, then $R = S^2 = SO(3)/SO(2) \cong (SU(2)/\mathbb{Z}_2)/SO(2)$. Notice that S^2 is not a group since it does not satisfy automorphism relation.

Nematic Liquid Crystal. Also a unit-vector but with no orientation, and $R = S^2/\mathbb{Z}_2 \cong \mathbb{RP}^2$.

Biaxial Nematics. A rectangular box with three 180-degree rotational axes, known as $D_2 = \{C_{2x}, C_{2y}, C_{2z}\}$ symmetry group, and $R = SO(3)/D_2$, where $SO(3)$ is for parameter to fix its orientation relative to some coordinates, one may use Euler-angle for rough understanding.

Superfluid ^{3}He (A-phase). *Dipole-locked.* As two unit-vectors with locked relative direction, and there's no symmetric operation, then $R = SO(3) \cong \mathbb{RP}^3$. *Dipole-free.* $f(\boldsymbol{r}) = \hat{n}(\boldsymbol{r})\hat{e}(\boldsymbol{r})$, where $\hat{n}$ for direction of first vector, while $\hat{e}$ for second one rotated from $\hat{n}$, and they are unrelated. However, the case where both two vectors change sign should be treated as the same, then we claim that $G = S^2 \times SO(3)$, while $H = \{[R(\hat{n} \perp \hat{z}, \pi), R(\hat{z}, \pi)]\}$.

3.2 Fundamental Group Description

We may generalize the idea of transforming classifying singularities into classifying loops as in the planar spin example. Say now for any point defects in two-dimension or line defects in three-dimension as shown in Fig. 3.4, we want to figure out the topological far-field property assigned to any order parameter configuration on loop surrounding in real space as contour C, which is invariant upon continuous deformation, and could be evaluated by property of loop itself in order parameter space R, and that property turns out to be the first homotopy group $\pi_1(R)$, where 1 refers to S^1 as image of $f : C \longrightarrow R$.

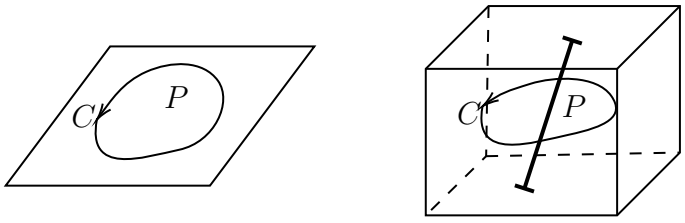


Figure 3.4: Point defects in 2-D and line defects in 3-D.

Claim 2. *Classification of defects equals classification of freely homotopic loops, which are equivalent to conjugate classes of* $\pi_1(R)$.

And then combination law of defects emerges as product of conjugate classes. One central task in this section is to prove this claim, and the only requirement here is that R is (path-)connected, as $\forall x, y \in R$, $\exists f(t) \in C[0,1]$, s.t. $f(0) = x$ and $f(1) = y$, where $C[a,b]$ denotes continuous function on set. And if there should be more than one disjoint components, where each of them is connected, one just put classification on every piece and unify by group product.

3.2.1 The fundamental group at a point

Loop is a closed path. Let $I = [0,1]$, then continuous map $f : I \longrightarrow R$ with $f(0) = f(1) = x$ describes a loop at point x. Denote all loops at x as set $L(R,x)$.

Based homotopy refers to homotopy between loops at point. Given loops f and g, homotopic relation $f \sim g$ stands if and only if there exists a continuous family of loops $h : I \times I \to R$ as $h_t(z)$, s.t. $h_0(z) = f(z)$, $h_1(z) = g(z)$ while $h_t(0) = x = h_t(1)$, $\forall z, t \in I$. Such a relation is transitive as

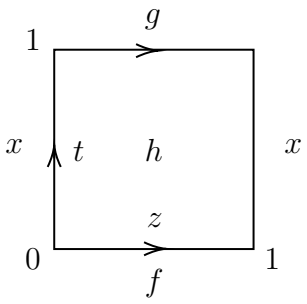


Figure 3.5: Homotopy relation h between f and g at point x.

$$f \sim g,\ g \sim k \Rightarrow f \sim k.$$

Based loop product refers to a binary function operation. Given loops f and g, we define

$$f \circ g(z) \equiv \begin{cases} f(2z), & 0 \le z \le 1/2 \\ g(2z-1), & 1/2 \le z \le 1 \end{cases}, \tag{3.7}$$

which simply gives us a 'bigger' loop firstly goes through f, back to x and through g. However, this law could not be imposed for group structure as it is not associative in the sense that $f \circ g \circ h \neq f \circ (g \circ h)$.

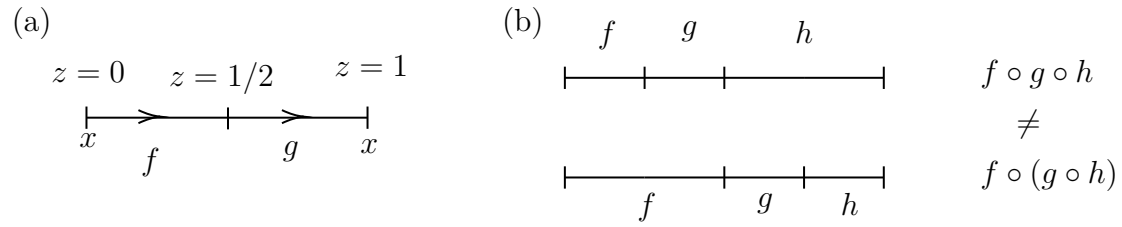


Figure 3.6: (a) $f \circ g$. (b) Illustration for non-association.

Homotopy class product. Denote $\alpha = [f] \equiv \{f' : f' \sim f, f' \in L(R,x)\}$ as homotopy class of loop f at x. Clearly, $L(R,x) = \alpha \cup \beta \cup \cdots$ as disjoint unions of different homotopy classes. Denote any f in $[f]$ as a representative for class. Then we define product of based classes as

$$\alpha \circ \beta = [f] \circ [g] = [f \circ g], \tag{3.8}$$

i.e., product of homotopy classes is homotopy class of representative product. This stands from theorem as

Theorem 3.2.1. *If $f \sim f'$, $g \sim g'$, then $f \circ g \sim f' \circ g'$,*

which is trivial and we omit the proof. Under such definition, we have: (i) no need for $[f \circ g](z = 1/2) = x$ as the product loop could be continuously deformed to the case that middle point doesn't touch x; (ii)

$$[f] \circ [g] \circ [h] = [f] \circ ([g] \circ [h]) \tag{3.9}$$

as associative law, which also comes from trivial theorem as

Theorem 3.2.2. *Loops differ only in path parameterization are homotopic.*

Fundamental group at point. Now take homotopy class product as combination law, we claim that $\pi_1(R,x) \equiv \{\alpha\}$ is a group known as the fundamental group considering that: (i) identity as $[e]$ with $e(z) \equiv x$, since $[e] \circ [f] = [e \circ f] = [f]$; (ii) inverse as $\alpha^{-1} = [f]^{-1} \equiv [f^{-1}]$, where we define

$$f^{-1}(z) = f(1-z) \tag{3.10}$$

as loop f going in inverse way, and this holds since $f \circ f^{-1} \sim e$ given homotopy

$$h_t(z) = \begin{cases} f(2zt), & 0 \le z \le \frac{1}{2} \\ f(2t(1-z)), & \frac{1}{2} \le z \le 1 \end{cases}.$$

Use planar spin as an example, a single singularity could be classified by loop C surrounding (at point), which maps into $R = S^1$ as a circle $\{f\} = S^1$. As we have showed, the time that latter S^1 surrounds on the former gives us winding number as topological index, loops with the same index are in the same class, which is equivalent to figuring out whether two spin configurations could be continuously deformed into each other, and that is described by

$$\pi_1(S^1) \cong \mathbb{Z}, \tag{3.11}$$

where by $\mathbb{Z}$ we refer to additive group of integers as an Abelian group.

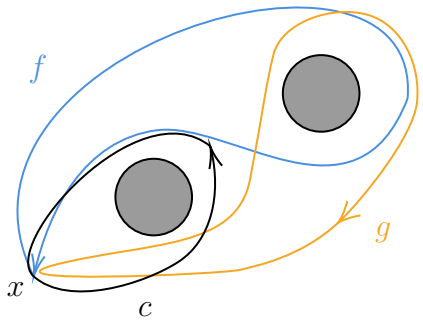


Figure 3.7: Figure-eight space.

However, $\pi_1(R, x)$ needs not to be Abelian, an important example is the figure-eight space with two holes on a plane as in Fig. 3.7. Clearly, due to the second hole, f and g surrounding the first one are not based homotopic while they are obviously freely homotopic. To let them base-connected, we need to introduce the mediator as loop c, and one could verify that $c \circ f \circ c^{-1} \sim g$, plus $f \nsim g$ one see more clear the non-Abelian nature as $f \circ c \nsim c \circ f$ (so that $[f \circ c] \neq [c \circ f]$).

There exists also trivial fundamental group. One example is ordinary spin in 2-D. Now $\pi_1(R) = \pi_1(S^2)$, but any loop on sphere surface could be contracted to a point intuitively (we do not fight against strange curves), and $\pi_1(S^2) = [e] \equiv 0$ as we represent a trivial group.

3.2.2 Fundamental group in R

Clearly we are not satisfied with the point-based description, and one simple argument for that is the freely-homotopic loops in figure-eight example, who are assigned with clearly the same topological index with the hole and thus should be attributed into the same physical class, while they are just classified into two based homotopy classes.

For a connected order parameter space R, we have

Theorem 3.2.3 (Path Isomorphism)**.**

$$\pi_1(R, x) \cong \pi_1(R, y). \tag{3.12}$$

Proof. Given path $c : I \to R$ as $c(z)$, we may define path product

$$c_1 \circ c_2(z) \equiv \begin{cases} c_1(2z), & 0 \leq z \leq \frac{1}{2} \\ c_2(2z-1), & \frac{1}{2} \leq z \leq 1 \end{cases},$$

if $c_1(1) = c_2(0)$, together with inverse path

$$c^{-1}(z) \equiv c(1-z).$$

Clearly, $c \circ c^{-1} \sim e$.

Now for $c(0) = x$ and $c(1) = y$, we could see that loop f at y is transferred to loop at x by $c \circ f \circ c^{-1}$, as shown in Fig. 3.8, so does the homotopy relation as if $f \overset{h_t}{\sim} g$,

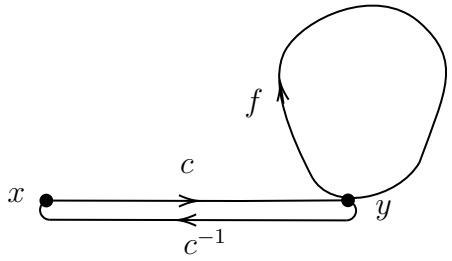


Figure 3.8: Loop f at y and path from x to y.

then $cfc^{-1} \overset{ch_tc^{-1}}{\sim} cgc^{-1}$, with $c \circ h_t \circ c^{-1}$ understood as all homotopic family of loops are transferred by c, then we say $\alpha_y = [f]$ is transferred to $\alpha_x = [cfc^{-1}]$. This indeed gives us a map as

$$c : \pi_1(R, y) \longrightarrow \pi_1(R, x),\ c(\alpha_y) \equiv \alpha_x, \tag{3.13}$$

and it's bijective by introducing inverse map as

$$c^{-1} : \pi_1(R, x) \longrightarrow \pi_1(R, y),\ c^{-1}(\alpha_x) \equiv [c^{-1} \circ cfc^{-1} \circ c] = [f] = \alpha_y, \tag{3.14}$$

which tells that $\forall \alpha_x$, c^{-1} transfers it to some α_y then c is onto, and $c^{-1}c = \mathbb{I}$ tells one-to-one. c is further a homomorphism since

$$cfc^{-1} \circ cgc^{-1} = c(fg)c^{-1} \Rightarrow c([f])c([g]) = c([fg]).$$

And we conclude that c induces an isomorphism between $\pi_1(R, x)$ and $\pi_1(R, y)$ known as path-isomorphism.

□

By theorem above we have shown that there exists one abstract fundamental group, and all based-group is its isomorphic copy. While copies at different base-points are also isomorphic, we shall show that in non-Abelian case, different path induces different isomorphism.

Theorem 3.2.4. $\exists c_1(\alpha_y) \neq c_2(\alpha_y) \Longleftrightarrow \pi_1(R)$ *is non-Abelian.*

Proof. $\Rightarrow$ If so, $\forall f \in \alpha_y$, $c_1 f c_1^{-1} \nsim c_2 f c_2^{-1}$, i.e., $(c_1^{-1}c_2)f(c_1^{-1}c_2)^{-1} \nsim f$, then by letting $g = c_1^{-1}c_2$ as loop at y, we have $gf \neq fg \Rightarrow [f] \circ [g] \neq [g] \circ [f]$, i.e., $\pi_1(R, y)$ thus $\pi_1(R)$ is non-Abelian.
$\Leftarrow$ If so, choose $fg \nsim gf$ and replace g with its homotopic loop $\bar{g}$ crossing both x and y, then one could decompose this into $c_1^{-1}c_2$ and show $c_1(\alpha_y) \neq c_2(\alpha_y)$ easily. □

However, we claim that

Theorem 3.2.5. *Conjugate class is invariant under different path-isomorphism.*

Proof. Given path c_1 and c_2 from x to y, since $c_1 f c_1^{-1} = c_1 c_2^{-1} c_2 f c_2^{-1} c_2 c_1^{-1}$, for any $f \in \alpha_y$, then by letting $k = c_1 c_2^{-1}$ as loop at x, we find

$$[k] \circ c_2(\alpha_y) \circ [k]^{-1} = c_1(\alpha_y),$$

i.e., different paths induces isomorphisms differ by conjugate relation depending on both paths. What's more, given conjugate class $C(\alpha) = \{\beta\alpha\beta^{-1} : \beta \in \pi_1(R)\}$ with base point omitted,

$$[k] \circ C(\alpha) \circ [k]^{-1} = \{[k] \circ \beta\alpha\beta^{-1} \circ [k]^{-1} : \beta \in \pi_1(R)\}$$

$$= \{([k]\beta) \circ \alpha \circ ([k]\beta)^{-1} : \beta \in \pi_1(R)\} \equiv C(\alpha),\ \forall k,$$

as an inner-automorphism. And we reach the theorem. □

One may notice that conjugate class degenerates back to group element in Abelian case, and all path-isomorphism leaves exact bijection between based fundamental group, which just corresponds to our theorem.

To connect the topological invariant as conjugate class to our desired continuous-deformation-invariant freely-homotopic loops, we further need

Theorem 3.2.6. *There exists a bijection between conjugate classes in $\pi_1(R)$ and classes of freely-homotopic loops.*

Proof. The picture is just based on the figure-eight example. Equivalently, we may prove the bijection by 'two loops are freely-homotopic to each other if and only if they belong to the same conjugate class'.
$\Leftarrow$ Let $\overset{F}{\sim}$ be freely-homotopic, then by assuming $f, g \in C(\alpha)$ (not a good sign, but we accept it), thus $f = b_1 a b_1^{-1}$ and $g = b_2 a b_2^{-1}$, and one gets $f = (b_1 b_2^{-1}) g (b_1 b_2^{-1})^{-1}$, i.e., $f \sim bgb^{-1}$ with $b = b_1 b_2^{-1}$, however, one may easily deform bgb^{-1} continuously back to g by tracing the starting point along b, then $g \overset{F}{\sim} bgb^{-1}$ and $f \overset{F}{\sim} g$.

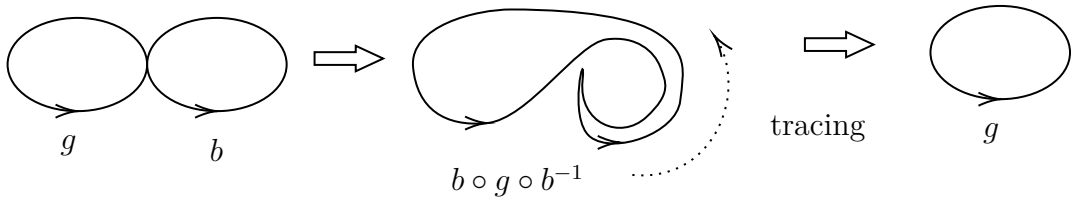


Figure 3.9: Continuous contraction of bgb^{-1} to g.

$\Rightarrow$ If $f \overset{F}{\sim} g$, there then exists a free homotopy h_t where $h_t(0)$ compose a path from f

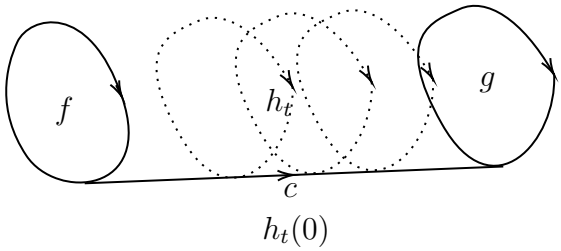


Figure 3.10: Free homotopy h_t, $h_t(0)$ compose c.

to g varying t. Clearly then $f \overset{F}{\sim} cgc^{-1}$ since $g \overset{F}{\sim} cgc^{-1}$, and they start at the same point. This free homotopy, however, must also be a based homotopy inherited from h_t as $f \overset{F}{\sim} g' = cgc^{-1}$ with $h'_t(0)$ tied to the point. Then $[f] = [cgc^{-1}] \in \pi_1(R, f(0))$ as path isomorphism. Now since that preserves conjugate class, we say f must be in the conjugate class of $\pi_1(R, f(0))$ mapped by c from $\pi_1(R, g(0))$ including g. □

Now we have a chain bijection as

Conjugate classes of $\pi_1(R)$ $\longleftrightarrow$ Freely homotopic loops $\longleftrightarrow$ Topological defects,

where the first stuff is one topological invariant. From which we could infer combination of defects could now be described by the multiplication table of conjugate classes. And especially for the case when π_1 is Abelian, conjugate classes are just its group elements, multiplication of elements equals addition.

In our discussion, R is always connected. However, as we have mentioned, order parameter space with pieces could be treated independently as theorem claims

Theorem 3.2.7.

$$\pi_1(R_1 \times R_2) \cong \pi_1(R_1) \times \pi_1(R_2). \tag{3.15}$$

The latter $\times$ refers to group direct product while the first one is the topological product of spaces, which conserves continuity as topological. The theorem is trivial in sense that map $S^1 \to R_1 \times R_2$ is viewed as pair-map $(S^1 \to R_1, S^1 \to R_2)$.

3.2.3 Fundamental theorem for π_1

Knowing the relation is not enough, we need a way to calculate the fundamental group. And that leads us to the fundamental theorem for fundamental group.

Lemma 3.2.8. *For continuous group consisted of separated connected pieces as $G = G_0 \cup G_1 \cup \cdots$, where G_0 is the piece with identity e, G_0 is the normal subgroup of G.*

Proof. Subgroup: closeness. Since G_0 is connected and continuous, there exist continuous maps from I to G_0 as continuous sequences $\{a_t\}$ and $\{b_t\}$, starting from $a_0 = e = b_0$ and end at $a_1 = a, b_1 = b$. Again due to continuity, mapped sequence $\{a_t b_t^{-1}\}$ should also be continuous starting from e, and its end ab^{-1} must lie somewhere in G_0, otherwise disrespecting continuity.

Normal subgroup: $cac^{-1} \in G_0$, $\forall c \in G$. Still using a_t, since now continuous ca_tc^{-1} starts again at $cec^{-1} = e$, its end cac^{-1} must lie in G_0, too. □

Lemma 3.2.9. *Given continuous group above, $G_{i\neq 0}$ are (left) cosets of G_0.*

Proof. Need G_i contains one and only one coset, for each i.
One: Given a in G_i, for any $h \in G_0$, since h is connected with e, then ah is connected with $ae = a \in G_i$, then $aG_0 \subset G_i$.
Only one: $\forall a_1 \in G_i$, $\exists a_t, t \in I$ as path in G_i, s.t. $a_0 = a, a_1 = a_1$; however, $a^{-1}a_t$ must lie in G_0 since $a^{-1}a_{t=0} = e$, then $a^{-1}a_1 \in G_0$, i.e., $a_1 \in aG_0, \forall a_1 \in G_i$, then $G_i \subset aG_0$. □

We conclude that $G_i = g_iG_0$, $\forall g_i \in G_i$ as a representative. Now we could write the quotient group for G as cosets, which equal disconnected pieces:

$$G/G_0 = \{gG_0 : g \in G\} = \{G_i\}_{i=0}. \tag{3.16}$$

Notice that $\forall g_i \in G_i$, we have $G_i = g_iG_0$, then in this sense all points in piece G_i are equivalent; or more topologically, this property emerges as G_i is connected thus any pair points in G_i are path-connected, thus they could be transferred (deformed) into each other continuously. In this way we could introduce the zeroth homotopy group as

$$\pi_0(G) = \{[g_i]\} \cong \{G_i\} = G/G_0. \tag{3.17}$$

Notice that for continuous group which is a whole piece,

$$\pi_0(G) = e \equiv 0$$

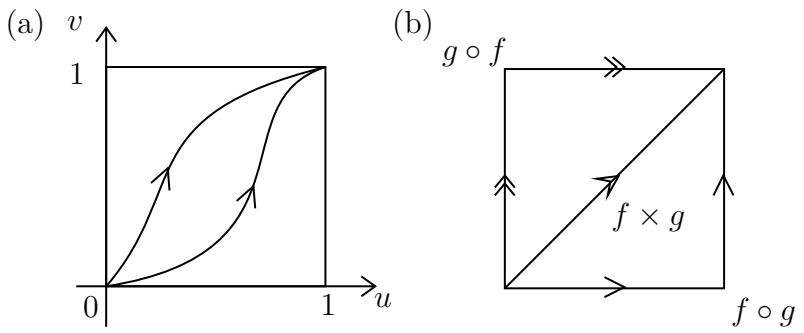


Figure 3.11: (a) (u, v) map. (b) Three special maps.

is trivial.

With all previous knowledge, we could already proceed, however, speaking of continuous group, there is a special property when order parameter space itself is also a continuous group.

Theorem 3.2.10. *$\pi_1(R)$ is Abelian if $R = G$ as a topological group.*

Proof. To proof this, we may firstly introduce, other than loop product $f \circ g$ for identity-based fundamental group $\pi_1(G, e) \cong \pi_1(R)$, a group product (on loop) as

$$f \times g(z) \equiv f(z)g(z), \ z \in I,$$

and essentially this a loop at e. For our purpose it is adequate by proving $f \circ g \sim g \circ f$, and firstly we define a more general map than group product as $I \times I \to G$ as $(u, v) \to f(u)g(v)$, whose legality is protected by continuity. As shown in Fig. 3.11(a), due to continuity, any loops given by curves shown are homotopic. What's more, there exist three special maps as two follow the edges while the third along diagonal line as $f(u)g(v = u) = f \times g(u)$, while bottom-up edges give $f(u) = f(u)g(0) \to f(1)g(v) = g(v)$ as $f \circ g$, and up-top give $g(v) = f(0)g(v) \to f(u)g(1) = f(u)$ as $g \circ f$, respectively. Then $f \circ g \sim g \circ f$. □

Notice that some continuous spaces like figure-eight space is continuous but with a non-Abelian fundamental group, and we conclude that such spaces could not be parameter space for any continuous group.

Now we could move on and reach first peak. Given a simple example as warm-up, we may consider once again planar spin with $R = S^1 \cong T(1)/\mathbb{Z}$ as 1-D translational group mod integer group, and amazingly one find $\pi_1(R) \cong \mathbb{Z}$.

Theorem 3.2.11 (Fundamental Theorem on Fundamental Group)**.** *Given simply connected topological group G together with its subgroup H, which is also continuous with well-defined normal subgroup H_0 containing e,*

$$\pi_1(G/H) \cong H/H_0. \tag{3.18}$$

Proof. The key lies in proving that H_0 serves as the homotopic relation. Notice that by definition simple connectedness means $\pi_1(G) = 0$, and continuous H serves as group with disjoint pieces as in lemma.

• We firstly show the correspondence between different pieces of H and homotopic loops in G/H. The points are defining loops and their product.

Due to continuity, path $g(z)$ in G induces path in G/H by $K(z) = g(z)H$, and this map is clearly onto in sense that given K, one could manually set paths between points

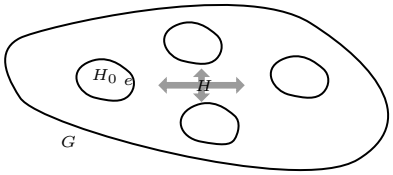


Figure 3.12: Illustration for G, H and H_0.

mapped to g in cosets, which is achievable for G's continuity. Now we choose $g(0)$ and $g(1)$ are both in H that are not necessarily the same point (i.e., element) in G, however, they both represent identity H in coset space G/H as $K(0) = g(0)H = H = g(1)H = K(1)$. Then such path in G induces loop in G/H. We could further modify this by redefining $g(z)g^{-1}(1)$ as $g(z)$, and the new path in G would end at $e \in H_0$, while starting from any piece in H. Now given loops K_1, K_2, we have corresponding g_1, g_2 starting from, say

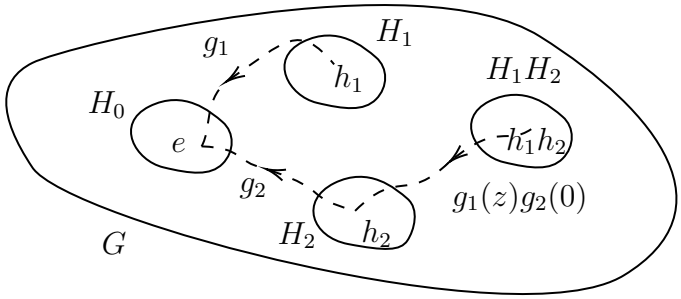


Figure 3.13: Illustration for loop product.

$h_1 \in H_1$ and $h_2 \in H_2$, respectively. Now since $g_1(1)g_2(0) = eh_2 = h_2 \in H_2$, $g_1(0)g_2(0) = h_1h_2 \in H_1H_2$, where one may notice that $H_1H_2 = H_i \subset H$ as elements in quotient group, then one immediately see that path $g_1(z)g_2(0)$ starts and ends in H and still maps to loop $K_1(z)K_2(0)$ in G/H, as shown in Fig. 3.13. However, $K_2(0) = h_2H = H$ and we claim that $K_1(z) = K_1(z)K_2(0)$, then union-path $g_1(z)g_2(0) \to g_2(z)$ in G induces loop product $K_1 \circ K_2$ in G/H.

On the other hand, essentially by repetition, one observes that every given path g_i starts from one piece $g_i(0) \in H_i \subset H$, and the defined product $g_i(z)g_j(0) \to g_j(z)$ starts $g_i(0)g_j(0) \in H_iH_j$, which is another piece of H due to closeness of quotient group, and we could claim that there exists a correspondence between loop K_i in G/H and piece H_i as element in H/H_0, where the link is given by starting point of path g_i in G.

• Next we show such correspondence is an isomorphism. Homomorphism is shown as the correspondence between loop product and group multiplication, and we only need to show bijection where onto is obvious and then we require one-to-one as :

$$g_1(0) \ \& \ g_2(0) \in H_i \Longleftrightarrow K_1 \sim K_2.$$

$\Rightarrow$ Given paths in Fig. 3.14, one may establish another path $c \subset H_i$ from $g_2(0)$ to $g_1(0)$, then $g_2^{-1} \circ c \circ g_1$ forms a loop in G, and due to simple connectedness, it is trivial. Then one could define area surrounded by this loop as $h_t(z)$, which represents the homotopy between g_1 and g_2, then $K_1 \sim K_2$.

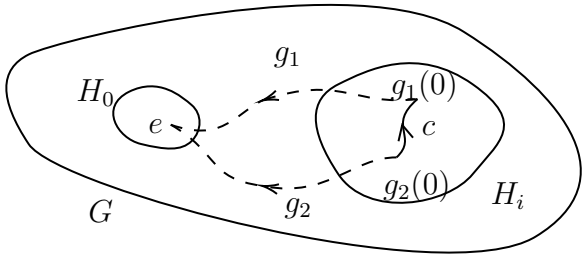


Figure 3.14: One-to-one correspondence.

$\Leftarrow$ Inversely, given $K_1 \sim K_2$, we could construct homotopy K_t. For each t, one may give two arbitrary paths as $K_t(z) = f(z)H = g(z)H$, then path $f^{-1}(z)g(z) \in H$ for any z. Notice $f^{-1}(1)g(1) = e \in H_0$, then this path lies in H_0 totally, especially $f^{-1}(0)g(0) \in H_0$, i.e., $g(0) \in f(0)H_0$, and they live in the same coset piece $H_i \equiv f(0)H_0$. Notice that t is arbitrary, then by continuity of K_t we complete the proof. □

In short, we establish isomorphism relation between elements in fundamental group and pieces of its subgroup. More suggestively, we may write

$$\pi_1(G/H) \cong H/H_0 \cong \pi_0(H). \tag{3.19}$$

Notice that there are special cases: (i) H is discrete so that $H_0 = e$, and $\pi_1(G/H) \cong H$; (ii) H is connected as $H = H_0$, then $\pi_1(G/H)$ is trivial.

Ordinary spin. $R = S^2$ as connected, then $\pi_1(R) = 0$. On the other hand, we have $S^2 = SO(3)/SO(2) = (SU(2)/\mathbb{Z}_2)/SO(2)$, with universal covering group simply connected as $G = SU(2) \cong S^3$. To find isotopy group under $SU(2)$, we may employ 3-unit vector $\hat{s} = R(\hat{n}, \theta)\hat{z}$ to spin as $\hat{s} \cdot \boldsymbol{\sigma} = U^\dagger(\hat{n}, \theta)\sigma_z U(\hat{n}, \theta)$, where $R \in SO(3)$ and $U(\hat{n}, \theta) = \exp(i\theta\hat{n} \cdot \boldsymbol{\sigma}/2) \in SU(2)$, and clearly, the isotopy subgroup H for representative σ_z is given by

$$U(\hat{z}, \theta) = \begin{pmatrix} e^{i\theta/2} & \\ & e^{-i\theta/2} \end{pmatrix},$$

which is connected and isomorphic to $U(1)$, as $S^2 = SU(2)/U(1)$, then

$$\pi_1(S^2) = 0.$$

Nematics. Now $R = \mathbb{RP}^2 = S^2/\mathbb{Z}_2 = [(SU(2)/\mathbb{Z}_2)/SO(2)]/\mathbb{Z}_2$. Still we have $G = SU(2)$, but apart from $U(\hat{z}, \theta)$ lifted from $R(\hat{z}, \theta)$, we also have a flip $z \leftrightarrow -z$ done by lifted $U(\hat{n} \perp \hat{z}, \pi) = U(\hat{y}, \pi)U(\hat{z}, \theta)$ with $\hat{n} = R(\hat{z}, \theta)\hat{y}$, and explicitly,

$$U(\hat{y}, \pi)U(\hat{z}, \theta) = i\sigma_y \begin{pmatrix} e^{i\theta/2} & \\ & e^{-i\theta/2} \end{pmatrix} = \begin{pmatrix} & e^{-i\theta/2} \\ -e^{i\theta/2} & \end{pmatrix},$$

which reveals that it is not connected to $U(\hat{z}, \theta)$ by continuous varying parameter. Then we conclude that $H = \{U(\hat{z}, \theta), i\sigma_y U(\hat{z}, \theta)\}$ known as $D_{\infty h}$ lifted from $D_\infty = \{R(\hat{z}, \theta), R(\hat{y}, \pi)R(\hat{z}, \theta)\}$ and $H_0 = \{U(\hat{z}, \theta)\}$ as its normal subgroup, then

$$\pi_1(\mathbb{RP}^2) \cong \pi_1(G/H) \cong H/H_0 = \{\mathbb{I}, i\sigma_y\}\{U(\hat{z}, \theta)\} \cong \mathbb{Z}_2.$$

Examples are shown in Fig. 3.15, where the left two panels show topologically equivalent

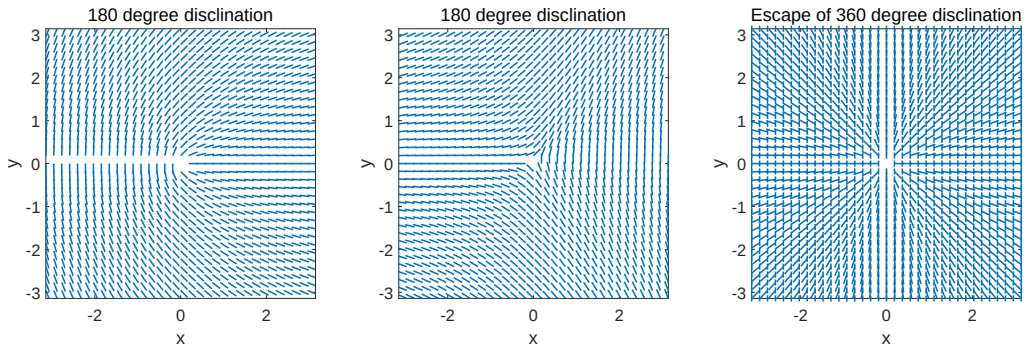


Figure 3.15: Topological defects of nematics (point in 2-D and line in 3-D). First two panels show non-trivial 180° disclination, while the third shows seemingly 360° defect, which in fact is trivial from the escape.

180° disclination as the only kind of non-trivial defect in nematics. While it is possible to form configuration on the right panel like $n = 1$ defect in planar spin, it is unstable in fact and could be transformed into trivial alignment by 3rd-dim escape as the rod rotate, say around axis in plane and vertical to the rod, and become perpendicular to the page.

Another question immediately emerges as the classification for nematics limited to two-dimension. Now similarly, $R = S^1/\mathbb{Z}_2 \cong \mathbb{RP}^1$, and just like planar spin, universal covering group for S^1 is $G = T(1)$ with $S^1 = T(1)/\mathbb{Z}$, then the lift of $\mathbb{Z}_2$ as equivalence between φ and $\varphi + \pi$ becomes $H = \{T_{n\pi} : n \in \mathbb{Z}\}$, which is discrete and yields

$$\pi_1(\mathbb{RP}^1) \cong \mathbb{Z}.$$

Biaxial nematics. Still, we choose $G = SU(2)$ as universal covering group of $SO(3)$, and the lift of $D_2 = \{C_{2x}, C_{2y}, C_{2z}\} = \{\mathbb{I}, R(\hat{x}, \pi), R(\hat{y}, \pi), R(\hat{z}, \pi)\}$ becomes

$$H = \{\pm\mathbb{I}, \pm U(\hat{x}, \pi), , \pm U(\hat{y}, \pi), , \pm U(\hat{z}, \pi)\} \cong \{\pm 1, \pm i\sigma_x, \pm i\sigma_y, \pm i\sigma_z\} \cong Q,$$

where $Q = \{\pm 1, \pm i, , \pm j, \pm k\}$ is known as the quaternion group with $i^2 = j^2 = k^2 = ijk = -1$ and cyclic $ij = k$. Then correspondingly, we have

$$i \leftrightarrow i\sigma_x, \ j \leftrightarrow i\sigma_z, \ k \leftrightarrow i\sigma_y.$$

Since Q is discrete, we conclude that

$$\pi_1(SO(3)/D_2) \cong \pi_1(SU(2)/Q) \cong Q.$$

Notice that this is a non-Abelian group, and classification of defects corresponds to conjugate classes, which could be found as

$$C_0 = \{\mathbb{I}\}, \ \bar{C}_0 = \{-\mathbb{I}\}, \ C_i = \{\pm i\sigma_i\}, i = x, y, z.$$

Superfluid ^3He. Dipole-locked. Similarly, by covering $SO(3)$ with simply connected $G = SU(2) \cong S^3$, we find $\mathbb{I}$ is lifted to $H = \{\pm\mathbb{I}\}$, which is discrete and we have

$$\pi_1(SO(3)) \cong \pi_1(S^3/\mathbb{Z}_2) \cong \mathbb{Z}_2.$$

Dipole-free. When $G = S^2 \times SO(3)$, $H = \{[R(\hat{n} \perp \hat{z}, \pi), R(\hat{z}, \pi)]\}$; when $G = SO(3) \times SO(3)$, $H = \{[R(\hat{z}, \theta), \mathbb{I}], [R(\hat{n} \perp \hat{z}, \pi), R(\hat{z}, \pi)]\}$, $\theta \in [0, 2\pi]$; when cover with $G =$

$SU(2) \times SU(2)$, we lift $H = \{[U(\hat{z}, \theta), \pm\mathbb{I}], [U(\hat{x}, \pi)U(\hat{z}, \theta), \pm U(\hat{z}, \theta)]\},\ \theta \in [0, 4\pi]$. Then by defining $g = \{[U(\hat{x}, \pi), U(\hat{z}, \pi)]\}$, H could be split into four pieces with normal subgroup $H_0 = \{[U(\hat{z}, \theta), \mathbb{I}]\}$ and $H_1 = gH_0,\ H_2 = g^2H_0,\ H_3 = g^3H_0$. Then

$$\pi_1(R) \cong \{e, g, g^2, g^3\} \cong \mathbb{Z}_4.$$

3.3 Second Homotopy Group

Similar question about classifying point defects in two-dimension could be extended as how to classify that in 3-D. Now, instead of a loop, a closed surface could serve as a geometrical object that totally surrounds the defect. Then by using the continuity, order parameter on this surface must also be continuous, and since the surrounding surface is closed, the mapped values of f in R must also form a closed two-dimensional object, known as 2-sphere S^2, and we shall deal with the second homotopy group $\pi_2(R)$ and figure out the correspondence between its elements and physical defects.

3.3.1 $\pi_2(R, x)$

2-sphere is also described by a continuous map $f : I \times I \to R$ as $f(u, v)$ with $f(u, 0) = f(u, 1) = x = f(0, v) = f(1, v)$.

Based product of 2-spheres are defined with combination of one-dimension

$$f \circ g(u, v) \equiv \begin{cases} f(2u, v), & 0 \le u \le \frac{1}{2} \\ g(2u - 1, v), & \frac{1}{2} \le u \le 1 \end{cases}. \tag{3.20}$$

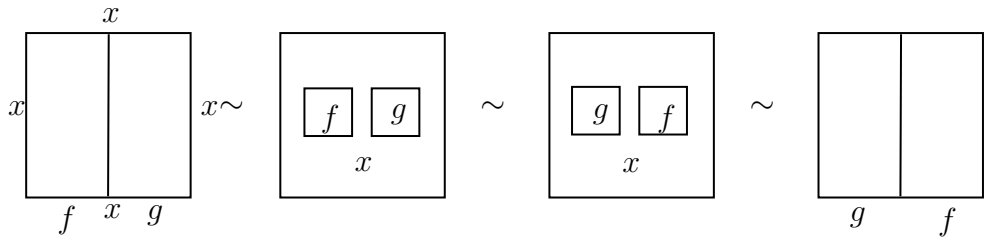
Second homotopy group at point could be defined similarly using homotopy class as $\pi_2(R, x) = \{[f]\}$, where identity $e : f(u, v) \equiv x$, and inverse $f^{-1}(u, v) \equiv f(1 - u, v)$. And similarly, homotopy now is a three-parameter continuous function between two 2-spheres.

Theorem 3.3.1. *$\pi_2(R, x)$ is always Abelian.*

Proof. The thought that why π_1 could be non-Abelian is that there are geometrical or topological obstacles that one loop could not overcome in a connected 2-D space. However, S^2 lives at least in 3-D and has enough space and flexibility to slide from un-commutative situation.

For mathematical proof one may refer to 'Eckmann-Hilton trick', however, here we may stick on some naive thought.

To show $[f] \circ [g] = [g] \circ [f]$ one needs $f \circ g \sim g \circ f$. As shown in the process, we



could exchange two squares as they may flow in the extra dimension, in contrast to the un-exchangeable 1-D case. □

3.3.2 $\pi_2(R)$

Theorem 3.3.2 (Path Isomorphism)**.**

$$\pi_2(R, x) \cong \pi_2(R, y). \tag{3.21}$$

Proof. At this time, instead the close relation between loop and path, we need to connect 2-sphere with a path, which one may think as a balloon plus string. As shown in Fig.

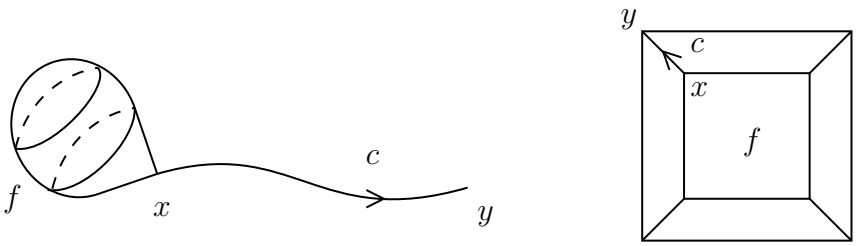


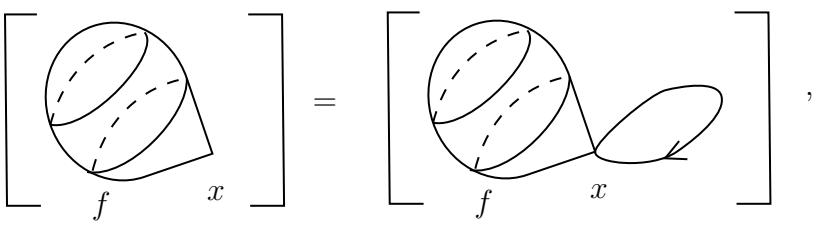
Figure 3.16: (a) Balloon-string. (b) Formal construction of c.

3.16(a), we may construct a union function $c(f)$ to describe this geometrical structure, while its formal construction may follow Fig. 3.16(b) as contracting (u, v) parameter space for f, and loops in (u, v) space outside f contract to the curve c from x to y, the innermost going into x and the outermost to y. And what this $c(f)$ has done is put S^2 at x forward to S^2 at y.

Then if $f \sim g$ at x, we have $c(f) \sim c(g)$ at y since we can similarly shrink the side edge of homotopy block (as 3-D cube in parameter space) to path c, which is always possible since the operation just gives an extension of a line from x to y without affecting homotopy at x. The inverse is also true by viewing f as $c^{-1}[c(f)]$ with only forward and back strings which could be continuously contracted to x and re-parametrization. It is also ready to see $c(f)c(g) \sim c(fg)$ and we conclude that c induces an isomorphism between $\pi_2(R, x)$ and $\pi_2(R, y)$, and we obtain a unique abstract Abelian group as $\pi_2(R)$. □

Now recall that we need conjugate classes for π_1 because different loops induces different isomorphism if and only if π_1 is non-Abelian, and conjugate class just provides a way to re-identify invariants by conjugate relation as an inner-automorphism. Now for π_2, since it is Abelian, we have no conjugate class discussion. However, we could still show that given different paths c_1 and c_2, the formed loop $c_1 c_2^{-1}$ still gives an automorphism on π_2, which could further classify the homotopy group. If this automorphism only contains identity operation as a trivial one, then we could say that each element in $\pi_2(R)$ corresponds to one kind of physical defect as freely-homotopic 2-spheres, and R is said to be 2-simple.

To make the point clear, by 'identity' proposed we mean that



,

i.e.,

$$\text{balloon} \sim (\text{balloon} + \text{loop}) \text{ at } x,$$

and if so, we could decompose the loop as $c_1 c_2^{-1}$ stated, and $c_1(f_x)$ is balloon at y while $c_2^{-1}(g_y)$ is balloon at x. Essentially, $c_2^{-1}(c_1(f))$ leads f at x to another f' as balloon f plus loop $c_1 c_2^{-1}$ at x, due to the identity, they are homotopic and the isomorphism does not change element, and there exists not only isomorphism, but somehow equivalence. On the other hand, if there's no such equivalence, i.e., balloon plus loop is homotopic to another balloon which is in different homotopy class with f, then we say the automorphism induced by loop contains non-trivial operation, and this further classifies $\pi_2(R)$ into disjoint automorphism classes, which corresponds to actual physical defect (notice that disjoint is trivial due to transitivity).

This classification could be spoken generally as action of π_1 on π_2. We say in this way by noticing that for π_1, it is conjugate classes that are invariant under different path-isomorphism, by inner-automorphism; then now we also claim that it is the conjugate classes of π_1 that induce and compose the automorphism classes of π_2, which is independent of base points. Further explanation relies on defect combination process, which may beyond the outline of this essay.

3.3.3 Fundamental theorem for π_2

Theorem 3.3.3 (Fundamental Theorem on Second Homotopy Group)**.** *Given simply connected compact Lie group G and its subgroup H, whose normal subgroup H_0 is again the connected piece containing identity e, we have*

$$\pi_2(G/H) \cong \pi_1(H_0). \tag{3.22}$$

Proof. The reason for the strong requirement comes from a theorem by Cartan in 1936, who claims that $\pi_2(G) = 0$ when G is a compact Lie group (though not necessary), and also $\pi_1(G) = 0$ as definition for simply connectedness.

Meanwhile, we could replace H_0 with H as exact sequence tells us in next section. This is merely because since all pieces in H are cosets of H_0, we may write $H \cong A_n \times H_0$ where A_n is isomorphic to finite discrete quotient group, and

$$\pi_1(H) \cong \pi_1(A_n \times H_0) \cong \pi_1(A_n) \times \pi_1(H_0) = \pi_1(H_0),$$

since fundamental group for discrete group is trivial.

Just like in π_1 case, we need to specify what is a sphere in G/H, and how does that relates to loop in H_0. We firstly show the correspondence between $S^2 \subset G/H$ and a square map into G. Say given $g : I \times I \to G$ as $g(u, v)$, then $K(u, v) \equiv g(u, v)H$ gives a square map into G/H, and to make K a sphere as $K(u, 0) = K(u, 1) = H = K(0, v) = K(1, v)$, we need correspondingly the mapped edges of g to live in H, and more importantly, due to the connectedness of the boundary, its mapped values must lie in some piece $H_i \subset H$, and clearly, we could re-define $g(u, v) \to g(u, v)h_i^{-1}$ where $H_i = h_i H_0$, which maintains mapped boundary of K to still live in $H = h_i^{-1} H$ while pull the mapped boundary of g back to H_0 as $H_0 = g_i^{-1} H_i = H_i g_i^{-1}$ from property of normal subgroup. The inverse is also true as we intuitively imagine for S^2 given by K to G/H, we pull representative points back to G and establish connection surfaces manually among points. We talk in a rather superficial way, while the actual proof for this lies in 'covering homotopy theorem', which beyond the frame here.

Then, like we establish representative point in H_i as beginning point of path in G as loop in G/H, we also establish correspondence between loop in H_0 (arbitrary H_i in fact, but we have pulled it back since they are just equivalent) as the boundary of square map to G, which is again a 2-sphere in G/H, with base point $H_0 \subset H$ (in G/H description, this is point in point, and we just claim like this for emphasize). Then still we need to show the isomorphism. Let us denote K as 2-sphere square map into G/H at H, and g as square map into G with boundary map l into H_0.

• Bijection is given by map from K to l.

One-to-one: one needs that whenever $l_1 \sim l_2$, we have $K_1 \sim K_2$. As shown in Fig.

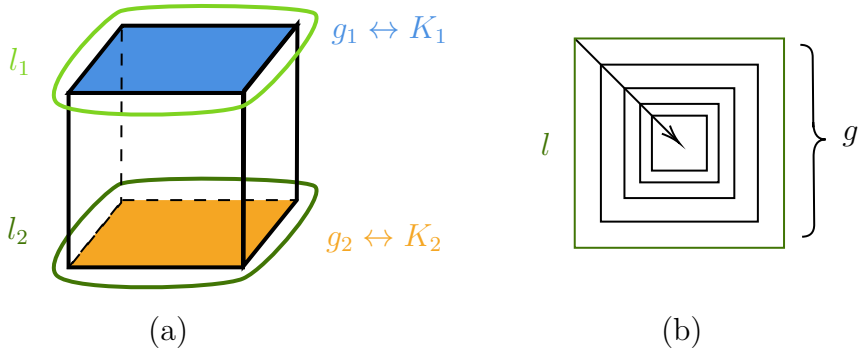


Figure 3.17: (a) One-to-one. (b) Onto.

3.17(a), given g_1 and g_2, with boundary l_1 and l_2 respectively; if $l_1 \sim l_2$ in H_0, there exists homotopy shown as the sides of cube, and all sides are taken into H_0; now top plus bottom plus sides compose a 2-sphere in G, which is shrinkable since $\pi_2(G) = 0$ as required. The contraction process is roughly a 3-cube as shown, which is continuous and provides as the homotopy between g_1 and g_2 in G; also since all sides are in H_0, we could take this cube-homotopy into coset space G/H as homotopy between 2-spheres K_1 and K_2.

Onto: one needs $\forall l$, $\exists K$. The construction is shown in Fig. 3.17(b). Given $l \subset G$, $\pi_1(G) = 0$ makes l contractible, and the process again forms a continuous square map with boundary as l in H_0. Then this map, when carried into G/H, is exactly a 2-sphere K.

• Homomorphism. We introduce the so-called special canonical form of square map g

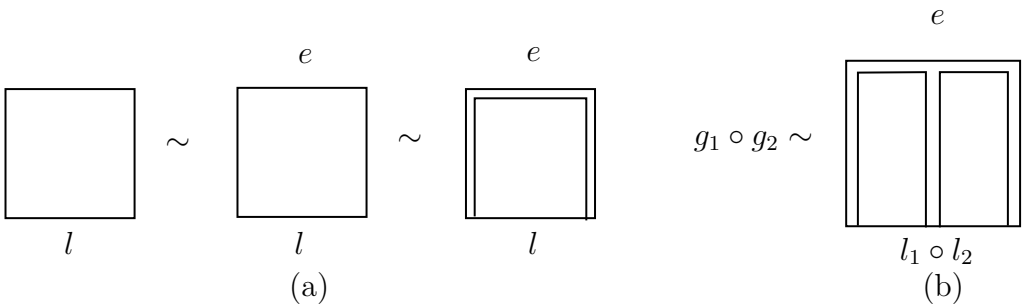


Figure 3.18: (a) Special canonical form of g. (b) Homomorphism.

in G, as shown in Fig. 3.18(a), where we firstly deform the boundary map l to attach identity e, and then expand this point continuously until only one edge is non identity, which might formally be $g(u, v = 0)$ as a special form of l. Then it becomes clear that

under this form, the combination between g_1 and g_2 induces a combination of l_1 and l_2 as shown in Fig. 3.18(b). □

Then we complete the proof. In fact, intuitive picture behind this is quite vivid, as one imagine that S^2 in G/H just gives an open mouth pocket in G, and the mouth as boundary map lies exactly in H_0 after some pull-back. Then the correspondence is rather clear.

3.3.4 Automorphism classes

$$\begin{array}{ccccccc} \pi_1(G/H) & \cong & \pi_0(H) & \cong & H/H_0 & \longrightarrow & h_i \longrightarrow g(z)\subset G \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \quad \swarrow \\ \pi_2(G/H) & \cong & \pi_1(H) & \cong & \pi_1(H_0) & \longrightarrow & l \end{array} \quad . \tag{3.23}$$

As shown above, due to isomorphism chain, the action of $\pi_1(G/H)$ on $\pi_2(G/H)$ could be viewed equivalently as action of representative $h_i \in H_i \subset H$ on $l \subset H_0$, which would show itself by figuring out the action of path $g(z)$ in G with $g(1) = e$, $g(0) = h_i$ as loop in G/H. We then may define a combined map

$$F(z) \equiv g(z)l,\ z \in I.$$

Since $l \subset H_0 \subset H$, $F(z)$ then lies in some coset $g(z)H$ for any z. Essentially, $F(1) = l \subset H$, $F(0) = h_i l \subset H$, then $F(z)$ represents a loop in G/H, which is attached to the balloon. However, this balloon-loop as pocket-path in G starts from $F(0) \subset H_i$, and to pull it back, one may re-define $F(z) = g(z)lg(0)^{-1}$ since due to property of normal subgroup, new $F(0) = g(0)lg(0)^{-1} = h_i l h_i^{-1} = l' \subset H_0$, but this is just the desired automorphism: l' needs not to be homotopic with l for non-Abelian $\pi_1(H_0)$ case; and this form of transformation is just the conjugate relation, and we claim that automorphism classes of $\pi_2(G/H)$ is just conjugate classes of $\pi_1(H)$.

Automorphism classes of $\pi_2(R)$ $\longleftrightarrow$ Freely homotopic spheres $\longleftrightarrow$ Topological defects.

It is quite a coincidence that fundamental theorem make equal to $\pi_2(G/H)$ and $\pi_1(H)$ while again classify the former using the latter. And there are two special cases: (i) H is Abelian, where conjugate classes vanish and G/H is 2-simple; (ii) H is discrete, then $H_0 = \{e\}$ and $\pi_2(G/H) \cong \pi_1(H_0) = 0$. Examples below are talked in 3-D.

Planar spin. $R = S^1 \cong T(1)/\mathbb{Z}$, discrete H, trivial point defect in 3-D.

Ordinary spin. $R = S^2 = SU(2)/U(1)$, $H_0 = H = U(1) \cong S^1$, $\pi_2(R) = \pi_2(S^2) \cong \pi_1(S^1) \cong \mathbb{Z}$, physical examples might be Gaussian law for electric field and magnetic monopole.

Nematics. $R = \mathbb{RP}^2 = SU(2)/H$, $H = \{U(\hat{z},\theta), i\sigma_y U(\hat{z},\theta)\}$ and $H_0 = \{U(\hat{z},\theta)\}$, then $\pi_2(R) = \pi_2(\mathbb{RP}^2) \cong \pi_1(U(1)) \cong \mathbb{Z}$. However, conjugate relation tells us that $i\sigma_y U(\hat{z},\theta)(i\sigma_y)^{-1} = U(\hat{z},-\theta)$, and thus relates n with $-n$, then true point defects are classified by N as set for natural numbers. This is easily understandable since point defects for nematics are just like those for ordinary spins, which again is similar as electric field with Gaussian law, where vector field are aligned perpendicular to a 2-sphere. But now the 'vector' for nematics has no direction, and we identify positive and negative n.

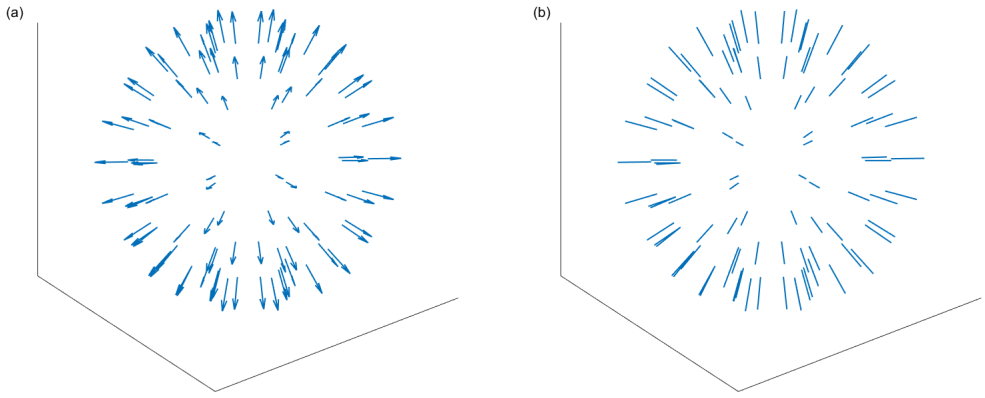


Figure 3.19: Point defects in 3-D. (a) Ordinary spin with $n = 1$. (b) Nematics with $n = 1$.

Biaxial nematics. $H \cong Q$ is discrete and thus no defects. The simple reason for this is: point defects for order parameter like rectangular box could not form configuration above, since the rectangular surface on a 2-sphere is just like a tangent field, and would again create unavoidable singularities.

Superfluid ^3He. *Dipole-locked.* $H \cong \mathbb{Z}_2$ and trivial, similar argument as biaxial nematics. *Dipole-free.* $H \cong \mathbb{Z}_4 \times H_0$, $H_0 = \{U(\hat{z}, \theta), \mathbb{I}\} \cong U(1)$, then $\pi_2(R) \cong \mathbb{Z}$. But one may further find that $g^n U(\hat{z}, \theta) g^{-n} = U(\hat{z}, (-1)^n \theta)$, then we also have equivalent n and $-n$, then defect as N.

3.4 Higher Homotopy Group and Exact Sequence

Finally here we come to give a systematic definition for general homotopy groups, and we could use one exact homotopy sequence to recover the fundamental theorems.

3.4.1 Higher homotopy groups

Consider general N-dimensional real space where our ordered medium lives, and if there is a d-dim defect, we shall use n-sphere surrounding with $n \equiv N - d - 1$ to describe it, which then induces also an S^n in order parameter space R due to continuity and gives us nth homotopy group $\pi_n(R)$.

Similarly, we define a based-n-sphere as a map $f : I^n \to R$ as $f(u_1, u_2, \cdots, u_n)$ with all (super-)surfaces identified to be point x as $f(\cdots, u_i = 0, \cdots) \equiv x \equiv f(\cdots, u_i = 1, \cdots)$, $\forall i$. Correspondingly, in coset space, all those boundaries of n-cube are mapped to identity H as a conventional choice.

We may also define based n-sphere product by sticking u_1, or equivalently described, gluing boundaries, and homotopy as some $(n+1)$-cube. Then similarly, we define based-homotopy classes $[f]$ and $\pi_n(R, x) = \{[f]\}$, which is also Abelian as $n \geq 2$, since it contains more degrees of freedom for sphere escaping.

What is more, we also have path-isomorphism between different based homotopy groups, which serves us the abstract $\pi_n(R) \cong \pi_n(R, x)$. And by further representing $R = G/H$, the relation to physical topological defect as freely-homotopic n-spheres are again classified by automorphism classes induced by conjugate relation in H, as conjugate

classes of $\pi_1(H)$. If the automorphism only contains identity, R is called n-simple, and element in nth homotopy group shares a bijection with different physical defects.

3.4.2 Relative homotopy group

Relative homotopy group, compared to absolute one discussed as $\pi_n(R)$, is only well-defined when $n \geq 2$. Basically, given space R, base point x and subset $x \in A \subset R$, we may define $\pi_n(R, A, x)$ as group for homotopy class $[f]$, where essentially we also let $f : I^n \to R$. The difference, however, appears on the boundary as we let $f(\cdots, u_i = 0, \cdots) \equiv x \equiv f(\cdots, u_i = 1, \cdots)$, for all i but $i = 1$, where we only require $f(u_1, \cdots) \in A$. We could also similarly define product between such maps as

$$f \circ g(\boldsymbol{u}) \equiv \begin{cases} f(2u_1, \cdots), & 0 \leq u_1 \leq \frac{1}{2} \\ g(2u_1 - 1, \cdots), & \frac{1}{2} \leq u_1 \leq 1 \end{cases}.$$

Notice that if $A = \{x\}$, relative becomes absolute.

Intuitively f gives us a map from n-cube I^n to a pocket in R whose mouth containing x is located in A mapped from some (say, let it be the 'bottom' one) $(n-1)$-dim surface of the cube, and essentially since the boundary of the mouth is mapped into x, it exactly offers us an $(n-1)$-sphere mapped from an $(n-1)$-cube, which again allows us to define (absolute) homotopy group $\pi_{n-1}(A, x)$. Also, the product between elements in $\pi_n(R, A, x)$ is defined in the way that $(n-1)$-surfaces into A are connected, which again also serves as $(n-1)$-sphere product for $\pi_{n-1}(A, x)$, and this essentially leads to a homomorphism by boundary map.

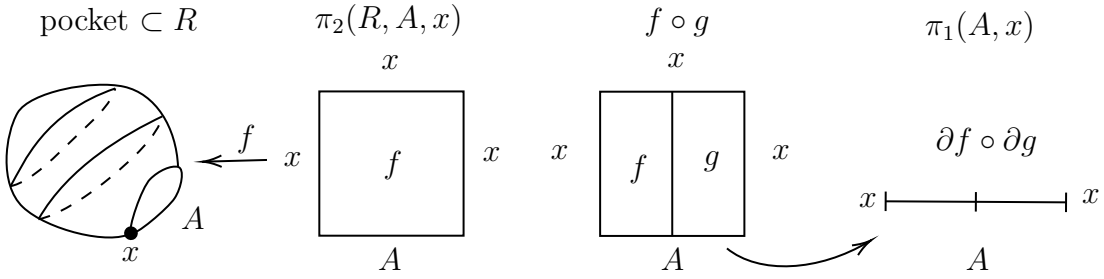


Figure 3.20: $n = 2$.

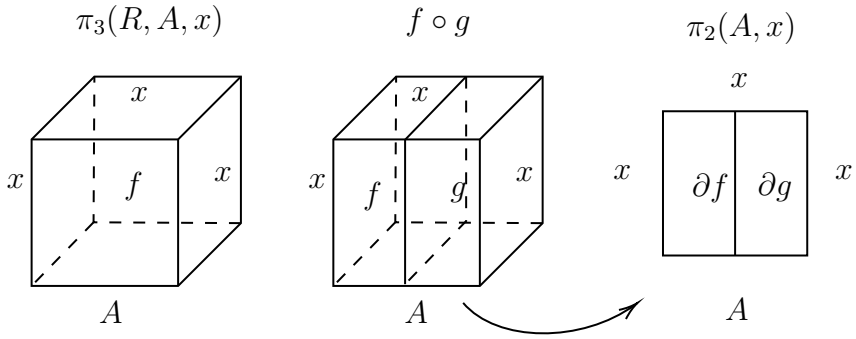


Figure 3.21: $n = 3$.

Discussion above is abstract, and let's look at concrete examples as shown in Fig. 3.20 and 3.21, where relative pocket map together with their product are shown, so does the

induced homomorphism. One may now gain observation that due to the homomorphism, the combination law for n-dim pockets are just law for $(n-1)$-sphere product, and in this way we say that $\pi_n(R, A, x)$ is Abelian when $n \geq 3$, but not necessarily for $n = 2$.

At this time, we may also gain some insight why $\pi_1(R, A, x)$ is just a set rather than group. When $n \geq 2$, the bottom surface of n-cube mapped into A always contains a boundary mapped as x, which makes it possible to talk about $\pi_{n-1}(A, x)$; however, for $n = 1$, the 'bottom' just maps as some point in A without x from boundary, then no $\pi_0(A, x)$ stuff. On the other hand, the situation is relaxed when $R = G$ as a group, $A = H$ as a subgroup and $x = e$, at which time we let

$$f \circ g(z) \equiv \begin{cases} f(0)g(2z), & 0 \leq z \leq \frac{1}{2} \\ f(2z-1), & \frac{1}{2} \leq z \leq 1 \end{cases}.$$

As shown in Fig. 3.22, by letting $f(0) = h_1 \in H, f(1) = e$, we find $f(0)g(0) = h_1 h_2 \in$

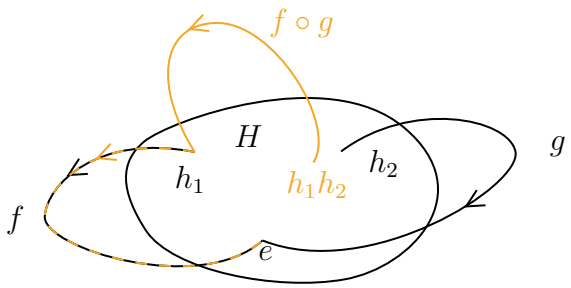


Figure 3.22: Product law for $\pi_1(G, H, e)$.

H due to closure, with $f(0)g(1) = h_1$, and the product is well-defined. Through this multiplication law, we also have natural homomorphism $\pi_1(G, H, e) \xrightarrow{\partial} \pi_0(H) \cong H/H_0$.

3.4.3 Exact homotopy sequence

We firstly claim that boundary map above as

$$\partial_n : \pi_n(R, A, x) \to \pi_{n-1}(A, x) \tag{3.24}$$

is only homomorphism with unclear bijection, where n is for dimension signal. It's not one-to-one since non-homotopic n-cubes may share the same base in A, and it's not necessarily onto since there could be complicate S^{n-1} can't be base of n-pocket.

To proceed, we need two extra inclusion maps as homomorphisms:
(i) Since $A \subset R$, we have

$$\alpha_n : \pi_n(A, x) \to \pi_n(R, x). \tag{3.25}$$

It is clearly not bijective, since spheres extended outside A make it not onto by being non-homotopic to spheres inside A, while geometrical structure of A makes it possible that non-homotopic spheres in A might be homotopic in R, which leads to not one-to-one.
(ii) Since $x \in A$, we have

$$\beta_n : \pi_n(R, x) \to \pi_n(R, A, x), \tag{3.26}$$

i.e., the mouth of pocket happens to close, or the base happens to be just x. Clearly it is not bijective, where not onto is easy as long as mouth opens (and being non-homotopic),

and mouth of pocket makes it possible to escape from some singularities and let non-homotopic spheres in R to be homotopic pockets.

Now we could construct a chain of homomorphisms

$$\begin{array}{ccccccc}
\pi_n(A,x) & \xrightarrow{\alpha_n} & \pi_n(R,x) & \xrightarrow{\beta_n} & \pi_n(R,A,x) & \xrightarrow{\partial_n} & \\
\pi_{n-1}(A,x) & \xrightarrow{\alpha_{n-1}} & \pi_{n-1}(R,x) & \xrightarrow{\beta_{n-1}} & \pi_n(R,A,x) & \xrightarrow{\partial_{n-1}} & \\
\cdots\cdots & & & & & & \\
\pi_1(A,x) & \xrightarrow{\alpha_1} & \pi_1(R,x) & \xrightarrow{\beta_1} & \pi_1(R,A,x) & \xrightarrow{\partial_1} & \\
\pi_0(A) & \xrightarrow{\alpha_0} & \pi_0(R). & & & &
\end{array} \tag{3.27}$$

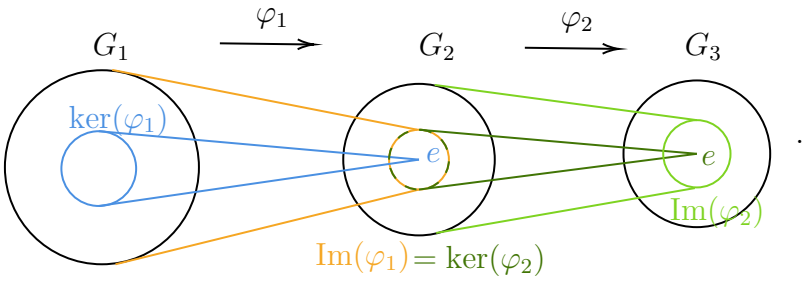
One needs to be aware that β_1, ∂_1 and α_0 need $(R, A, x) = (G, H, e)$ if we want them to be group homomorphism.

Claim 3. *This homomorphism chain is also an exact sequence.*

By exact sequence, we mean a mapping/operating sequence among certain objects like

$$\cdots \longrightarrow G_1 \xrightarrow{\varphi_1} G_2 \xrightarrow{\varphi_2} G_3 \longrightarrow \cdots,$$

which satisfies that



(3.28)

For our purpose, we give a rough proof.

Proof. We are not that ambitious so we only use intuitive examples, but without loss of generality.
(i) $\mathrm{Im}(\partial_n) = \ker(\alpha_{n-1})$. Related piece of chain for $n = 3$ is shown in Fig. 3.23, notice that identity in $\pi_2(R, x)$ denotes S^2 that could be continuously contracted to point x in R, while 3-pocket in $\pi_3(R, A, x)$ naturally offers such a homotopy between top and bottom surface, while the boundary map and inclusion map also exactly transfer such 2-sphere. One may doubt since α_2 is not onto, but all contractible 2-spheres belong to the same homotopy class and is thus the same element in $\pi_2(R)$.
(ii) $\mathrm{Im}(\beta_n) = \ker(\partial_n)$. Related piece of chain for $n = 3$ is shown in Fig. 3.24, then this is rather direct since inclusion map β_3 gives us base-image to be just homotopy class $[x]$, which is again kernel for $\pi_2(R, x)$ by projection/boundary map.
(iii) $\mathrm{Im}(\alpha_n) = \ker(\beta_n)$. Related piece of chain for $n = 2$ is shown in Fig. 3.25. Still, we define identity for relative homotopy group as pockets homotopic to point, and since β_2 is not necessarily one-to-one, we have to be more careful. As shown in Fig. 3.26, we have

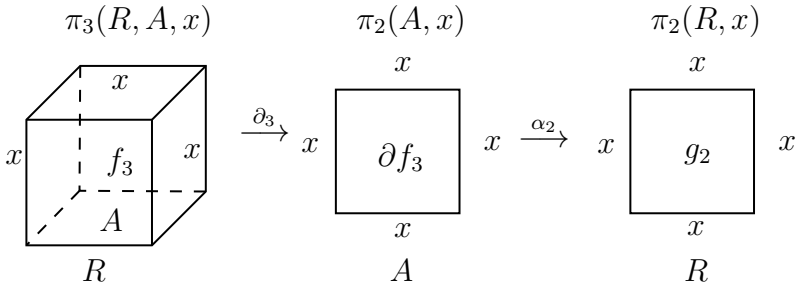


Figure 3.23:

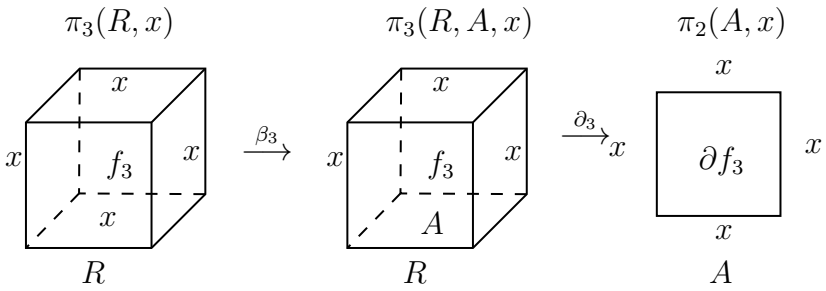


Figure 3.24:

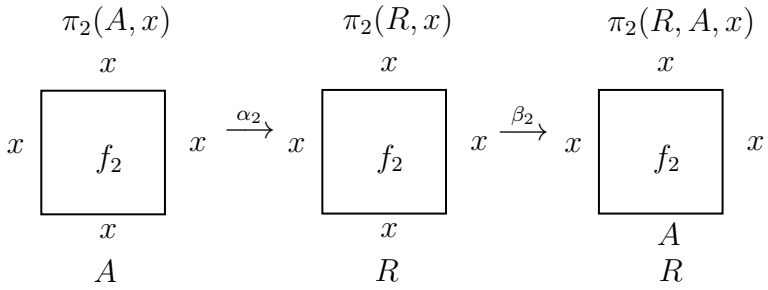


Figure 3.25:

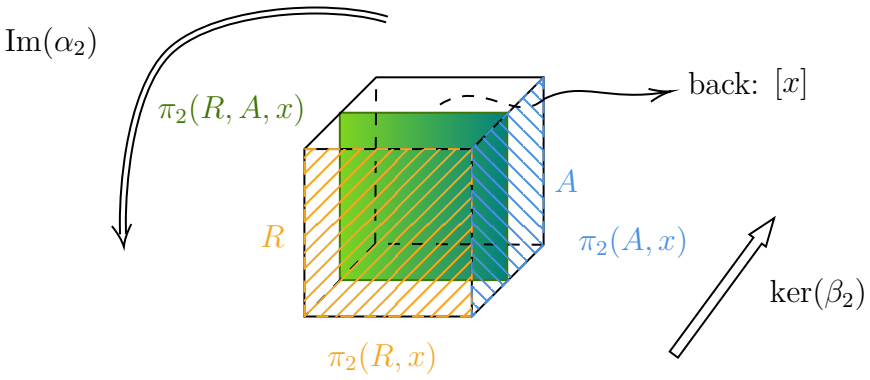


Figure 3.26:

to ask help from $(n+1)$-cube, where we set the bulk and one side to be R, while one neighbouring side to be A, and all other sides and all edges to be x.

Now consider β_2. Say we give $[f_2] \in \pi_2(R, x)$, and by inclusion map β it belongs to $\pi_2(R, A, x)$, then as $\ker(\beta_2)$ arrow shows, this parametrization offers us a homotopy as $f_2 \sim x$, which peer $\ker(\beta_2)$ off $\pi_2(R, x)$. We then further consider the rotational-parametrization, which serves us a homotopy between inclusion-α_2-mapped elements from $\pi_2(A, x)$ to $\pi_2(R, x)$ as $\mathrm{Im}(\alpha_2)$, and those serve as $\ker(\beta_2)$ mentioned before. Then we could say that $\mathrm{Im}(\alpha_2) = \ker(\beta_2)$. □

3.4.4 Restore of fundamental theorem

Let $(R, A, x) \to (G, H, e)$. By ignoring e as some base point, we also need to identify $\pi_n(G/H)$ and $\pi_n(G, H)$, which is geometrically achievable since the former pull all surfaces of n-cube while the latter does the same, but just fix all expect one as identity. Using this identification, we have our exact homomorphism sequence

$$\cdots \longrightarrow \pi_n(G) \longrightarrow \pi_n(G/H) \longrightarrow \pi_{n-1}(H) \longrightarrow \pi_{n-1}(G) \longrightarrow \cdots . \tag{3.29}$$

By assuming enough good property on G as $\pi_n(G) = 0 = \pi_{n-1}(G)$, we have essentially

$$0 \longrightarrow \pi_n(G/H) \longrightarrow \pi_{n-1}(H) \longrightarrow 0, \tag{3.30}$$

which leads to the conclusion that

$$\pi_n(G/H) \cong \pi_{n-1}(H) \tag{3.31}$$

as fundamental theorem.

Chapter 4

Group Extension

4.1 Group Extension

A group extension of Q by N refers an exact sequence of groups

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \longrightarrow 1 \tag{4.1}$$

where N is a normal subgroup of G and Q is the quotient group G/N. The group G is called an extension of Q by N. From the view of set, G can be identified with the set of pairs (n, q) with $n \in N$ and $q \in Q$, i.e., direct product $N \times Q$. However, considering the inner structure, we can say mostly generally that G is a fiber bundle over Q with fiber N. Such kind of extension is not unique, and its classification is a nontrivial problem in group theory.

The non-unique (right-)inverse of projection π is called a section s of the extension, with $\pi(s(q)) = q$, and one may think s as some 'deformed copy' of Q in G. Given a section, the element $g \in G$ can be uniquely written as $g = \iota(n)s(q)$, for some $n \in N$ and $q \in Q$. One may think that $\iota(n)$ moves along the fiber N and $s(q)$ moves along the base Q.

An extension splits if there exists a section $s : Q \to G$ which is a group homomorphism. Under the condition, $G \cong N \rtimes_\omega Q$ is a semi-direct product over another homomorphism $\omega : Q \to \mathrm{Aut}(N)$. The isomorphism is defined by $g = \Psi(n, q) = \iota(n)s(q)$, with

$$\Psi(n_1, q_1)\Psi(n_2, q_2) = \Psi((n_1, q_1) \cdot_\omega (n_2, q_2)) = \Psi(n_1\omega_{q_1}(n_2), q_1q_2),$$

as a semi-direct product. The map $q \mapsto \omega_q$ is defined by $\iota(\omega_q(n)) = s(q)\iota(n)s(q)^{-1}$, and is a homomorphism in this special case since $s(q)$ is a group homomorphism, i.e., $s(q_1)s(q_2) = s(q_1q_2)$. The isomorphism above is then clear by

$$\begin{aligned}
\Psi(n_1, q_1)\Psi(n_2, q_2) &= \iota(n_1)s(q_1)\iota(n_2)s(q_2) \\
&= \iota(n_1)s(q_1)\iota(n_2)s(q_1)^{-1}s(q_1)s(q_2) \\
&= \iota(n_1\omega_{q_1}(n_2))s(q_1q_2) \\
&= \Psi(n_1\omega_{q_1}(n_2), q_1q_2).
\end{aligned}$$

A central extension is a special case of group extension, where the normal subgroup N is mapped into center elements (commute with all group elements) in G. Due to this property, we see that the above ω function is always trivial as $\iota(\omega_q(n)) = s(q)\iota(n)s(q)^{-1} =$

$s(q)s(q)^{-1}\iota(n) = \iota(n)$, and ι is an injection isomorphism. Use a different notation, the central extension is defined by the exact sequence

$$1 \longrightarrow A \xrightarrow{\iota} \tilde{G} \xrightarrow{\pi} G \longrightarrow 1, \tag{4.2}$$

for some abelian group A with $\iota(A) \subset Z(\tilde{G})$. For instance, the familiar sequence

$$1 \longrightarrow \mathbb{Z}_2 \xrightarrow{\iota} \mathrm{SU}(2) \xrightarrow{\pi} \mathrm{SO}(3) \longrightarrow 1$$

that describes the double cover of SO(3) by SU(2) is a central extension, and is a non-split, non-trivial extension. This comes from the fact that SU(2) is simply connected while SO(3) is not, thus there is no global section. Indeed, this extension comes from the first homology group $H_1(\mathrm{SO}(3), \mathbb{Z}) \cong \mathbb{Z}_2$, with the identity representing the trivial extension $O(3) \cong \mathbb{Z}_2 \times SO(3)$.

4.2 Projective Representation

A representation sends a group to a subgroup of the general linear group $GL(d, \kappa)$ (d dimension, κ field) by homomorphism. A projective representation is a generalization of representation, where the homomorphism is relaxed to a projective homomorphism, something nearly homomorphism. Denote $\rho : G \to GL(n, \kappa)$, we have the combination role

$$\rho(g_1)\rho(g_2) = f(g_1, g_2)\rho(g_1 g_2), \tag{4.3}$$

with $f(g_1, g_2)$ a phase factor (cocycle, more generally some number), i.e., a function $f : G \times G \to \kappa^*$ (κ^* is the multiplicative group of field κ without zero element).

Under certain conditions, the phase factor can be absorbed into the representation, and the projective representation can be reduced to a representation. To do so, one introduces $\tilde{\rho}(g) = b(g)\rho(g)$ with $b(g)$ a number factor, and has

$$\begin{aligned}\tilde{\rho}(g_1)\tilde{\rho}(g_2) &= b(g_1)b(g_2)\rho(g_1)\rho(g_2) \\ &= b(g_1)b(g_2)f(g_1, g_2)\rho(g_1 g_2) \\ &= b(g_1)b(g_2)f(g_1, g_2)b(g_1 g_2)^{-1}\tilde{\rho}(g_1 g_2),\end{aligned}$$

and immediately, one sees that the projective representation is equivalent to the representation if there exists $b(g)$ satisfying the cocycle condition

$$f(g_1, g_2) = \frac{b(g_1 g_2)}{b(g_1)b(g_2)}, \quad \forall g_1, g_2 \in G. \tag{4.4}$$

For instance, we know there is a two-to-one homomorphism from SU(2) to SO(3), but if we want to do that inversely from SO(3) to SU(2), the best we can do is to find a projective representation, like using Euler angles with patching conditions.

In quantum mechanics, Born's rule brings the abstract Hilbert space into the physically measurable probability space, and that is where symmetry group acts on the quantum states. Most generally, we first have a pair of bijective map acting on space of physical states $s_1 : \mathcal{S} \to \mathcal{S}$, with $\mathcal{S}$ the space of physical density matrices, and acting on the space of observables $s_2 : \mathcal{O} \to \mathcal{O}$, with $\mathcal{O}$ the space of observables. The measurable symmetry property means that $P_{s_1(\rho), s_2(O)} = P_{\rho, O}$, with $P_{\rho,O}$ giving the probability of measuring observable O in state ρ. In detail, we have $P_{\rho,O}(\lambda) = \mathrm{Tr}(P_O(\lambda)\rho)$, with $P_O(\lambda)$

the projector of observable O with eigenvalue λ. The set of such map is the group of *quantum automorphisms*. However, due to the linear property of trace operation and the convexity of density matrices, we indeed only need to take care of behavior the extreme points of the convex set of density matrices, i.e., the pure states, which form dimension-one projection operators that can be identified with the projective Hilbert space $\mathbb{P}\mathcal{H} = (\mathcal{H} - \{0\})/\mathbb{C}^* = \mathbb{CP}^{N-1}$, for $\dim_{\mathbb{C}}(\mathcal{H}) = N$.

With the above observation, we see that any symmetry group of a quantum system is a subgroup of the quantum automorphism group $\mathrm{Aut}(QM)$, which maintains the Born's rule by preserving the inner product of the Hilbert space $\mathrm{Tr}(P_1P_2) = |\langle\psi_1|\psi_2\rangle|^2/\|\psi_1\|^2\|\psi_2\|^2$ (one thinks observable to be expanded under chosen basis). For instance, when $N = 2$ as a qubit system, we have $\mathbb{P}\mathcal{H} = \mathbb{CP}^1 = S^2$, meaning that projectors can be identified with points on Bloch sphere through $P_n = \frac{1}{2}(1 + \boldsymbol{n}\cdot\boldsymbol{\sigma})$. Since $\mathrm{Tr}(P_{n_1}P_{n_2}) = \frac{1}{2}(1 + \boldsymbol{n}_1\cdot\boldsymbol{n}_2)$, the symmetry group is just the group of isometries of Bloch sphere, i.e., $\mathrm{Aut}_{\mathrm{qtm}}(\mathbb{P}\mathcal{H}) = O(3)$, which essentially preserves the distance on the Bloch sphere, or equivalently the quantum metric (notice that $\boldsymbol{n}_1\cdot\boldsymbol{n}_2$ is determined by the angle between them, which equals to the length of geodesic between their ends on Bloch sphere). In general for N-dimensional quantum system, the symmetry group is the group of isometries of $\mathbb{CP}^{N-1}$, which is the group of unitary transformations that preserve its Fubini-Study metric (quantum metric). There is one thing interesting and important that the projector space has abandoned the global $U(1)$ phase due to the ket-bra structure, for instance, when $N = 2$ the states indeed lives on S^3 while the projectors lives on S^2, and there exists a $U(1)$ bundle over S^2 that is the phase factor by the famous Hopf fibration $S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$.

However, projectors do not share linear space structure — it is bilinear in states, and is hard to work with. On the other hand and luckily, we have a fundamental theorem in quantum mechanics known as Wigner's theorem, which states that any symmetry of the quantum system can be represented by a unitary or anti-unitary operator acting on the Hilbert space. The sentence means that $\mathrm{Aut}(QM)$ is a quotient of a certain group $\mathrm{Aut}(\mathcal{H})$, which is the group of unitary or anti-unitary operators on the Hilbert space that preserves the norms. Anti-unitary operators, or $\mathbb{C}$-anti-linear operators, can be written as a combination of unitary operators and complex conjugation, and the group of anti-unitary operators is isomorphic to the unitary group $U(\mathcal{H})$, which leads to the exact sequence

$$1 \longrightarrow U(\mathcal{H}) \longrightarrow \mathrm{Aut}(\mathcal{H}) \xrightarrow{\phi} \mathbb{Z}_2 \longrightarrow 1, \tag{4.5}$$

with map $\phi(g) = 1$ if g is unitary and $\phi(g) = -1$ if g is anti-unitary. Like we know in quantum mechanics, the symmetry operators act on the Hilbert space in a linear way, e.g., it sends ket $|\psi\rangle$ to $g|\psi\rangle$, and for the projection operators, we can define its transformation under symmetry operation (which we are familiar with) through a homomorphism π from $\mathrm{Aut}(\mathcal{H})$ to $\mathrm{Aut}(QM)$ as

$$\pi(g) : P \mapsto gPg^\dagger, \tag{4.6}$$

where one needs to carefully treat the adjoint in anti-unitary case, and $g^\dagger = g^{-1}$. This form of transformation naturally preserves the Born's rule. Clearly, through this 'projection' map, information of a global $U(1)$ phase in the $\mathrm{Aut}(\mathcal{H})$ becomes invisible, or redundant, and that is our fiber,

$$1 \longrightarrow U(1) \longrightarrow \mathrm{Aut}(\mathcal{H}) \xrightarrow{\pi} \mathrm{Aut}(QM) \longrightarrow 1. \tag{4.7}$$

Due to this redundancy, when we want to represent the symmetry group operator $g \in G \subset \mathrm{Aut}(QM)$ (in reality, only its subgroup) over quantum mechanic system in the

mathematically touchable Hilbert space $U(g) \in \mathrm{Aut}(\mathcal{H})$, the best we can do is to find a projective representation such that

$$U(g_1)U(g_2) = c(g_1, g_2)U(g_1g_2), \tag{4.8}$$

where $c(g_1, g_2)$ is a phase factor that can not be removed generally. Notice that in this we indeed embed $\rho : G \to \mathrm{Aut}(QM)$, and its corresponding symmetry group in Hilbert space is given by a pullback construction

$$\begin{array}{ccccccccc} 1 & \longrightarrow & U(1) & \longrightarrow & \tilde{\mathrm{Aut}}(\mathcal{H}) & \xrightarrow{\tilde{\pi}} & G & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \tilde{\rho}} & & \downarrow{\scriptstyle \rho} & & \\ 1 & \longrightarrow & U(1) & \longrightarrow & \mathrm{Aut}(\mathcal{H}) & \xrightarrow{\pi} & \mathrm{Aut}(QM) & \longrightarrow & 1 \end{array}, \tag{4.9}$$

where $\tilde{\mathrm{Aut}}(\mathcal{H}) = \{\tilde{U}(g) = (g, U(g)) | \pi(U(g)) = \rho(g)\}$, and $\tilde{\pi}(\tilde{U}(g)) = g$, $\tilde{\rho}(\tilde{U}(g)) = U(g)$. $U(1)$ is the kernel of $\tilde{\pi}$, since $\tilde{\pi}(\tilde{U}(g)) = \mathbb{1}_G$ requires that $\pi(U(g)) = \rho(\mathbb{1}_G) = \mathbb{1}_{\mathrm{Aut}(QM)}$, and the elements $U(g)$ satisfying this condition just form the kernel of π, i.e., $U(1)$. In this way, the above line is also an exact sequence, and illustrates the group extension of a symmetry group G by $U(1)$ to form its projective representation in Hilbert space. Notice that we should call $\tilde{\mathrm{Aut}}(\mathcal{H})$ as the quantum symmetry group, which shares a direct homomorphism representation in Hilbert space by $\tilde{\rho}$.

Since matrix multiplication is associative, the phase factor $c(g_1, g_2)$ satisfies the cocycle condition

$$c(g_1, g_2)c(g_1g_2, g_3) = c(g_2, g_3)c(g_1, g_2g_3), \tag{4.10}$$

from $U(g_1)U(g_2)U(g_3) = U(g_1)(U(g_2)U(g_3))$.

From quantum mechanics, we know that eigen-subspace of a Hamiltonian (a quantum system) forms an invariant subspace of symmetry operation, this is just the projective representations of symmetry group acting on the Hilbert space.

4.3 Classifying Central Extension

As discussed, if an extension splits, then bigger group becomes semi-direct product of the smaller group and the quotient group, but if the extension is central, then the bigger group becomes a direct product of the smaller group and the quotient group $\tilde{G} \cong A \times G$, $\iota(a)s(g) \to (a, g)$, since the normal subgroup A is mapped into the center of $\tilde{G}$.

However, for the non-split extension, the classification is a non-trivial problem. Since whether an extension splits or not is determined by the existence of a section that is a group homomorphism, we may consider how far the section is from being a group homomorphism, and this leads to the consideration of quantity $s(g_1)s(g_2)(s(g_1g_2)^{-1})$. As $\pi(s(g_1)s(g_2)(s(g_1g_2)^{-1})) = g_1g_2(g_1g_2)^{-1} = 1$, the quantity is in the kernel of π, thus is identified with the image of ι, which is just the fiber. Thus, there exists vertical movement on the fiber that prevents the section from being a group homomorphism, to describe which we can define a 2-cocycle function $f_s : G \times G \to A$ by

$$\iota(f_s(g_1, g_2)) = s(g_1)s(g_2)(s(g_1g_2)^{-1}), \tag{4.11}$$

or equivalently,

$$s(g_1)s(g_2) = \iota(f_s(g_1, g_2))s(g_1g_2), \tag{4.12}$$

which already shares a similar form with the phase factor in projective representation. By definition,

$$\begin{aligned} s(g_1g_2g_3) &= \iota(f_s(g_1g_2,g_3))^{-1}s(g_1g_2)s(g_3) = \iota(f_s(g_1g_2,g_3))^{-1}\iota(f_s(g_1,g_2))^{-1}s(g_1)s(g_2)s(g_3) \\ &= \iota(f_s(g_1,g_2g_3))^{-1}s(g_1)s(g_2g_3) = \iota(f_s(g_1,g_2g_3))^{-1}\iota(f_s(g_2,g_3))^{-1}s(g_1)s(g_2)s(g_3), \end{aligned}$$

from which we read the cocycle identity

$$f(g_1,g_2)f(g_1g_2,g_3) = f(g_2,g_3)f(g_1,g_2g_3). \tag{4.13}$$

By letting $g_2 = g_3 = 1, g_1 = g$, we have $f(g,1)f(g,1) = f(1,1)f(g,1)$, thus $f(g,1) = f(1,1)$, while by $g_1 = g_2 = 1, g_3 = g$, we have $f(1,g) = f(1,1)$. Based on this, when $g_1 = g_3 = g^{-1}, g_2 = g$, we have $f(g^{-1},g)f(1,g^{-1}) = f(g,g^{-1})f(g^{-1},1)$, and $f(g^{-1},g) = f(g,g^{-1})$.

The 2-cocycle functions form an Abelian group $Z^2(G,A)$, which is a subgroup of the 2-chain Abelian group $C^2(G,A)$ that contains *all* functions $f : G \times G \to A$. They are Abelian due to the Abelian nature of A.

In different sections, the 2-cocycle functions may differ by a coboundary. Consider a new section $s'(g) = \iota(t(g))s(g)$, with $t : G \to A$ a function such that $\iota(t(g))$ determines a fiber movement at every point between two sections by G, then by definition

$$\begin{aligned} \iota(f_{s'}(g_1,g_2)) &= s'(g_1)s'(g_2)(s'(g_1g_2))^{-1} \\ &= \iota(t(g_1))s(g_1)\iota(t(g_2))s(g_2)(s(g_1g_2))^{-1}(\iota(t(g_1g_2)))^{-1} \\ &= \iota(f_s(g_1,g_2))\iota(t(g_1))\iota(t(g_2))(\iota(t(g_1g_2)))^{-1}, \end{aligned}$$

and

$$f_{s'}(g_1,g_2) = f_s(g_1,g_2)t(g_1)t(g_2)t(g_1g_2)^{-1}, \tag{4.14}$$

where we have used the fact that ι maps to center. The inverse is also true, that if another f' differs from a 2-cocycle f in this way (differ by a coboundary), then f' also forms a 2-cocycle function, for another section. Thus the *differ by a coboundary* condition defines an equivalence relation. Notice that the relations of different sections, i.e., t functions are fixed for a given extension, then each extension should somwhow be classified by the 2-cocycle functions in its single section. We can thus define

Definition 4.3.1. *The second cohomology group $H^2(G,A)$ is the set of equivalence classes of 2-cocycle functions under the relation of differ by a coboundary.*

By discussion above, the homology class of the 2-cocycles does not depend on the choice of section in one extension.

Theorem 4.3.1. *Isomorphism classes of central extensions of G by A are in one-to-one correspondence with the elements of $H^2(G,A)$.*

Proof. Let us denote $\tilde{G}$ for abbreviation of central extension $1 \to A \to \tilde{G} \to G \to 1$, and let $\varepsilon(G,A) = \{G\}$ as set of all extensions, while $\tilde{\varepsilon}(G,A) = \{[\tilde{G}]\}$ as set of all isomorphism classes of extensions. What we need to prove is that there exists an isomorphism between $\tilde{\varepsilon}(G,A)$ and $H^2(G,A)$.

To achieve this, we firstly define a map $\Psi : \varepsilon(G,A) \to H^2(G,A)$ by letting $\Psi(\tilde{G}) = [f_s]$. Notice that this map is well-defined since the 2-cocycle function class $[f_s]$ is independent of the choice of section. We need then to show that $\Psi(\tilde{G}) = \Psi([\tilde{G}])$ only depends on

the isomorphism class of $\tilde{G}$. To do so, we need to introduce isomorphism between two extensions $\tilde{G}$ and $\tilde{G}'$ as

$$\begin{array}{ccccccccc} 1 & \longrightarrow & A & \xrightarrow{\iota} & \tilde{G} & \xrightarrow{\tilde{\pi}} & G & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \mathrm{id}} & & \\ 1 & \longrightarrow & A & \xrightarrow{\iota'} & \tilde{G}' & \xrightarrow{\pi'} & G & \longrightarrow & 1 \end{array},$$

where ψ is an isomorphism between $\tilde{G}$ and $\tilde{G}'$. Consider that $s' = \psi \circ s$, $\iota' = \psi \circ \iota$, we have

$$\begin{aligned} s'(g_1)s'(g_2) &= \psi(s(g_1))\psi(s(g_2)) = \psi(s(g_1)s(g_2)) = \psi(\iota(f_s(g_1,g_2))s(g_1g_2)) \\ &= \psi(\iota(f_s(g_1,g_2)))\psi(s(g_1g_2)) = \iota'(f_s(g_1,g_2))s'(g_1g_2), \end{aligned}$$

thus isomorphic extensions have the same 2-cocycle function (class). This allows us to define $\tilde{\Psi} : \tilde{\varepsilon}(G,A) \to H^2(G,A)$ as a map from isomorphism classes of extensions to equivalence classes of 2-cocycles.

To show the one-to-one correspondence, we need to find the inverse map. The idea is to construct extension from a 2-cocycle function. To do this, we let the extension $\tilde{G}$ to be $A \times G$ as a set, with group operation defined by

$$(a_1, g_1)(a_2, g_2) = (a_1 a_2 f(g_1, g_2), g_1 g_2),$$

this law is valid, with associativity coming from the cocycle identity. For the 2-cocycle class that differs by a coboundary defined by $t(g)$, we have the extension isomorphism defined by $(a, g) \to (at(g)^{-1}, g)$, thus we also establish the inverse map $\tilde{\Psi}^{-1} : H^2(G,A) \to \tilde{\varepsilon}(G,A)$. □

Thus, the central extension of a group G by an abelian group A is classified by the second cohomology group $H^2(G,A)$, an Abelian group with multiplicative law

$$(f_1 \cdot f_2)(g_1, g_2) = f_1(g_1, g_2) f_2(g_1, g_2),$$

i.e., $[f_1] \cdot [f_2] = [f_1 \cdot f_2]$. Its identity element is the class of trivial 2-cocycle function $[f_0]$ with $f_0(g_1, g_2) \equiv 1$, and correspondingly the trivial extension $[A \times G]$.

Compare with the projective representation, we find that it is a special case of central extension, where the normal subgroup A is $U(1)$, and the quotient group G is the symmetry group $\mathrm{Aut}(QM)$, and the statement that one symmetry can only be projectively represented in Hilbert space is just the statement that the extension is non-split.

When cocycle is non-trivial? Given cocycle f by coboundary function $t(g)$, we have $f^{(1)}(1,1) = f(1,1)\frac{t(1)t(1)}{t(1\cdot 1)} = f(1,1)t(1)$. We choose $t(1) = f(1,1)^{-1}$ and $t(g \neq 1) = 1$, such that $f^{(1)}(g,1) = f^{(1)}(1,g) = f^{(1)}(1,1) = 1$. For $g \neq 1$, we have $f^{(1)}(g,g^{-1}) = f(g,g^{-1})t(g)t(g^{-1})/t(1) = f(g,g^{-1})f(1,1)$. Now we do a second transformation $\tilde{t}(g)$ with $\tilde{t}(1) = 1$, and $f^{(2)}(g,g^{-1}) = f^{(1)}(g,g^{-1})\frac{\tilde{t}(g)\tilde{t}(g^{-1})}{\tilde{t}(g\cdot g^{-1})} = f^{(1)}(g,g^{-1})\tilde{t}(g)\tilde{t}(g^{-1})$. For non-involution g, i.e., $g^2 \neq 1$, we can always choose $\tilde{t}(g) = f^{(1)}(g,g^{-1})^{-1}\tilde{t}(g^{-1})^{-1}$ such that $f^{(2)}(g,g^{-1}) = 1$. Notice that this is achievable since we can separate $g \in S_1$ and $g^{-1} \in S_2$ as two non-intersecting sets. Then by identity $f^{(2)}(g^{-1},g) = f^{(2)}(g,g^{-1}) = 1$ stands for all $g \neq g^{-1}$. However, for $g^2 = 1$, we have $f^{(2)}(g,g) = f^{(1)}(g,g)(\tilde{t}(g))^2$, then only when $f^{(1)}(g,g) = (g')^2$ for some g', can we make $f^{(2)}(g,g) = 1$ by $\tilde{t}(g) = g'^{-1}$. Otherwise, $[f]$ must be non-trivial.

For instance, we consider extensions of $G = \mathbb{Z}_2 = \{1, \sigma_2\}$ by $A = \mathbb{Z}_2 = \{1, \sigma_1\}$. As stated, we can choose $f(1,1) = f(\sigma_2, 1) = f(1, \sigma_2) = 1$, and both $f(\sigma_2, \sigma_2) = 1$ or σ_1 satisfy the cocycle condition. However, since σ_1 is not a square, the cocycle with $f(\sigma_2, \sigma_2) = \sigma_1$ is non-trivial. To see this more clearly, we can firstly show that these two are not equivalent by a coboundary. Assuming they are, then we need

$$\begin{cases} \sigma_1 = 1 \cdot t(\sigma_2)t(\sigma_2)/t(1) = t^2(\sigma_2)t(1)^{-1} \equiv t(1)^{-1} \\ 1 = 1 \cdot t(1)t(\sigma_2)/t(\sigma_2) = t(1) \end{cases},$$

for $f(\sigma_2, \sigma_2)$ and $f(1, \sigma_2)$, which is contradictory. Thus we see there are two classes of extensitons. For $f(\sigma_2, \sigma_2) = 1$, the cocycle is trivial, corresponding to the trivial extension as a direct product $\tilde{G} \cong A \times G = \mathbb{Z}_2 \times \mathbb{Z}_2$. For $f(\sigma_2, \sigma_2) = \sigma_1$, we write down the group multiplication law generated by f as

$$\begin{cases} (1,1)(?,?) = (f(1,?)?,?) = (?,?) = (?,?)(1,1) \\ (\sigma_1,1)(\sigma_1,1) = (f(1,1)\sigma_1^2, 1) = (1,1) \\ (\sigma_1,1)(1,\sigma_2) = (f(1,\sigma_2)\sigma_1, \sigma_2) = (\sigma_1, \sigma_2) \\ (1,\sigma_2)(1,\sigma_2) = (f(\sigma_2,\sigma_2), \sigma_2^2) = (\sigma_1, 1) \\ (1,\sigma_2)(\sigma_1,1) = (f(\sigma_2,1)\sigma_1, \sigma_2) = (\sigma_1, \sigma_2) \end{cases}$$

and we see a map from $\tilde{G}$ to $\mathbb{Z}_4$ as $(1,1) \to 1$, $(1,\sigma_2) \to \omega$, $(\sigma_1, 1) \to \omega^2$ and $(\sigma_1, \sigma_2) \to \omega^3$, where ω is the fourth root of unity. The non trival extension follows as

$$1 \longrightarrow \mathbb{Z}_2 \xrightarrow{\iota} \mathbb{Z}_4 \xrightarrow{\pi} \mathbb{Z}_2 \longrightarrow 1.$$

Consider group commutator of $\tilde{G}$

$$\begin{aligned} [(a_1, g_1), (a_2, g_2)] &= (a_1 a_2 f(g_1, g_2), g_1 g_2)(a_1^{-1} a_2^{-1} f(g_1, g_1^{-1})^{-1} f(g_2, g_2^{-1})^{-1} f(g_1^{-1}, g_2^{-1}), g_1^{-1} g_2^{-1}) \\ &= (f(g_1, g_2) f(g_1, g_1^{-1})^{-1} f(g_2, g_2^{-1})^{-1} f(g_1^{-1}, g_2^{-1}) f(g_1 g_2, g_1^{-1} g_2^{-1}), g_1 g_2 g_1^{-1} g_2^{-1}), \end{aligned}$$

where one notices that $(a_1, g_1)^{-1} = (a_1^{-1} f(g_1, g_1^{-1})^{-1}, g_1^{-1})$ such that $(a_1, g_1)(a_1, g_1)^{-1} = (1,1)$. Then by

$$\begin{aligned} f(g_1^{-1}, g_2^{-1}) f(g_2 g_1, g_1^{-1} g_2^{-1}) &= f(g_2 g_1, g_1^{-1}) f(g_2, g_2^{-1}) \\ f(g_2 g_1, g_1^{-1}) f(g_2, g_1) &= f(g_1^{-1}, g_1) f(g_2 g_1, 1), \end{aligned}$$

with choice $f(1, g) = f(g, 1) = f(1,1) = 1$, we have

$$[(a_1, g_1), (a_2, g_2)] = \left(\frac{f(g_1, g_2)}{f(g_2, g_1)} \frac{f(g_1 g_2, g_1^{-1} g_2^{-1})}{f(g_2 g_1, g_1^{-1} g_2^{-1})}, g_1 g_2 g_1^{-1} g_2^{-1} \right).$$

When G is Abelian, this reduces to $[(a_1, g_1), (a_2, g_2)] = (f(g_1, g_2)/f(g_2, g_1), 1)$, and $\tilde{G}$ is Abelian iff $f(g_1, g_2) = f(g_2, g_1)$, i.e., f is symmetric.

4.4 Quantum Anomaly from Extension

In this section, we shall try to illustrate how the quantum anomaly arises from the (central) extension of symmetry group, using basically a simple example, that a particle on a ring.

4.4.1 Quantum symmetry

The classical action of the system reads

$$S = \int \frac{1}{2} m r^2 \dot{\phi}^2 \, \mathrm{d}t = \int \frac{I}{2} \dot{\phi}^2 \, \mathrm{d}t \,,$$

where $I = mr^2$ is the moment of inertia. Now we assume that a magnetic flux is threaded through the ring in the center, such that the particle does not experience the magnetic field but the vector potential $A = \frac{B}{2\pi} \mathrm{d}\phi$, where we use B to denote the flux (as an abbreviation) in the 1-form language (and we omit a r factor). As we know, the Lagrangian of a charged particle in magnetic field reads $e\dot{\boldsymbol{x}} \cdot \boldsymbol{A}$ ($\boldsymbol{A} = \frac{B}{2\pi r} \hat{\phi}$), which leads to an additional term in the action

$$S = \int \left(\frac{I}{2} \dot{\phi}^2 + e\dot{\boldsymbol{x}} \cdot \boldsymbol{A} \right) \mathrm{d}t = \int \left(\frac{I}{2} \dot{\phi}^2 + \frac{eB}{2\pi} \dot{\phi} \right) \mathrm{d}t \,. \tag{4.15}$$

This additional term is about the first order derivative of ϕ, and is a total derivative in the case, thus does not affect the classical equation of motion (EoM). Indeed, after a re-writing we see that this term equals

$$\int \frac{eB}{2\pi} \dot{\phi} \, \mathrm{d}t = \oint eA,$$

which represents a topological term, and especially, a θ-term here.

The symmetry of the classical system is clearly $O(2) = SO(2) \rtimes \mathbb{Z}_2$, a proper rotation $R(\alpha) : \phi \to \phi + \alpha$ and a parity/charge conjugation $P : \phi \to -\phi$ (call charge since it sends $e^{i\phi}$ to $e^{-i\phi}$), with semidirect product structure

$$R(\alpha)R(\beta) = R(\alpha + \beta), \quad P^2 = 1, \quad PR(\alpha)P = R(-\alpha).$$

Notice that the θ-term is non-invariant under parity, but since this term does not affect EoM, the classical system is still symmetric under parity. This also reflects that in classical mechanics, it is field rather than vector potential that is physical.

However, in quantum mechanics, the vector potential is physical by directly affecting the phase and coherence. To see this, we write down the conjugate (angular) momentum of ϕ as $l = \frac{\partial L}{\partial \dot{\phi}} = I\dot{\phi} + \frac{eB}{2\pi}$, and obtain the classical Hamiltonian from Legendre transformation as

$$H_{\mathrm{cl}} = \dot{\phi} l - L = \frac{1}{2I} \left(l - \frac{eB}{2\pi} \right)^2 . \tag{4.16}$$

As there's no ϕ dependence, $\dot{l} = 0$ or equivalently $\dot{\phi} = const$ is the EoM, and the system is completely free without further constraint. On the other hand, quantization requires $[\phi, l] = i\hbar$ and gives $l \to -i\hbar\partial_\phi$, and the quantum Hamiltonian reads

$$H_{\mathcal{B}} = \frac{\hbar^2}{2I} \left(-i\partial_\phi - \mathcal{B} \right)^2 , \quad \mathcal{B} = \frac{eB}{h}. \tag{4.17}$$

The quantization re-built the solution space of the system, which is now

$$\begin{cases} E_m = \frac{\hbar^2}{2I}(m - \mathcal{B})^2 \\ \Psi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \end{cases} , \quad m \in \mathbb{Z}. \tag{4.18}$$

Notice that the requirement $m \in \mathbb{Z}$ comes from the requirement of single-valuedness of the wave function, and leads to the quantization of angular momentum.

The energy spectrum, as we can directly see, is shifted by $\mathcal{B}$, and the topological term that has no influence classically indeed affects the system quantum mechanically. Due to the discreteness of m, at most values of $\mathcal{B}$, this shifted spectrum becomes non-degenerate with broken parity symmetry, and it is only when $2\mathcal{B} \in \mathbb{Z}$ that the spectrum becomes symmetric. Especially, when $2\mathcal{B}$ is even meaning $\mathcal{B}$ is an integer, there exists a sole non-degenerate state at $m = \mathcal{B}$, and other states are doubly-degenerate at $m = \mathcal{B} \pm n$. When $2\mathcal{B}$ is odd with $\mathcal{B}$ a half-integer, all states are doubly-degenerate at $m = \mathcal{B} \pm (n + \frac{1}{2})$. In any case, it is m and $2\mathcal{B} - m$ that shares the same energy, and the system is symmetric under parity defined in this way.

This single-valuedness has more constraint on taking proper choice of ϕ. As we can see, it does not matter in classical mechanics whether we take ϕ or $\phi + 2\pi$ since the EoM is invariant under this transformation, however, focusing on our quantum mechanical solution, this transformation is equivalent to $m \to m + 1$, and will shift all energy levels by 1. This is indeed a non-trivial transformation. Note that the shift $m \to m + 1$ is equivalent to shift $\mathcal{B} \to \mathcal{B} + 1$, indicating the periodicity 1 of $\mathcal{B}$. Such a transformation can be generated by a unitary operator

$$\begin{cases} U\Psi_m = \Psi_{m+1} \\ UH_{\mathcal{B}}U^{-1} = H_{\mathcal{B}+1} \end{cases}.$$

Now we want to implement the symmetry $O(2)$ in this quantum system. The proper rotation is realized through mapping $\rho(R(\alpha)) = \mathcal{R}(\alpha)$ with

$$\mathcal{R}(\alpha)\Psi_m = e^{im\alpha}\Psi_m,$$

resembling the shift of ϕ by α. The parity symmetry, which if we directly implement as $P : \phi \to -\phi$ with $\rho(P) = \mathcal{P}$, takes $\mathcal{P}H_{\mathcal{B}}\mathcal{P}^{-1} = H_{-\mathcal{B}}$. At first appearance, this does not look like a symmetry in quantum mechanics, since it is obviously non-commutative, but since the Hamiltonian is indeed written under a basis, such a transformation is acceptable as a symmetry as long as there exists a unitary equivalence between $H_{\mathcal{B}}$ and $H_{-\mathcal{B}}$. Generally, if $2\mathcal{B}$ is not an integer, such an equivalence does not exist, at which case the symmetry group is reduced to $SO(2)$. Alternatively, when $2\mathcal{B} \in \mathbb{Z}$, by the introduced unitary operator U, we have

$$U^{2\mathcal{B}}H_{-\mathcal{B}}U^{-2\mathcal{B}} = H_{\mathcal{B}},$$

which gives the equivalence between $H_{\mathcal{B}}$ and $H_{-\mathcal{B}}$. This relation relates the $-m$ state of $H_{-\mathcal{B}}$ with the m state of $H_{\mathcal{B}}$ by $U^{2\mathcal{B}}$. As shown, there exist states within the same eigen-space, whose quantum numbers are related by $m \leftrightarrow 2\mathcal{B} - m$, thus defining the parity operation $\rho(P) = \mathcal{P}$ as

$$\mathcal{P}\Psi_m = \xi_m\Psi_{2\mathcal{B}-m},$$

where $\xi_m = \pm 1$ is a phase factor. As with the classical case, the quantum symmetry does not share direct product structure, specifically, since

$$\begin{aligned} \mathcal{P}\mathcal{R}(\alpha)\mathcal{P}\Psi_m &= \xi_m\mathcal{P}\mathcal{R}(\alpha)\Psi_{2\mathcal{B}-m} = \xi_m e^{i(2\mathcal{B}-m)\alpha}\mathcal{P}\Psi_{2\mathcal{B}-m} \\ &= \xi_m\xi_{2\mathcal{B}-m}e^{i(2\mathcal{B}-m)\alpha}\Psi_m = e^{i2\mathcal{B}\alpha}\mathcal{R}(-\alpha)\Psi_m, \end{aligned}$$

where $\xi_m\xi_{2\mathcal{B}-m} = 1$ as $\mathcal{P}^2 = 1$, we obtain the combination law

$$\mathcal{P}\mathcal{R}(\alpha)\mathcal{P} = e^{i2\mathcal{B}\alpha}\mathcal{R}(-\alpha),$$

which hints a projective representation as a central extension of $O(2)$ by $U(1)$, since an additional phase factor in the law is introduced compared with the classical case. We are thus expecting

$$1 \longrightarrow U(1) \longrightarrow \mathcal{G}_B \longrightarrow O(2) \longrightarrow 1 \ .$$

For $\mathcal{B}$ an integer, we can indeed absorb the phase factor into the proper rotation by redefining $\mathcal{R}(\alpha) \to e^{i\mathcal{B}\alpha}\mathcal{R}(\alpha)$, which is not conflicting the 2π-periodicity of α, and in this case, the combination law comes back to the classical case, and the quantum symmetry group is still $O(2)$ with $\mathcal{G}_B = U(1) \times O(2)$.

The somewhat unusual case happens when $\mathcal{B}$ is a half-integer, where the phase factor can not be simply absorbed since $e^{i\mathcal{B}\alpha}$ is not 2π-periodic. Nevertheless, it is 4π-periodic, and this feature indeed reminds us the double cover of $SO(3)$ by $SU(2)$, and it is indeed the case. The physical motivation comes from the observation that when $\mathcal{B}$ is a half-integer, the angular momentum l is half-integer, and the system shall follow a half-integer spin representation. Under the circumstance, we introduce the spin group Spin(2) (which is indeed another $U(1)$ group), and writes down the double cover $\pi : \mathrm{Spin}(2) \to SO(2)$ by $\hat{R}(\hat{\alpha}) \mapsto R(2\hat{\alpha})$, where

$$\hat{R}(\hat{\alpha}) = \exp(\hat{\alpha}\sigma_1\sigma_2) = \cos(\hat{\alpha}) + \sin(\hat{\alpha})\sigma_3, \ \hat{\alpha} \in [0, 2\pi),$$

is element of Spin(2), and $R(\theta)$ is element of $SO(2)$. The key is to make another 2 factor such that the phase factor can be absorbed into the rotation part. And then, the parity operation is similarly realized $\hat{P} : \mathbb{Z}_2 \to \mathrm{Aut}(\mathrm{Spin}(2))$ by

$$\alpha(\hat{P}) : \hat{R}(\hat{\alpha}) \to (\hat{R}(\hat{\alpha}))^{-1} = \hat{R}(-\hat{\alpha}),$$

and the total group structure is given by the semi-direct product

$$\mathrm{Pin}^+(2) \cong \mathrm{Spin}(2) \rtimes_\alpha \mathbb{Z}_2. \tag{4.19}$$

Such a procedure can be generalized to higher dimensions, with the spin group $\mathrm{Spin}(d)$ as the double cover of $SO(d)$, and the pinor group $\mathrm{Pin}^+(d)$ as the double cover of $O(d)$. For half-integer $\mathcal{B}$, we have the generators of $\mathcal{G}_B$ group as

$$\begin{cases} z \in U(1) \\ \rho(\hat{R}(\hat{\alpha})) = e^{-i(2\mathcal{B})\hat{\alpha}}\mathcal{R}(2\hat{\alpha}), \ \hat{\alpha} \in [0, 2\pi) \\ \rho(\hat{P}) = \mathcal{P} \end{cases} .$$

To see what $\mathcal{G}_B$ is, we need to consider the commutative diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathbb{Z}_2 & \longrightarrow & \mathrm{Pin}^+(2) & \xrightarrow{\pi} & O(2) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow{\scriptstyle \rho} & & \downarrow{\scriptstyle \mathrm{id}} & & \\ 1 & \longrightarrow & U(1) & \longrightarrow & \mathcal{G}_B & \longrightarrow & O(2) & \longrightarrow & 1 \end{array} ,$$

with ρ an inclusion map, and $\pi(\mathrm{Pin}^+(2)) = \pi(\mathrm{Spin}(2)) \rtimes_\alpha \mathbb{Z}_2 = SO(2) \rtimes_\alpha \mathbb{Z}_2 = O(2)$. Note that when $\mathcal{B}$ is an integer, we can also construct such a diagram, but now ρ contains a kernel isomorphic to $\mathbb{Z}_2$, indicating the needless of the double cover. This classical-integer (or no), quantum-fraction from quantum symmetries with topological terms is very intriguing.

One more practical situation arises when we add further a potential in the system. Most generally, we have $U^g(\phi) = \sum_{n\in\mathbb{Z}} c_n \cos(n\phi) + s_n \sin(n\phi)$ as periodic potential. Nevertheless, this potential breaks the $O(2)$ symmetry completely and does not give to much insight. However, if we restrict to a special form $U^s(\phi) = \sum_{n\in\mathbb{Z}} u_n \cos(2n\phi)$, which reduces the $O(2)$ symmetry to $\mathbb{Z}_2 \times \mathbb{Z}_2$ generated by $P : \phi \to -\phi$ and $R(\pi) : \phi \to \phi + \pi$, we obtain the commutative diagram that describes the representation of the symmetry group in the Hilbert space

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathbb{Z}_2 & \longrightarrow & D_4 & \xrightarrow{\pi_1} & \mathbb{Z}_2 \times \mathbb{Z}_2 & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \iota} & & \\ 1 & \longrightarrow & \mathbb{Z}_2 & \longrightarrow & \mathrm{Pin}^+(2) & \xrightarrow{\pi_2} & O(2) & \longrightarrow & 1 \end{array} ,$$

where $D_4 \cong \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ is the dihedral group with generators, four-fold rotations and parity by $\rho(\hat{r}) = e^{iB\pi} R(\pi)$ and $\rho(\hat{P}) = P$, where $\hat{r}^2 = -1, \hat{P}^2 = 1$ and $\hat{P}\hat{r}\hat{P} = \hat{r}^{-1}$. Notice that we are unable to give a representation of D_4 in the Hilbert space.

4.4.2 Partition Function

Basically our starting point will be the quantity we are familiar with, the partition function $Z = \mathrm{Tr}_{\mathcal{H}} e^{-\beta H}$. Taking $\hbar = k_B = 1$, we have

$$Z = \sum_{m\in\mathbb{Z}} e^{-\frac{\beta}{2I}(m-\mathcal{B})^2}, \tag{4.20}$$

as clearly from the energy spectrum. Two interesting points again similarly arise. First is the formal periodicity of the partition function under integer shifts of $\mathcal{B}$, the theta angle in the system. The other is the intrinsic proper of partition function that as $T \to 0$ or equivalently $\beta \to +\infty$, only ground states (infrared (IR) behavior) matter, and for $\mathcal{B}$ integer, $Z = 1$ is non-interesting, while for $\mathcal{B}$ half-integer, Z comes from two copies of $e^{-\frac{\beta}{2I}\left(\frac{1}{2}\right)^2}$, reflecting the double degeneracy of the ground state — a Qbit.

Quantum anomaly often arises from the quantum field theories, which suits best with the language of path integral. In our case, the partition function can be written as $Z = \int_0^{2\pi} \mathrm{d}\phi \, \langle\phi|e^{-\beta H}|\phi\rangle$ as a summation over loop-amplitudes, while the generate amplitude as matrix element of the Euclidean time evolution operator reads

$$\langle\phi_2|e^{-t_E H}|\phi_1\rangle = \frac{1}{2\pi} \sum_{m\in\mathbb{Z}} e^{-\frac{t_E}{2I}(m-\mathcal{B})^2 + im(\phi_2-\phi_1)}, \tag{4.21}$$

where a complete relation is inserted with wavefunction $\langle\phi|m\rangle = \frac{1}{\sqrt{2\pi}} e^{im\phi}$. The Euclidean time propagator is an analytic continuation of e^{-itH} with $t \to z = -it_E$, and such a continuation requires the (Hermitian) Hamiltonian to be lower bounded in spectrum. The above expression is given under Hamiltonian formalism, while the 'Lagrangian' formalism that appears more frequently in path integral is also available

$$\begin{aligned} \langle\phi_2|e^{-\beta H}|\phi_1\rangle = Z(\phi_2, \phi_1|\beta) &= \int \mathrm{D}\phi\big|_{\phi(0)=\phi_1}^{\phi(-i\beta)=\phi_2} \exp\left(i \int_0^{-i\beta} \mathrm{d}t \left[\frac{I}{2}\dot{\phi}^2 + \mathcal{B}\dot{\phi} \right] \right) \\ &\overset{\tau=it}{=} \int \mathrm{D}\phi\big|_{\phi(0)=\phi_1}^{\phi(\beta)=\phi_2} \exp\left(- \int_0^{\beta} \mathrm{d}\tau \left[\frac{I}{2}\dot{\phi}^2 - i\mathcal{B}\dot{\phi} \right] \right), \end{aligned} \tag{4.22}$$

where in the second line $\dot{\phi} = \partial_\tau \phi$. Below we will again replace τ with t. The evaluation of this integral utilizes the instanton-fluctuation skill, where 'instanton' actually refers to the classical solution of the equation of motion $\ddot{\phi} = 0$ under boundary condition $\phi(0) = \phi_1, \phi(\beta) = \phi_2$ on a loop, which is

$$\phi_{cl}(t) = \phi_1 + \frac{\phi_2 - \phi_1 + 2\pi\omega}{\beta} t, \quad \omega \in \mathrm{Z}.$$

The winding number $\omega \in \pi_1(S^1)$, comes from the map of movement, from the lower $M = S^1$ as the spacetime manifold to the target space $\chi = S^1$, as the allowed configuration space of ϕ. $\pi_1(S^1)$ then classifies $\mathrm{Map}(M, S^1)$. This classical instanton path suffers from the quantum fluctuations $\phi(t) = \phi_{cl}(t) + \phi_q(t)$, where the fluctuation part is constrained that $\phi_q(0) = 0 = \phi_q(\beta)$, and is thus forced to be topologically trivial. Two points: as ϕ_{cl} solves the classical EoM (meaning it is the saddle point), the action splits into $S = S(\{\phi_{cl}\}) + S(\{\phi_q\})$; since ϕ_{cl} is certain, the path integral is done over ϕ_q only, and the instanton solutions appear as phase factors. The full path integral then reads

$$\begin{aligned} Z(\phi_2, \phi_1|\beta) &= Z_q \sum_{\omega\in\mathbb{Z}} \exp\left(- \int_0^\beta \mathrm{d}t \left[\frac{I}{2} \left(\frac{\phi_2 - \phi_1 + 2\pi\omega}{\beta} \right)^2 - i\mathcal{B} \frac{\phi_2 - \phi_1 + 2\pi\omega}{\beta} \right] \right) \\ &= Z_q \sum_{\omega\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}\left(\omega + \frac{\phi_2-\phi_1}{2\pi}\right)^2 + 2\pi i\mathcal{B}\left(\omega + \frac{\phi_2-\phi_1}{2\pi}\right)}, \end{aligned} \tag{4.23}$$

where the prefactor Z_q is the fluctuation determinant

$$Z_q = \int \mathrm{D}\phi_q \big|_{\phi_q(0)=0}^{\phi_q(\beta)=0} \exp\left(- \int_0^\beta \mathrm{d}t \frac{I}{2} \dot{\phi}_q^2 \right),$$

where the theta term vanishes due to the boundary condition. Through Gaussian integral, $Z_q = N \det{}'^{-1/2}(\mathcal{O})$, where $\mathcal{O} = -\frac{I}{2\beta}\frac{\mathrm{d}^2}{\mathrm{d}t^2}$ with boundary constraint $\phi(0) = 0 = \phi(1)$. Note that we have rescaled the integral interval, and the determinant is taken over the non-zero modes.

The operator $\mathcal{O}$ has eigenfunctions $\phi_n(t) = \sqrt{2}\sin(n\pi t)$ $(n \neq 0)$ with eigenvalues $\lambda_n = \frac{I}{2\beta}\pi^2 n^2$. Clearly, the determinant can not be directly computed, and we need to regularize it. Here it comes to the ζ-function regularization, where one notices

$$\frac{\mathrm{d}}{\mathrm{d}s} \lambda^{-s} \big|_{s=0} = -\ln \lambda,$$

and by defining operator-specified ζ-function as $\zeta_{\mathcal{O}}(s) = \sum_n \lambda_n^{-s}$, we have

$$\det(\mathcal{O}) = \prod_n \lambda_n = \exp(-\zeta'_{\mathcal{O}}(0)).$$

By evaluating the exponential first and then taking the $s \to 0$ limit (with $\mathrm{Re}(s) > 0$), one can obtain the regularized determinant for suitable class of operators. Now since

$$\begin{aligned} \zeta_{\mathcal{O}}(s) &= 2\sum_n \left(\frac{I}{2\beta}\pi^2 n^2 \right)^{-s} = 2\left(\frac{2\beta}{\pi^2 I} \right)^s \zeta(2s) \\ &= 2\left(\frac{2\beta}{\pi^2 I} \right)^s \left(-\frac{1}{2} + 2s \ln\left(\frac{1}{\sqrt{2\pi}} \right) + O(s^2) \right), \end{aligned}$$

where the ζ-function is defined as $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, we have

$$\begin{aligned}\det(\mathcal{O}) &= \exp\left(-2\left(\frac{2\beta}{\pi^2 I}\right)^s \ln\left(\frac{2\beta}{\pi^2 I}\right)\zeta(2s) - 2\left(\frac{2\beta}{\pi^2 I}\right)^s 2\ln\left(\frac{1}{\sqrt{2\pi}}\right)\right)\Big|_{s=0} \\ &= \exp\left(\ln\left(\frac{2\beta}{\pi^2 I}\right) - 4\ln\left(\frac{1}{\sqrt{2\pi}}\right)\right) = \frac{8\beta}{I}.\end{aligned}$$

Thus, the partition function reads

$$Z(\phi_2, \phi_1|\beta) = N\left(\frac{I}{\beta}\right)^{1/2} \sum_{\omega\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}\left(\omega+\frac{\phi_2-\phi_1}{2\pi}\right)^2 + 2\pi i\mathcal{B}\left(\omega+\frac{\phi_2-\phi_1}{2\pi}\right)},$$

with N composed of the normalization factor and the fluctuation determinant with some 2 and π factors. This classical-EoM based path integral can tell us something. One notices that β appears in an inverse way as compared with the Hamiltonian formalism, and now $\beta \to 0$ makes $1/\beta$ large, which suppresses all $m \neq 0$ instantons, and only the trivial sector with $m = 0$ matters. This corresponds to our intuition since now β is a measure of time passed, and for short time, the system is not able to climb the potential barrier. In the circumstance, we expect the partition function

$$Z(\phi_2, \phi_1|\beta) = Z_q e^{-\frac{I}{2\beta}(\phi_2-\phi_1)^2 + i\mathcal{B}(\phi_2-\phi_1)}(1 + O(e^{-1/\beta}))$$

will be the same with the propagator of the free particle on a straight line (notice that in the case $\mathcal{B}$ can be absorbed)

$$\langle x_2|e^{-\frac{\beta}{2m}\hat{p}^2}|x_1\rangle = \left(\frac{m}{2\pi\beta}\right)^{1/2} e^{-\frac{m}{2\beta}(x_2-x_1)^2},$$

with identification $x = \phi, m = I$. Thus $Z_q = \sqrt{\frac{I}{2\pi\beta}}$, and

$$Z(\phi_2, \phi_1|\beta) = \sqrt{\frac{I}{2\pi\beta}} e^{-\frac{I}{2\beta}(\phi_2-\phi_1)^2 + i\mathcal{B}(\phi_2-\phi_1)}(1 + O(e^{-1/\beta})). \tag{4.24}$$

The partition functions evaluated in two different ways should be the same, and indeed they are by Possion resummation formula of Riemann theta functions. Under the Hamiltonian formalism, we have

$$Z(\phi_2, \phi_1|\beta) = \frac{1}{2\pi}\sum_{m\in\mathbb{Z}} e^{-\frac{\beta}{2I}(m-\mathcal{B})^2 + im(\phi_2-\phi_1)} = \frac{1}{2\pi} e^{i\mathcal{B}(\phi_2-\phi_1)}\vartheta\begin{bmatrix}\theta \\ \phi\end{bmatrix}(0|\tau),$$

where $\tau = i\frac{\beta}{2\pi I}$, $\theta = -\mathcal{B}$ and $\phi = \frac{\phi_2-\phi_1}{2\pi}$. At the same time, the path integral formalism gives

$$\begin{aligned}Z(\phi_2, \phi_1|\beta) &= \sqrt{\frac{I}{2\pi\beta}}\sum_{\omega\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}\left(\omega+\frac{\phi_2-\phi_1}{2\pi}\right)^2 + 2\pi i\mathcal{B}\left(\omega+\frac{\phi_2-\phi_1}{2\pi}\right)} \\ &\overset{\omega\to-\omega}{=} \sqrt{\frac{I}{2\pi\beta}}\sum_{\omega\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}\left(\omega-\frac{\phi_2-\phi_1}{2\pi}\right)^2 - 2\pi i\mathcal{B}\left(\omega-\frac{\phi_2-\phi_1}{2\pi}\right)} \\ &= \sqrt{\frac{I}{2\pi\beta}}\vartheta\begin{bmatrix}\theta' \\ \phi'\end{bmatrix}(0|\tau'),\end{aligned}$$

where $\tau' = i\frac{2\pi I}{\beta} = -\frac{1}{\tau}$, $\theta' = -\frac{\phi_2-\phi_1}{2\pi} = -\phi$, $\phi' = -\mathcal{B} = \theta$. Then the Poisson resummation formula gives the modular transformation

$$\begin{aligned} Z(\phi_2,\phi_1|\beta) &= \frac{1}{2\pi} e^{i\mathcal{B}(\phi_2-\phi_1)} \vartheta \begin{bmatrix} \theta \\ \phi \end{bmatrix} (0|\tau) = \frac{1}{2\pi} e^{i\mathcal{B}(\phi_2-\phi_1)} \cdot (-i\tau)^{-1/2} e^{2\pi i\theta\phi} \vartheta \begin{bmatrix} -\phi \\ \theta \end{bmatrix} (0|-\frac{1}{\tau}) \\ &= \frac{1}{2\pi} \left(\frac{2\pi I}{\beta}\right)^{1/2} \vartheta \begin{bmatrix} \theta' \\ \phi' \end{bmatrix} (0|\tau') = Z(\phi_2,\phi_1|\beta). \end{aligned}$$

In the Hamiltonian formalism, the partition function ($\phi_1 = \phi_2$, integral just multiplies 2π) diverges in the high-temperature limit $\beta \to 0$, while in the path integral formalism, we can write

$$\begin{aligned} Z(S^1) &= \left(\frac{2\pi I}{\beta}\right)^{1/2} \sum_{\omega\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}\omega^2 + 2\pi i\mathcal{B}\omega} \\ &= \left(\frac{2\pi I}{\beta}\right)^{1/2} \left(1 + 2e^{-2\pi^2\frac{I}{\beta}} \cos(2\pi\mathcal{B}) + \cdots\right), \end{aligned}$$

and the divergence appears as a multiplicative factor, modified by the instanton effect.

4.4.3 Gauging and Anomaly

By gauging the symmetry we mean that the symmetry is promoted to a local one, and couples the dynamical variable in the system to a gauge field. Sometimes we further require dynamics of the gauge field, such that we can integrate out the gauge field and obtain an effective theory. If we do not need the latter procedure, the gauge field appears as an external background field.

In a mathematically more precise way, the two steps are: Identify the symmetry group with the structure group of a principle bundle, and we change the bordism category in the domain of the field theory functor from the category of manifolds to the category of manifolds with principal G-bundles with connection; Sum over isomorphism classes of principle bundles and integrate over the isomorphism classes of connections on those bundles.

Practically, we firstly gauge $SO(2)$ symmetry $\phi \to \phi + \alpha$ into a 'time'-local one

$$\phi(t) \to \phi(t) + \alpha(t).$$

As familiar from the Maxwell theory, in this case the original term $\int \dot{\phi}^2$ will not be invariant under the gauge transformation, and we need to introduce an (external) gauge field $A^{(e)}(t)$ to compensate the change. And by writing more instructively, we let $\Phi(t) = e^{i\phi(t)}$ and obtain

$$\begin{aligned} S &= \int \mathrm{d}t\ \frac{1}{2} I(\dot{\phi} + A^{(e)}(t))^2 + \mathcal{B}(\dot{\phi} + A^{(e)}(t)) \\ &= \int \mathrm{d}t\ \frac{1}{2} I\left(\Phi^{-1}(t)(-i\partial_t + A^{(e)}(t))\Phi(t)\right)^2 + \mathcal{B}\Phi^{-1}(t)(-i\partial_t + A^{(e)}(t))\Phi(t). \end{aligned} \tag{4.25}$$

This action is invariant under the gauge transformation

$$\begin{cases} \phi(t) \to \phi(t) + \alpha(t) \\ A^{(e)}(t) \to A^{(e)}(t) - \partial_t\alpha(t) \end{cases}.$$

Note that $\mathcal{A}^{(e)} \equiv A^{(e)}(t)\mathrm{d}t$ is the 1-form associated with the $SO(2)$ bundle over time manifold $M = S^1$, and it is a fictitious (rather than Maxwell) field. The gauge transformation in this language has a more instructive form

$$\begin{cases} \Phi(t) \to U(t)\Phi(t) \\ (\mathrm{d} - i\mathcal{A}^{(e)}) \to U(t)(\mathrm{d} - i\mathcal{A}^{(e)})U^{-1}(t) \end{cases},$$

where $U(t) = e^{i\alpha(t)}$ is the gauge transformation matrix.

For the parity/charge conjugation symmetry, which is a discrete one, there's no direct gauging, but we can see how it is realized. As discussed before, this symmetry transforms $\phi \to -\phi$, $H_{\mathcal{B}} \to H_{-\mathcal{B}}$ when $2\mathcal{B} \in \mathbb{Z}$, while in the gauged action Eq. (4.25), for its invariance we have to further implement $A^{(e)} \to -A^{(e)}$, i.e., under charge conjugation,

$$\begin{cases} \phi(t) \to -\phi(t) \\ \mathcal{B} \to -\mathcal{B} \\ A^{(e)}(t) \to -A^{(e)}(t) \end{cases}.$$

However, when we want to check the unitary equivalence, protected by the existence of periodicity-1 $\mathcal{B}$ dependence, we immediately find that the $\int \mathcal{B}\mathcal{A}^{(e)}$ term breaks this property, and the parity symmetry is spoiled. To compensate that, we introduce a new term, the Chern-Simons (CS) term in the action, and in Euclidean space, it reads

$$e^{-S} = \exp\left(-\left[\int \mathrm{d}t \left(\frac{1}{2}I(\dot{\phi} + A^{(e)})^2 - i\mathcal{B}(\dot{\phi} + A^{(e)}) + ikA^{(e)}\right)\right]\right), \tag{4.26}$$

where the last term is the Chern-Simons term at level k. In this sense, the periodicity is somehow restored as we can assign a synchronous change $(\mathcal{B}, r) \to (\mathcal{B} + r, k + r)$ where $r \in \mathbb{R}$. Nevertheless, the parity symmetry has to be taken into account again with the CS term. Basically, as we send $A^{(e)} \to -A^{(e)}$, we have to further send $k \to -k$ to keep the action invariant. And as we have shown, the unitary equivalence is related to the integer periodicity, i.e., we have some $N \in \mathbb{Z}$ such that

$$(-\mathcal{B} + N, -k + N) = (\mathcal{B}, k),$$

which leads to the further requirement that $k = \mathcal{B} = N/2$. This is a rather strong requirement.

Through the introduction of the CS term, we reconcile the requirements of invariance of action and unitary equivalence for parity symmetry. However, as we know, we introduce the external gauge field $A^{(e)}$ to compensate the gauge transformation of the symmetry, which means that $A^{(e)}$ itself is not gauge-invariant, and that seems to cause trouble in the CS term. To resolve this issue, we have to make it clear what is the impact of the gauge transformation on the CS term. First thing we should notice that although we did not labelled it explicitly, the integral in the action is done over the whole spacetime manifold, and especially for the θ term and CS term that are integrals of closed forms on this maximal chain. As one can guess, the CS term will almost be gauge-invariant, except when we meet some non-trivial gauge transformation. Next, we study the gauge transformation of gauge fields. Denote the spacetime manifold as M, we have the *space of gauge fields* $\mathscr{A} = \mathrm{Map}(M \to \mathbb{R})$ as set of all continuous functions on M. The gauge group G is some Lie group, and leads to *group of gauge transformation* $\mathscr{G} = \mathrm{Map}(M \to$

G). For common M, $\mathscr{A}$ is infinite-dimensional, while $\mathscr{G}$ is finite or infinite-dimensional depending on whether G is continuous (positive dimensional). What interests is the quotient space $\mathscr{A}/\mathscr{G}$, which is the space of gauge orbits, or gauge-inequivalent fields. Let us take examples.

Firstly when $M = [t_1, t_2]$ and $G = \mathbb{R}$. By gauge transformation $A(t) \to \tilde{A}(t) = A(t) - \partial_t \alpha(t)$. And if we do not impose constraints on the boundary, we can always assign $\partial_t \alpha(t) = A(t)$ to solve an α such that $\tilde{A}(t) = 0$. In this sense $\mathscr{A}/\mathscr{G}$ is a point. Indeed, when solving the differential equation, we find that we are indeed tuning the quantity

$$e^{i\int_{t_1}^{t_2} A(t)\mathrm{d}t} \to e^{-i\alpha(t_2)} e^{i\int_{t_1}^{t_2} A(t)\mathrm{d}t} e^{i\alpha(t_1)},$$

which can always be tuned to 1. More generally, we replace $e^{i\alpha(t)}$ with $g(t)$ as some group action.

Next we take also $M = \mathbb{R}$ with boundary condition $A(t) \to 0$ as $|t| \to \infty$. This condition also constrains $\alpha(t)$ to vanish at infinity. As a result, the Wilson line exponential $\int_{-\infty}^{+\infty} A\mathrm{d}t$ is gauge-invariant, allowed for any quantity. Thus $\mathscr{A}/\mathscr{G} = \mathbb{R}$.

Now, we take $M = S^1$ and $G = U(1)$, and the boundary condition is $A(t) = A(t + 2\pi)$. In this case, $\mathscr{G} = \mathrm{Map}(S^1 \to U(1))$, which we know to contain possible windings. We can denote the so-called *small gauge transformation* (SGT) group $\mathscr{G}_0$ with element $g(t) = e^{i\alpha(t)}$, where $\alpha(t)$ is single-valued, or β-periodic. Simply speaking, it does not contain any winding and can be represented in a single chart. Then

$$1 \longrightarrow \mathscr{G}_0 \longrightarrow \mathscr{G} \xrightarrow{\pi} \mathbb{Z} \longrightarrow 1,$$

where π is the winding number map, and we may call gauge transformation with winding as *large gauge transformation* (LGT). This extension splits since for $\omega \in \mathbb{Z}$, we can assign $s(\omega) = g_\omega$ with $g_\omega(t) = e^{2\pi i \omega t/\beta}$, and $g_\omega g_{\omega'} = g_{\omega+\omega'}$ is obvious. It is thus a central split extension with Abelian structure by addition. Now consider the Wilson loop $e^{i\oint_{S^1} A\mathrm{d}t}$, and by gauge transformation $A(t) \to A(t) - \partial_t \alpha(t)$, we see that by decomposing $\alpha(t) = 2\pi\omega t/\beta + \alpha_0(t)$, where $\alpha_0 \in \mathscr{G}_0$, $\omega \in \mathbb{Z}$ we have

$$\begin{aligned}\oint_{S^1} A\mathrm{d}t \to \oint_{S^1} A\mathrm{d}t - \oint_{S^1} \partial_t \alpha \mathrm{d}t \\ = \oint_{S^1} A\mathrm{d}t - \oint_{S^1} \partial_t \alpha_0 \mathrm{d}t - \oint_{S^1} \frac{2\pi\omega}{\beta}\mathrm{d}t \\ = \oint_{S^1} A\mathrm{d}t - 2\pi\omega.\end{aligned}$$

Then by exponential (with i factor), the LGT part vanishes and does not affect the Wilson loop/holonomy. Indeed, the holonomy is fully determined by the homogeneous part of A, the zero-frequency Fourier mode, since all other modes integrate to zero. Thus, we denote $\mathscr{A}^{\mathrm{red}} = \mathscr{A}/\mathscr{G}_0$, with element $A(t) \equiv \mu/\beta$. The LGT shift this value by $2\pi\omega/\beta$, while the Wilson loop is invariant under this shift. In this sense, we have

$$\mathscr{A}/\mathscr{G} \cong \mathscr{A}^{\mathrm{red}}/\mathbb{Z} = \{[\mu] = [\mu + 2\pi\omega] | \omega \in \mathbb{Z}\} \cong U(1).$$

Finally, we let $G = O(2) \cong SO(2) \rtimes \mathbb{Z}_2$. Since $SO(2) \cong U(1)$, most of the discussions above can be applied. Especially, by denoting upper $\mathcal{G}$ as $\mathcal{G}_{SO(2)}$, we expect $\mathcal{G}_{O(2)} = \mathcal{G}_{SO(2)} \rtimes \mathbb{Z}_2$, and the extension

$$1 \longrightarrow \mathscr{G}_0 \longrightarrow \mathcal{G}_{O(2)} \xrightarrow{\pi} \mathbb{Z} \rtimes \mathbb{Z}_2 \cong D_\infty \longrightarrow 1,$$

also splits. Similarly, D_∞ acts on μ by $s_\omega : \mu \to \mu + 2\pi\omega$, $\sigma : \mu \to -\mu$.

Now we can return to the CS term. As discussed, it does not change under $\mathscr{G}_0$, while when we meet LGT (without parity operation), it changes by $k \to k + k\omega$. This gives rise the factor

$$e^{ik \oint_{S^1} A\mathrm{d}t} \to e^{2\pi i k\omega} e^{ik \oint_{S^1} A\mathrm{d}t},$$

and the LGT invariance can be preserved iff k is an integer. Notice that by gauge invariance, we are not talking about the invariance of the action itself, but its exponential appeared in the partition function. If we simultaneously require invariance of partition function and related functional measure, then this exponential invariance is general.

Quantum Anomaly

One should have noticed the inconsistency: the requirement that the CS term is gauge-invariant is not compatible with the parity symmetry condition when $\mathcal{B}$ is a half-integer. In this sense, to preserve the gauge symmetry we have to admit an anomaly caused by the CS term, that is, some symmetry is broken by the quantum effect.

More formally, we view the coupling between dynamical variable and gauge field as a fibration $\mathcal{F}^{dyn} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}^{bck}$, and by integrating over the fiber $\mathcal{F}^{dyn}$, we obtain the effective action on some section about $\mathcal{F}^{bck}$

$$Z[\phi^{bck}] = \int_{\mathcal{F}^{dyn}} \mathcal{D}\phi^{dyn} e^{-S[\phi^{dyn}, \phi^{bck}]}.$$

This action is useful in the sense that it is about the external physical quantities that we can probe and measure, and the most prominent example is the response theory. Sometimes, we expect that there exists a group G acting on $\mathcal{F}$ that preserves the physical quantities (this is another fibration), but this is not always the case, as the partition function may not be invariant under the action of G. And this is the so-called quantum anomaly.

Sometimes, there exists subgroups $\mathcal{H}_{1,2} \subset \mathcal{G}$ that separately preserves the physical quantities, but not simultaneously. In this case, we say that the quantum anomaly is a mixed anomaly.

This is indeed the case for the parity symmetry and the CS term. To see it more clearly, we can consider the partition function after gauging. The partition function is

$$\begin{aligned}
Z(\mu) &= e^{ik\mu} Z_q \sum_{\omega\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}(\omega+\frac{\mu}{2\pi})^2 + 2\pi i\mathcal{B}(\omega+\frac{\mu}{2\pi})} \overset{\omega\to-\omega}{=} e^{ik\mu} Z_q \sum_{\omega\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}(\omega-\frac{\mu}{2\pi})^2 - 2\pi i\mathcal{B}(\omega-\frac{\mu}{2\pi})} \\
&= e^{ik\mu} Z_q \vartheta \begin{bmatrix} -\mu/2\pi \\ -\mathcal{B} \end{bmatrix} (0| -\frac{1}{\tau}) = e^{ik\mu} Z_q (-i\tau)^{1/2} e^{i\mu\mathcal{B}} \vartheta \begin{bmatrix} \mathcal{B} \\ -\mu/2\pi \end{bmatrix} (0|\tau) \\
&= e^{ik\mu} e^{i\mu\mathcal{B}} \frac{1}{2\pi} \sum_{m\in\mathbb{Z}} e^{-\frac{\beta}{2I}(m+\mathcal{B})^2 - i\mu(m+\mathcal{B})} \overset{m\to-m}{=} e^{ik\mu} \frac{1}{2\pi} \sum_{m\in\mathbb{Z}} e^{-\frac{\beta}{2I}(m-\mathcal{B})^2 + im\mu} \\
&= e^{ik\mu} \operatorname{Tr}\big(e^{-\beta H_{\mathcal{B}}} e^{i\mu Q}\big),
\end{aligned} \tag{4.27}$$

where Q is the $SO(2)$ charge operator $Q\Psi_m = m\Psi_m$. Clearly, when $k = \mathcal{B} \in \mathbb{Z}$, we have $Z(\mu) = Z(\mu+2\pi)$, maintaining the gauge invariance. On the other hand, as $k = \mathcal{B} \in \mathbb{Z}+\frac{1}{2}$, we have $Z(\mu) = -Z(\mu + 2\pi)$, and the gauge invariance is broken in price of preserving the parity symmetry. This is an example of mixed anomaly. In detail, we consider the IR

behavior by taking $\beta \to \infty$, and the partition function becomes (taking $\mathcal{B} = 1/2$)

$$Z \to \frac{e^{-\beta E_{\text{GS}}}}{2\pi} e^{i(k-\frac{1}{2})\mu} \left(e^{i\mu/2} + e^{-i\mu/2} + \cdots \right).$$

If we want to maintain gauge invariance, say we take $k = 0$, then the partition function becomes

$$Z \to \frac{e^{-\beta E_{\text{GS}}}}{2\pi} \left(1 + e^{-i\mu} + \cdots \right),$$

and the parity symmetry, which takes $\mu \to -\mu$, is broken. On the other hand, if we want to maintain the parity symmetry, we take $k = 1/2$, and the partition function becomes

$$Z \to \frac{e^{-\beta E_{\text{GS}}}}{2\pi} \left(e^{i\mu/2} + e^{-i\mu/2} + \cdots \right),$$

and the gauge invariance is broken.

To make the half-integer level CS term non-anomalous, we can view it as a $0+1$ boundary theory of a $1+1$ bulk theory. In this sense, we assign S^1 to be the boundary of a disk $\partial\Sigma = S^1$, and the Stokes' theorem applies as

$$e^{ik \oint_{S^1} A} = e^{ik \int_\Sigma F},$$

where $F = \mathrm{d}A$. Such an extending theory is anomaly-free, since the total spacetime geometry now becomes D_2, and a well-defined gauge transformation on it, when restricted to the boundary, is always trivial. One can think in this way: for pure S^1 as the boundary, we assign non-trivial gauge transformation that owns a non-trivial winding, and if we require continuity, this transformation function will continue to own such a winding when we shrink the boundary to the interior of the disk; however, this raises the problem that as we shrink the boundary to a point, the function has to meet the problem of multi-valuedness — since it does not know whether to take the value or the value plus $2\pi\times$ winding, and the continuity is broken. This contradiction tells us that the gauge transformation on the boundary has to be trivial, and the anomaly is resolved.

4.5 Heisenberg Extension

A Heisenberg extension, or a Heisenberg group, refers to a central extension of an Abelian group G,

$$1 \longrightarrow A \xrightarrow{\iota} \tilde{G} \xrightarrow{\pi} G \longrightarrow 1,$$

where the extended group $\tilde{G}$ is non-Abelian.

4.5.1 Fundamental Example

Possibly one the most fundamental relation in quantum mechanics can be expressed by the position-momentum commutation relation (the other Planck relation I guess)

$$[\hat{q}, \hat{p}] = i\hbar. \tag{4.28}$$

As operators, they act on the state by

$$\begin{cases} (\hat{q} \cdot \psi)(q) = \langle q|\hat{q}|\psi\rangle = q\psi(q) \\ (\hat{p} \cdot \psi)(q) = \langle q|\hat{p}|\psi\rangle = -i\hbar\partial_q \psi(q) \end{cases}.$$

Position generates shifts in momentum, and momentum generates shifts in position. This is the essence of the Heisenberg uncertainty principle. Such translational operators themselves form two groups with elements

$$U(\alpha) = e^{i\alpha\hat{p}}.\ V(\beta) = e^{i\beta\hat{q}},$$

which are separately Abelian groups isomorphic to $\mathbb{R}$.

However, when we want to group then together, we immediately encounter the non-commutativity of the position-momentum commutation relation. Basically, by consulting the Baker-Campbell-Hausdorff (BCH) formula

$$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\frac{1}{12}([X,[X,Y]]-[Y,[X,Y]])+\cdots},$$

we can find that the group multiplication of $U(\alpha)$ and $V(\beta)$ is

$$U(\alpha)V(\beta) = e^{i\hbar\alpha\beta}V(\beta)U(\alpha), \tag{4.29}$$

which contains a phase factor, and a phase factor only, thanks to the central property of the commutator $[\hat{q},\hat{p}] = i\hbar$. Or let us derive it using its definition,

$$\begin{aligned}((U(\alpha)V(\beta))\cdot\psi)(q) &= (V(\beta)\cdot\psi)(q+\hbar\alpha) = e^{i\beta(q+\hbar\alpha)}\psi(q+\hbar\alpha),\\ ((V(\beta)U(\alpha))\cdot\psi)(q) &= e^{i\beta q}(U(\alpha)\cdot\psi)(q) = e^{i\alpha q}\psi(q+\hbar\beta),\end{aligned}$$

and the phase factor is obvious.

Defining central extension

$$1 \longrightarrow U(1) \longrightarrow \mathrm{Heis}(\mathbb{R}\times\mathbb{R}) \longrightarrow \mathbb{R}\times\mathbb{R} \longrightarrow 1, \tag{4.30}$$

it is easy to find the corresponding cocycle

$$(z_1,(\alpha_1,\beta_1))\cdot(z_2,(\alpha_2,\beta_2)) = (z_1z_2e^{\frac{i}{2}\hbar(\alpha_1\beta_2-\alpha_2\beta_1)},(\alpha_1+\alpha_2,\beta_1+\beta_2)),$$

i.e.,

$$f((\alpha_1,\beta_1),(\alpha_2,\beta_2)) = e^{\frac{i}{2}\hbar(\alpha_1\beta_2-\alpha_2\beta_1)} = e^{\frac{i}{2}\hbar v_1^T J v_2},$$

where $v = (\alpha,\beta)^T$ and $J = \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$, under the choice of section $s(\alpha,\beta) = e^{i(\alpha\hat{p}+\beta\hat{q})}$. There certainly exists more obvious choices of section, like $\tilde{s}(\alpha,\beta) = U(\alpha)V(\beta)$, which leads to the cocycle $\tilde{f}((\alpha_1,\beta_1),(\alpha_2,\beta_2)) = e^{-i\hbar\beta_1\alpha_2}$. The two cocycles are related by a coboundary, $f_{12} = \tilde{f}_{12}g_1g_2g_{12}^{-1}$, with $g_i = g(\alpha_i,\beta_i) = e^{-\frac{i}{2}\hbar\alpha_i\beta_i}$.

The reason we prefer to use the first definition is that $\omega(v_1,v_2) = v_1^TJv_2$ defines a symplectic form ω on $\mathbb{R}^2$, as we have the invariance of ω under sympletic transformations $\omega(Av_1,Av_2) = \omega(v_1,v_2)$ for any symplectic matrix $A \in \mathrm{Sp}(2,\mathbb{R})$, satisfying that $A^TJA = J$. Such a definition calls us back to the Poisson bracket in classical mechanics, which is also a symplectic form. In this way, we may say that the Poisson bracket is lifted to a geometric obstruction in representing position and momentum operators simultaneously in quantum mechanics. Such a form can be easily generalized to more degrees of freedom, with $[\hat{q}^i,\hat{p}_j] = i\hbar\delta^i_j$, $\boldsymbol{v} = (\boldsymbol{\alpha},\boldsymbol{\beta})^T$, $s(\boldsymbol{v}) = \exp(i(\boldsymbol{\alpha}\cdot\boldsymbol{p}+\boldsymbol{\beta}\cdot\boldsymbol{q}))$, $f(\boldsymbol{v}_1,\boldsymbol{v}_2) = \exp\big(i\frac{\hbar}{2}\omega(\boldsymbol{v}_1,\boldsymbol{v_2})\big)$ and $\omega(\boldsymbol{v}_1,\boldsymbol{v}_2) = \boldsymbol{v}_1^TJ\boldsymbol{v}_2$ with $J = \begin{pmatrix}0 & \mathbb{I}_n\\ -\mathbb{I}_n & 0\end{pmatrix}$.

When we want to add more ingredients, like the magnetic field, we need to specialize on simpler case like Landau Level problem, and the magnetic translational groups emerge. Under the symmetric gauge, we have

$$H = \frac{1}{2m}\left[\left(p_1 + \frac{eBx_2}{2}\right)^2 + \left(p_2 - \frac{eBx_1}{2}\right)^2\right] = \frac{1}{2m}(\tilde{p}_1^2 + \tilde{p}_2^2),$$

where the canonical momentum $\tilde{p} = p - eA$ is defined. This momentum fails to be a symmetry of the Hamiltonian as $[\tilde{p}_1, \tilde{p}_2] \neq 0$. Nevertheless, we have the magnetic translation operators $\pi = p + eA$, such that $[\pi_i, \tilde{p}_j] = 0$ and thus $[\pi, H] = 0$. These two new translators are both symmetry operators of the Hamiltonian, and again, separate Abelian groups. However, since $[\pi_1, \pi_2] = -i\hbar eB \neq 0$, the union group of them is non-Abelian, similar to the above position-momentum case (notice that this similarity is also the reason we solve it like a harmonic oscillator). Thus, by defining

$$U(a_1) = e^{ia_1\pi_1/\hbar},\ V(a_2) = e^{ia_2\pi_2/\hbar},$$

we have the group multiplication

$$U(a_1)V(a_2) = e^{ia_1a_2eB/\hbar}V(a_2)U(a_1). \tag{4.31}$$

Especially when a_i form lattice constants, we obtain $UV = e^{2\pi i\theta}VU$, where $\theta = \Phi/\Phi_0$ is magnetic flux fraction with $\Phi = Ba_1a_2$ the flux and $\Phi_0 = h/e$ the flux quantum. This algebra $\mathcal{A}_\theta$ applies of non-commutative geometry to the quantum Hall effect, and the Hall conductance is quantized as $\sigma_{xy} = \frac{e^2}{h}\frac{1}{\theta}$.

4.5.2 Commutator Function

As we have shown in Section 4.3, for an Abelian G, the group $\tilde{G}$ commutator is

$$[(a_1, g_1), (a_2, g_2)] = \left(\frac{f(g_1, g_2)}{f(g_2, g_1)}, 1\right),$$

which leads to the natural definition of the commutator function $\kappa : G \times G \to A$ as

$$\kappa(g_1, g_2) = f(g_1, g_2)f(g_2, g_1)^{-1}. \tag{4.32}$$

There are several immediate features of the commutator function:

- *Gauge invariance*: obvious by performing 'differ by a coboundary' transformation.
- *Skew*: $\kappa(g_1, g_2) = \kappa(g_2, g_1)^{-1}$, which is obvious.
- $\kappa(1, g) = \kappa(g, 1) = 1$ following from the cocycle condition $f(1, g) = f(g, 1) = f(1, 1)$.
- *Alternating*: $\kappa(g, g) = 1$.
- $\tilde{G}$ is Abelian iff cocycle f is symmetric.
- *Bimultiplicative*: $\kappa(g_1g_2, g_3) = \kappa(g_1, g_3)\kappa(g_2, g_3)$, $\kappa(g_1, g_2g_3) = \kappa(g_1, g_2)\kappa(g_1, g_3)$.

This last one needs some trick: to prove it (say the first one), we expand it as

$$\begin{aligned}
& f(g_1g_2,g_3)f(g_3,g_2)f(g_3,g_1) = f(g_1,g_3)f(g_2,g_3)f(g_3,g_1g_2) \\
\Leftrightarrow & f(g_1,g_2)f(g_1g_2,g_3)f(g_3,g_2)f(g_3,g_1) = f(g_1,g_3)f(g_2,g_3)f(g_1,g_2)f(g_3,g_1g_2) \\
\Leftrightarrow & f(g_2,g_3)f(g_1,g_2g_3)f(g_3,g_2)f(g_3,g_1) = f(g_1,g_3)f(g_2,g_3)f(g_3,g_1)f(g_3g_1,g_2) \\
\Leftrightarrow & f(g_1,g_2g_3)f(g_3,g_2) = f(g_1,g_3)f(g_1g_3,g_2),
\end{aligned}$$

which is a cocycle identity.

It is also interesting to notice that the skew property can be derived using the bimultiplicative property and alternating property, as

$$\begin{aligned}
& \kappa(g_1,g_2)\kappa(g_2,g_1) = 1 \\
\Leftrightarrow & \kappa(g_1,g_2)\kappa(g_2,g_2)\kappa(g_2,g_1)\kappa(g_1,g_1) = 1 \\
\Leftrightarrow & \kappa(g_1g_2,g_2)\kappa(g_2g_1,g_1) = 1 \\
\Leftrightarrow & \kappa(g_1g_2,g_1g_2) = 1.
\end{aligned}$$

We call κ to be non-degenerate when $\forall g_1 \in G, g_1 \neq 1$, $\exists g_2$ s.t. $\kappa(g_1,g_2) \neq 1$. Or equivalently, we say that map $\kappa(g_1,) : G \to A$ is always non-trivial. In this case, the center of $\tilde{G}$ is minimal, i.e., $Z(\tilde{G}) = A$ with center elements $(a_i, 1)$, since any other element $(a_i, g_1 \neq 1)$ meets its un-commutable partner (a_i, g_2). On the other hand, if $g_1 \neq 1$ satisfies that $\kappa(g_1,g_2) = 1$ for all g_2, then g_1 is also in the center of $\tilde{G}$, or more strictly, $Z(\tilde{G}) = A \times (1, g_1)$. Such an idea can be extended to the whole group G.

Definition 4.5.1. *A Heisenberg extension a central extension of an Abelian group with* ***non-degenerate*** *commutator function.*

For the position-momentum case, the commutator function is $\kappa(v_1,v_2) = e^{i\hbar\omega(v_1,v_2)}$ with $\omega(v_1,v_2) = v_1^T J v_2$, which is non-degenerate.

For central extensions by $U(1)$, there exists a class of G, formed by the direct products of: cyclic group $\mathbb{Z}_n$, lattice $\mathbb{Z}^d$, vector spaces $\mathbb{R}^p$ and torus $\mathbb{T}^m$, that satisfies the exact sequence

$$1 \longrightarrow \pi_1(G) \xrightarrow{\iota} \mathrm{Lie}(G) \xrightarrow{\exp} G \xrightarrow{\pi=\pi_0} \pi_0(G) \longrightarrow 1 \ ,$$

where $\pi_1(G) \cong \mathbb{Z}^m$, which is then embedded in the Lie algebra $\mathrm{Lie}(G) \cong \mathbb{R}^p \times \mathbb{R}^m$. The exponential map then acts on the $\mathbb{R}^m$ part non-trivially and gives $\mathbb{R}^m \to U(1)^m \cong \mathbb{T}^m$, with kernel $\mathbb{Z}^m \cong \pi_1(G)$. The final projection map equals to the zeroth homotopy group $\pi_0(G) = \mathbb{Z}_n \times \mathbb{Z}^d$, as a image by $G/\operatorname{Im}(\exp)$.

Claim 4. *The isomorphism classes of such topological G (non-discrete) by $U(1)$ are in one-to-one correspondence with bimultiplicative continuous maps κ.*

In other words, κ corresponds one-to-one with cocycle classes in $H^2(G, U(1))$ when G is a topological group. This is evident considering the gauge invariance of κ that given $[f]$, κ is fixed.

4.5.3 Pontryagin Duality

As seen from the basic example, the Heisenberg group is born when we have a dual pair of operators with non-trivial but central commuting relations. And the Heisenberg group

reflects the non-commutativity of the (section of) dual pair by a cocycle/commutator function that takes the non-commutative pairs in two non-commuting operators generated Abelian group into a desired phase factor, as a projective representation, reflecting the non-compatibility of the accompanied Hilbert spaces of the operators.

Generalizing the above idea and we naturally meet with the Pontryagin duality, which is a generalization of the Fourier transform to locally compact Abelian groups. For a locally compact Abelian group S, its Pontryagin dual group $\hat{S}$ is defined as the group of all continuous characters $\chi : S \to U(1)$ by $\chi(s) \in U(1)$, or equivalently, $\hat{S} \cong \hom(S, U(1))$. As a group, the element multiplication law is defined as $(\chi_1 \cdot \chi_2)(s) = \chi_1(s)\chi_2(s)$. Following the idea, we expect to find a Heisenberg extension of $S \times \hat{S}$ by $U(1)$,

$$1 \longrightarrow U(1) \longrightarrow \mathrm{Heis}(S \times \hat{S}) \longrightarrow S \times \hat{S} \longrightarrow 1, \tag{4.33}$$

and following the position-momentum example, a similar cocycle exists as $f((s_1, \chi_1), (s_2, \chi_2)) = \frac{1}{\chi_1(s_2)}$, which brings the natural commutator function $\kappa((s_1, \chi_1), (s_2, \chi_2)) = \chi_2(s_1)/\chi_1(s_2)$.

To give more relations, we need to find the Hilbert spaces that S and $\hat{S}$ act on (To construct the extension, one is enough). For the former, we assume the existence of a Haar measure such that $\mathcal{H}_S = L^2(S)$ is square-integrable complex space, e.g. we have inner product

$$\begin{aligned}\langle \psi_1 | \psi_2 \rangle &= \int_{\mathbb{R}} \mathrm{d}x \psi_1^*(x)\psi_2(x), \ S = \mathbb{R}, \\ &= \frac{1}{2\pi} \int_0^{2\pi} \mathrm{d}\theta \psi_1^*(e^{i\theta})\psi_2(e^{i\theta}), \ S = \mathbb{T}^1,\end{aligned}$$

etc. The group elements of S and $\hat{S}$ can thus be similarly represented as translation and multiplication operators as

$$\begin{aligned}(T_{s_0} \cdot \psi)(s) &= \psi(s + s_0) \\ (M_{\chi_0} \cdot \psi)(s) &= \chi_0(s)\psi(s),\end{aligned} \tag{4.34}$$

where one notice that they preserve the group structure of S and $\hat{S}$ (There seems to be some interchange, try to think: $\hat{p}$ generate S, then element of S is labelled by position). Especially, this representation is that of $\mathrm{Heis}(S \times \hat{S})$, since

$$\begin{aligned}&(T_{s_0} M_{\chi_0} \cdot \psi)(s) = (M_{\chi_0} \cdot \psi)(s + s_0) = \chi_0(s + s_0)\psi(s + s_0) = \chi_0(s_0)\chi_0(s)\psi(s + s_0), \\ &(M_{\chi_0} T_{s_0} \cdot \psi)(s) = \chi_0(s)(T_{s_0} \cdot \psi)(s) = \chi_0(s)\psi(s + s_0),\end{aligned}$$

and gives

$$T_{s_0} M_{\chi_0} = \chi_0(s_0) M_{\chi_0} T_{s_0}, \tag{4.35}$$

this almost comes back to the position-momentum case.

Theorem 4.5.1. *(Stone-von Neumann-Mackey theorem) Up to isomorphism, there is a unique irreducible representation of the Heisenberg group Heis$(S \times \hat{S})$ such that $A = U(1)$ acts by scalar multiplication.*

Proof. In human language, there exists a unique subgroup such that all the κ factors are degenerate, where $U(1)$ exists as a member of direct product. One obvious subgroup is $U(1) \times S$ or $U(1) \times \hat{S}$. Further, one immediately finds that due to the non-degeneracy of κ, no more elements can be added to this subgroup. This maximal, and Abelian subgroup is what we need. Clearly, the two possibilities we give are isomorphic, and the uniqueness is proved.

We may consider the subgroup $\{1\} \times \{1\} \times \hat{S}$, whose element can be identified with M_χ, which commute and are simultaneously diagonalizable with complete basis ψ_α as simultaneous eigenstates as $M_\chi \psi_\alpha = \lambda_\alpha(\chi)\psi_\alpha$, and as $\lambda_\alpha(\chi) \in U(1)$, this map $\chi \to \lambda_\alpha$ is a character of $\hat{S}$, thus by Pontryagin duality, $\lambda_\alpha(\chi) = \chi(s_\alpha)$ for some $s_\alpha \in S$, i.e., $M_\chi \psi_\alpha = \chi(s_\alpha)\psi_\alpha$. While for subgroup $\{1\} \times S \times \{1\}$, we consider $T_s \cdot \psi_{\alpha_0}$ with some α_0, and by Heisenberg relations, we obtain

$$M_\chi(T_s \cdot \psi_{\alpha_0}) = \chi(s_{\alpha_0} - s)(T_s \cdot \psi_{\alpha_0}).$$

This means that $\{T_s \cdot \psi_{\alpha_0}\}_s$ is a copy of the representation. Thus the representation is unique up to isomorphism. □

For example, let us again consider the position-momentum case, where $S = \mathbb{R}^n$. For $v = (\alpha, \beta) \in \mathbb{R}^{2n}$ we have the section $s(v) = \exp(i(\alpha\hat{q} + \beta\hat{p}))$. Now consider representation of $\mathrm{Heis}(\mathbb{R}^n \times \hat{\mathbb{R}}^n)$ on $\mathcal{H} = L^2(\mathbb{R}^n)$. For $\psi_{1,2} \in \mathcal{H}$, we define the Wigner function as

$$\begin{aligned} W(\psi_1, \psi_2)(v) &= \langle s(v)\psi_1 | \psi_2 \rangle \\ &= \int_{\mathbb{R}^n} \mathrm{d}^n q \ \langle \psi_1 | e^{-i(\alpha\hat{q} + \beta\hat{p})} | q \rangle \langle q | \psi_2 \rangle \\ &= \int_{\mathbb{R}^n} \mathrm{d}^n q \ e^{-i\frac{1}{2}\alpha\beta} \langle \psi_1 | e^{-i\beta\hat{p}} e^{-i\alpha\hat{q}} | q \rangle \langle q | \psi_2 \rangle \\ &= \int_{\mathbb{R}^n} \mathrm{d}^n q \ e^{-i\frac{1}{2}\alpha\beta} e^{-i\alpha q} \psi_1^*(q + \beta)\psi_2(q) \\ &= \int_{\mathbb{R}^n} \mathrm{d}^n q \ \psi_1^*(q + \beta/2) e^{-i\alpha q} \psi_2(q - \beta/2). \end{aligned}$$

We then have

$$\begin{aligned} \|W(\psi_1, \psi_2)\|^2 &= \int_{\mathbb{R}^{2n}} \mathrm{d}^{2n} v \ |W(\psi_1, \psi_2)(v)|^2 \\ &= \int \mathrm{d}^n\alpha \mathrm{d}^n\beta \mathrm{d}^n q_1 \mathrm{d}^n q_2 \ e^{i\alpha(q_1 - q_2)} \psi_1(q_1 + \beta/2)\psi_2^*(q_1 - \beta/2)\psi_1^*(q_2 + \beta/2)\psi_2(q_2 - \beta/2) \\ &= \int \mathrm{d}^n\beta \mathrm{d}^n q \ \psi_1(q + \beta/2)\psi_2^*(q - \beta/2)\psi_1^*(q + \beta/2)\psi_2(q - \beta/2) \\ &= \|\psi_1\|^2 \|\psi_2\|^2. \end{aligned}$$

From this formula we can show that $\mathcal{H}$ is irreducible. Since if it is reducible with $\mathcal{H}_0 \subset \mathcal{H}$ invariant, we can then take $\psi_1 \in \mathcal{H}_0$, $\psi_\perp \notin \mathcal{H}_0$ such that

$$W(\psi_1, \psi_\perp)(v) = \langle s(v)\psi_1 | \psi_\perp \rangle = 0,$$

holds for any v, implying invariance $s(v)\psi_1 \in \mathcal{H}_0$, while $\|W(\psi_1, \psi_\perp)\|^2 = \|\psi_1\|^2 \|\psi_\perp\|^2 \neq 0$, which is a contradiction.

Consider $\hat{\hat{S}} \cong S$, we have

$$\mathrm{Heis}(S \times \hat{S}) \cong \mathrm{Heis}(\hat{S} \times \hat{\hat{S}}),$$

where for the latter, we represent over $\hat{\mathcal{H}} = L^2(\hat{S})$. Similarly, we have translation and multiplication operators $\hat{T}_\chi$ and $\hat{M}_s$, acting on $\hat{\psi}$ as

$$\begin{aligned} (\hat{T}_{\chi_0} \cdot \hat{\psi})(\chi) &= \hat{\psi}(\chi_0 \cdot \chi), \\ (\hat{M}_{s_0} \cdot \hat{\psi})(\chi) &= \chi(s_0)\hat{\psi}(\chi). \end{aligned} \tag{4.36}$$

These two representations are isomorphic, and the isomorphism is given by the unitary Fourier transform $\mathcal{S} : \mathcal{H} \to \hat{\mathcal{H}}$,

$$\begin{aligned} \mathcal{S}T_s\mathcal{S}^{-1} &= \hat{M}_s, \\ \mathcal{S}M_\chi\mathcal{S}^{-1} &= \hat{T}_\chi. \end{aligned} \tag{4.37}$$

In the representation way, we have

$$\begin{aligned} \mathcal{S} : \psi \mapsto \hat{\psi}(\chi) &= \int \mathrm{d}s\ \chi^{-1}(s)\psi(s), \\ \mathcal{S}^{-1} : \hat{\psi} \mapsto \psi(s) &= \int \mathrm{d}\chi\ \chi(s)\hat{\psi}(\chi). \end{aligned} \tag{4.38}$$

4.5.4 More Heisenberg Extensions

Heisenberg extension over a ring

Treating both A and G as some field(s) and allowing ring structure (additional multiplication) leads to more general Heisenberg extensions. Let R be the commutative ring with identity, then by considering

$$M(a,b,c) = \begin{pmatrix} 1 & a & c \\ & 1 & b \\ & & 1 \end{pmatrix},$$

we have the matrix multiplication law

$$M(a,b,c)M(a',b',c') = M(a+a', b+b', c+c'+ab'),$$

that gives us the Heisenberg extension

$$1 \longrightarrow R \xrightarrow{\iota} \mathrm{Heis}(R \times R) \xrightarrow{\pi} R \otimes R \longrightarrow 1, \tag{4.39}$$

where $\iota : c \to M(0,0,c)$, $\pi : M(a,b,c) \to (a,b)$ and the cocycle reads $f((a,b),(a',b')) = ab'$. The distributive law of multiplication ensures the cocycle identity. The commutator functions also forms a symplectic form $\kappa((a,b),(a',b')) = ab' - a'b$.

For instance, by $\mathbb{R} = \mathbb{Z}_n$, and defining generators

$$U = M(1,0,0),\ V = M(0,1,0),\ q = M(0,0,1),$$

we have $U^m = M(m,0,0)$ and so on. Moreover,

$$UV = qVU,\ [q,U] = 0 = [q,V],$$

and $\mathrm{Heis}(\mathbb{Z}_n \times \mathbb{Z}_n)$ over $\mathbb{Z}_n$ is defined.

More generally, we may consider any bilinear function as our cocycle $f((a,b),(a',b')) = c(a,b')$, and $\kappa((a,b),(a',b')) = c(a,b') - c(a',b)$.

Nontrivial $U(1)$ bundle over symplectic tori

Consider an n-torus $\mathbb{T}^n = \mathbb{R}^2/\Gamma \cong U^n(1)$, where Γ is an integral lattice. Assume that we have an integer-values symplectic form on Γ, which is a bilinear, anti-symmetric, non-degenerate map $\Omega : \Gamma \times \Gamma \to \mathbb{Z}$. Due to the anti-symmetric nature, we are always

able to find a set of ordered basis $\{\gamma_1, \cdots, \gamma_n\}$ such that Ω is represented in the matrix form $\Omega(\gamma_i, \gamma_j)$ as

$$\begin{pmatrix} & d_1 \\ -d_1 & \end{pmatrix} \otimes \begin{pmatrix} & d_2 \\ -d_2 & \end{pmatrix} \otimes \cdots$$

with all d_i integers. We may impose the non-degenerate condition by requiring that $n = 2m$ is even, and all d_i are non-zero.

We can extend Ω to a bilinear form on $V = \Gamma \otimes \mathbb{R} \cong \mathbb{R}^{2m}$ by $\Omega : V \times V \to \mathbb{R}$. If the requirement d_i are non-zero is satisfied, this a symplectic form, and we can always find an invertible matrix S to bring $S^T \Omega S = J$. A commutator function can be defined as

$$\kappa(v_1, v_2) = e^{2\pi i \Omega(v_1, v_2)},$$

which defines the Heisenberg extension

$$1 \longrightarrow U(1) \longrightarrow \text{Heis}(V, \Omega) \longrightarrow V \longrightarrow 0, \tag{4.40}$$

with cocycle function $f(v_1, v_2) = e^{i\pi\Omega(v_1, v_2)}$ (notice the lack of 2) and the combination law

$$(z_1, v_1) \cdot (z_2, v_2) = (z_1 z_2 e^{i\pi\Omega(v_1, v_2)}, v_1 + v_2).$$

As with the position-momentum case, this extension does not split.

We now want to pullback this sequence on the torus by inclusion map $\iota : \Gamma \to V$. Notice that we are embedding the lattice, not the torus itself. We claim that the pullback sequence splits, and for that purpose we consider the section $s(\gamma) = (\epsilon_\gamma, \gamma)$, with the homomorphism requirement

$$s(\gamma_1 + \gamma_2) = s(\gamma_1) s(\gamma_2),$$

which reads

$$(\epsilon_{\gamma_1+\gamma_2}, \gamma_1 + \gamma_2) = (\epsilon_{\gamma_1}\epsilon_{\gamma_2} e^{i\pi\Omega(\gamma_1, \gamma_2)}, \gamma_1 + \gamma_2),$$

i.e., $\epsilon_{\gamma_1}\epsilon_{\gamma_2} = e^{-i\pi\Omega(\gamma_1, \gamma_2)}\epsilon_{\gamma_1+\gamma_2}$. Such a section exists naturally with the choice $\epsilon_\gamma = \exp\Big(-i\pi \sum_{i<j} n_i n_j \Omega_{ij}\Big)$, where $\gamma = \sum n_i \gamma_i$, and $\Omega_{ij} = \Omega(\gamma_i, \gamma_j)$. We do need $i < j$ since Ω is anti-symmetric. To examine the correctness, we check

$$\begin{aligned} -\frac{1}{i\pi} \ln \frac{\epsilon_{\gamma_1}\epsilon_{\gamma_2}}{e^{-i\pi\Omega(\gamma_1, \gamma_2)}\epsilon_{\gamma_1+\gamma_2}} &= -\sum_{i<j} n_i^1 n_j^1 \Omega_{ij} - \sum_{i<j} n_i^2 n_j^2 \Omega_{ij} + \sum_{i<j} (n_i^1 + n_i^2)(n_j^1 + n_j^2)\Omega_{ij} - \Omega(\gamma_1, \gamma_2) \\ &= \sum_{i<j} (n_i^1 n_j^2 + n_i^2 n_j^1)\Omega_{ij} - \sum_{i,j} n_i^1 n_j^2 \Omega_{ij} \\ &= \sum_{i>j} -n_i^1 n_j^2 \Omega_{ij} + n_j^2 n_i^1 \Omega_{ji} = -2 \sum_{i>j} n_i^1 n_j^2 \Omega_{ij}, \end{aligned}$$

which is always an even integer and thus the section is well-defined. Since the sequence splits, its corresponding extension is an Abelian group $s(\Gamma) \in \text{Heis}(V, \Omega)$, then the quotient space

$$P(\mathbb{T}^n, \Omega, \epsilon) = \text{Heis}(V, \Omega)/s(\Gamma),$$

is the quotient set $U(1) \times V/\Gamma$ with the equivalence relation

$$(z, v) \sim (z, v) \cdot (\epsilon_\gamma, \gamma) = (z\epsilon_\gamma e^{i\pi\Omega(v, \gamma)}, \gamma), \ \forall \gamma \in \Gamma.$$

This is a $U(1)$ bundle over the symplectic torus $\mathbb{T}^n$ with projection map $\pi : P(\mathbb{T}^n, \Omega, \epsilon) \to \mathbb{T}^n$.

Notice that the group structure of $s(\gamma)$ is protected by the integer value of lattice sites γ and Ω, thus when embedded in $\mathrm{Heis}(V, \Omega)$, other elements are not able to generally commute with $s(\gamma)$ under this (extended) chosen section and commutator function. Put it more straightforwardly, $s(\gamma)$ is not a normal subgroup and P is not a group, only a bundle.

We can view $\mathrm{Heis}(V, \Omega)$ as a principle $U(1)$ bundle over V, and construct a natural connection on bundle P. The parallel transport of (z, v_0) over a straight path $\mathscr{P}_{v_0,\omega} = \{v_0 + t\omega | 0 \le t \le 1\}$ in V is defined by a left action of $(1, \omega)$

$$U(\mathscr{P}_{v_0,\omega}) : \ (z, v_0) \to (1, \omega) \cdot (z, v_0) = (z e^{i\pi\Omega(v_0,\omega)}, v_0 + \omega).$$

Then considering a small loop at v_0 by

$$\mathscr{P}_{v_0,\omega_1,\omega_2} = \mathscr{P}_{v_0+\omega_2,-\omega_2} \mathscr{P}_{v_0+\omega_1+\omega_2,-\omega_1} \mathscr{P}_{v_0+\omega_1,\omega_2} \mathscr{P}_{v_0,\omega_1},$$

and one obtains the holonomy

$$U(\mathscr{P}_{v_0,\omega_1,\omega_2}) : \ (z, v_0) \to = (z e^{2\pi i\Omega(\omega_1,\omega_2)}, v_0),$$

where Ω can be viewed as the curvature of the connection. Descend the connection and curvature to the $U(1)$ principle bundle $P(\mathbb{T}^n, \Omega, \epsilon)$, the cohomology class of $[\Omega]$ is the first Chern class of the bundle, characterized by integers $\boldsymbol{d} = (d_1, d_2, \cdots)$ that obstructs topological triviality.

4.5.5 Automorphisms of Heisenberg Extensions

An automorphism $\alpha \in \mathrm{Aut}(G)$ is sympletic if it preserves the commutator function

$$\alpha^* \kappa(g_1, g_2) = \kappa(\alpha(g_1), \alpha(g_2)) = \kappa(g_1, g_2), \tag{4.41}$$

by a pullback α^*. Such sympletic transformations are relevant to canonical transformations in physics.

We are interested in whether such automorphism can be lifted to the Heisenberg group, i.e., whether such a commutative diagram exists

$$\begin{array}{ccc} \tilde{G} & \xrightarrow{\pi} & G \\ \downarrow{\scriptstyle \tilde{\alpha}} & & \downarrow{\scriptstyle \alpha} \\ \tilde{G} & \xrightarrow{\pi} & G \end{array},$$

meaning that

$$\pi(\tilde{\alpha}(\tilde{g})) = \alpha(\pi(\tilde{g})). \tag{4.42}$$

Let us call $\tilde{\alpha}$ as T_α, then

$$\pi(T_\alpha(a, g)) = \alpha(\pi(a, g)) = \alpha(g),$$

and we are led to

$$T_\alpha(a, g) = (\xi_\alpha(a, g), \alpha(g)), \tag{4.43}$$

where the map $\xi_\alpha : A \times G \to A$, such that the projection gives the result. Then for T_α to be an automorphism, we require that

$$T_\alpha((a_1, g_1) \cdot (a_2, g_2)) = T_\alpha(a_1, g_1) \cdot T_\alpha(a_2, g_2), \tag{4.44}$$

which holds, in spirit of the Abelian group nature of A and G, iff

$$\xi_\alpha(a_1 + a_2 + f(g_1, g_2), g_1 + g_2) = \xi_\alpha(a_1, g_1) + \xi_\alpha(a_2, g_2) + f(\alpha(g_1), \alpha(g_2)). \tag{4.45}$$

For this general equation that needs to be solved, we specialize $g_1 = 0$, $g_2 = g$ and choose normalized cocycle function $f(0, g) = 0 = f(g, 0)$, then by noting that α is automorphism thus $\alpha(0) = 0$, we have

$$\xi_\alpha(a_1 + a_2, g) = \xi_\alpha(a_1, 0) + \xi_\alpha(a_2, g). \tag{4.46}$$

Now we have two observations:

- For $a_2 = 0$: $\xi_\alpha(a, g) = \xi_\alpha(a, 0) + \xi_\alpha(0, g)$.
- For $g = 0$: $\xi_\alpha(a_1 + a_2, 0) = \xi_\alpha(a_1, 0) + \xi_\alpha(a_2, 0)$, thus $\xi_\alpha(, 0) \in \mathrm{Aut}(A)$.

Especially for the second one, we can make the assumption for simplification that $\xi_\alpha(a, 0) = a$. Therefore, by renaming $\xi_\alpha(0, g) = \tau_\alpha(g)$, we obtain

$$T_\alpha(a, g) = (a + \tau_\alpha(g), \alpha(g)). \tag{4.47}$$

Utilizing this simplified equation, we have

$$\begin{aligned} T_\alpha((a_1, g_1) \cdot (a_2, g_2)) &= T_\alpha(a_1 + a_2 + f(g_1, g_2), g_1 + g_2) \\ &= (a_1 + a_2 + \tau_\alpha(g_1 + g_2) + f(g_1, g_2), \alpha(g_1) + \alpha(g_2)) \\ T_\alpha(a_1, g_1) \cdot T_\alpha(a_2, g_2) &= (a_1 + \tau_\alpha(g_1), \alpha(g_1)) \cdot (a_2 + \tau_\alpha(g_2), \alpha(g_2)) \\ &= (a_1 + a_2 + \tau_\alpha(g_1) + \tau_\alpha(g_2) + f(\alpha(g_1), \alpha(g_2)), \alpha(g_1) + \alpha(g_2)), \end{aligned}$$

and the requirement that T_α is a homomorphism becomes

$$(\alpha^* f - f)(g_1, g_2) = \tau_\alpha(g_1 + g_2) - \tau_\alpha(g_1) - \tau_\alpha(g_2), \tag{4.48}$$

where the pullback cocycle follows similarly as $\alpha^* f(g_1, g_2) = f(\alpha(g_1), \alpha(g_2))$. Most strictly, we can require that f is invariant under pullback with $\alpha^* f = f$ by $\tau_\alpha \equiv 0$, and we directly lift the automorphism to the Heisenberg group. On the other hand, the above equation offers more chance for α to be lifted to a 'symmetry' of the Heisenberg group, that $\alpha^* f - f$ is a trivializable cocycle (a coboundary). Indeed, by taking

$$\tilde{f}(g_1, g_2) = f(g_1, g_2) + t(g_1, g_2) - t(g_1) - t(g_2),$$

one can check that the solution becomes

$$\tilde{\tau}_\alpha(g) = \tau_\alpha(g) + (\alpha^* t - t)(g),$$

i.e., the existence of the solution is gauge invariant. Thus, the equation tells that α can be lifted when the homology class $[f] \in H^2(G, A)$ is preserved under the pullback action α^*. Let us denote the set of such automorphisms as $\mathrm{Aut}_0(G) \subset \mathrm{Aut}(G)$. Notice

that $\alpha \in \mathrm{Aut}_0(G)$ is automatically symplectic, since the commutator function $\kappa(g_1, g_2) = f(g_1, g_2) - f(g_2, g_1)$ is gauge invariant.

In the above solution, we choose a special case, and one should expect that $\mathrm{Aut}_0(G)$ is a subgroup of the symplectic automorphisms of G, and its lift is only a subset of $\mathrm{Aut}(\tilde{G})$. To illustrate the latter, considering

$$\begin{aligned} T_{\alpha_1} \circ T_{\alpha_2}(a, g) &= T_{\alpha_1}(a + \tau_{\alpha_2}(g), \alpha_2(g)) = (a + \tau_{\alpha_2}(g) + \tau_{\alpha_1}(\alpha_2(g)), \alpha_1(\alpha_2(g))), \\ T_{\alpha_1\circ\alpha_2}(a, g) &= (a + \tau_{\alpha_1\circ\alpha_2}(g), \alpha_1(\alpha_2(g))), \end{aligned}$$

tells us that if we want to have $T_{\alpha_1} \circ T_{\alpha_2} = T_{\alpha_1\circ\alpha_2}$, we need to have

$$\tau_{\alpha_1\circ\alpha_2}(g) = \tau_{\alpha_1}(\alpha_2(g)) + \tau_{\alpha_2}(g),$$

or equivalently,

$$\tau_{\alpha_1\circ\alpha_2} = \alpha_2^* \tau_{\alpha_1} + \tau_{\alpha_2}. \tag{4.49}$$

It is under this condition that the lifted $\mathrm{Aut}_0(G)$ form a subgroup of $\mathrm{Aut}((\tilde{G}))$ that is isomorphic to $\mathrm{Aut}_0(G)$. Thus, generally the lifted $\mathrm{Aut}_0(G) \subset \mathrm{Aut}(\tilde{G})$ does not follow a standard group product law, but instead, it will also own some cocycle structure, i.e., $\mathrm{Aut}_0(G)$ is lifted by an extension.

To illustrate that, we introduce that for $l \in \hom(G, A)$, the automorphism

$$P_l(a, g) = (a + l(g), g), \tag{4.50}$$

and gives us

$$T_{\alpha_1} \circ T_{\alpha_2} = T_{\alpha_1\circ\alpha_2} \circ P_{l_{\alpha_1,\alpha_2}}, \tag{4.51}$$

by $l_{\alpha_1,\alpha_2}(g) = \tau_{\alpha_1}(\alpha_2(g)) + \tau_{\alpha_2}(g) - \tau_{\alpha_1\circ\alpha_2}(g)$. This l_{α_1,α_2} is a true homomorphism from G to A as

$$\begin{aligned} l_{\alpha_1,\alpha_2}(g_1 + g_2) &= \tau_{\alpha_1}(\alpha_2(g_1 + g_2)) + \tau_{\alpha_2}(g_1 + g_2) - \tau_{\alpha_1\circ\alpha_2}(g_1 + g_2) \\ &= (\alpha_1^* f - f)(\alpha_2(g_1), \alpha_2(g_2)) + \tau_{\alpha_1}(\alpha_2(g_1)) + \tau_{\alpha_1}(\alpha_2(g_2)) \\ &+ (\alpha_2^* f - f)(g_1, g_2) + \tau_{\alpha_2}(g_1) + \tau_{\alpha_2}(g_2) \\ &- ((\alpha_1 \circ \alpha_2)^* f - f)(g_1, g_2) - \tau_{\alpha_1\circ\alpha_2}(g_1) - \tau_{\alpha_1\circ\alpha_2}(g_2) \\ &= \tau_{\alpha_1}(\alpha_2(g_1)) + \tau_{\alpha_2}(g_1) - \tau_{\alpha_1\circ\alpha_2}(g_1) + \tau_{\alpha_1}(\alpha_2(g_2)) + \tau_{\alpha_2}(g_2) - \tau_{\alpha_1\circ\alpha_2}(g_2) \\ &= l_{\alpha_1,\alpha_2}(g_1) + l_{\alpha_1,\alpha_2}(g_2). \end{aligned}$$

Furthermore,

$$P_{l_1} \circ P_{l_2}(a, g) = P_{l_1}(a + l_2(g), g) = (a + l_2(g) + l_1(g), g) = P_{l_1+l_2}(a, g),$$

and also,

$$\begin{aligned} T_\alpha \circ P_l(a, g) &= T_\alpha(a + l(g), g) = (a + l(g) + \tau_\alpha(g), \alpha(g)) \\ &= (a + (\alpha^{-1})^* l(\alpha(g)) + \tau_\alpha(g), \alpha(g)) = P_{(\alpha^{-1})^* l} \circ T_\alpha(a, g). \end{aligned}$$

Synthetically, we group the relations as

$$T_{\alpha_1} \circ T_{\alpha_2} = T_{\alpha_1\circ\alpha_2} \circ P_{l_{\alpha_1,\alpha_2}}, \tag{4.52a}$$

$$P_{l_1} \circ P_{l_2} = P_{l_1+l_2}, \tag{4.52b}$$

$$T_\alpha \circ P_l = P_{(\alpha^{-1})^* l} \circ T_\alpha, \tag{4.52c}$$

while T_α and P_l are themselves automorphisms and homomorphisms, respectively. These relations then tell that T_α represents α in $\mathrm{Aut}(\tilde{G})$ as a section, with cocycle induced by P_l (notice that the third relation always allow us to write in that way), and we obtain an extension of $\mathrm{Aut}_0(G)$ by $\hom(G, A)$ as

$$1 \longrightarrow \hom(G, A) \longrightarrow \widetilde{\mathrm{Aut}(\tilde{G})} \longrightarrow \mathrm{Aut}_0(G) \longrightarrow 1, \tag{4.53}$$

where $\widetilde{\mathrm{Aut}(\tilde{G})} \subset \mathrm{Aut}(\tilde{G})$. Finally, we can replace $\hom(G, A)$ to $\hom(G, A) \rtimes \mathrm{Aut}(A)$ to give the full automorphism group of $\tilde{G}$ lifted from $\mathrm{Aut}(G)$.

Inequivalent representations

Before closing the section, let us come back to the elementary example of the position-momentum case, but this time with a specific system of realization — the quantum harmonic oscillator. Recall that the Heisenberg group is generated over the phase space $\mathbb{R} \oplus \mathbb{R}$ with commutator function as a sympletic form defined by J. This relates us with the matrix group $Sp(2, \mathbb{R})$, and by assuming $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we need

$$A^T J A = (ad - bc) J = J,$$

which constrains $\det(A) = 1$, thus coincides with the special-linear group $SL(2, \mathbb{R})$. The automorphism group of the Heisenberg group is then the group of symplectic transformations $Sp(2, \mathbb{R})$.

Now since $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$, their Lie algebras are also the same $\mathfrak{sp}(2, \mathbb{R}) = \mathfrak{sl}(2, \mathbb{R})$. For the latter, we consider infinitesimal $A = 1 + \epsilon m + O(\epsilon^2)$, and notice that

$$\det(A) = 1 + \epsilon \operatorname{Tr}(m) + O(\epsilon^2) = 1,$$

thus $\operatorname{Tr}(m) = 0$. To explain this expansion, we notice that when a_i are eigenvalues of A, we have characteristic polynomial $\det(\lambda - A) = \prod_i (\lambda - a_i)$, thus

$$\det(1 + A) = (1 + a_1) \cdots (1 + a_n) = 1 + \operatorname{Tr}(A) + \cdots + \det(A),$$

and especially for $n = 2$, we have $\det(1 + A) = 1 + \operatorname{Tr}(A) + \det(A)$. Then we consider basis of $\mathfrak{sp}(2, \mathbb{R})$ as

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \ f = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \ h = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

that satisfies

$$[h, e] = -2e, \ [e, f] = h, [h, f] = 2f.$$

The exponential map can be calculated in principle using BCH formula.

Now we consider to represent these operators on $\mathcal{H} = L^2(\mathbb{R})$, and we have

$$\hat{e} = \frac{i}{2\hbar} \hat{p}^2, \ \hat{f} = \frac{i}{2\hbar} \hat{q}^2, \ \hat{h} = \frac{i}{2\hbar} (\hat{q}\hat{p} + \hat{p}\hat{q}),$$

which is indeed a representation. For this check, we may use

$$\begin{aligned}
[AB, C] &= ABC - ACB + ACB - CAB = A[B, C] + [A, C]B, \\
[A, CD] &= ACD - CAD + CAD - CDA = [A, C]D + C[A, D], \\
[AB, CD] &= A[B, CD] + [A, CD]B = A([B, C]D + C[B, D]) + (A[C, D] + [A, D]C)B \\
&= A[B, C]D + AC[B, D] + A[C, D]B + [A, D]CB, \\
&= [AB, C]D + C[AB, D] = (A[B, C] + [A, C]B)D + C(A[B, D] + [A, D]B) \\
&= A[B, C]D + [A, C]BD + CA[B, D] + C[A, D]B,
\end{aligned}$$

etc.

However, what interests is the one-parameter subgroup

$$e^{\theta(e+f)} = \cos\theta + \sin\theta(e + f),$$

as $e+f = i\sigma_y$. This is a 2π-periodic subgroup. On the other hand, as $\hat{e}+\hat{f} = \frac{i}{2}(\hat{p}^2+\hat{q}^2) = i(a^\dagger a + \frac{1}{2})$ where $a = \frac{1}{\sqrt{2}}(\hat{q} + i\hat{p})$, we obtain that

$$e^{\theta(\hat{e}+\hat{f})} = e^{i\theta(a^\dagger a+\frac{1}{2})},$$

and since $\hat{n} = a^\dagger a$ is the number operator that only takes non-negative integer values, the exponential operator is 4π periodic due to the zero-point shift. Thus, the group generated by $\hat{e}$, $\hat{f}$ and $\hat{h}$ is not isomorphic to the one generated by e, f and h, and it is indeed a double cover of $Sp(2, \mathbb{R})$, known as metaplectic group.

Notice that in any finite-dimensional representation of $\mathfrak{sp}(2, \mathbb{R})$, $\rho(e) + \rho(f) = iJ_2$, which correspond to integer angular momentum and always contain (imaginary) integer eigenvalues. Then we must conclude that there are no finite-dimensional representations of the metaplectic group, which is a familiar result to us, as there's no finite-dimensional matrix representation of position and momentum operators. The metaplectic group is not a matrix group.

4.6 Twisted Cohomology

Here, we shall consider the extension of a group G (that is not necessarily Abelian) by an Abelian group A, which is not necessarily central. This refers to the short exact sequence

$$1 \longrightarrow A \longrightarrow \tilde{G} \longrightarrow G \longrightarrow 1,$$

where in spirit of the exactness, we still have $G \cong \tilde{G}/A$, based on the fact that A is a **normal subgroup** of $\tilde{G}$. However, we no longer require that A is **central** in $\tilde{G}$. Notice that the latter is much stronger, requiring the conjugate action of any element in $\tilde{G}$ over any element in A to be invariant, i.e., $\tilde{g}a\tilde{g}^{-1} = a$; while the former only requires the conjugate action of any element in $\tilde{G}$ over any element in A to be in A, i.e., $\tilde{g}a\tilde{g}^{-1} \in A$, and the conjugated element can change, or more vividly, the conjugate action can cause vertical movement in the fiber, thus causing the *twist*.

The starting point now comes back to the map $\omega : G \to \mathrm{Aut}(A)$, which is not necessarily trivial, but forms a homomorphism. We understand this step by step. Firstly, since A is a normal subgroup, we expect a map $\tilde{\omega} : \tilde{G} \to \mathrm{Aut}(A)$ by conjugate action as

$$\iota(\tilde{\omega}_{\tilde{g}}(a)) = \tilde{g}\iota(a)\tilde{g}^{-1},$$

and notice that $\tilde{\omega} : \tilde{g} \mapsto \tilde{\omega}_{\tilde{g}}$ is naturally a homomorphism, as

$$\iota(\tilde{\omega}_{\tilde{g}\tilde{g}'}(a)) = \tilde{g}\tilde{g}'\iota(a)(\tilde{g}\tilde{g}')^{-1} = \tilde{g}\iota(\tilde{\omega}_{\tilde{g}'}(a))\tilde{g}^{-1} = \iota(\tilde{\omega}_{\tilde{g}}(\tilde{\omega}_{\tilde{g}'}(a))),$$

i.e., $\tilde{\omega}_{\tilde{g}\tilde{g}'} = \tilde{\omega}_{\tilde{g}}\tilde{\omega}_{\tilde{g}'}$. At the same time, due to the Abelian nature of A (thus $\iota(A)$), we see that $\tilde{\omega}_{\tilde{g}}$ depends only on the equivalence class $\tilde{G}/\iota(A)$, since

$$\iota(\tilde{\omega}_{\tilde{g}\iota(a_0)}(a)) = \tilde{g}\iota(a_0)\iota(a)(\tilde{g}\iota(a_0))^{-1} = \tilde{g}\iota(a)\tilde{g}^{-1} = \iota(\tilde{\omega}_{\tilde{g}}(a)).$$

Now since $\tilde{G}/\iota(A) \cong \tilde{G}/A \cong G$, we see that this isomorphism defines ω as a homomorphism from G to $\mathrm{Aut}(A)$, as $\omega_g = \tilde{\omega}_{[\tilde{g}]}$.

More explicitly, we define section-dependent $\omega_{g,s}$ such that $\iota(\omega_{g,s}(a)) = s(g)\iota(a)s(g)^{-1}$, and only to find that $\omega_{g,s}$ is independent of the section s. This is because as we change section $\hat{s}(g) = \iota(t(g))s(g)$ (order matters), we see that

$$\begin{aligned}\iota(\omega_{g,\hat{s}}(a)) &= \hat{s}(g)\iota(a)\hat{s}(g)^{-1} = \iota(t(g))s(g)\iota(a)s(g)^{-1}\iota(t(g))^{-1} \\ &= \iota(t(g))\iota(\omega_{g,s}(a))\iota(t(g))^{-1} = \iota(t(g)\cdot\omega_{g,s}(a)\cdot t(g)^{-1}) \\ &= \iota(\omega_{g,s}(a)),\end{aligned}$$

where we used the fact that A is Abelian for the last equality, and we thus have a well-defined ω_g. To see the homomorphism property, we notice that counting on the 2-cocycle function $s(g_1)s(g_2) = \iota(f_s(g_1,g_2))s(g_1g_2)$, we have

$$\begin{aligned}\iota(\omega_{g_1}\circ\omega_{g_2}(a)) &= s(g_1)\iota(\omega_{g_2}(a))s(g_1)^{-1} = s(g_1)s(g_2)\iota(a)s(g_2)^{-1}s(g_1)^{-1} \\ &= \iota(f_s(g_1,g_2))s(g_1g_2)\iota(a)s(g_1g_2)^{-1}\iota(f_s(g_1,g_2))^{-1} \\ &= \iota(f_s(g_1,g_2))\iota(\omega_{g_1g_2}(a))\iota(f_s(g_1,g_2)^{-1}) \\ &= \iota(f_s(g_1,g_2)\cdot\omega_{g_1g_2}(a)\cdot f_s(g_1,g_2)^{-1}) \\ &= \iota(\omega_{g_1g_2}(a)),\end{aligned}$$

where again we used the fact that A is Abelian for the last equality, and we thus have a well-defined group homomorphism $\omega_{g_1}\circ\omega_{g_2} = \omega_{g_1g_2}$.

Utilizing this homomorphism, we calculate

$$\begin{aligned}s(g_1)s(g_2)s(g_3) &= \iota(f_s(g_1,g_2))s(g_1g_2)s(g_3) = \iota(f_s(g_1,g_2))\iota(f_s(g_1g_2,g_3))s(g_1g_2g_3) \\ &= s(g_1)\iota(f_s(g_2,g_3))s(g_2g_3) = s(g_1)\iota(f_s(g_2,g_3))s(g_1)^{-1}s(g_1)s(g_2g_3) \\ &= \iota(\omega_{g_1}(f_s(g_2,g_3)))\iota(f_s(g_1,g_2g_3))s(g_1g_2g_3),\end{aligned}$$

which gives us the *twisted* 2-cocycle condition

$$f_s(g_1,g_2)f_s(g_1g_2,g_3) = \omega_{g_1}(f_s(g_2,g_3))f_s(g_1,g_2g_3). \tag{4.54}$$

It is the non-central nature of A that causes the twist, while it is the normality of A that allows the twisted relation to be well-defined. Now similarly, we pick up several cases by solving the twisted 2-cocycle condition in special cases:

- $g_2 = g_3 = 1$, $g_1 = g$. We have $f_s(g,1)f_s(g,1) = \omega_g(f_s(1,1))f_s(g,1) \Rightarrow f_s(g,1) = \omega_g(f_s(1,1))f_s(g,1)$, and $\omega_g(f_s(1,1)) = f_s(g,1)$.
- $g_1 = g_2 = 1$, $g_3 = g$. We have $f_s(1,1)f_s(1,g) = \omega_1(f_s(1,g))f_s(1,g) \Rightarrow f_s(1,g) = \omega_1(f_s(1,g))f_s(1,g)$, and $f_s(1,1) = f_s(1,g)$ (note ω_1 is identity).

- $g_1 = g_3 = g^{-1}$, $g_2 = g$. We have $f_s(g^{-1}, g) f_s(1, g^{-1}) = \omega_{g^{-1}}(f_s(g, g^{-1})) f_s(g^{-1}, 1)$.

Now, since again we would like to classify the extensions, we again consider the equivalence relation on the 2-cocycles, that by considering two sections that can be related by $\hat{s}(g) = \iota(t(g))s(g)$ (note that this can either refer to sections in the same $\tilde{G}$ or different extensions), we have

$$\begin{aligned}\iota(f_{\hat{s}}(g_1, g_2)) &= \hat{s}(g_1)\hat{s}(g_2)\hat{s}(g_1 g_2)^{-1} = \iota(t(g_1))s(g_1)\iota(t(g_2))s(g_2)s(g_1 g_2)^{-1}\iota(t(g_1 g_2))^{-1} \\ &= \iota(t(g_1))s(g_1)\iota(t(g_2))s(g_1)^{-1}s(g_1)s(g_2)s(g_1 g_2)^{-1}\iota(t(g_1 g_2))^{-1} \\ &= \iota(t(g_1))\iota(\omega_{g_1}(t(g_2)))\iota(f_s(g_1, g_2))\iota(t(g_1 g_2)^{-1}),\end{aligned}$$

we are led to

$$\begin{aligned}f_{\hat{s}}(g_1, g_2) &= t(g_1)\omega_{g_1}(t(g_2)) f_s(g_1, g_2) t(g_1 g_2)^{-1} \\ &= t(g_1)\omega_{g_1}(t(g_2)) t(g_1 g_2)^{-1} f_s(g_1, g_2),\end{aligned} \tag{4.55}$$

where for the Abelian A case, the sequence does not matter here. This defines a *twisted* differ by a coboundary condition, or twisted gauge transformation. From this equation, we consider:

- $g_1 = g_2 = 1$. We have $f_{\hat{s}}(1, 1) = t(1) f_s(1, 1)$, and by $t(1) = f_s(1, 1)^{-1}$, $f_{\hat{s}}(1, 1) = 1$.
- $g_1 = g, g_2 = 1$. We have $f_{\hat{s}}(g, 1) = \omega_g(t(1)) f_s(g, 1)$.

Now let us come back to the first relation by cocycle condition $\omega_g(f(1, 1)) = f(g, 1)$ that stands for any section, and if we choose the defined $\hat{s}$, we see that $f_{\hat{s}}(g, 1) = \omega_g(f_{\hat{s}}(1, 1)) = 1$, thus we may choose this special section for simplicity. Under this choice, we have $f_{\hat{s}}(1, 1) = f_{\hat{s}}(1, g) = f_{\hat{s}}(g, 1) = 1$, and the third case above becomes $f_{\hat{s}}(g^{-1}, g) = \omega_{g^{-1}}(f_{\hat{s}}(g, g^{-1}))$.

At the same time, we need to assign a group structure of $\tilde{G}$ from the 2-cocycle, and it reads

$$(a_1, g_1) \cdot (a_2, g_2) = (a_1 \omega_{g_1}(a_2) f(g_1, g_2) f(g_1, 1), g_1 g_2), \tag{4.56}$$

where an additional twist factor is added to the fiber part. To prove this, we need to construct the inverse and the associativity. To find the inverse, we note that we must let $g_2 = g_1^{-1}$, thus

$$(a_1, g) \cdot (a_2, g^{-1}) = (a_1 \omega_g(a_2) f(g, g^{-1}) f(g, 1), 1),$$

and we are led to $\omega_g(a_2) = a_1^{-1} f(g, 1)^{-1} f(g, g^{-1})^{-1}$. The inverse is given by

$$(a, g)^{-1} = (\omega_{g^{-1}}(a^{-1}) f(1, 1)^{-1} \omega_{g^{-1}}(f(g, g^{-1})^{-1}), g^{-1}), \tag{4.57}$$

where we used $\omega_g(f(1, 1)) = f(g, 1)$. This is the true inverse since

$$\begin{aligned}(a, g) \cdot (a, g)^{-1} &= (a, g) \cdot (\omega_{g^{-1}}(a^{-1}) f(1, 1)^{-1} \omega_{g^{-1}}(f(g, g^{-1})^{-1}), g^{-1}) \\ &= (a \cdot a^{-1} \omega_g(f(1, 1))^{-1} f(g, g^{-1})^{-1} f(g, g^{-1}) f(g, 1), g g^{-1}) = (1, 1), \\ (a, g)^{-1} \cdot (a, g) &= (\omega_{g^{-1}}(a^{-1}) f(1, 1)^{-1} \omega_{g^{-1}}(f(g, g^{-1})^{-1}), g^{-1}) \cdot (a, g) \\ &= (\omega_{g^{-1}}(a^{-1}) f(1, 1)^{-1} \omega_{g^{-1}}(f(g, g^{-1})^{-1}) \omega_{g^{-1}}(a) f(g^{-1}, g) f(g^{-1}, 1), g^{-1} g) \\ &= (\omega_{g^{-1}}(f(g, g^{-1}))^{-1} f(1, 1)^{-1} f(g^{-1}, g) f(g^{-1}, 1), g^{-1} g) = (1, 1),\end{aligned}$$

where in the last line we used $f(1,1) = f(1,g^{-1})$ and the third case above. For the associativity, we notice that

$$
\begin{aligned}
&(a_1,g_1)\cdot(a_2,g_2)\cdot(a_3,g_3)\\
=&(a_1\omega_{g_1}(a_2)f(g_1,g_2)f(g_1,1),g_1g_2)\cdot(a_3,g_3)\\
=&(a_1\omega_{g_1}(a_2)f(g_1,g_2)f(g_1,1)\omega_{g_1g_2}(a_3)f(g_1g_2,g_3)f(g_1g_2,1),g_1g_2g_3)\\
=&(a_1f(g_1,1)\omega_{g_1}(a_2)\omega_{g_1g_2}(a_3)\omega_{g_1}(f(g_2,g_3))f(g_1,g_2g_3)f(g_1,g_2)^{-1}\omega_{g_1}(f(g_2,1))f(g_1,g_2),g_1g_2g_3)\\
=&(a_1\omega_{g_1}(a_2\omega_{g_2}(a_3)f(g_2,g_3)f(g_2,1))f(g_1,g_2g_3)f(g_1,1),g_1g_2g_3)\\
=&(a_1,g_1)\cdot(a_2\omega_{g_2}(a_3)f(g_2,g_3)f(g_2,1),g_2g_3)\\
=&(a_1,g_1)\cdot((a_2,g_2)\cdot(a_3,g_3)).
\end{aligned}
$$

The above procedure, although solid, seems a bit tedious, and indeed, as we stick on the section with $f(1,1)=1$, we can simplify the group structure to

$$(a_1,g_1)\cdot(a_2,g_2) = (a_1\omega_{g_1}(a_2)f(g_1,g_2),g_1g_2), \tag{4.58}$$

and the inverse becomes

$$(a,g)^{-1} = (\omega_{g^{-1}}(a^{-1}f(g,g^{-1})^{-1}),g^{-1}). \tag{4.59}$$

(In the above way we are choosing $(1,1)$ as identity, while we can choose $(f(1,1)^{-1},1)$ as identity such that we do not need the tedious calculation.)

Now again we claim (and the proof follows similarly) that the equivalence relation as twist gauge transformation can be identified with the isomorphism of the extension. Thus, the *twisted cohomology group* $H^{2+\omega}(G,A)$, the Abelian group of equivalence classes of 2-cocycles under the twisted gauge transformation, classifies the equivalence classes of extensions of G by A that induces ω, denoted as $\mathrm{Ext}^{\omega}(G,A)$.

4.6.1 More Structures on Triviality

The trivial element in $H^{2+\omega}(G,A)$ where $[f]=1$, corresponds to sections that are homomorphisms and leads to representative group laws as

$$(a_1,g_1)\cdot(a_2,g_2) = (a_1\omega_{g_1}(a_2),g_1g_2), \tag{4.60}$$

which we identify as the equivalence class of semi-direct products $G\rtimes_{\omega}A$. Nevertheless, even for this trivial case with $f=1$, further classification structure emerges as we have different ways to perform the trivialization, or equivalently, different coboundary/gauge transformations that preserve the triviality. For $f=1$ before and after transformation, this requires that

$$t(g_1g_2) = t(g_1)\omega_{g_1}(t(g_2)), \tag{4.61}$$

and is known as *twisted homomorphism*. It gets the name since, if we consider a central extension with $\omega=\mathrm{id}$, then t function preserving the triviality is just a homomorphism. From the view of homotopy, it is the different ways of the trivialization, the procedure viewed as a whole object, that are classified.

Let us introduce terms:

- 1-cochain group $C^1(G,A)$ has element as map $t: G\to A$.

- (Twisted) 1-cocycle group $Z^{1+\omega}(G,A) \subset C^1(G,A)$ has element satisfying the twisted homomorphism $t(g_1g_2) = t(g_1)\omega_{g_1}(t(g_2))$.
- Twisted cohomology group $H^{1+\omega}(G,A) \subset Z^{1+\omega}(G,A)$ contains the equivalence classes of 1-cocycles under the inclusion section transformation.

To make the third point clear, we consider that in the trivial splitting case, section s is group homomorphism, and we can produce new section with the inclusion map as $\tilde{s}(g) = \iota(a)s(g)\iota(a)^{-1}$, which preserves the homomorphic property trivially. Re-write the new section

$$\begin{aligned}\tilde{s}(g) &= \iota(a)s(g)\iota(a)^{-1} = \iota(a)s(g)\iota(a)^{-1}s(g)^{-1}s(g)\\ &= \iota(a)\iota(\omega_g(a^{-1}))s(g) = \iota(a\omega_g(a^{-1}))s(g),\end{aligned}$$

we see that the related 1-cocycle is $t_a(g) = a\omega_g(a^{-1})$, and

$$\begin{aligned}t_a(g_1g_2) &= a\omega_{g_1g_2}(a^{-1}) = a\omega_{g_1}(a^{-1})\omega_{g_1}(a)\omega_{g_1g_2}(a^{-1})\\ &= a\omega_{g_1}(a^{-1})\omega_{g_1}(a\omega_{g_2}(a^{-1})) = t_a(g_1)\omega_{g_1}(t_a(g_2)).\end{aligned}$$

We then see that any $t \cdot t_a$ is also a 1-cocycle in the sense $t \cdot t_a(g) = t(g)t_a(g)$, and equivalence relation is given by $t \sim t \cdot t_a$. This is the equivalence relation since it gives isomorphism between sections and does not change the splitting.

Example: Euclidean Group and Crystallographic Group

For instance, let us consider the Euclidean group

$$0 \longrightarrow \mathbb{R}^d \xrightarrow{\iota} \mathrm{Euc}(d) \xrightarrow{\pi} O(d) \longrightarrow 1, \tag{4.62}$$

where $\mathrm{Euc}(d)$ is the automorphism group of affine space A^d ($\mathbb{R}^d$ without origin), including translations represented by $\mathbb{R}^d$ and rotations by $O(d)$. The inclusion map ι is the translation $\iota(v) = T_v$ by $T_v(p) = p + v$. The section is chosen such that for any $p \in \mathbb{A}^d$, we have section $R \mapsto s_p(R) \in \mathrm{Euc}(d)$ with transformation $s_p(R) : p + v \to p + Rv$, meaning that we define rotations with p as the center. For our purpose, we need to consider the automorphism function $\omega : O(d) \to \mathrm{Aut}(\mathbb{R}^d)$, and we see that by definition

$$\begin{aligned}T_{\omega_R(v)}(p) &= p + \omega_R(v)\\ &= s_p(R)T_v s_p(R)^{-1}(p) = s_p(R)T_v(p) = s_p(R)(p+v) = p + Rv,\end{aligned}$$

thus we have $\omega_R(v) = Rv$, which is a homomorphism. To find the 1-cocycle defined by the difference of sections $s_{p'}(R) = T_{t(R)}s_p(R)$, we calculate

$$\begin{aligned}T_{t(R)}s_p(R)(p+v) = T_{t(R)}(p+Rv) = p + t(R) + Rv,\\ s_{p'}(R)(p+v) = s_{p'}(R)(p' + (p - p' + v)) = p' + R(p-p') + Rv,\end{aligned}$$

from which we identify that $t(R) = (1-R)(p'-p) = (1-R)\delta p$ where $\delta p = p' - p$. Notice that $t(R) \in \mathbb{R}^d$ and follows addition law, and the twisted homomorphism follows as

$$t(R_1) + \omega_{R_1}(t(R_2)) = (1 - R_1) + R_1(1 - R_2) = 1 - R_1R_2 = t(R_1R_2).$$

Notice that $t_{\delta p}(R) = \delta p - \omega_R(\delta p)$, and is always trivial following the above equivalence relation. All splittings are equivalent.

If we restrict the sequence as

$$0 \longrightarrow \mathbb{Z}^d \xrightarrow{\iota} G \xrightarrow{\pi} \{1, \sigma\} \longrightarrow 1, \tag{4.63}$$

we have $\{1, \sigma\} \cong \mathbb{Z}_2 \subset O(d)$ with $\sigma = -\mathbf{1}$. Then $\omega : \mathbb{Z}_2 \to \mathbb{Z}^d$ is a homomorphism by $\omega_\sigma(\boldsymbol{n}) = -\boldsymbol{n}$. The section $s_{\boldsymbol{n}}(1) = \{0|\mathbf{1}\}$, and $s_{\boldsymbol{n}}(\sigma) = \{\boldsymbol{n}|\sigma\}$. Following the above deduction,

$$s^2_{\boldsymbol{n}}(\sigma)(\boldsymbol{n}_0) = s_{\boldsymbol{n}}(\sigma)(\boldsymbol{n} - \boldsymbol{n}_0) = \boldsymbol{n} - (\boldsymbol{n} - \boldsymbol{n}_0) = \boldsymbol{n}_0,$$

and we have $s^2_{\boldsymbol{n}}(\sigma) = 1$. Now that $t_{\boldsymbol{a}}(\sigma) = \boldsymbol{a} - \omega_\sigma(\boldsymbol{a}) = 2\boldsymbol{a} \in 2\mathbb{Z}^d$, and the splittings are equivalent when $\boldsymbol{n} - \boldsymbol{n}' \in 2\mathbb{Z}^d$, thus $H^{1+\omega}(\mathbb{Z}_2, \mathbb{Z}^d) \cong \mathbb{Z}^d/2\mathbb{Z}^d \cong (\mathbb{Z}_2)^d$.

More generally, we consider crystallographic groups over a crystal $C \subset \mathbb{A}^d$, with translation group $L(C) \subset \mathbb{R}^d$, point group $P(C) \subset O(d)$ and full automorphism group $G(C) \subset \mathrm{Euc}(d)$ forming

$$0 \longrightarrow L(C) \longrightarrow G(C) \longrightarrow P(C) \longrightarrow 1.$$

When this sequence does not split, the crystallographic group $G(C)$ is said to be nonsymmorphic. Notice that this is not the whole story for physical electrons on a lattice, since when we consider system with magnetism or spin-orbital coupling, the symmetry group is enlarged to magnetic space group or magnetic point group due to the participation of spin-rotational symmetry, and the sequence may split.

4.6.2 Anti-Unitary Symmetry

Anti-unitary symmetry can induce non-central extensions with twisted cocycle. For instance, we consider the case where we have in G a time-reversal symmetry, and there exists a homomorphism $\tau : G \to \{\pm 1\} \cong \mathbb{Z}_2$, indicating whether the orientation of time is preserved or reversed.

The anti-unitarity of time reversal is reflected when we consider its quantum implementation and its automorphism action over the $U(1)$, and especially we expect a non-trivial one. It is unwasteful to give some discussion on this point.

Non-trivial automorphism The automorphism $\mu : G \to G$ forms a group $\mathrm{Aut}(G)$. Especially, there exists a bunch of such operations provided by the conjugation homomorphism $I : G \to \mathrm{Aut}(G)$, where $I(a) : g \mapsto aga^{-1}$, and we will call the automorphism group generated by these operations as the inner automorphism group $\mathrm{Inn}(G)$, with $I(a) \circ I(b) = I(ab)$ the group law. Notice that for $a \in Z(G)$ the center, $I(a)$ is trivial, thus we have

$$1 \longrightarrow Z(G) \longrightarrow G \xrightarrow{I} \mathrm{Inn}(G) \longrightarrow 1, \tag{4.64}$$

indicating that $\mathrm{Inn}(G) \cong G/Z(G)$. Moreover, $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$, since $\forall \phi \in \mathrm{Aut}(G)$, we have

$$\phi \circ I(a) \circ \phi^{-1}(g) = \phi(a\phi^{-1}(g)a^{-1}) = \phi(a)g\phi(a)^{-1} = I(\phi(a))(g),$$

thus indicating the normality $\phi \circ I(a) \circ \phi^{-1} = I(\phi(a))$. Therefore, we have another short exact sequence

$$1 \longrightarrow \mathrm{Inn}(G) \longrightarrow \mathrm{Aut}(G) \longrightarrow \mathrm{Out}(G) \longrightarrow 1, \tag{4.65}$$

where $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$ is the outer automorphism group. We can indeed write

$$1 \longrightarrow Z(G) \longrightarrow G \longrightarrow \mathrm{Aut}(G) \longrightarrow \mathrm{Out}(G) \longrightarrow 1.$$

Especially notice that when G is Abelian, $\mathrm{Inn}(G) = 1$ and $\mathrm{Out}(G) = \mathrm{Aut}(G)$ are all non-trivial.

Now let us consider $\mathrm{Aut}(U(1)) \cong \mathrm{Out}(U(1)) \cong \mathbb{Z}_2$ with the non-trivial automorphism $z \to z^* = z^{-1}$. Thus, we assign a classical symmetry group G homomorphism $\omega : G \to \mathrm{Aut}(U(1))$ with

$$\omega_g(z) = \begin{cases} z, & \tau(g) = 1 \\ z^{-1}, & \tau(g) = -1 \end{cases}. \tag{4.66}$$

At the core of essence, we can let $G \equiv M_2 = \{1, \bar{T}\} \cong \mathbb{Z}_2$, where $\bar{T}^2 = 1$, meaning that twice reversal comes back (classically). Then as we have done, to represent this group in Hilbert space (quantum mechanically), we need to consider the projective representation as a (possibly twisted thus non-central) extension of M_2 by $U(1)$, where the homomorphism $\omega : M_2 \to \mathrm{Aut}(U(1)) \cong \mathbb{Z}_2$ needs specification. Of course, we can establish a central extension by $\omega_{\bar{T}} = 1$, but then 'time-reversal' becomes unitary, which contradicts our spirit. Hence, we consider the anti-unitary case with a non-trivial outer automorphism by time-reversal: $\omega_{\bar{T}} : z \to z^{-1}$. Notice that under this ω, we need to classify the equivalence classes of extensions of M_2 by $U(1)$ using

$$H^{2+\omega}(\mathbb{Z}_2, U(1)) \cong \mathbb{Z}_2. \tag{4.67}$$

To see this, we need to solve the twisted cocycle identity to get the 2-cocycle function, and again as we discussed, we fix gauge by $f(\bar{T}, 1) = f(1, \bar{T}) = f(1, 1) = 1$. Then only $f(\bar{T}, \bar{T})$ needs consideration, and by $g_1 = g_2 = g_3 = \bar{T}$, we have

$$f(\bar{T}, \bar{T}) = \omega_{\bar{T}}(f(\bar{T}, \bar{T})) = f(\bar{T}, \bar{T})^*,$$

thus $f(\bar{T}, \bar{T}) = \pm 1$. These two choices are not related by gauge transformation, since by gauge fixing $t(1) = 1$, and we have

$$\tilde{f}(\bar{T}, \bar{T}) = t(\bar{T})\omega_{\bar{T}} f(\bar{T}, \bar{T})/t(\bar{T}) = |t(\bar{T})|^2 f(\bar{T}, \bar{T}) = f(\bar{T}, \bar{T}),$$

is gauge invariant.

The two choices of $f(\bar{T}, \bar{T}) = \pm 1$ correspond to the two extensions of M_2 by $U(1)$ as

$$1 \longrightarrow U(1) \longrightarrow M_2^{\pm} \xrightarrow{\pi} M_2 \longrightarrow 1, \tag{4.68}$$

with group laws

$$\begin{aligned}
(z_1, 1) \cdot (z_2, 1) &= (z_1 \omega_1(z_2) f(1, 1), 1) = (z_1 z_2, 1), \\
(z_1, \bar{T}) \cdot (z_2, 1) &= (z_1 \omega_1(z_2) f(\bar{T}, 1), \bar{T}) = (z_1 z_2, \bar{T}), \\
(z_1, 1) \cdot (z_2, \bar{T}) &= (z_1 \omega_{\bar{T}}(z_2) f(1, \bar{T}), 1) = (z_1 z_2^{-1}, \bar{T}), \\
(z_1, \bar{T}) \cdot (z_2, \bar{T}) &= (z_1 \omega_{\bar{T}}(z_2) f(\bar{T}, \bar{T}), \bar{T}) = (\pm z_1 z_2^{-1}, 1).
\end{aligned}$$

This group structure is a bit tedious. To make it clearer, let us consider lifting $\bar{T}$ to $\tilde{T} \in M_2^{\pm}$, and since $\pi(\tilde{T}^2) = 1$ (seen from the last law), we have $\tilde{T}^2 = z \in U(1)$. Meanwhile,

$$\begin{aligned}
\tilde{T} z &= \tilde{T}\tilde{T}^2 = \tilde{T}^2 \tilde{T} = z\tilde{T} \\
&= z^{-1}\tilde{T},
\end{aligned}$$

where the second line follows from the third law above, we have $z^2 = 1$ thus $z = \pm 1$. This leads to the two possible group structures of $M_2^{\pm}$ as

$$M_2^{\pm} = \{z\tilde{T}|\tilde{T}z = z^{-1}\tilde{T}, \tilde{T}^2 = \pm 1\}. \tag{4.69}$$

M_2^+ corresponds to the identity element of $H^{2+\omega}(\mathbb{Z}_2, U(1))$, and by treating $U(1) \cong SO(2)$, we see that $\tilde{T}$ is just a reflection, thus $M_2^+ \cong O(2) = SO(2) \rtimes \mathbb{Z}_2$. M_2^- corresponds to the non-trivial element of $H^{2+\omega}(\mathbb{Z}_2, U(1))$, and is another double cover of $O(2)$ known as $\mathrm{Pin}^-(2)$ group. The double cover is induced by the map $\pi^- : M_2^- \to O(2)$ by

$$\pi^-(\tilde{T}) = P,\ \pi^-(z = e^{i\alpha}) = R(2\alpha),$$

and the law $PRP = R^{-1}$ parallels with $\tilde{T}z\tilde{T} = -z^{-1}$, while $\pi^-(-1) = R(2\pi) = 1$.

Notice that for system with obvious spin-orbital coupling, we have $T^2 = (-1)^{2j}$ for spin-j particles, with spin-j obtained by tensor product of $2j$-spin-1/2 representations, known as Clebsch-Gordan decomposition.

A magnetic crystallographic group

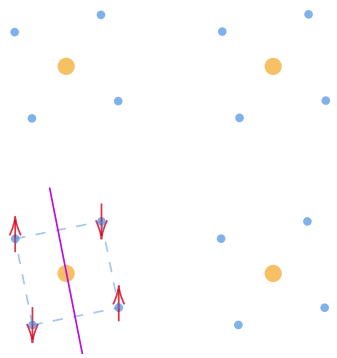


Figure 4.1: A symmorphic crystal structure with D_4 symmetry. Origin dots denote fictitious cell location, blue dots denote atoms, red arrows denote spin, and purple curve denotes a reflection line.

We may consider crystal

$$C = (\mathbb{Z}^2 + (\delta_1, \delta_2)) \cup (\mathbb{Z}^2 + (-\delta_2, \delta_1)) \cup (\mathbb{Z}^2 + (-\delta_1, -\delta_2)) \cup (\mathbb{Z}^2 + (\delta_2, -\delta_1)),$$

as shown in Fig. 4.1. The point group without magnetism is $P(C) = D_4 = \{P, R|R^4 = 1, P^2 = 1, PRP = R^{-1}\}$, and $G(C) = \mathbb{Z}^2 \rtimes D_4$. Assuming intra-cell Ising antiferromagnetic order dictated by the red arrows, and the new crystal becomes $\hat{C} = \hat{C}_+ \cup \hat{C}_-$, where

$$\hat{C}_+ = (\mathbb{Z}^2 + (\delta_1, \delta_2)) \times \{+S\} \cup (\mathbb{Z}^2 + (-\delta_1, -\delta_2)) \times \{+S\},$$
$$\hat{C}_- = (\mathbb{Z}^2 + (-\delta_2, \delta_1)) \times \{-S\} \cup (\mathbb{Z}^2 + (\delta_2, -\delta_1)) \times \{-S\}.$$

Notice that $\hat{C}$ is a subset of $\mathbb{R}^2 \times \mathcal{S}$ with $\mathcal{S} = \{S, -S\}$. Denote $\mathbb{Z}_2 = \{1, \sigma\}$ acting on $\mathcal{S}$ only with σ a spin (time) reversal, then the new symmetry group, the *magnetic crystallographic group*, $\widehat{G(C)} \subset \mathrm{Euc}(2) \times \mathbb{Z}_2$, and by quotient translations,

$$0 \longrightarrow \mathbb{Z}^2 \longrightarrow \widehat{G(C)} \longrightarrow \widehat{P(C)} \longrightarrow 1, \tag{4.70}$$

we obtain the *magnetic point group* $\widehat{P(C)}$. The elements are

$$\{(1,1),(R,\sigma),(R^2,1),(R^3,\sigma),(P,\sigma),(PR,1),(PR^2,\sigma),(PR^3,1)\},$$

and is directly isomorphic to D_4. In this group, we can assign $\tau(\cdot,\sigma)=-1$, and

$$1\longrightarrow\mathbb{Z}_2\times\mathbb{Z}_2\longrightarrow\widehat{P(C)}\xrightarrow{\tau}\mathbb{Z}_2\longrightarrow 1. \tag{4.71}$$

Notice that the sequence does not split, as there is no non-trivial automorphism of $\mathbb{Z}_2$ and $\widehat{P(C)}$ is non-Abelian.

From the Dirac equation, we know that not only time-reversal, but there exists another anti-unitary symmetry, the particle-hole symmetry. As we have seem. based on the assumption that the spacetime is time-orientable, the physical symmetry group is equipped with $\tau: G\to\mathbb{Z}_2$, indicating whether the orientation of time is preserved or reversed. However, in the above, we see that in principle, the time-reversal symmetry can be a unitary one (if we do not consider quantum dynamics), while Wigner's theorem states that we can always use grading $\phi: G\to\mathbb{Z}_2$ to tell whether the represented $\rho(g)$ on the Hilbert space is unitary or anti-unitary. In this sense, we find two possible anti-unitary symmetries, bi-graded by τ and ϕ, generally.

We should consider quantum dynamics, however. We consider unitary operators $U(t_1,t_3)=U(t_1,t_2)U(t_2,t_3)$, simply generated by time-independent Hamiltonian

$$U(t_1,t_2)=U(t_1-t_2)=e^{-iH(t_1-t_2)}.$$

The represented symmetry $\rho: G\to\mathrm{Aut}(\mathcal{H})$ is a symmetry of dynamics of $\forall g\in G$,

$$\rho(g)U(t)\rho(g)^{-1}=U(\tau(g)t). \tag{4.72}$$

Taking into account the possible anti-unitarity of ϕ, we have

$$\begin{aligned}\rho(g)U(t)\rho(g)^{-1}&=\sum_n\frac{1}{n!}\rho(g)(-it)^nH^n\rho(g)^{-1}\\&=\sum_n\frac{1}{n!}(-i\phi(g)t)^n(\rho(g)H\rho(g)^{-1})^n\\&=\exp\bigl(-it\phi(g)\rho(g)H\rho(g)^{-1}\bigr)=U(\phi(g)\rho(g)H\rho(g)^{-1}),\end{aligned}$$

then we are led to

$$\phi(g)\rho(g)H\rho(g)^{-1}=\tau(g)H,$$

or equivalently,

$$\rho(g)H\rho(g)^{-1}=\phi(g)\tau(g)H. \tag{4.73}$$

Very generally, we see that ϕ and τ are unrelated. However, by defining the third homomorphism $\chi: G\to\mathbb{Z}_2$ by $\chi(g)=\phi(g)\tau(g)=\pm1$, we see that

$$\phi\cdot\tau\cdot\chi=1, \tag{4.74}$$

emerges as a constraint. We may call χ the chirality.

4.7 General Extensions

After the extension by Abelian group, where it can be central or twisted, we finally come back to the general case

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q \longrightarrow 1,$$

where both N and Q are not necessarily Abelian.

From the discussion about twisted homology, we see that the most important quantities that determine an extension are two maps

$$\begin{cases} \omega_s : Q \to \mathrm{Aut}(N) \\ f_s : Q \times Q \to N \end{cases} \Longrightarrow \begin{cases} \iota(\omega_{s,q}(n)) = s(q)\iota(n)s(q)^{-1} \\ s(q_1)s(q_2) = \iota(f_s(q_1,q_2))s(q_1q_2) \end{cases}, \tag{4.75}$$

where we have labelled the dependence on section s, since unlike the Abelian case, these maps, automorphism map and 2-cocycle, will generally depend on the choice of section. This can be seen from the repeated calculations

$$\begin{aligned} \iota(\omega_{q_1} \circ \omega_{q_2}(n)) &= s(q_1)\iota(\omega_{q_2}(n))s(q_1)^{-1} = s(q_1)s(q_2)\iota(n)s(q_2)^{-1}s(q_1)^{-1} \\ &= \iota(f_s(q_1,q_2))s(q_1q_2)\iota(n)s(q_1q_2)^{-1}\iota(f_s(q_1,q_2))^{-1} \\ &= \iota(f_s(q_1,q_2))\iota(\omega_{q_1q_2}(n))\iota(f_s(q_1,q_2)^{-1}) \\ &= \iota(f_s(q_1,q_2)\cdot\omega_{q_1q_2}(n)\cdot f_s(q_1,q_2)^{-1}), \end{aligned}$$

from which we can only tell that

$$\omega_{s,q_1} \circ \omega_{s,q_2} = I(f_s(q_1,q_2)) \circ \omega_{s,q_1q_2}, \tag{4.76}$$

where $I(n \in N) \in \mathrm{Aut}(N)$ is the conjugate inner automorphism. Thus generally ω is no longer a section-independent map, and especially it is not a homomorphism (and ω_1 is not necessarily identity). Be careful that we still have the homomorphism map from G to $\mathrm{Aut}(N)$, but it is not class-dependent as N is not necessarily Abelian, and can not be identified with ω.

Exactly the same, we have the *twisted* 2-cocycle condition

$$f_s(q_1,q_2)f_s(q_1q_2,q_3) = \omega_{q_1}(f_s(q_2,q_3))f_s(q_1,q_2q_3), \tag{4.77}$$

and similar deductions

$$\begin{cases} \omega_q(f_s(1,1)) = f_s(q,1) \\ f_s(1,1) = \omega_1(f_s(1,q)) \\ f_s(q^{-1},q)f_s(1,q^{-1}) = \omega_{q^{-1}}(f_s(q,q^{-1}))f_s(q^{-1},1) \end{cases}.$$

The two maps completely determine one extension, where we have group multiplication law in G as

$$(n_1,q_1)\cdot_{f,\omega}(n_2,q_2) = (n_1\omega_{q_1}(n_2)f(q_1,q_2), q_1q_2). \tag{4.78}$$

Proof. To verify this, we need to carefully check the identity element that turns out to be $(f(1,1)^{-1},1)$, since

$$\begin{aligned} (n,q)\cdot(f(1,1)^{-1},1) &= (n\omega_q(f(1,1)^{-1})f(q,1), q\cdot 1) = (n,q) \\ (f(1,1)^{-1},1)\cdot(n,q) &= (f(1,1)^{-1}\omega_1(n)f(1,q), 1\cdot q) = (n,q), \end{aligned}$$

where for the second line, one notices marvelously that

$$\omega_1(f(1,1)^{-1}\omega_1(n)f(1,q)) = f(1,1)^{-1}\omega_1 \circ \omega_1(n)f(1,1) = \omega_1(n),$$

and since $\omega_1 \in \mathrm{Aut}(N)$ (an isomorphism), we have to conclude that

$$f(1,1)^{-1}\omega_1(n)f(1,q) = n. \tag{4.79}$$

Then we need to find the inverse element, satisfying that

$$\begin{aligned}(n,q)^{-1}\cdot(n,q) &= (n',q^{-1})\cdot(n,q)\\ &= (n'\omega_{q^{-1}}(n)f(q^{-1},q),q^{-1}q) \quad = (f(1,1)^{-1},1),\end{aligned}$$

which leads to that $n'\omega_{q^{-1}}(n)f(q^{-1},q) = f(1,1)^{-1}$ and

$$(n,q)^{-1} = (f(1,1)^{-1}f(q^{-1},q)^{-1}\omega_{q^{-1}}(n^{-1}),q^{-1}).$$

We need further verification as

$$\begin{aligned}(n,q)\cdot(n,q)^{-1} &= (n,q)\cdot(f(1,1)^{-1}f(q^{-1},q)^{-1}\omega_{q^{-1}}(n^{-1}),q^{-1})\\ &= (n\omega_q(f(1,1)^{-1}f(q^{-1},q)^{-1}\omega_{q^{-1}}(n^{-1}))f(q,q^{-1}),qq^{-1})\\ &= (n\omega_q(f(1,1)^{-1})\omega_q(f(q^{-1},q)^{-1})\omega_q\circ\omega_{q^{-1}}(n^{-1})f(q,q^{-1}),1)\\ &= (nf(q,1)^{-1}\omega_q(f(q^{-1},q)^{-1})f(q,q^{-1})\omega_1(n^{-1})f(q,q^{-1})^{-1}f(q,q^{-1}),1)\\ &= (nf(1,q)^{-1}f(q,q^{-1})^{-1}f(q,q^{-1})\omega_1(n^{-1}),1)\\ &= (nf(1,q)^{-1}\omega_1(n^{-1}),1)\\ &= (nn^{-1}f(1,1)^{-1},1) = (f(1,1)^{-1},1),\end{aligned}$$

where the third relation and the marvelous identity has been used.

The final stuff is the associative relation, that

$$\begin{aligned}&(n_1,q_1)(n_2,q_2)(n_3,q_3) = (n_1\omega_{q_1}(n_2)f(q_1,q_2),q_1q_2)(n_3,q_3)\\ =&(n_1\omega_{q_1}(n_2)f(q_1,q_2)\omega_{q_1q_2}(n_3)f(q_1q_2,q_3),q_1q_2q_3)\\ =&(n_1\omega_{q_1}(n_2)f(q_1,q_2)f(q_1,q_2)^{-1}\omega_{q_1}\circ\omega_{q_2}(n_3)f(q_1,q_2)f(q_1q_2,q_3),q_1q_2q_3)\\ =&(n_1\omega_{q_1}(n_2)\omega_{q_1}\circ\omega_{q_2}(n_3)\omega_{q_1}(f(q_2,q_3))f(q_1,q_2q_3),q_1q_2q_3)\\ =&(n_1\omega_{q_1}(n_2\omega_{q_2}(n_3)f(q_2,q_3))f(q_1,q_2q_3),q_1q_2q_3)\\ =&(n_1,q_1)(n_2\omega_{q_2}(n_3)f(q_2,q_3),q_2q_3) = (n_1,q_1)[(n_2,q_2)(n_3,q_3)].\end{aligned}$$

□

With the construction of a single group, then in spirit of extension, where we have multiple possibility for the formation of the new group, we again want to make statement, and again it is about classifying extensions by considering equivalent sections determined by (ω_s, f_s) pairs. And it is the time for our 1-cocycle $t : Q \to N$ to play the role, which induces equivalent section relation

$$\tilde{s}(q) = \iota(t(q))s(q). \tag{4.80}$$

Then following the same repeated deduction,

$$\begin{aligned}\iota(\omega_{q,\tilde{s}}(n)) &= \tilde{s}(q)\iota(n)\tilde{s}(q)^{-1} = \iota(t(q))s(q)\iota(n)s(q)^{-1}\iota(t(q))^{-1}\\ &= \iota(t(q))\iota(\omega_{g,s}(n))\iota(t(q))^{-1} = \iota(t(q)\cdot\omega_{g,s}(n)\cdot t(q)^{-1}),\\ \iota(f_{\tilde{s}}(q_1,q_2)) &= \tilde{s}(q_1)\tilde{s}(q_2)\tilde{s}(q_1q_2)^{-1} = \iota(t(q_1))s(q_1)\iota(t(q_2))s(q_2)s(q_1q_2)^{-1}\iota(t(q_1q_2))^{-1}\\ &= \iota(t(q_1))s(q_1)\iota(t(q_2))s(q_1)^{-1}s(q_1)s(q_2)s(q_1q_2)^{-1}\iota(t(q_1q_2))^{-1}\\ &= \iota(t(q_1))\iota(\omega_{q_1}(t(q_2)))\iota(f_s(q_1,q_2))\iota(t(q_1q_2)^{-1}),\end{aligned}$$

we obtain

$$\omega_{\tilde{s},q} = I(t(q)) \circ \omega_{s,q}, \tag{4.81a}$$
$$f_{\tilde{s}}(q_1, q_2) = t(q_1)\omega_{q_1}(t(q_2)) f_s(q_1, q_2) t(q_1 q_2)^{-1}. \tag{4.81b}$$

Reversely, if we have (ω, f) and t such that a new pair (ω', f') is constructed in the way that

$$\begin{cases} \omega'_q = I(t(q)) \circ \omega_q \\ f'(q_1, q_2) = t(q_1)\omega_{q_1}(t(q_2)) f(q_1, q_2) t(q_1 q_2)^{-1} \end{cases},$$

we will be led to another well-defined extension as they satisfy the twisted ω combination law and twisted 2-cocycle condition. This transformation thus generalizes the coboundary transformation in the Abelian case and thus defines the equivalence relation of pair (ω, f). Under the situation, we can indeed write down the isomorphism between two extensions $\varphi : G \to G'$ as

$$\varphi(n, q) = (nt(q)^{-1}, q), \tag{4.82}$$

and as

$$\begin{aligned}
\varphi(n_1, q_2) \cdot_{f',\omega'} \varphi(n_2, q_2) &= (n_1 t(q_1)^{-1}, q_1) \cdot_{f',\omega'} (n_2 t(q_2)^{-1}, q_2) \\
&= (n_1 t(q_1)^{-1} \omega'_{q_1}(n_2 t(q_2)^{-1}) f'(q_1, q_2), q_1 q_2) \\
&= (n_1 t(q_1)^{-1} t(q_1) \omega_{q_1}(n_2 t(q_2)^{-1}) t(q_1)^{-1} t(q_1) \omega_{q_1}(t(q_2)) f(q_1, q_2) t(q_1 q_2)^{-1}, q_1 q_2) \\
&= (n_1 \omega_{q_1}(n_2) f(q_1, q_2) t(q_1 q_2)^{-1}, q_1 q_2) \\
&= \varphi(n_1 \omega_{q_1}(n_2) f(q_1, q_2), q_1 q_2) \\
&= \varphi((n_1, q_1) \cdot_{f,\omega} (n_2, q_2)),
\end{aligned}$$

the group isomorphism is well-defined.

Now let us consider the below diagram

$$\begin{array}{ccccccccc}
1 & \longrightarrow & N & \xrightarrow{\iota} & G & \xrightarrow{\pi} & Q & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle I} & & \downarrow{\scriptstyle \psi} & \swarrow{\scriptstyle \omega} & \downarrow{\scriptstyle \bar{\omega}} & & \\
1 & \longrightarrow & \mathrm{Inn}(N) & \longrightarrow & \mathrm{Aut}(N) & \xrightarrow{\tilde{\pi}} & \mathrm{Out}(N) & \longrightarrow & 1
\end{array} . \tag{4.83}$$

Here we try to think that there exists a homomorphism $\bar{\omega} : Q \to \mathrm{Out}(N)$, by mapping $q \mapsto \omega_q \in \mathrm{Aut}(N)$ to $\bar{\omega}_q$ with projection $\tilde{\pi}$. Possibly a better picture is, assume a section $\tilde{s}$ dual to $\tilde{\pi}$ lifts $\bar{\omega}_q$ to ω_q, then since $\omega_{q_1} \circ \omega_{q_2} = I(f(q_1, q_2)) \circ \omega_{q_1 q_2}$, we identify $I(f(q_1, q_2))$ as the 2-cocycle , while exactly, $I(f(q_1, q_2)) \in \mathrm{Inn}(N)$ as $f(q_1, q_2) \in N$. This gives us the extension in the second line, where $I : N \to \mathrm{Inn}(N)$, and ψ is the pre-mentioned homomorphism from G to $\mathrm{Aut}(N)$ by $\iota(\psi_g(n)) = g\iota(n)g^{-1}$.

Now we ask that given $\bar{\omega} : Q \to \mathrm{Out}(N)$, can we construct the extension G? The answer is generally no, as a simple pullback extension gives Q extended by $\mathrm{Inn}(N)$, while there could be possible obstruction when we want to incorporate center $Z(N)$ into the extension ($\mathrm{Inn}(N) \cong N/Z(N)$). This relates to $H^{3+\bar{\omega}}(Q, Z(N))$.

Nevertheless, we may assume that we do have such an extension, and we want to know how much constraints does $\bar{\omega}$ give such that the extension $\mathrm{Ext}^{\bar{\omega}}(Q, N)$ is classified. Assume that two such extensions exist, with lifted $\omega_q^{(1)}$ and $\omega_q^{(2)}$. Consider them as element of bundle $\mathrm{Aut}(N)$, they can be related by the decomposition

$$\omega_q^{(1)} = I(t(q)) \circ \omega_q^{(2)},$$

with $t : Q \to N$, and note that we simply use $I(t(q)) \in \mathrm{Inn}(N)$, while t is not necessarily the gauge transformation. Now that since

$$\begin{aligned}
&\omega^{(1)}_{q_1} \circ \omega^{(1)}_{q_2}(n) = t(q_1)\omega^{(2)}_{q_1}\left(t(q_2)\omega^{(2)}_{q_2}(n)t(q_2)^{-1}\right)t(q_1)^{-1} \\
=&t(q_1)\omega^{(2)}_{q_1}(t(q_2))\omega^{(2)}_{q_1} \circ \omega^{(2)}_{q_2}(n)\omega^{(2)}_{q_1}(t(q_2))^{-1}t(q_1)^{-1} \\
=&t(q_1)\omega^{(2)}_{q_1}(t(q_2))f^{(2)}(q_1,q_2)\omega^{(2)}_{q_1q_2}(n)f^{(2)}(q_1,q_2)^{-1}\omega^{(2)}_{q_1}(t(q_2))^{-1}t(q_1)^{-1} \\
=&\left(t(q_1)\omega^{(2)}_{q_1}(t(q_2))f^{(2)}(q_1,q_2)t(q_1q_2)^{-1}\right)\omega^{(1)}_{q_1q_2}(n)\left(t(q_1)\omega^{(2)}_{q_1}(t(q_2))f^{(2)}(q_1,q_2)t(q_1q_2)\right)^{-1} \\
=&\hat{f}^{(2)}(q_1,q_2)\omega^{(1)}_{q_1q_2}(n)\hat{f}^{(2)}(q_1,q_2)^{-1},
\end{aligned}$$

where $\hat{f}^{(2)}(q_1,q_2) = t(q_1)\omega^{(2)}_{q_1}(t(q_2))f^{(2)}(q_1,q_2)t(q_1q_2)^{-1}$, while

$$\omega^{(1)}_{q_1} \circ \omega^{(1)}_{q_2}(n) = f^{(1)}(q_1,q_2)\omega^{(1)}_{q_1q_2}(n)f^{(1)}(q_1,q_2)^{-1},$$

we can, at the most general ground, say that

$$\hat{f}^{(2)}(q_1,q_2) = f^{(1)}(q_1,q_2)\zeta(q_1,q_2), \tag{4.84}$$

where a torsor map $\zeta : Q \times Q \to Z(N)$ is introduced. This tells us that apart from a gauge transformation, a torsor map is also needed to relate two equivalent extensions. It is intuitive to see that if $f^{(1)}$ and $\hat{f}^{(2)}$ are both 2-cocycles, then ζ also is, meaning that it satisfies the twisted 2-cocycle identity, thus

$$\zeta \in H^{2+\bar{\omega}}(Q, Z(N)). \tag{4.85}$$

We see that as long as ζ is non-trivial, a direct change of a f gives an inequivalent extension, thus $\mathrm{Ext}^{\bar{\omega}}(Q,N)$ is a torsor for $H^{2+\bar{\omega}}(Q,Z(N))$, meaning that any two elements in $\mathrm{Ext}^{\bar{\omega}}(Q,N)$ is uniquely related by a torsor map ζ.

4.8 Group Cohomology

For any group G and an Abelian group A, we can (as we have shown in the twisted case) equip a homomorphism $\omega : G \to \mathrm{Aut}(A)$ with $\omega_{g_1} \circ \omega_{g_2} = \omega_{g_1g_2}$.

Definition 4.8.1. *An n-cochain is a function $\phi : G^{\times n} \to A$. The set of all n-cochains is denoted by $C^n(G,A)$, the Abelian cochain group.*

Note that $n \geq 0$, and when $n = 0$, we directly assign $\phi_0 = a \in A$ for some a. The Abelian nature is reflected as $(\phi^1_n + \phi^2_n)(\boldsymbol{g}) = \phi^1_n(\boldsymbol{g}) + \phi^2_n(\boldsymbol{g})$, where $\boldsymbol{g} = (g_1, g_2, \cdots, g_n)$.

The differential (coboundary) operation is defined by the homomorphism $\mathrm{d} : C^n(G,A) \to C^{n+1}(G,A)$

$$\begin{aligned}
(\mathrm{d}\phi)(g_1,g_2,\cdots,g_{n+1}) = &\omega_{g_1}(\phi(g_2,g_3,\cdots,g_{n+1})) + \sum_{i=1}^{n}(-1)^i\phi(g_1,\cdots,g_ig_{i+1},\cdots,g_{n+1}) \\
&+ (-1)^{n+1}\phi(g_1,g_2,\cdots,g_n).
\end{aligned} \tag{4.86}$$

For instance, we have

- For $n = 0$: $(\mathrm{d}\phi_0)(g) = \omega_g(a) - a$, which is just 1-coboundary;

- For $n=1$: $(\mathrm{d}\phi_1)(g_1,g_2)=\omega_{g_1}(\phi_1(g_2))-\phi_1(g_1g_2)+\phi_1(g_1)$, which is 2-coboundary.
- For $n=2$: $(\mathrm{d}\phi_2)(g_1,g_2,g_3)=\omega_{g_1}(\phi_2(g_2,g_3))-\phi_2(g_1g_2,g_3)+\phi_2(g_1,g_2g_3)-\phi_2(g_1,g_2)$, which is 3-coboundary.
- $\cdots$

There it comes one of the central equation in group cohomology, or more generally in homological algebra

$$\mathrm{d}^2=0, \tag{4.87}$$

or let us emphasize it again

$$\mathrm{d}(\mathrm{d}\phi)=0. \tag{4.88}$$

This equation ensures the clossure of the cochain complex sequence

$$\cdots\to C^n\xrightarrow{\mathrm{d}_n}C^{n+1}\xrightarrow{\mathrm{d}_{n+1}}C^{n+2}\to\cdots$$

by $\mathrm{d}_{n+1}\circ\mathrm{d}_n=0$. However, this sequence is not exact in general, and the obstruction to achieve exactness is depicted by the cohomology group. Before proving the theorem, let us again provide some examples

- For $n=0$, we have
$$\begin{aligned}(\mathrm{d}^2\phi_0)(g_1,g_2)&=\omega_{g_1}(\mathrm{d}\phi_0(g_2))-\mathrm{d}\phi_0(g_1g_2)+\mathrm{d}\phi_0(g_1)\\&=\omega_{g_1}(\omega_{g_2}(a)-a)-(\omega_{g_1g_2}(a)-a)+(\omega_{g_1}(a)-a)=0.\end{aligned}$$
- For $n=1$, we have
$$\begin{aligned}(\mathrm{d}^2\phi_1)(g_1,g_2,g_3)&=\omega_{g_1}(\mathrm{d}\phi_1(g_2,g_3))-\mathrm{d}\phi_1(g_1g_2,g_3)+\mathrm{d}\phi_1(g_1,g_2g_3)-\mathrm{d}\phi_1(g_1,g_2)\\&=\omega_{g_1}(\omega_{g_2}(\phi_1(g_3)))-\omega_{g_1}(\phi_1(g_2g_3))+\omega_{g_1}(\phi_1(g_2))\\&\quad-\omega_{g_1g_2}(\phi_1(g_3))+\phi_1(g_1g_2g_3)-\phi_1(g_1g_2)\\&\quad+\omega_{g_1}(\phi_1(g_2g_3))-\phi_1(g_1g_2g_3)+\phi_1(g_1)\\&\quad-\omega_{g_1}(\phi_1(g_2))+\phi_1(g_1g_2)-\phi_1(g_1)=0,\end{aligned}$$
with terms in the same color canceling each other.
- $\cdots$

Proof. Before giving a flowery proof, let us first prove the theorem in a more direct way. We have

$$\begin{aligned}&(\mathrm{d}^2\phi)(g_1,g_2,\cdots,g_{n+2})\\=&\omega_{g_1}(\mathrm{d}\phi(g_2,\cdots,g_{n+2}))+\sum_{m=1}^{n+1}(-1)^m\mathrm{d}\phi(g_1,\cdots,g_mg_{m+1},\cdots,g_{n+2})+(-1)^n\mathrm{d}\phi(g_1,\cdots,g_{n+1})\\=&\omega_{g_1g_2}(\phi(g_3,\cdots,g_{n+2}))+\sum_{p=1}^{n}(-1)^p\omega_{g_1}(\phi(g_2,\cdots,g_{p+1}g_{p+2},\cdots,g_{n+2}))+(-1)^{n+1}\omega_{g_1}\phi(g_2,\cdots,g_{n+2})\\&+\left(-\omega_{g_1g_2}(\phi(g_3,\cdots,g_{n+2}))+\sum_{m=2}^{n+1}(-1)^m\omega_{g_1}(\phi(g_2,\cdots,g_mg_{m+1},\cdots,g_{n+2}))\right)+\sum_m\sum_p\cdots\\&+\left(\sum_{m=1}^{n}(-1)^m(-1)^{n+1}\phi(g_1,\cdots,g_mg_{m+1},\cdots,g_{n+1})+(-1)^{2(n+1)}\phi(g_1,\cdots,g_n)\right)\\&+(-1)^n\left(\omega_{g_1}(\phi(g_2,\cdots,g_{n+1}))+\sum_{p=1}^{n}(-1)^p\phi(g_1,\cdots,g_pg_{p+1},\cdots,g_{n+1})+(-1)^{n+1}\phi(g_1,\cdots,g_n)\right),\end{aligned}$$

where terms in the same color cancel each other. Especially note that we depart the $\sum_m$ terms into three parts separately. Now the only term left is the double-summation term, which reads

$$\begin{aligned}
\sum_m \sum_p \cdots \sim & \sum_{m=1}^{n+1}(-1)^m \sum_{p=1}^{n}(-1)^p \phi(g_1, \cdots, g_m g_{m+1}, \cdots, g_p g_{p+1}, \cdots, g_{n+1}) \text{ roughly} \\
= & -\left(-\phi(g_1 g_2 g_3, \cdots, g_{n+2}) + \sum_{p=2}^{n}(-1)^p \phi(g_1 g_2, \cdots, g_{p+1} g_{p+2}, \cdots, g_{n+2})\right) \\
& + \sum_{m=2}^{n}(-1)^m \left(\sum_{p<m-1}(-1)^p \phi(g_1, \cdots, g_p g_{p+1}, \cdots, g_m g_{m+1}, \cdots, g_{n+2})\right. \\
& + (-1)^{m-1}\phi(g_1, \cdots, g_{m-1} g_m g_{m+1}, \cdots, g_{n+2}) \\
& + (-1)^m \phi(g_1, \cdots, g_m g_{m+1} g_{m+2}), \cdots, g_{n+2} \\
& \left. + \sum_{p>m}(-1)^p \phi(g_1, \cdots, g_m g_{m+1}, \cdots, g_{p+1} g_{p+2}, \cdots, g_{n+2})\right) \\
& + (-1)^{n+1}\left(\sum_{p=1}^{n-1}(-1)^p \phi(g_1, \cdots, g_p g_{p+1}, \cdots, g_{n+1} g_{n+2}) + (-1)^n \phi(g_1, \cdots, g_n g_{n+1} g_{n+2})\right) \\
= & \, 0,
\end{aligned}$$

where terms in the same color cancel each other, as one can verify. Notice that the double-summation term in this expression cancels due to sign mismatch, where feeling of braiding emerges.

A more delicate proof exists by introducing the homogeneous n-cochains as $\varphi : G^{\times(n+1)} \to A$ with property

$$\varphi(hg_0, hg_1, \cdots, hg_n) = \omega_h \varphi(g_0, g_1, \cdots, g_n). \tag{4.89}$$

Denote homogeneous cochain group by $\mathcal{C}^n(G, A)$, and this is also an Abelian group but with $n+1$ arguments. The differential operation on homogeneous cochain is defined as $\delta : \mathcal{C}^n(G, A) \to \mathcal{C}^{n+1}(G, A)$ by

$$\delta\varphi(g_0, \cdots, g_{n+1}) = \sum_{i=0}^{n+1}(-1)^i \varphi(g_0, \cdots, \hat{g}_i, \cdots, g_{n+1}), \tag{4.90}$$

where the hat denotes the omission of the argument. The differential operation δ is a derivation with $\delta^2 = 0$, which can be verified by direct calculation. Now we relate the homogeneous cochain group with the cochain group by an isomorphism $\psi : \mathcal{C}^n(G, A) \to C^n(G, A)$, which states that $\phi_n = \psi(\varphi_n)$ by

$$\phi_n(g_1, \cdots, g_n) = \psi(\varphi_n)(g_1, \cdots, g_n) = \varphi_n(1, g_1, g_1 g_2, \cdots, g_1 g_2 \cdots g_n).$$

In this sense, we have

$$\mathrm{d}\psi = \psi\delta, \tag{4.91}$$

or the commutative diagram

$$\begin{array}{ccc}
\mathcal{C}^n(G, A) & \xrightarrow{\delta} & \mathcal{C}^{n+1}(G, A) \\
\downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \psi} \\
C^n(G, A) & \xrightarrow{\mathrm{d}} & C^{n+1}(G, A).
\end{array}$$

It is straightforward to verify that

$$\begin{aligned}(\mathrm{d}\phi)(g_1,\cdots,g_{n+1}) &= \mathrm{d}(\psi(\varphi))(g_1,\cdots,g_{n+1}) = \psi(\delta\varphi)(g_1,\cdots,g_{n+1})\\ &= \delta\varphi(1,g_1,\cdots,g_1\cdots g_{n+1})\\ &= \varphi(g_1,g_1g_2,\cdots,g_1g_2\cdots g_{n+1}) + \sum_{i=2}^{n+2}(-1)^{i+1}\varphi(1,\cdots,\widehat{\prod_{j=1}^{i-1}g_j},\cdots,g_1\cdots g_{n+1})\\ &= \omega_{g_1}(\varphi(1,g_2,\cdots,g_2\cdots g_n)) + \sum_{i=2}^{n+2}(-1)^{i+1}\omega_{g_1}\varphi(1,\cdots,\widehat{\prod_{j=1}^{i-1}g_j},\cdots,g_1\cdots g_n)\\ &= \omega_{g_1}(\phi(g_2,\cdots,g_{n+1})) + \sum_{i=1}^{n}(-1)^i\phi(g_1,\cdots,g_ig_{i+1},\cdots,g_{n+1})\\ &+ (-1)^{n+1}\phi(g_1,\cdots,g_n),\end{aligned}$$

which is indeed our coboundary operation. Thus

$$\mathrm{d}^2\phi = \mathrm{d}(\mathrm{d}\psi(\varphi)) = \mathrm{d}\psi(\delta\varphi) = \psi\delta^2\varphi = 0.$$

□

Having seen the structure, finally we come to the definition of the cohomology group.

Definition 4.8.2. *The (ω-twisted) n-cocycle group is defined as a subgroup of n-cochain group $Z^{n+\omega}(G,A)\subset C^n(G,A)$ by cochains that are closed, i.e., $\mathrm{d}\phi_n = 0$.*

Definition 4.8.3. *The n-coboundary group is defined as a subgroup of n-cochain group $B^{n+\omega}(G,A)\subset C^n(G,A)$ by cochains that are exact, i.e., $\phi_n = \mathrm{d}\phi_{n-1}$.*

Now we notice that since exact cochains are naturally closed by $\mathrm{d}^2=0$, the coboundary group is a subgroup of cocycle group, i.e., $B^{n+\omega}(G,A)\subset Z^{n+\omega}(G,A)$. Furthermore, due to the Abelian nature, we can define

Definition 4.8.4. *The n-th cohomology group is defined as the quotient group $H^{n+\omega}(G,A) = Z^{n+\omega}(G,A)/B^{n+\omega}(G,A)$.*

Now we can draw

$$\begin{array}{ccccccccc}\cdots & \longrightarrow & C^{n-1}(G,A) & \xrightarrow{\mathrm{d}_{n-1}} & C^n(G,A) & \xrightarrow{\mathrm{d}_n} & C^{n+1}(G,A) & \longrightarrow & \cdots\\ & & \downarrow\mathrm{d} & & & & & & \\ 0 & \longrightarrow & B^{n+\omega}(G,A) & \longrightarrow & Z^{n+\omega}(G,A) & \longrightarrow & H^{n+\omega}(G,A) & \longrightarrow & 0,\end{array}$$

where we would better not treat two lines to be parallel. We see that $\mathrm{Im}(\mathrm{d}_{n-1}) = B^{n+\omega}$, while $\ker(\mathrm{d}_n) = Z^{n+\omega}$, and they are not equal in general, and this allows us to re-write the cohomology group as

$$H^n(C^*,\mathrm{d}_*) = \frac{\ker(\mathrm{d}_n)}{\mathrm{Im}(\mathrm{d}_{n-1})}. \tag{4.92}$$

The name cohomology is due to the fact that the cohomology group is dual to the homology group, which is defined by the chain group $C_n(G,A)$ with group homomorphism

(boundary operator) $\partial : C_n(G,A) \to C_{n-1}(G,A)$, such that $\partial^2 = \partial_n \circ \partial_{n+1} = 0$ (a boundary has no boundary), giving us the chain complex

$$\cdots \longrightarrow C_{n+1}(G,A) \xrightarrow{\partial_{n+1}} C_n(G,A) \xrightarrow{\partial_n} C_{n-1}(G,A) \longrightarrow \cdots,$$

which is also not exact obstructed from the homology group

$$H_n(C_*, \partial_*) = \frac{\ker(\partial_n)}{\mathrm{Im}(\partial_{n+1})}. \tag{4.93}$$

The chain and cochain complex are dual, in one sense that giving (C_n, ∂_n) the chain complex, a cochain complex (C^n, d_n) can be defined by $C^n = \mathrm{Hom}(C_n, A)$ with $\mathrm{d}_n = \partial_n^*$, where ∂_n^* is the dual of ∂_n.

Example. Let us consider the simplest case, where $G = \mathbb{Z}_2 = \{1, \sigma\}$ and $A = \mathbb{Z}$. Since $\mathrm{Aut}(\mathbb{Z}) = \{\mathrm{id}, \mathcal{P}\} = \mathbb{Z}_2$ with identity and reflection, we have two possible ω maps $\omega_\sigma = \mathrm{id}$ or $\mathcal{P}$. The former is trivial (T) and the latter is isomorphism ($\mathcal{I}$). We consider $H^{1+\omega}(\mathbb{Z}_2, \mathbb{Z})$. To do this we first find 1-boundary, and by letting $\phi_0 = n_0$, we have

$$\begin{aligned}
(\mathrm{d}\phi_0)(1) &= \omega_1(n_0) - n_0 = n_0 - n_0 = 0, \\
(\mathrm{d}\phi_0)(\sigma) &= \omega_\sigma(n_0) - n_0 = \begin{cases} 0, & \omega_\sigma = \mathrm{id} \\ -n_0 - n_0 = -2n_0, & \omega_\sigma = \mathcal{P} \end{cases},
\end{aligned}$$

a and the 1-coboundary group is trivial, $B^{1+T} = 0$ for $\omega = T$ and $B^{1+\mathcal{I}} = 2\mathbb{Z}$ for $\omega = \mathcal{I}$. Next, we find 1-cocycle by considering its differential's action on $\mathbb{Z}_2^{\times 2}$:

$$\begin{aligned}
(\mathrm{d}\phi_1)(1,1) &= \omega_1(\phi_1(1)) - \phi_1(1) + \phi_1(1) = \phi_1(1), \\
(\mathrm{d}\phi_1)(1,\sigma) &= \omega_1(\phi_1(\sigma)) - \phi_1(\sigma) + \phi_1(1) = \phi_1(1), \\
(\mathrm{d}\phi_1)(\sigma,1) &= \omega_\sigma(\phi_1(1)) - \phi_1(\sigma) + \phi_1(\sigma) = \omega_\sigma(\phi_1(1)), \\
(\mathrm{d}\phi_1)(\sigma,\sigma) &= \omega_\sigma(\phi_1(\sigma)) - \phi_1(1) + \phi_1(\sigma),
\end{aligned}$$

and we see that for some ϕ_1 to be 1-cocycle, we have to make equations stand as

$$\begin{cases} \phi_1(1) = 0 \\ \omega_\sigma(\phi_1(\sigma)) + \phi_1(\sigma) = 0 \end{cases}.$$

Now if $\omega = T$, then the second equation tells that $\phi_1(\sigma) = 0$, and thus $\phi_1 = 0$, which makes the 1-cocycle group trivial $Z^{1+T} = 0$. On the other hand, if $\omega = \mathcal{I}$, then the second equation is automatically satisfied, such that we can take $\phi_1(\sigma)$ to be any integer, thus we identify $Z^{1+\mathcal{I}} = \mathbb{Z}$. Finally, we have

$$\begin{aligned}
H^{1+T}(\mathbb{Z}_2, \mathbb{Z}) &= Z^{1+T}(\mathbb{Z}_2, \mathbb{Z}) / B^{1+T}(\mathbb{Z}_2, \mathbb{Z}) = 0, \\
H^{1+\mathcal{I}}(\mathbb{Z}_2, \mathbb{Z}) &= Z^{1+\mathcal{I}}(\mathbb{Z}_2, \mathbb{Z}) / B^{1+\mathcal{I}}(\mathbb{Z}_2, \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2,
\end{aligned}$$

with non-trivial cohomology group for a nontrivial twisted $\omega = \mathcal{I}$, where non-triviality indicates that odd and even mapping of cocycles are not equivalent. This is the difference between $\mathbb{Z} \times \mathbb{Z}_2$ and $\mathbb{Z} \rtimes \mathbb{Z}_2$.

We may summarize a little of the group cohomology.

- $H^{0+\omega}$: 0 cohomology is just 0 cocycle, $\mathrm{d}\phi_0(g) = \omega_g(a) - a = 0$, defining fixed points of ω.

- $H^{1+\omega}$ is related to the non-triviality of trivial splitting extensions

$$0 \to A \to A \rtimes G \to G \to 0.$$

- $H^{2+\omega}$ is in one-to-one correspondence with the equivalence classes of the group extensions $\mathrm{Ext}^{\omega}(G, A)$

$$0 \to A \to \tilde{G} \to G \to 0,$$

where one notices that the 2-cocycle nature of f is represented by $\mathrm{d}f = 0$, i.e., the 2-cocycle relation as

$$(\mathrm{d}f)(q_1, q_2, q_3) = \omega_{q_1}(f(q_2, q_3)) - f(q_1 q_2, q_3) + f(q_1, q_2 q_3) - f(q_1, q_2) = 0.$$

- $H^{3+\omega}$ tells when we can construct general extension of Q by N with canonical homomorphism $\bar{\omega} : Q \to \mathrm{Out}(N)$. By the suppositional sequence

$$\begin{array}{ccccccccc} 1 & \longrightarrow & N & \xrightarrow{\iota} & ? & \xrightarrow{\pi} & Q & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle I} & & \downarrow{\scriptstyle \psi} & \swarrow{\scriptstyle \omega} & \downarrow{\scriptstyle \bar{\omega}} & & \\ 1 & \longrightarrow & \mathrm{Inn}(N) & \longrightarrow & \mathrm{Aut}(N) & \xrightarrow{\tilde{\pi}} & \mathrm{Out}(N) & \longrightarrow & 1 \end{array},$$

one has to admit $\omega_{q_1} \circ \omega_{q_2} = I(f(q_1, q_2)) \circ \omega_{q_1 q_2}$ for some f function, and with this relation, one deduces that (prove $I(\omega_{q_1}(f(q_2, q_3))) = \omega_{q_1} \circ I(\omega_{q_1}(f(q_2, q_3))) \circ \omega_{q_1}^{-1}$ by acting on $n \in N$)

$$I(\omega_{q_1}(f(q_2, q_3)) f(q_1, q_2 q_3)) = I(f(q_1, q_2) f(q_1 q_2, q_3)),$$

which tells that upon a center $Z(N)$, we have the twisted cocycle relation as

$$f(q_1, q_2) f(q_1 q_2, q_3) = z(q_1, q_2, q_3) \omega_{q_1}(f(q_2, q_3)) f(q_1, q_2 q_3).$$

Utilizing this as the definition of z, one may show the (twisted) 3-cocycle nature of z, thus relating to $H^{3+\bar{\omega}}(Q, Z(N))$, meaning that it is only when z is a coboundary, one can construct the extension, and the classification of this trivialization is again related to $H^{2+\bar{\omega}}(Q, Z(N))$. For the former, we may make an analogy with the 2-cocycle, where the equivalence relation is reflected by the 2-coboundary from differential of 1-cochain, thus enlightening us to transform $\tilde{f}(q_1, q_2) = b(q_1, q_2) f(q_1, q_2)$, where $b : Q \times Q \to Z(N)$ makes the transformation well-defined (order does not matter). Then from the 2-cocycle relation (or equivalently, stated as the definition of z), we have (note that we did not require section s to change, thus ω is unaffected)

$$\begin{aligned} \tilde{z}(q_1, q_2, q_3) &= \tilde{f}(q_1, q_2) \tilde{f}(q_1 q_2, q_3) \tilde{f}(q_1, q_2 q_3)^{-1} \omega_{q_1}(\tilde{f}(q_2, q_3))^{-1} \\ &= b(q_1, q_2) b(q_1 q_2, q_3) b(q_1, q_2 q_3)^{-1} \omega_{q_1}(b(q_2, q_3))^{-1} z(q_1, q_2, q_3), \end{aligned}$$

and sees that the equivalence relation is just modular a 3-coboundary

$$z(q_1, q_2, q_3) = \tilde{z}(q_1, q_2, q_3) + (\mathrm{d}b)(q_1, q_2, q_3).$$

To exactly verify $\mathrm{d}z = 0$, one may write down each (of the five) term and use the definition to prove it.

Chapter 5

Category: An Introduction

5.1 Categories

Definition 5.1.1. *A category $\mathcal{C}$ consists of the following data:*

1. *A set of objects, denoted by* $\mathrm{Ob}(\mathcal{C})$.
2. *A collection* $\mathrm{Mor}(\mathcal{C})$ *of sets* $\hom(X, Y)$*, for* ***any*** *two objects* $X, Y \in \mathrm{Ob}(\mathcal{C})$*. Here, the element of set* $\phi \in \hom(X, Y)$ *is called a morphism from* X *to* Y*, denoted as arrows*
$$X \xrightarrow{\phi} Y.$$
3. *A composition law:*
$$\hom(X, Y) \times \hom(Y, Z) \to \hom(X, Z),$$
$$(\phi_1, \phi_2) \mapsto \phi_2 \circ \phi_1.$$

Such that the following properties hold:

- *A morphism* ϕ *uniquely determines its source* X *and target* Y*, i.e.,* $\hom(X, Y)$ *are disjoint for distinct ordered pairs* (X, Y).
- $\forall X \in \mathrm{Ob}(\mathcal{C})$ *there exists a distinguished morphism* 1_X *or* $\mathrm{id}_X \in \hom(X, X)$*, satisfying that*
$$1_X \circ \phi = \phi, \quad \psi \circ 1_X = \psi,$$
for any $\phi \in \hom(Y, X)$ *and* $\psi \in \hom(X, Y)$, $\forall Y \in \mathrm{Ob}(\mathcal{C})$.
- *The composition law is associative,* $(\psi_1 \circ \psi_2) \circ \psi_3 = \psi_1 \circ (\psi_2 \circ \psi_3)$.

Be aware of the following notations:

- The sequential in composition is mainly because we draw arrow right-wise while defining morphism action left-wise
$$X \xrightarrow{\phi_{YX}} Y \xrightarrow{\phi_{ZY}} Z \Rightarrow X \xrightarrow{\phi_{ZY} \circ \phi_{YX}} Z,$$
or equivalently $Y = \phi_{YX}(X), Z = \phi_{ZY}(Y) = \phi_{ZY} \circ \phi_{YX}(X)$.
- For a (X, Y) pair, the morphisms for a **set** $\hom(X, Y)$.

- 1_X is unique, since if we have another $1'_X$ satisfying the same property, then

$$1'_X = 1'_X \circ 1_X = 1_X,$$

where the left equality is due to the property of 1_X and the right is due to the property of $1'_X$.

- A morphism $\phi \in \hom(X, Y)$ is invertible if there exists a morphism $\psi \in \hom(Y, X)$ such that

$$\phi \circ \psi = 1_Y, \text{ and } \psi \circ \phi = 1_X.$$

Such two objects X, Y are called isomorphic.

- Especially notice the invertible morphism in set $\hom(X, X)$ other than 1_X is called automorphism.

- The category with only one object and all morphisms invertible is called a group.

As one can see, the most prominent feature of a category is it does not require a specific feature of objects or morphisms, but only a general possible relation between them, while we know that any specific object is defined by its own special relations, thus in this way we can tell that category studies relation of relations.

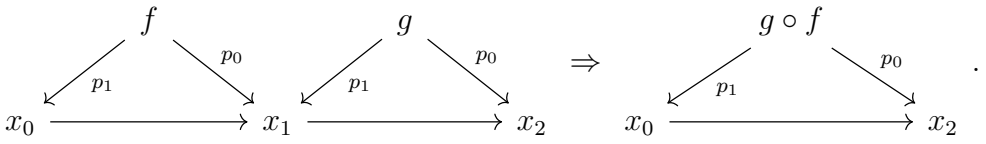
We can introduce more ingredients to enrich the category. Let us call $\mathcal{C}_0 = \mathrm{Ob}(\mathcal{C})$, $\mathcal{C}_1 = \mathrm{Mor}(\mathcal{C})$ and equip them with additional two maps: $p_0 : \mathcal{C}_1 \to \mathcal{C}_0$ and $p_1 : \mathcal{C}_1 \to \mathcal{C}_0$, such that for $x_0 \xrightarrow{f} x_1$, we have

$$p_0(f) = x_1, \quad p_1(f) = x_0,$$

i.e., p_0 is the range map while p_1 is the domain map. The category can then be defined by its composition law indicated by composable morphisms as

$$\mathcal{C}_2 = \{(f, g) \in \mathcal{C}_1 \times \mathcal{C}_1 | p_0(f) = p_1(g)\},$$

which is equivalent to drawing



Sometimes we denote the category as $\mathcal{C}_{1p_1} \times_{p_0} \mathcal{C}_1$ as a fiber product. As a category, we have the following properties:

- $\forall x \in \mathcal{C}_0$, $\exists 1_x$ s.t. $1_x f = f$ and $g 1_x = g$ for all suitable composable $f, g \in \mathcal{C}_1$.

- Associativity: $(f_1 \circ f_2) \circ f_3 = f_1 \circ (f_2 \circ f_3)$ (or $(hg)f = h(gf)$ in arrow sequence).

Now we list some examples of categories:

- **Set**: The category of sets, where objects are sets and morphisms are maps of sets.

- **Top**: The category of topological spaces, where objects are topological spaces and morphisms are continuous functions.

- **TopH**: The category of topological spaces with homotopy classes of continuous functions as morphisms.
- **Manifold**: The category of manifolds with suitable maps. Like topological manifolds with continuous maps, smooth manifolds with smooth maps, etc.
- **Bord**(n): The bordism category of boundaryless n-dimensional manifolds with bordisms as morphisms. A bordism from an n-manifold M_1 to an n-manifold M_2 is an $(n+1)$-manifold Y with a disjoint union of M_1 and M_2 as its boundary, i.e., $\partial Y = (\partial Y)_{in} \sqcup (\partial Y)_{out}$ with diffeomorphisms $\theta_1 : (\partial Y)_{in} \to M_1$ and $\theta_2 : (\partial Y)_{out} \to M_2$. The bordism category in important in topological field theory.
- **Group**: The category of groups, where objects are groups and morphisms are group homomorphisms. We can also require the morphisms to be isomorphisms, and the category is quite different.
- **AB**: The (sub)category of Abelian groups.
- **G-Set**: The category of sets with a group action, where objects are sets with a group action and morphisms are equivariant maps. Specifically, the G-action is a group homomorphism $G \to \mathrm{Aut}(X)$ by $g \cdot x$ with $g \in G$ and $x \in X$. The morphism is a map $f : X \to Y$ such that $f(g \cdot x) = g \cdot f(x)$.
- **Vect**$_k$: The category of vector spaces over a field κ, where objects are vector spaces and morphisms are linear transformations.
- **Ring**: The category of rings, where objects are rings and morphisms are ring homomorphisms.
- **Mod**$_R$: The category of modules over a ring R, where objects are modules and morphisms are module homomorphisms.
- **Cat**: The category of small categories (2-category), where objects are categories (of sets, rather than proper classes, meaning that it is composed of sets while is not itself a set) and morphisms are functors.

It is necessary to clarify about the category of categories by clarifying the definition of a functor between two categories $\mathcal{C}_1$ and $\mathcal{C}_2$ (differ from the above alternative definition!).

Definition 5.1.2. *A functor $F : \mathcal{C}_1 \to \mathcal{C}_2$ between two categories consists of the following data:*

1. *A map $F_{\mathrm{obj}} : \mathrm{Ob}(\mathcal{C}_1) \to \mathrm{Ob}(\mathcal{C}_2)$.*
2. *A map $F_{\mathrm{mor}} : \mathrm{Mor}(\mathcal{C}_1) \to \mathrm{Mor}(\mathcal{C}_2)$.*

Such that the following properties hold:

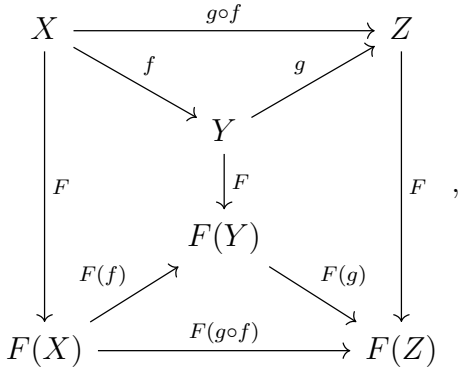
- *For $X \xrightarrow{f} Y$, one has $F_{\mathrm{obj}}(X) \overset{F_{\mathrm{mor}}(f)}{\longleftrightarrow} F_{\mathrm{obj}}(Y)$.*
- *For any $X \in \mathrm{Ob}(\mathcal{C}_1)$, $F_{\mathrm{mor}}(1_X) = 1_{F_{\mathrm{obj}}(X)}$.*

Basically, a functor is a map between two categories that preserves the structure of the categories. The left-right arrow of the functor indicate that there exists two possible (but definitely certain specifically) ways to preserve the composition laws:

- *Covariant functor:* $F_{\text{mor}}(f_1 \circ f_2) = F_{\text{mor}}(f_1) \circ F_{\text{mor}}(f_2)$.
- *Contravariant functor:* $F_{\text{mor}}(f_1 \circ f_2) = F_{\text{mor}}(f_2) \circ F_{\text{mor}}(f_1)$.

Notice that there could be cases where there are functors both ways between two categories, and the two functors are called adjoint functors, where the terminology is borrowed from the adjoint operators in linear algebra, like in a Hilbert space. Under such a situation, we may say that the two categories are in some sense 'the same'.

We can draw the commutative diagram for a covariant functor as



where we have abbreviated F_{obj} and F_{mor} as F for simplicity. A better way to view this diagram is to view it as a triangular prism, with X, Y, Z composing one base face of the prism, $F(X), F(Y), F(Z)$ composing the other face, and F arrows as the height edge. Introducing more objects with their morphisms will generally make the diagram like a 'bordism' between two 'manifolds' of objects, and the functor is like a 'diffeomorphism' between them, while if we insists to draw only neighboring morphisms between objects in a plane, we obtain a polygon prism.

Why do not we draw that for a contravariant functor? This is essentially because once we draw it with the original $\mathcal{C}_1$, we find contradictory arrows that are not constituent of a commutative diagram. To address and solve this, we have to introduce the opposite category.

Definition 5.1.3. *The opposite category $\mathcal{C}^{op}$ of a category $\mathcal{C}$ is defined by reversing the direction of all morphisms, i.e., it contains data:*

1. $\text{Ob}(\mathcal{C}_{^{op}}) = \text{Ob}(\mathcal{C})$.
2. $\text{Mor}(\mathcal{C}_{^{op}}) = \text{Mor}(\mathcal{C})$.
3. *For $X \xrightarrow{f} Y$, we have $X \xleftarrow{f^{op}} Y$.*

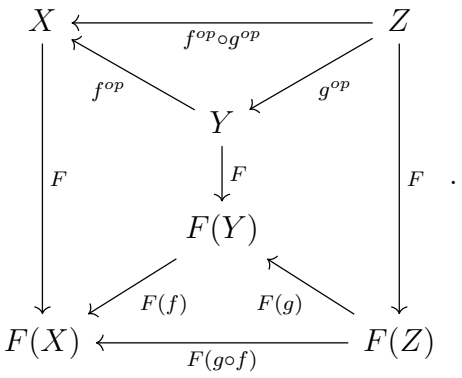
Such that the composition law is also reversed

$$f^{op} \circ_{\mathcal{C}^{op}} g^{op} = (g \circ_{\mathcal{C}} f)^{op}.$$

Notice that we can not draw the commutative diagram from category to its opposite category, like that for a contravariant functor, and we can only compare to reveal

$$X_1 \xrightarrow{f} X_2 \xrightarrow{g} X_3 \Rightarrow X_1 \xrightarrow{g\circ f} X_3,$$
$$X_1 \xleftarrow{f^{op}} X_2 \xleftarrow{g^{op}} X_3 \Rightarrow X_1 \xleftarrow{f^{op}\circ g^{op}} X_3.$$

It is interesting to notice that due to the reversal of the arrows, the composition law seems more 'natural' in the opposite category. We can then draw the commutative diagram for a contravariant functor from $\mathcal{C}_1^{op}$ to $\mathcal{C}_2$ as



Here are some examples of functors:

- The identity functor $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$, where $\mathrm{id}_{\mathcal{C}}(X) = X$ and $\mathrm{id}_{\mathcal{C}}(f) = f$.
- The forgetful functor U : **Group** $\to$ **Set**, where we forget the group structure and only keep the underlying set. This concept can be generalized to other categories.
- Obvious embedding functor, like **AB** $\to$ **Group**.
- Fix a group G, we then have a functor F_G : **Set** $\to$ **Group** by observing that for any set X, we can define a group $F_G(X) = \mathrm{Maps}[X \to G]$ (possible think X as a manifold and that we attach a group space at each point x and the global group configuration live in the fiber). We then get a contravariant functor, obtain from $X_1 \xrightarrow{f} X_2$ to $F_G(X_2) \xleftarrow{F_G(f)} F_G(X_1)$. We recognize $F_G(f) = f^*$ as the pullback of f, like that for differential forms. Explicitly, for $\Psi : X_2 \to G$ we have $F_G(f)(\Psi) = f^*\Psi = \Psi \circ f : X_1 \to G$. This functor is important in certain non-linear sigma models as quantum field theories with target space G.
- Fix a set Y in the category **G-Set**, then we have a functor $F_{G,Y}$: **G-Set** $\to$ **G-Set** by observing that for any G-set X, we can define another G-set $F_{G,Y}(X) = \mathrm{Maps}[X \to Y]$, since if $\Psi : X \to Y$ acts by $\Psi(x) = y$, we then have $g \cdot \Psi$ acts by $(g \cdot \Psi)(x) = \Psi(gx) = g \cdot y$. Thus we again obtain a contravariant functor, from $X_1 \xrightarrow{f} X_2$ to $F_{G,Y}(X_1) \xleftarrow{F_{G,Y}(f)} F_{G,Y}(X_2)$. This can be checked as

$$\begin{aligned}[g \cdot (F_{G,Y}(f)(\Psi_2))](x_1) &= (f^*\Psi_2)(g \cdot x_1) = \Psi_2(f(g \cdot x)) \\ &= \Psi_2(g \cdot f(x)) = g \cdot \Psi_2(f(x)) = f^*(g \cdot \Psi_2)(x_1),\end{aligned}$$

  and hence $\Psi_2 \to g \cdot \Psi_2$ is actually a G-set. Notice that $F_{G,Y}$ is the functor, and $F_{G,Y}(f)$ is the mapped morphism, $F_{G,Y}(f)(\Psi_2)$ pulls $\Psi_2 \in \mathrm{Maps}[X_2 \to Y]$ back to some $\Psi_1 \in \mathrm{Maps}[X_1 \to Y]$, which thus acts on X_1 to Y, and it is a G-set.
- As Sir. Atiyah has suggested, the topological field theory can be described by a functor Z : **Bord** $\to$ **Vect**$_{\mathbb{C}}$, where the objects are n-dimensional manifolds and morphisms are $(n+1)$-dimensional bordisms, and the functor maps a manifold to a vector space, and a bordism to a linear transformation.

To introduce more structures, we on the one hand can introduce functor chains, and see the composition law of functors like that for morphisms. Essentially, we have

$$\begin{aligned}\text{Covariant} \circ \text{Covariant} &= \text{Covariant},\\ \text{Contravariant} \circ \text{Contravariant} &= \text{Covariant},\\ \text{Covariant} \circ \text{Contravariant} &= \text{Contravariant},\\ \text{Contravariant} \circ \text{Covariant} &= \text{Contravariant}.\end{aligned}$$

While on the other hand, like that for two categories $\mathcal{C}_1$ and $\mathcal{C}_2$ we introduce a functor beteen them, for two functors between two categories $F_1 : \mathcal{C}_1 \to \mathcal{C}_2$, $F_2 : \mathcal{C}_1 \to \mathcal{C}_2$, we may also introduce another relation between them, called a natural transformation.

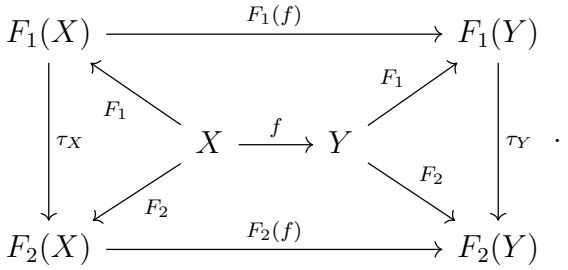
Definition 5.1.4. *A natural transformation between the two (covariant) functors $\tau : F_1 \to F_2$ consists of the following data:*

1. *For each object $X \in \mathcal{C}_1$, a morphism $\tau_X : F_1(X) \to F_2(X)$.*
2. *For each morphism $f : X \to Y$ in $\mathcal{C}_1$, we have $\tau_Y \circ F_1(f) = F_2(f) \circ \tau_X$.*

In terms of the commutative diagram, we have

$$\begin{array}{ccc} F_1(X) & \xrightarrow{F_1(f)} & F_1(Y) \\ \downarrow{\scriptstyle \tau_X} & & \downarrow{\scriptstyle \tau_Y} \\ F_2(X) & \xrightarrow{F_2(f)} & F_2(Y) \end{array},$$

or we include the objects X and Y in the diagram as



Notice that we can also construct natural transformations between two contravariant functors, and the only difference is that the arrows are reversed. Nevertheless, we are unable to construct a natural transformation between a covariant functor and a contravariant functor, since the arrows are contradictory. We can also introduce chains of natural transformations, and the composition law is similar to that for functors with triangular prism-like diagrams.

With the introduction of natural transformations, we can define $\mathbf{Funct}(\mathcal{C}_1, \mathcal{C}_2)$ as the category of functors between two categories $\mathcal{C}_1$ and $\mathcal{C}_2$. The objects are functors and the morphisms are natural transformations. With this category, we can then introduce the concept of an equivalence of categories.

Definition 5.1.5. *Two categories $\mathcal{C}_1$ and $\mathcal{C}_2$ are equivalent if there exist functors $F : \mathcal{C}_1 \to \mathcal{C}_2$ and $G : \mathcal{C}_2 \to \mathcal{C}_1$ such that $F \circ G \simeq \mathrm{id}_{\mathcal{C}_2}$ and $G \circ F \simeq \mathrm{id}_{\mathcal{C}_1}$, where $\mathrm{id}_{\mathcal{C}} \in \boldsymbol{Funct}(\mathcal{C}, \mathcal{C})$.*

As indicated in the beginning, the category with only one object and all morphisms invertible is called a group. We can then generalize the concept of a group to a groupoid.

Definition 5.1.6. *A groupoid is a category in which every morphism is invertible.*

Note that an object in a groupoid is not necessarily a group, but $\forall X \in \mathrm{Ob}(\mathbf{GroupOid})$, $\hom(X, X)$ is a group, the automorphism group of X. We can provide some examples of groupoids:

- Any equivalence relation on a set X gives rise to a groupoid, where the objects are elements of X and the set $\hom(a, b)$ is non-empty with a single element iff $a \sim b$. The composition law naturally emerges from the transitivity of the equivalence relation.
- Any specific solution to a time evolution problem in quantum mechanics defines a groupoid. The objects are states in the associated Hilbert space, and the morphisms are unitary time-translational operators, as $\psi(t_1) \overset{U(t_2,t_1)}{\longrightarrow} \psi(t_2)$, $U(t_3, t_2)U(t_2, t_1) = U(t_3, t_1)$. Note that $\hom(\psi(t), \psi(t)) = \mathrm{id}$ is trivial.
- Let X be a topological space. The fundamental groupoid $\pi_{\leq 1}(X)$ of X is a groupoid with objects as points in X and morphisms as homotopy classes of paths between points $x \xrightarrow{f} x'$. $\hom(x, x) = \pi_1(X, x)$, point based fundamental group.
- The gauge theory in physics can be described by a groupoid, where the objects are connections and the morphisms are gauge transformations.

5.2 Categorical Group Cohomology

Definition 5.2.1. *The associated simplicial space $|\mathcal{C}|$ of a category $\mathcal{C}$ is a simplicial set with n-simplices as sequences of composable morphisms in $\mathcal{C}$.*

The higher simplices are made by gluing the lower simplices as (we are using the alternative definition of category)

- 0-simplices = objects
- 1-simplices = $\Delta_1(f)$ associated to morphisms $f : x_0 \to x_1 \in \mathcal{C}_1$
- 2-simplices = $\Delta_2(f_1, f_2)$ associated to composable morphisms

$$(f_1, f_2) \in \mathcal{C}_2 \equiv \{(f_1, f_2) \in \mathcal{C}_1 \times \mathcal{C}_1 | p_0(f_1) = p_1(f_2)\}$$

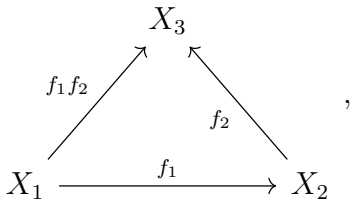
- 3-simplices = $\Delta_3(f_1, f_2, f_3)$ associated to composable morphisms

$$(f_1, f_2, f_3) \in \mathcal{C}_3 \equiv \{(f_1, f_2, f_3) \in \mathcal{C}_1 \times \mathcal{C}_1 \times \mathcal{C}_1 | p_0(f_1) = p_1(f_2), p_0(f_2) = p_1(f_3)\}$$

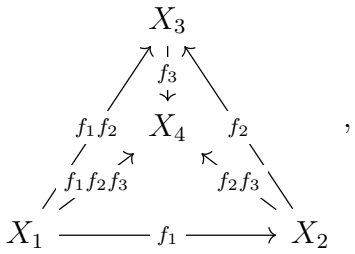
- $\cdots$
- n-simplices = $\Delta_n(f_1, \cdots, f_n)$ associated to composable morphisms

$$(f_1, \cdots, f_n) \in \mathcal{C}_n \equiv \{(f_1, \cdots, f_n) \in \mathcal{C}_1^{\times n} | p_0(f_i) = p_1(f_{i+1}), i = 1, \cdots, n-1\}$$

There are infinitely higher simplices, as we can always glue the lower simplices to form a higher one. For instance, we have 2-simplex and 3-simplex as



,



,

where we have used the arrow composition law such that $f_1 f_2 = f_2 \circ f_1$. The 3-simplex is like looking down a tetrahedron from the top. Considering the orientation of the arrows, we can write down the boundary map from n-simplex to $(n-1)$-simplex as

$$\begin{aligned}
\partial_1\Delta_1(f_1) &= X_2 - X_1, \\
\partial_2\Delta_2(f_1, f_2) &= \Delta_1(f_2) - \Delta_1(f_2 f_1) + \Delta_1(f_1), \\
\partial_3\Delta_3(f_1, f_2, f_3) &= \Delta_2(f_2, f_3) - \Delta_2(f_1 f_2, f_3) + \Delta_2(f_1, f_2 f_3) - \Delta_2(f_1, f_2), \\
&\cdots \\
\partial\Delta_{n+1}(f_1, \cdots, f_{n+1}) &= \Delta_n(f_2, \cdots, f_{n+1}) + \sum_{i=1}^{n} (-1)^i \Delta_n(f_1, \cdots, f_i f_{i+1}, \cdots, f_{n+1}) \\
&\quad + (-1)^{n+1} \Delta_n(f_1, \cdots, f_n),
\end{aligned}$$

where the generalization is protected by the choice of the orientation of the arrows.

Now we consider the group, and we replace every object with a single group G. We also simply replace $f_i \to g_i$. We define $(n+1)$ maps from $G^{\times n}$ to $G^{\times(n-1)}$ by

$$\begin{aligned}
d^0(g_1, \cdots, g_n) &= (g_2, \cdots, g_n), \\
d^i(g_1, \cdots, g_n) &= (g_1, \cdots, g_i g_{i+1}, \cdots, g_n), \\
d^n(g_1, \cdots, g_n) &= (g_1, \cdots, g_n),
\end{aligned}$$

where $i = 1, \cdots, n-1$. We also define the $(n+1)$ face maps of n-simplex to its boundary as

$$d_i(t_0, \cdots, t_{n-1}) = (t_0, \cdots, 0, \cdots, t_{n-1}),$$

where 0 occupies the i-th position. Now we identify

$$(\sqcup_{n=0}^{\infty} \Delta_n \times G^{\times n}) / \sim,$$

via the equivalence relation

$$(d_i(\boldsymbol{t}), \boldsymbol{g}) \sim (\boldsymbol{t}, d^i(\boldsymbol{g})),$$

as the classifying space BG of the group G. It is characterized by the property that for any space X, the set of homotopy classes of maps $[X, BG]$ is in one-to-one correspondence with the set of isomorphism classes of principal G-bundles over X.

Appendix A

Supplementary Materials

A.1 Elementary Group Theory

A group G is a set of elements $\{g\}$ assigned with a combination law (multiplication or addition for Abelian case) that is closed and associative, with sole identity e and inverse element given. A group with finite elements is said to be a finite group and the number of elements is its order. A group with countable elements is said to be discrete.

To make different structures clear, let us bring laws here:

- A *set* in sense of axiomatic set theory is a collection of distinct objects under ZFC axioms.
- A *magma* is a set with a binary operation defined on it, i.e., G with $\cdot : G \times G \to G$.
- A *semigroup* is a magma with associative law, i.e., $\forall g_1, g_2, g_3 \in G$, $(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$.
- A *monoid* is a semigroup with identity element e, i.e., $\forall g \in G$, $e \cdot g = g \cdot e = g$.
- A *group* G is a monoid with inverse element for each element, i.e., $\forall g \in G$, $\exists g^{-1} \in G$ s.t. $g \cdot g^{-1} = g^{-1} \cdot g = e$.
- An *Abelian group* is a group with commutative law, i.e., $\forall g_1, g_2 \in G$, $g_1 \cdot g_2 = g_2 \cdot g_1$, and $\cdot$ degrades into addition $+$.
- A *ring* R is an Abelian group with an additional multiplication operation that is associative and distributive over addition, i.e., $\forall g_1, g_2, g_3 \in G$, $g_1 \cdot (g_2 + g_3) = g_1 \cdot g_2 + g_1 \cdot g_3$ and $(g_1 + g_2) \cdot g_3 = g_1 \cdot g_3 + g_2 \cdot g_3$.
- A *unital ring* is a ring with a multiplicative identity 1_g, i.e., $\forall g \in G$, $1_g \cdot g = g \cdot 1_g = g$.
- A *field* F (κ) is a unital ring with multiplicative inverse for each non-zero element, i.e., $\forall g \in G$, $g \neq 0 \Rightarrow \exists g^{-1} \in G$ s.t. $g \cdot g^{-1} = g^{-1} \cdot g = 1_g$.
- A *module* Mod is a ring with a scalar multiplication over another ring, i.e., consider R_1 and R_2 as two rings, then a finite multiplication-sum $m = \sum_i r_1^i r_2^i \in$ Mod.
- An *algebra* A is a ring with a scalar multiplication over a field, i.e., A is a vector space with bilinear multiplication defined on it, i.e., $\forall v_1, v_2 \in A$, $\alpha \in \kappa$, $(\alpha v_1) \cdot v_2 = v_1 \cdot (\alpha v_2)$. Possibly we also say that an algebra is a vector space with a ring structure.

- A *vector space* V is a set with a vector addition and scalar multiplication over a field.

A continuous group, aka topological group since its parameter space is some topological space, is defined as the existence of two limits: for $\lim_{n\to+\infty} g_n = g$, $\lim_{n\to+\infty} h_n = h$ with $\{g_n\}, g, \{h_n\}, h$ all belong to G, we have

$$\begin{cases} \lim_{n\to+\infty} g_n h_n = gh \\ \lim_{n\to+\infty} g_n^{-1} = g^{-1} \end{cases} \iff \lim_{n\to+\infty} g_n h_n^{-1} = gh^{-1}.$$

An element in continuous group induces transformation among points in a topological space. A smooth group (smooth parameterization), i.e. a continuous group with smooth multiplication and inverse, is said to be a Lie group which is also equivalent to a differential manifold, whose tangent space at identity serves as the vector space for elements in Lie algebra, which are known to be generators for one Lie group. A continuous group is said to be compact if it is in some sense closed and bounded.

A subgroup H is some subset of G which is also a group. A left/right coset is defined as set gH/Hg.

Theorem A.1.1. *$\forall g_1, g_2 \in G$, either $g_1H = g_2H$ or $g_1H \cap g_2H = \emptyset$.*

Proof. Firstly, if $g \notin H$, $gH \cap H = \emptyset$ for H's closeness. Since otherwise, $\exists h_1, h_2 \in H$, s.t. $gh_1 = h_2 \Rightarrow g = h_1^{-1}h_2 \in H$ as contradiction. Then, if $g_1H \cap g_2H \neq \emptyset$ while $g_1H \neq g_2H$, i.e., $\exists h_1, h_2, h_3 \in H$, s.t. $g_1h_1 = g_2h_2 \Rightarrow g_2^{-1}g_1 = h_2h_1^{-1}$ while $g_1h_3 \neq g_2h_i \Rightarrow g_2^{-1}g_1 \neq h_ih_3^{-1}, \forall h_i \in H$, one immediately sees the contradiction as one says $g_2^{-1}g_1$ belongs to H while the other says not. □

This theorem holds similarly for right coset. On the other hand, no single element g in G could be excluded from all cosets since $ge = eg = g$, then we conclude that cosets forms a disjoint cover for one group. The space of cosets is denoted as G/H.

A normal subgroup/invariant subgroup is the subgroup satisfies that $gH = Hg, \forall g$. At this time, its coset space also possesses group structure as

$$\begin{cases} \text{multiplication:} & (g_1H)(g_2H) = g_1g_2H \\ \text{identity:} & H/eH \\ \text{inverse:} & (gH)^{-1} = g^{-1}H \end{cases},$$

and is known as quotient/factor group G/H.

The direct product between groups is also a group defined as

$$G \times K = \{(g, k) : g \in G, k \in K\}, \tag{A.1}$$

with multiplication as $(g_1, k_1)(g_2, k_2) = (g_1g_2, k_1k_2)$.

Two elements $a, b \in G$ are conjugate if $\exists g \in G$, s.t. $b = gag^{-1}$; the conjugate class of one element is defined as

$$C(a) = \{gag^{-1} : g \in G\}. \tag{A.2}$$

Theorem A.1.2. *G could be partitioned into disjoint conjugate classes.*

Proof. Similar to cosets, two conjugate classes are either the same or disjoint. Say now $C(a) \cap C(b) \neq \emptyset$, then $\exists g_1, g_2 \in G$, s.t. $g_1 a g_1^{-1} = g_2 b g_2^{-1}$ i.e., $a = (g_1^{-1} g_2) b (g_1^{-1} g_2)^{-1}$, then any conjugate member of a induced by g would be conducted to that of b, and since g runs over G, one concludes $C(a) = C(b)$. Also, for any $a \in G$, there at least exists $eae^{-1} = a$ so that $C(a)$ at least contains a. And in this sense the theorem is proved. □

One may notice that $C(e) = e$ so that $G = e \cup C(a) \cup C(b) \cdots$. For Abelian group, every element is itself a conjugate class. What is more, a normal subgroup H must be some union of conjugate classes of G, otherwise, say $C(a) \cap H \neq \emptyset$ but $C(a) \subsetneq H$, then $\exists a_1, a_2 \in C(a)$, $a_1 \in H$ while $a_2 \notin H$; however $a_2 = g a_1 g^{-1}$, which means that under transformation g, a_1 is transformed out of H, which is impossible since $gHg^{-1} = H, \forall g \in G$.

There exists a map between two groups or one group and its representation known as homomorphism, which satisfies the group structural condition as

$$\varphi : G \longrightarrow K, \varphi(g) = k \tag{A.3}$$

with

$$\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2), \tag{A.4}$$

with property

$$\begin{cases} \varphi(g) = \varphi(ge) = \varphi(g)\varphi(e), \ \forall g \in G \Rightarrow \varphi(e) = e \\ e = \varphi(g^{-1}g) = \varphi(g^{-1})\varphi(g) \Rightarrow \varphi(g^{-1}) = \varphi(g)^{-1} \end{cases} .$$

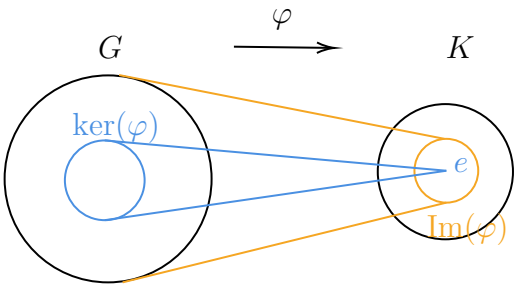


Figure A.1: Homomorphism between groups.

An isomorphism is a bijective homomorphism which means that the map is not only injective (one-to-one) but also surjective (onto), or simply speaking, two structures linked by isomorphism are just the same, denoted as

$$G \cong K. \tag{A.5}$$

Clearly, $g : G \longrightarrow G$ by $gG = G$ when $g \in G$ is an isomorphism known as automorphism, as it just permutes elements in G.

We now have an important theorem as

Theorem A.1.3 (First Isomorphism Theorem/The Fundamental Homomorphism Theorem)**.** *Given homomorphism $\varphi : G \longrightarrow K$, denote elements in G that map onto identity into K as* $\ker(\varphi)$*, while the values of φ in K as* $\mathrm{Im}(\varphi)$*, shown in Fig. A.1. Then,*

$$G/\ker(\varphi) \cong \mathrm{Im}(\varphi). \tag{A.6}$$

Proof. We need firstly to approve that $\ker(\varphi)$ is a normal subgroup of G, i.e., need

$$g\ker(\varphi) = \ker(\varphi)g,\ \forall g \in G,$$

or equivalently,

$$\forall a \in \ker(\varphi),\ \exists a' \in \ker(\varphi),\ \text{s.t. } ga = a'g \Leftrightarrow gag^{-1} = a',\ \forall g \in G.$$

Now since

$$\varphi(gag^{-1}) = \varphi(g)\varphi(a)\varphi(g^{-1}) = \varphi(g)e\varphi(g)^{-1} = e,$$

then $gag^{-1} \in \ker(\varphi)$ and there must exist some a' s.t. $gag^{-1} = a'$. Subgroup property is trivial as

$$\begin{cases} \varphi(e) = e,\ \text{i.e., } e \in \ker(\varphi) \\ \varphi(g^{-1}) = \varphi(g)^{-1} = e^{-1} = e,\ \forall g \in \ker(\varphi),\ \text{i.e., } g^{-1} \in \ker(\varphi) \\ \varphi(gk) = \varphi(g)\varphi(k) = e,\ \forall g, k \in \ker(\varphi),\ \text{i.e., closeness} \end{cases}.$$

Now denote $H = G/\ker(\varphi) = \{g\ker(\varphi)\}$ as the quotient group, and let map $\hat{\varphi} : H \longrightarrow \mathrm{Im}(\varphi)$ with $\hat{\varphi}(g\ker(\varphi)) = \varphi(g)$. Firstly, $\hat{\varphi}$ is a homomorphism since

$$\hat{\varphi}(g_1\ker(\varphi))\hat{\varphi}(g_2\ker(\varphi)) = \varphi(g_1)\varphi(g_2) = \varphi(g_1g_2) = \hat{\varphi}(g_1g_2\ker(\varphi)).$$

Meanwhile, $\hat{\varphi}$ is bijective. Onto: $\forall k \in \mathrm{Im}(\varphi)$, $\exists g \in G$, s.t. $\hat{\varphi}(g\ker(\varphi)) = \varphi(g) = k$; one-to-one: firstly again, we show that for $g\ker(\varphi) \neq \ker(\varphi)$,

$$\varphi(g) = \hat{\varphi}(g\ker(\varphi)) \neq \hat{\varphi}(\ker(\varphi)) = \varphi(e) = e,$$

otherwise $g \in \ker(\varphi)$ and $g\ker(\varphi) = \ker(\varphi)$. Then for $g_1\ker(\varphi) \neq g_2\ker(\varphi)$, i.e., $g_2^{-1}g_1\ker(\varphi) \neq \ker(\varphi)$, we have

$$\hat{\varphi}(g_2^{-1}g_1\ker(\varphi)) \neq \hat{\varphi}(\ker(\varphi)) \Rightarrow \hat{\varphi}(g_1\ker(\varphi)) \neq \hat{\varphi}(g_2\ker(\varphi)).$$

And to conclude, $\hat{\varphi}$ induces group isomorphism between H and $\mathrm{Im}(\varphi)$, i.e., $G/\ker(\varphi) \cong \mathrm{Im}(\varphi)$. □

Equivalently, one can write $G \cong \ker(\varphi) \times \mathrm{Im}(\varphi)$, and one yields $\dim(G) = \dim(\ker(\varphi)) + \dim(\mathrm{Im}(\varphi))$ by viewing G and K to be linear vector space and φ to be transformation.

We now have two theorems about exact homomorphism sequences.

Theorem A.1.4.

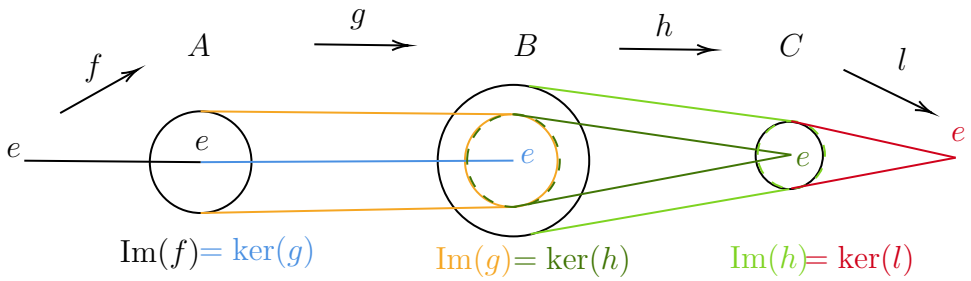
$$0 \xrightarrow{\alpha} G \xrightarrow{\beta} K \xrightarrow{\varphi} 0 \Rightarrow G \cong K. \tag{A.7}$$

Proof. $\ker(\beta) = \mathrm{Im}(\alpha) = e$ makes β one-to-one. Otherwise, $\exists g_1 \neq g_2 \in G$, s.t. $\beta(g_1) = \beta(g_2)$, and by homomorphism $\beta(g_1g_2^{-1}) = e$ but $g_1g_2^{-1} \neq e$, which is contradictory.

$\mathrm{Im}(\beta) = \ker(\varphi) = K$ makes β onto, obviously. Then with homomorphism we say β induces an isomorphism $G \cong K$. □

Theorem A.1.5.

$$0 \xrightarrow{f} A \xrightarrow{g} B \xrightarrow{h} C \xrightarrow{l} 0 \Rightarrow C \cong B/A. \tag{A.8}$$



Proof. As shown above, $\ker(g) = \mathrm{Im}(f) = e$ makes g one-to-one, and $\mathrm{Im}(h) = \ker(l) = C$ makes h onto. We also have $\mathrm{Im}(g) = \ker(h)$. On the other hand, $g : A \to \mathrm{Im}(g)$ is onto, then $A \cong \mathrm{Im}(g)$. Meanwhile, by first isomorphism theorem we have $B/\ker(h) \cong \mathrm{Im}(h)$, i.e., $B/A \cong C$. □

The most famous example of group homomorphism in physics might be between special unitary group of degree 2 and special/proper rotational group of degree 3 as

$$SU(2)/\mathbb{Z}_2 \cong SO(3), \tag{A.9}$$

where $\mathbb{Z}_2$ is cyclic group of order 2, as an example of $\mathbb{Z}_n$, which is isomorphic to additive group of integers modulo n. The isomorphic relation above reveals that there exists a two-to-one homomorphism between two groups, as element in $SU(2)$ could be represented as

$$u(\hat{n},\theta) = \exp(i\boldsymbol{S}\cdot\hat{n}\theta/\hbar) = \exp(i\boldsymbol{\sigma}\cdot\hat{n}\theta/2) = \cos\left(\frac{\theta}{2}\right)\mathbb{I} + i\sin\left(\frac{\theta}{2}\right)\boldsymbol{\sigma}\cdot\hat{n}, \tag{A.10}$$

from which one reads $u(\hat{n},\theta+2\pi) = -u(\hat{n},\theta)$, while for element in $SO(3)$ as $R(\hat{n},\theta)$, one has $R(\hat{n},\theta+2\pi) = R(\hat{n},\theta)$, i.e., homomorphism φ from $SU(2)$ to $SO(3)$ has equivalent image elements θ and $\theta+2\pi$ when $\theta \in [0,4\pi)$ for any axis, and this indeed tells us that $\ker(\varphi) = \{\mathbb{I}, -\mathbb{I}\} \cong \mathbb{Z}_2$. Then by first isomorphism theorem we have the relation above.

On the other hand, representation above could be written under pseudo-basis $(\sigma_0, \boldsymbol{\sigma})$ as identity and Pauli matrices, then any 2×2 special unitary matrix could be expressed as

$$u = a_0\sigma_0 + i\boldsymbol{a}\cdot\boldsymbol{\sigma}, \tag{A.11}$$

with real unit-4-vector satisfies that

$$a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1,$$

which is geometrically 3-sphere S^3 embedded in $\mathbb{R}^4$. Then by mod $\mathbb{Z}_2$ we may assign points inversion-symmetric to be identical, i.e., $(a_0,\boldsymbol{a}) \Leftrightarrow -(a_0,\boldsymbol{a})$. By this operation, we may imagine that 'half' of the 3-sphere is cut-off while points on 'equator' of the remaining half are viewed identical if on the same 'diameter', and this gives us topological space for $SO(3)$ as real-projective space of dimension 3 as $\mathbb{RP}^3$. The explanation may be more viewable for $\mathbb{RP}^2$, as half of 3-ball's surface with points on the equator identified if on the same diameter, and this could be continuously deformed into a square with parallel edges proceeding inversely as shown in Fig. A.2(c), while (a) and (b) correspond to a 2-torus T^2 and Klein bottle, respectively.

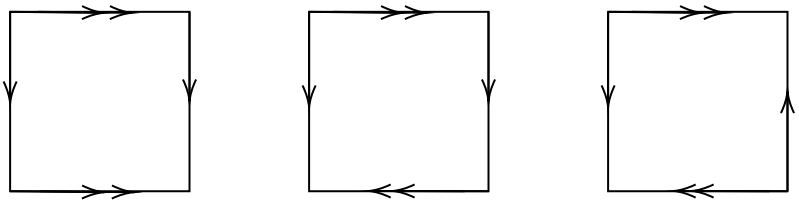


Figure A.2: Different boundary taken for a square.

A.2 Poisson Summation Formula

Consider a well-behaved complex function $f : \mathbb{R} \to \mathbb{C}$ that decays sufficiently fast at infinity that the function $F(x) = \sum_{n\in\mathbb{Z}} f(x+n)$ exists. Since $F(x+m) = F(x)$ for any integer m, F is a periodic function with period 1, thus defining function $F : \mathbb{R}/\mathbb{Z} = U(1) \to \mathbb{C}$. We can thus decompose F in terms of the irreducible representations of $U(1)$, which are the characters $\chi_m(x) = e^{2\pi imx}$, where $m \in \mathbb{Z}$. We have

$$F(x) = \sum_{m\in\mathbb{Z}} \hat{F}(m)\chi_m(x) = \sum_{m\in\mathbb{Z}} \hat{F}(m)e^{2\pi imx},$$

which is indeed the Fourier series of F. The Fourier coefficients $\hat{f}(m)$ are given by

$$\begin{aligned}\hat{F}(m) &= \int_0^1 e^{-2\pi imx}F(x)\mathrm{d}x = \int_0^1 e^{-2\pi imx}\sum_{n\in\mathbb{Z}} f(x+n)\mathrm{d}x \\ &= \int_{-\infty}^{\infty} e^{-2\pi imx}f(x)\mathrm{d}x = \hat{f}(m).\end{aligned}$$

Thus, by taking $x = 0$, we have the Poisson summation formula

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{m\in\mathbb{Z}} \hat{f}(m), \tag{A.12}$$

where $\hat{f}$ is the Fourier transform of f.

For instance, consider $f(n) = e^{2\pi in(t-t')}$ is character itself, and its Fourier transform is

$$\hat{f}(m) = \int_{-\infty}^{\infty} e^{-2\pi imx}e^{2\pi ix(t-t')}\mathrm{d}x = \delta_{\mathbb{R}}(t-t'-m),$$

and the Poisson summation formula gives

$$\sum_{n\in\mathbb{Z}} e^{2\pi in(t-t')} = \sum_{m\in\mathbb{Z}} \delta_{\mathbb{R}}(t-t'-m) = \delta_{\mathbb{R}/\mathbb{Z}}(t-t'), \tag{A.13}$$

which is the standard orthogonality relation of the characters of $U(1)$.

For another example, we consider the Riemann theta function

$$\vartheta\begin{bmatrix}\theta\\ \phi\end{bmatrix}(z|\tau) = \sum_{n\in\mathbb{Z}} e^{i\pi\tau(n+\theta)^2+2\pi i(n+\theta)(z+\phi)},$$

where $\theta, \phi \in \mathbb{R}$, $z, \tau \in \mathbb{C}$ and $\operatorname{Im}\tau > 0$. The Fourier transform of the Riemann theta function is given by a Gaussian integral

$$\begin{aligned}
\hat{v}_m \begin{bmatrix} \theta \\ \phi \end{bmatrix} (z|\tau) &= \int_{-\infty}^{\infty} e^{-2\pi i m x} e^{i\pi\tau(x+\theta)^2 + 2\pi i(x+\theta)(z+\phi)} \mathrm{d}x \\
&= \int_{-\infty}^{\infty} e^{i\pi\tau x^2 + 2\pi i(-m+\tau\theta+z+\phi)x + i\pi\theta(\tau\theta+2(z+\phi))} \mathrm{d}x \\
&= e^{i\pi\theta(\tau\theta+2(z+\phi))} \int_{-\infty}^{\infty} e^{-(i\pi\tau)\left(\left[x+\frac{-m+\tau\theta+z+\phi}{\tau}\right]^2 - \frac{(-m+\tau\theta+z+\phi)^2}{\tau^2}\right)} \mathrm{d}x \\
&= \sqrt{\frac{\pi}{-i\pi\tau}} e^{-i\pi\frac{(-m+\tau\theta+z+\phi)^2}{\tau} + i\pi\theta(\tau\theta+2(z+\phi))} \\
&= (-i\tau)^{-1/2} e^{-i\pi\frac{z^2}{\tau}} e^{2\pi i\theta\phi} e^{-\frac{i\pi}{\tau}(m-\phi)^2 + 2\pi i(\frac{z}{\tau}+\theta)(m-\phi)}
\end{aligned}$$

which leads to the modular transformation of the Riemann theta function given by the Poisson summation formula as

$$\begin{aligned}
\vartheta \begin{bmatrix} \theta \\ \phi \end{bmatrix} (z|\tau) &= \sum_m \hat{v}_m \begin{bmatrix} \theta \\ \phi \end{bmatrix} (z|\tau) \\
&= (-i\tau)^{-1/2} e^{-i\pi\frac{z^2}{\tau}} e^{2\pi i\theta\phi} \sum_m e^{-\frac{i\pi}{\tau}(m-\phi)^2 + 2\pi i(\frac{z}{\tau}+\theta)(m-\phi)} \\
&= (-i\tau)^{-1/2} e^{-i\pi\frac{z^2}{\tau}} e^{2\pi i\theta\phi} \vartheta \begin{bmatrix} -\phi \\ \theta \end{bmatrix} (\frac{z}{\tau}| -\frac{1}{\tau}).
\end{aligned}$$

A.3 Spin and Pin Groups

The groups are defined using the Clifford algebras, generated by $\{\gamma_i, \gamma_j\} = 2Q_{ij}$ with Q an invertible $d \times d$ matrix. Given vector v, we define $\gamma(v) = v^i\gamma_i$. For $Q_{ij} = \delta_{ij}$, we have the group $\mathrm{Pin}^+(d) = \{\pm\gamma(v_1)\cdots\gamma(v_r)\}$. Here r is an arbitrary positive integer, and it can be easily verified that by simplification, only terms with $d, \cdots, 1, 0$ numbers of γ's are left. For instance, take $d = 1$ such that there is only one γ, then $\mathrm{Pin}^+(1) = \kappa$ depending on the field. For $d = 2$, we can take $\gamma_1 = \sigma_1$ and $\gamma_2 = \sigma_2$ as the Pauli matrices, and since $\sigma_1^2 = 1 = \sigma_2^2$, $\sigma_1\sigma_2 = -\sigma_2\sigma_1$, then most generally,

$$\pm\gamma(v_1)\cdots\gamma(v_r) = \begin{cases} a + b\sigma_1\sigma_2, & r \text{ even} \\ a\sigma_1 + b\sigma_2, & r \text{ odd} \end{cases}.$$

The group $\mathrm{Pin}^-(d)$ is defined similarly with $Q_{ij} = -\delta_{ij}$. The spin group $\mathrm{Spin}(d)$ is the subgroup of $\mathrm{Pin}^+(d)$ with r even, and for instance, when $d = 2$, $\pm\gamma(v_1)\cdots\gamma(v_r)\big|_{r \text{ even}} = a + b\sigma_1\sigma_2 \sim \cos\alpha + i\sin\alpha\sigma_3 = e^{i\alpha\sigma_3}$, which is the $\mathrm{Spin}(2)$ group. Similar verification shows that $\mathrm{Spin}(3) \cong SU(2)$ group.

The exact sequence

$$1 \to \mathbb{Z}_2 \to \mathrm{Pin}^+(d) \xrightarrow{\pi} O(d) \to 1$$

defines the projection (also suitable for $\mathrm{Pin}^-(d)$) by

$$\gamma(\pi(g)\omega) = (-1)^r g\gamma(\omega)g^{-1}.$$

The idea comes from the formula

$$-\gamma(v)\gamma(\omega)\gamma(v)^{-1} = \gamma(R_v(\omega)).$$

To see what happens, we notice that $\gamma_i^2 = 1$ gives $\gamma_i^{-1} = \gamma_i$, thus

$$\gamma(v)^{-1} = \frac{1}{\sum_i v^i\gamma_i} = \frac{\sum_i v^i\gamma_i}{\sum_i v^iv^j\gamma_i\gamma_j} = \frac{\sum_i v^i\gamma_i}{\sum_i (v^i)^2} = \sum_i v^i\gamma_i = \gamma(v),$$

where we choose real unit v. Then by expansion,

$$\begin{aligned}\gamma(v)\gamma(\omega)\gamma(v)^{-1} &= \sum_{ijk} v^i\gamma_i\omega^j\gamma_j v^k\gamma_k = \sum_{ijk}(-1)^{1+\delta_{ij}}v^iv^k\omega^j\gamma_j\gamma_i\gamma_k \\ &= \sum_{ij}(-1)^{1+\delta_{ij}}(v^i)^2\omega^j\gamma_j = \sum_j\left((v^j)^2 - \sum_{i\neq j}(v^i)^2\right)\omega^j\gamma_j \\ &= -\gamma(R_v(\omega)),\end{aligned}$$

where in the third equality, we used the anticommutation relation of γ's. To make the meaning clear, we may consider $d = 3$ and $v = (1, 0, 0)$, thus

$$\gamma(v)\gamma(\omega)\gamma(v)^{-1} = \omega^1\gamma_1 - \omega^2\gamma_2 - \omega^3\gamma_3 = -\gamma((-\omega^1, \omega^2, \omega^3)),$$

thus, $R_v(\omega)$ represents a reflection of ω with respect to the plane perpendicular to v. Thus, by letting $g = \gamma(v)$, we see that

$$g\gamma(\omega)g^{-1} = \gamma(v)\gamma(\omega)\gamma(v)^{-1} = -\gamma(R_v(\omega)) = (-1)^1\gamma(\pi(\gamma(v))\omega),$$

where we identify $\pi(\gamma(v))\omega = R_v(\omega)$. For g with r products of $\gamma(v_i)$, this generalizes, since all elements in $O(d)$ can be represented by a series product of reflections. This way of constructing the projection reflects the double cover nature of $\mathrm{Spin}(d)$ over $SO(d)$, and takes analogy with the double cover of $SU(2)$ over $SO(3)$

$$U(\boldsymbol{v}\cdot\boldsymbol{\sigma})U^\dagger = R(\boldsymbol{v})\cdot(\boldsymbol{\sigma}),$$

where $U \in SU(2)$ and $R \in SO(3)$. This map contains $\mathbb{Z}_2$ kernel by identifying $U \sim -U$.

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